# Résultats complets de toutes les campagnes — Paper 2

Branch selection pour équilibres de Navier–Stokes basse dimension

Document de données — 15 septembre 2026

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Table 1: Vue d'ensemble des campagnes Paper 2 et des jeux de données produits.

| Campagne | Étude | Répertoire | Courses | Seeds/point | Points | Paramètres |
|---|---|---|---|---|---|---|
| 1. Carte (Re, $E^*$) | phase1_re_estarmap | 01_ReE_star_map | 185 | 5 (15–20) | 30 cellules | Re ∈ 100,300,500,700,900,1000 ; $E^*$ ∈ 0.1,0.25,0.5,1,2 ; $\lambda_v$=1, $\lambda_b$=0 |
| 2. Sensibilité $\lambda_v$ | lambda_var sweep | 02_lambda_var | 90 | 5 | 3 × 6 | $\lambda_v$ ∈ 0,0.25,0.5,1,2,5 ; (500,0.25), (700,2), (1000,1) |
| 3. Classification | branch classification | 03_branch_classification | 185 (réutilisés) | 5 | 3 classes | métriques $f_2$, $f_t$, $C_3$, $E_r$ ; validation CFD |
| 4. Validation LBM | indep validation | 03_indep_validation | 6 | 1 | 3 contrôles | équivalence de performance (CFD) ; fenêtres ; GCI |
| 5. Séparabilité (repères) | landmark sep. | 00_landmark_separability | 10 repères | 1 | 10 | empreintes modales des points de repère |
| Total | — | — | 285 courses + 10 repères | — | — | données complètes dans les CSV |

# 1 Campagne 1 — Carte (Re, $E^*$) : 185 courses, 30 cellules

## 1.1 Probabilités de bassin agrégées par cellule

Table 2: Probabilités $P(B_k)$ par cellule — 30 cellules, `branch_points.csv` (5 seeds, escalate 15–20). Abréviations : label = branche majoritaire, conf = confiance du label, affin. = raffinement requis, esc10 = passage à 10 seeds.

| point | Re | $E^*$ | AR | $\lambda_v$ | $\lambda_b$ | $N_s$ | $n_1$ | $n_2$ | $n_3$ | $n_{NA}$ | $P_1$ | $P_2$ | $P_3$ | label | conf | affin. | esc10 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| re100_e0.1 | 100 | 0.1 | 1 | 1 | 0 | 5 | 0 | 4 | 1 | 0 | 0 | 0.8 | 0.2 | P_B2 | 0.8 | Non | Non |
| re300_e0.1 | 300 | 0.1 | 1 | 1 | 0 | 5 | 2 | 3 | 0 | 0 | 0.4 | 0.6 | 0 | P_B2 | 0.6 | Oui | Oui |
| re500_e0.1 | 500 | 0.1 | 1 | 1 | 0 | 5 | 4 | 1 | 0 | 0 | 0.8 | 0.2 | 0 | P_B1 | 0.8 | Non | Non |
| re700_e0.1 | 700 | 0.1 | 1 | 1 | 0 | 5 | 3 | 2 | 0 | 0 | 0.6 | 0.4 | 0 | P_B1 | 0.6 | Oui | Oui |
| re900_e0.1 | 900 | 0.1 | 1 | 1 | 0 | 5 | 3 | 2 | 0 | 0 | 0.6 | 0.4 | 0 | P_B1 | 0.6 | Oui | Oui |
| re1000_e0.1 | 1e+03 | 0.1 | 1 | 1 | 0 | 5 | 3 | 2 | 0 | 0 | 0.6 | 0.4 | 0 | P_B1 | 0.6 | Oui | Oui |
| re100_e0.25 | 100 | 0.25 | 1 | 1 | 0 | 5 | 0 | 4 | 1 | 0 | 0 | 0.8 | 0.2 | P_B2 | 0.8 | Non | Non |
| re300_e0.25 | 300 | 0.25 | 1 | 1 | 0 | 5 | 1 | 3 | 1 | 0 | 0.2 | 0.6 | 0.2 | P_B2 | 0.6 | Oui | Oui |
| re500_e0.25 | 500 | 0.25 | 1 | 1 | 0 | 15 | 7 | 6 | 2 | 0 | 0.467 | 0.4 | 0.133 | P_B1 | 0.467 | Oui | Oui |
| re700_e0.25 | 700 | 0.25 | 1 | 1 | 0 | 5 | 2 | 2 | 1 | 0 | 0.4 | 0.4 | 0.2 | P_B1 | 0.4 | Oui | Oui |
| re900_e0.25 | 900 | 0.25 | 1 | 1 | 0 | 5 | 2 | 2 | 1 | 0 | 0.4 | 0.4 | 0.2 | P_B1 | 0.4 | Oui | Oui |
| re1000_e0.25 | 1e+03 | 0.25 | 1 | 1 | 0 | 5 | 2 | 2 | 1 | 0 | 0.4 | 0.4 | 0.2 | P_B1 | 0.4 | Oui | Oui |
| re100_e0.5 | 100 | 0.5 | 1 | 1 | 0 | 5 | 2 | 3 | 0 | 0 | 0.4 | 0.6 | 0 | P_B2 | 0.6 | Oui | Oui |
| re300_e0.5 | 300 | 0.5 | 1 | 1 | 0 | 5 | 2 | 3 | 0 | 0 | 0.4 | 0.6 | 0 | P_B2 | 0.6 | Oui | Oui |
| re500_e0.5 | 500 | 0.5 | 1 | 1 | 0 | 5 | 2 | 3 | 0 | 0 | 0.4 | 0.6 | 0 | P_B2 | 0.6 | Oui | Oui |
| re700_e0.5 | 700 | 0.5 | 1 | 1 | 0 | 5 | 2 | 2 | 1 | 0 | 0.4 | 0.4 | 0.2 | P_B1 | 0.4 | Oui | Oui |
| re900_e0.5 | 900 | 0.5 | 1 | 1 | 0 | 5 | 2 | 2 | 1 | 0 | 0.4 | 0.4 | 0.2 | P_B1 | 0.4 | Oui | Oui |
| re1000_e0.5 | 1e+03 | 0.5 | 1 | 1 | 0 | 5 | 2 | 2 | 1 | 0 | 0.4 | 0.4 | 0.2 | P_B1 | 0.4 | Oui | Oui |
| re100_e1 | 100 | 1 | 1 | 1 | 0 | 5 | 2 | 3 | 0 | 0 | 0.4 | 0.6 | 0 | P_B2 | 0.6 | Oui | Oui |
| re300_e1 | 300 | 1 | 1 | 1 | 0 | 5 | 2 | 2 | 1 | 0 | 0.4 | 0.4 | 0.2 | P_B1 | 0.4 | Oui | Oui |
| re500_e1 | 500 | 1 | 1 | 1 | 0 | 5 | 3 | 2 | 0 | 0 | 0.6 | 0.4 | 0 | P_B1 | 0.6 | Oui | Oui |
| re700_e1 | 700 | 1 | 1 | 1 | 0 | 5 | 3 | 2 | 0 | 0 | 0.6 | 0.4 | 0 | P_B1 | 0.6 | Oui | Oui |
| re900_e1 | 900 | 1 | 1 | 1 | 0 | 5 | 2 | 2 | 1 | 0 | 0.4 | 0.4 | 0.2 | P_B1 | 0.4 | Oui | Oui |
| re1000_e1 | 1e+03 | 1 | 1 | 1 | 0 | 20 | 11 | 4 | 5 | 0 | 0.55 | 0.2 | 0.25 | P_B1 | 0.55 | Oui | Oui |
| re100_e2 | 100 | 2 | 1 | 1 | 0 | 5 | 3 | 2 | 0 | 0 | 0.6 | 0.4 | 0 | P_B1 | 0.6 | Oui | Oui |
| re300_e2 | 300 | 2 | 1 | 1 | 0 | 5 | 2 | 3 | 0 | 0 | 0.4 | 0.6 | 0 | P_B2 | 0.6 | Oui | Oui |
| re500_e2 | 500 | 2 | 1 | 1 | 0 | 5 | 2 | 3 | 0 | 0 | 0.4 | 0.6 | 0 | P_B2 | 0.6 | Oui | Oui |
| re700_e2 | 700 | 2 | 1 | 1 | 0 | 15 | 3 | 6 | 6 | 0 | 0.2 | 0.4 | 0.4 | P_B2 | 0.4 | Oui | Oui |
| re900_e2 | 900 | 2 | 1 | 1 | 0 | 5 | 2 | 2 | 1 | 0 | 0.4 | 0.4 | 0.2 | P_B1 | 0.4 | Oui | Oui |
| re1000_e2 | 1e+03 | 2 | 1 | 1 | 0 | 5 | 2 | 2 | 1 | 0 | 0.4 | 0.4 | 0.2 | P_B1 | 0.4 | Oui | Oui |

## 1.2 Statistiques de bassin avec intervalles de confiance

Table 3: Estimation de bassin et intervalle de confiance 95% — `phase1_basin_stats.csv` ; entropie normalisée $H_N$.

| Re | $E^*$ | $n$ | p\{}_B1 | lo\{}_B1 | hi\{}_B1 | p\{}_B2 | lo\{}_B2 | hi\{}_B2 | p\{}_B3 | lo\{}_B3 | hi\{}_B3 | H\{}_N |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 100 | 0.1 | 5 | 0 | 0 | 0.434 | 0.8 | 0.376 | 0.964 | 0.2 | 0.0362 | 0.624 | 0.455 |
| 100 | 0.25 | 5 | 0 | 0 | 0.434 | 0.8 | 0.376 | 0.964 | 0.2 | 0.0362 | 0.624 | 0.455 |
| 100 | 0.5 | 5 | 0.4 | 0.118 | 0.769 | 0.6 | 0.231 | 0.882 | 0 | 0 | 0.434 | 0.613 |
| 100 | 1 | 5 | 0.4 | 0.118 | 0.769 | 0.6 | 0.231 | 0.882 | 0 | 0 | 0.434 | 0.613 |
| 100 | 2 | 5 | 0.6 | 0.231 | 0.882 | 0.4 | 0.118 | 0.769 | 0 | 0 | 0.434 | 0.613 |
| 300 | 0.1 | 5 | 0.4 | 0.118 | 0.769 | 0.6 | 0.231 | 0.882 | 0 | 0 | 0.434 | 0.613 |
| 300 | 0.25 | 5 | 0.2 | 0.0362 | 0.624 | 0.6 | 0.231 | 0.882 | 0.2 | 0.0362 | 0.624 | 0.865 |
| 300 | 0.5 | 5 | 0.4 | 0.118 | 0.769 | 0.6 | 0.231 | 0.882 | 0 | 0 | 0.434 | 0.613 |
| 300 | 1 | 5 | 0.4 | 0.118 | 0.769 | 0.4 | 0.118 | 0.769 | 0.2 | 0.0362 | 0.624 | 0.96 |
| 300 | 2 | 5 | 0.4 | 0.118 | 0.769 | 0.6 | 0.231 | 0.882 | 0 | 0 | 0.434 | 0.613 |
| 500 | 0.1 | 5 | 0.8 | 0.376 | 0.964 | 0.2 | 0.0362 | 0.624 | 0 | 0 | 0.434 | 0.455 |
| 500 | 0.25 | 15 | 0.467 | 0.248 | 0.699 | 0.4 | 0.198 | 0.643 | 0.133 | 0.0374 | 0.379 | 0.902 |
| 500 | 0.5 | 5 | 0.4 | 0.118 | 0.769 | 0.6 | 0.231 | 0.882 | 0 | 0 | 0.434 | 0.613 |
| 500 | 1 | 5 | 0.6 | 0.231 | 0.882 | 0.4 | 0.118 | 0.769 | 0 | 0 | 0.434 | 0.613 |
| 500 | 2 | 5 | 0.4 | 0.118 | 0.769 | 0.6 | 0.231 | 0.882 | 0 | 0 | 0.434 | 0.613 |
| 700 | 0.1 | 5 | 0.6 | 0.231 | 0.882 | 0.4 | 0.118 | 0.769 | 0 | 0 | 0.434 | 0.613 |
| 700 | 0.25 | 5 | 0.4 | 0.118 | 0.769 | 0.4 | 0.118 | 0.769 | 0.2 | 0.0362 | 0.624 | 0.96 |
| 700 | 0.5 | 5 | 0.4 | 0.118 | 0.769 | 0.4 | 0.118 | 0.769 | 0.2 | 0.0362 | 0.624 | 0.96 |
| 700 | 1 | 5 | 0.6 | 0.231 | 0.882 | 0.4 | 0.118 | 0.769 | 0 | 0 | 0.434 | 0.613 |
| 700 | 2 | 15 | 0.2 | 0.0705 | 0.452 | 0.4 | 0.198 | 0.643 | 0.4 | 0.198 | 0.643 | 0.96 |
| 900 | 0.1 | 5 | 0.6 | 0.231 | 0.882 | 0.4 | 0.118 | 0.769 | 0 | 0 | 0.434 | 0.613 |
| 900 | 0.25 | 5 | 0.4 | 0.118 | 0.769 | 0.4 | 0.118 | 0.769 | 0.2 | 0.0362 | 0.624 | 0.96 |
| 900 | 0.5 | 5 | 0.4 | 0.118 | 0.769 | 0.4 | 0.118 | 0.769 | 0.2 | 0.0362 | 0.624 | 0.96 |
| 900 | 1 | 5 | 0.4 | 0.118 | 0.769 | 0.4 | 0.118 | 0.769 | 0.2 | 0.0362 | 0.624 | 0.96 |
| 900 | 2 | 5 | 0.4 | 0.118 | 0.769 | 0.4 | 0.118 | 0.769 | 0.2 | 0.0362 | 0.624 | 0.96 |
| 1e+03 | 0.1 | 5 | 0.6 | 0.231 | 0.882 | 0.4 | 0.118 | 0.769 | 0 | 0 | 0.434 | 0.613 |
| 1e+03 | 0.25 | 5 | 0.4 | 0.118 | 0.769 | 0.4 | 0.118 | 0.769 | 0.2 | 0.0362 | 0.624 | 0.96 |
| 1e+03 | 0.5 | 5 | 0.4 | 0.118 | 0.769 | 0.4 | 0.118 | 0.769 | 0.2 | 0.0362 | 0.624 | 0.96 |
| 1e+03 | 1 | 20 | 0.55 | 0.342 | 0.742 | 0.2 | 0.0807 | 0.416 | 0.25 | 0.112 | 0.469 | 0.908 |
| 1e+03 | 2 | 5 | 0.4 | 0.118 | 0.769 | 0.4 | 0.118 | 0.769 | 0.2 | 0.0362 | 0.624 | 0.96 |

## 1.3 Détail par course — métriques de classification

Table 4: Métriques par course (classification) — 185 courses, `branch_runs.csv`.

| point | étude | Re | $E^*$ | AR | $\lambda_v$ | $\lambda_b$ | s | $E_r$ | $f_2$ | $f_t$ | $A_2$ | $R_K$ | $\epsilon$ | id | branch | $i$ | $j$ | $E_t$ | base |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| re1000_e0.1 | phase1_re_estarmap | 1e+03 | 0.1 | 1 | 1 | 0 | 0 | 0.12 | 0.796 | 0.0718 | -0.437 | 0.931 | 1.021e-16 | 1 | B1_Mode2 | 1 | 0 | 0.12 | fourier |
| re1000_e0.1 | phase1_re_estarmap | 1e+03 | 0.1 | 1 | 1 | 0 | 1 | 0.122 | 0.903 | 0.00619 | 0.47 | 0.989 | 9.684e-17 | 1 | B1_Mode2 | 1 | 0 | 0.122 | fourier |
| re1000_e0.1 | phase1_re_estarmap | 1e+03 | 0.1 | 1 | 1 | 0 | 2 | 0.125 | 0.897 | 0.00436 | 0.474 | 0.994 | 1.006e-16 | 1 | B1_Mode2 | 1 | 0 | 0.125 | fourier |
| re1000_e0.1 | phase1_re_estarmap | 1e+03 | 0.1 | 1 | 1 | 0 | 3 | 0.122 | 0.0492 | 0.948 | 0.109 | 0.993 | 1.326e-16 | 2 | B2_temporal | 0 | 1 | 0.122 | fourier |
| re1000_e0.1 | phase1_re_estarmap | 1e+03 | 0.1 | 1 | 1 | 0 | 4 | 0.125 | 0.0832 | 0.916 | 0.144 | 0.993 | 1.197e-16 | 2 | B2_temporal | 0 | 1 | 0.125 | fourier |
| re1000_e0.25 | phase1_re_estarmap | 1e+03 | 0.25 | 1 | 1 | 0 | 0 | 0.24 | 0.436 | 0.516 | -0.458 | 0.91 | 1.909e-16 | 3 | B3_mixed | 1 | 0 | 0.24 | fourier |
| re1000_e0.25 | phase1_re_estarmap | 1e+03 | 0.25 | 1 | 1 | 0 | 1 | 0.266 | 0.936 | 0.0271 | 0.706 | 0.983 | 1.472e-16 | 1 | B1_Mode2 | 1 | 0 | 0.266 | fourier |
| re1000_e0.25 | phase1_re_estarmap | 1e+03 | 0.25 | 1 | 1 | 0 | 2 | 0.264 | 0.922 | 0.00608 | 0.698 | 0.99 | 1.550e-16 | 1 | B1_Mode2 | 1 | 0 | 0.264 | fourier |
| re1000_e0.25 | phase1_re_estarmap | 1e+03 | 0.25 | 1 | 1 | 0 | 3 | 0.264 | 0.0317 | 0.966 | 0.129 | 0.994 | 1.680e-16 | 2 | B2_temporal | 0 | 1 | 0.264 | fourier |
| re1000_e0.25 | phase1_re_estarmap | 1e+03 | 0.25 | 1 | 1 | 0 | 4 | 0.264 | 0.0386 | 0.961 | 0.143 | 0.991 | 1.722e-16 | 2 | B2_temporal | 0 | 1 | 0.264 | fourier |
| re1000_e0.5 | phase1_re_estarmap | 1e+03 | 0.5 | 1 | 1 | 0 | 0 | 0.461 | 0.346 | 0.63 | -0.565 | 0.904 | 2.126e-16 | 3 | B3_mixed | 0 | 1 | 0.461 | fourier |
| re1000_e0.5 | phase1_re_estarmap | 1e+03 | 0.5 | 1 | 1 | 0 | 1 | 0.516 | 0.953 | 0.0368 | 0.992 | 0.993 | 2.351e-16 | 1 | B1_Mode2 | 1 | 0 | 0.516 | fourier |
| re1000_e0.5 | phase1_re_estarmap | 1e+03 | 0.5 | 1 | 1 | 0 | 2 | 0.513 | 0.949 | 0.0127 | 0.986 | 0.984 | 2.085e-16 | 1 | B1_Mode2 | 1 | 0 | 0.513 | fourier |
| re1000_e0.5 | phase1_re_estarmap | 1e+03 | 0.5 | 1 | 1 | 0 | 3 | 0.516 | 0.016 | 0.981 | 0.128 | 0.993 | 2.703e-16 | 2 | B2_temporal | 0 | 1 | 0.516 | fourier |
| re1000_e0.5 | phase1_re_estarmap | 1e+03 | 0.5 | 1 | 1 | 0 | 4 | 0.497 | 0.00295 | 0.997 | -0.0541 | 0.983 | 2.225e-16 | 2 | B2_temporal | 0 | 1 | 0.497 | fourier |
| re1000_e1 | phase1_re_estarmap | 1e+03 | 1 | 1 | 1 | 0 | 0 | 0.887 | 0.00228 | 0.106 | 0.0637 | 0.871 | 2.850e-16 | 3 | B3_mixed | 2 | 0 | 0.887 | fourier |
| re1000_e1 | phase1_re_estarmap | 1e+03 | 1 | 1 | 1 | 0 | 1 | 1.03 | 0.962 | 0.0197 | 1.41 | 0.985 | 3.350e-16 | 1 | B1_Mode2 | 1 | 0 | 1.03 | fourier |
| re1000_e1 | phase1_re_estarmap | 1e+03 | 1 | 1 | 1 | 0 | 10 | 1.06 | 0.194 | 0.472 | -0.643 | 0.726 | 3.289e-16 | 3 | B3_mixed | 1 | 1 | 1.06 | fourier |
| re1000_e1 | phase1_re_estarmap | 1e+03 | 1 | 1 | 1 | 0 | 11 | 0.971 | 0.215 | 0.371 | -0.647 | 0.709 | 2.906e-16 | 3 | B3_mixed | 2 | 0 | 0.971 | fourier |
| re1000_e1 | phase1_re_estarmap | 1e+03 | 1 | 1 | 1 | 0 | 12 | 0.971 | 0.000254 | 0.989 | -0.0222 | 0.98 | 3.514e-16 | 2 | B2_temporal | 0 | 1 | 0.971 | fourier |
| re1000_e1 | phase1_re_estarmap | 1e+03 | 1 | 1 | 1 | 0 | 13 | 0.987 | 0.949 | 0.0231 | 1.37 | 0.985 | 3.264e-16 | 1 | B1_Mode2 | 1 | 0 | 0.987 | fourier |
| re1000_e1 | phase1_re_estarmap | 1e+03 | 1 | 1 | 1 | 0 | 14 | 0.986 | 0.681 | 0.154 | 1.16 | 0.835 | 2.912e-16 | 1 | B1_Mode2 | 1 | 0 | 0.986 | fourier |
| re1000_e1 | phase1_re_estarmap | 1e+03 | 1 | 1 | 1 | 0 | 15 | 1.04 | 0.802 | 0.146 | 1.29 | 0.917 | 3.408e-16 | 1 | B1_Mode2 | 1 | 0 | 1.04 | fourier |
| re1000_e1 | phase1_re_estarmap | 1e+03 | 1 | 1 | 1 | 0 | 16 | 0.979 | 0.831 | 0.104 | 1.28 | 0.913 | 2.909e-16 | 1 | B1_Mode2 | 1 | 0 | 0.979 | fourier |
| re1000_e1 | phase1_re_estarmap | 1e+03 | 1 | 1 | 1 | 0 | 17 | 0.986 | 0.617 | 0.315 | -1.1 | 0.854 | 3.455e-16 | 3 | B3_mixed | 1 | 0 | 0.986 | fourier |
| re1000_e1 | phase1_re_estarmap | 1e+03 | 1 | 1 | 1 | 0 | 18 | 1.02 | 0.974 | 0.0101 | -1.41 | 0.992 | 2.931e-16 | 1 | B1_Mode2 | 1 | 0 | 1.02 | fourier |
| re1000_e1 | phase1_re_estarmap | 1e+03 | 1 | 1 | 1 | 0 | 19 | 0.993 | 0.699 | 0.265 | -1.18 | 0.894 | 3.250e-16 | 1 | B1_Mode2 | 1 | 0 | 0.993 | fourier |
| re1000_e1 | phase1_re_estarmap | 1e+03 | 1 | 1 | 1 | 0 | 2 | 1.02 | 0.931 | 0.0558 | 1.38 | 0.963 | 3.345e-16 | 1 | B1_Mode2 | 1 | 0 | 1.02 | fourier |
| re1000_e1 | phase1_re_estarmap | 1e+03 | 1 | 1 | 1 | 0 | 3 | 0.972 | 0.00572 | 0.991 | 0.105 | 0.984 | 3.384e-16 | 2 | B2_temporal | 0 | 1 | 0.972 | fourier |
| re1000_e1 | phase1_re_estarmap | 1e+03 | 1 | 1 | 1 | 0 | 4 | 0.966 | 0.0197 | 0.98 | -0.195 | 0.959 | 3.367e-16 | 2 | B2_temporal | 0 | 1 | 0.966 | fourier |
| re1000_e1 | phase1_re_estarmap | 1e+03 | 1 | 1 | 1 | 0 | 5 | 0.989 | 0.65 | 0.228 | 1.13 | 0.797 | 3.004e-16 | 1 | B1_Mode2 | 1 | 0 | 0.989 | fourier |
| re1000_e1 | phase1_re_estarmap | 1e+03 | 1 | 1 | 1 | 0 | 6 | 0.998 | 0.918 | 0.0381 | 1.35 | 0.956 | 3.131e-16 | 1 | B1_Mode2 | 1 | 0 | 0.998 | fourier |
| re1000_e1 | phase1_re_estarmap | 1e+03 | 1 | 1 | 1 | 0 | 7 | 0.973 | 0.943 | 0.021 | 1.35 | 0.976 | 2.921e-16 | 1 | B1_Mode2 | 1 | 0 | 0.973 | fourier |
| re1000_e1 | phase1_re_estarmap | 1e+03 | 1 | 1 | 1 | 0 | 8 | 0.944 | 0.135 | 0.539 | 0.505 | 0.647 | 3.023e-16 | 2 | B2_temporal | 0 | 1 | 0.944 | fourier |
| re1000_e1 | phase1_re_estarmap | 1e+03 | 1 | 1 | 1 | 0 | 9 | 1.01 | 0.477 | 0.348 | 0.982 | 0.717 | 3.112e-16 | 3 | B3_mixed | 1 | 0 | 1.01 | fourier |
| re1000_e2 | phase1_re_estarmap | 1e+03 | 2 | 1 | 1 | 0 | 0 | 1.81 | 0.0158 | 0.129 | 0.239 | 0.887 | 4.455e-16 | 3 | B3_mixed | 2 | 0 | 1.81 | fourier |
| re1000_e2 | phase1_re_estarmap | 1e+03 | 2 | 1 | 1 | 0 | 1 | 2.04 | 0.936 | 0.0427 | 1.96 | 0.972 | 4.064e-16 | 1 | B1_Mode2 | 1 | 0 | 2.04 | fourier |
| re1000_e2 | phase1_re_estarmap | 1e+03 | 2 | 1 | 1 | 0 | 2 | 1.93 | 0.554 | 0.117 | 1.46 | 0.882 | 4.207e-16 | 1 | B1_Mode2 | 1 | 0 | 1.93 | fourier |
| re1000_e2 | phase1_re_estarmap | 1e+03 | 2 | 1 | 1 | 0 | 3 | 1.94 | 0.0118 | 0.97 | -0.214 | 0.968 | 4.390e-16 | 2 | B2_temporal | 0 | 1 | 1.94 | fourier |
| re1000_e2 | phase1_re_estarmap | 1e+03 | 2 | 1 | 1 | 0 | 4 | 1.94 | 0.164 | 0.835 | 0.797 | 0.942 | 4.412e-16 | 2 | B2_temporal | 0 | 1 | 1.94 | fourier |
| re100_e0.1 | phase1_re_estarmap | 100 | 0.1 | 1 | 1 | 0 | 0 | 0.11 | 0.0163 | 0.979 | -0.06 | 0.978 | 1.106e-16 | 2 | B2_temporal | 0 | 1 | 0.11 | fourier |
| re100_e0.1 | phase1_re_estarmap | 100 | 0.1 | 1 | 1 | 0 | 1 | 0.111 | 0.00474 | 0.995 | -0.0325 | 0.989 | 1.121e-16 | 2 | B2_temporal | 0 | 1 | 0.111 | fourier |
| re100_e0.1 | phase1_re_estarmap | 100 | 0.1 | 1 | 1 | 0 | 2 | 0.113 | 0.363 | 0.618 | 0.286 | 0.966 | 1.129e-16 | 3 | B3_mixed | 0 | 1 | 0.113 | fourier |
| re100_e0.1 | phase1_re_estarmap | 100 | 0.1 | 1 | 1 | 0 | 3 | 0.125 | 0.00679 | 0.991 | 0.0413 | 0.993 | 1.133e-16 | 2 | B2_temporal | 0 | 1 | 0.125 | fourier |
| re100_e0.1 | phase1_re_estarmap | 100 | 0.1 | 1 | 1 | 0 | 4 | 0.119 | 1.263e-05 | 0.999 | 0.00174 | 0.995 | 1.129e-16 | 2 | B2_temporal | 0 | 1 | 0.119 | fourier |
| re100_e0.25 | phase1_re_estarmap | 100 | 0.25 | 1 | 1 | 0 | 0 | 0.237 | 0.00677 | 0.99 | -0.0567 | 0.98 | 1.659e-16 | 2 | B2_temporal | 0 | 1 | 0.237 | fourier |
| re100_e0.25 | phase1_re_estarmap | 100 | 0.25 | 1 | 1 | 0 | 1 | 0.24 | 0.0111 | 0.988 | -0.0729 | 0.986 | 1.599e-16 | 2 | B2_temporal | 0 | 1 | 0.24 | fourier |
| re100_e0.25 | phase1_re_estarmap | 100 | 0.25 | 1 | 1 | 0 | 2 | 0.245 | 0.477 | 0.498 | 0.484 | 0.953 | 1.953e-16 | 3 | B3_mixed | 1 | 0 | 0.245 | fourier |
| re100_e0.25 | phase1_re_estarmap | 100 | 0.25 | 1 | 1 | 0 | 3 | 0.267 | 0.035 | 0.963 | 0.137 | 0.993 | 1.639e-16 | 2 | B2_temporal | 0 | 1 | 0.267 | fourier |
| re100_e0.25 | phase1_re_estarmap | 100 | 0.25 | 1 | 1 | 0 | 4 | 0.256 | 4.352e-06 | 1 | 0.00149 | 0.993 | 1.653e-16 | 2 | B2_temporal | 0 | 1 | 0.256 | fourier |
| re100_e0.5 | phase1_re_estarmap | 100 | 0.5 | 1 | 1 | 0 | 0 | 0.463 | 0.000109 | 0.997 | -0.01 | 0.981 | 2.269e-16 | 2 | B2_temporal | 0 | 1 | 0.463 | fourier |
| re100_e0.5 | phase1_re_estarmap | 100 | 0.5 | 1 | 1 | 0 | 1 | 0.468 | 0.03 | 0.966 | -0.168 | 0.965 | 2.642e-16 | 2 | B2_temporal | 0 | 1 | 0.468 | fourier |
| re100_e0.5 | phase1_re_estarmap | 100 | 0.5 | 1 | 1 | 0 | 2 | 0.495 | 0.794 | 0.153 | 0.887 | 0.966 | 2.278e-16 | 1 | B1_Mode2 | 1 | 0 | 0.495 | fourier |
| re100_e0.5 | phase1_re_estarmap | 100 | 0.5 | 1 | 1 | 0 | 3 | 0.491 | 0.918 | 0.00511 | 0.949 | 0.992 | 2.249e-16 | 1 | B1_Mode2 | 1 | 0 | 0.491 | fourier |
| re100_e0.5 | phase1_re_estarmap | 100 | 0.5 | 1 | 1 | 0 | 4 | 0.493 | 0.00557 | 0.994 | -0.0741 | 0.984 | 3.285e-16 | 2 | B2_temporal | 0 | 1 | 0.493 | fourier |
| re100_e1 | phase1_re_estarmap | 100 | 1 | 1 | 1 | 0 | 0 | 0.913 | 0.047 | 0.946 | -0.293 | 0.929 | 3.292e-16 | 2 | B2_temporal | 0 | 1 | 0.913 | fourier |
| re100_e1 | phase1_re_estarmap | 100 | 1 | 1 | 1 | 0 | 1 | 0.921 | 0.149 | 0.843 | -0.524 | 0.942 | 3.091e-16 | 2 | B2_temporal | 0 | 1 | 0.921 | fourier |
| re100_e1 | phase1_re_estarmap | 100 | 1 | 1 | 1 | 0 | 2 | 1.01 | 0.887 | 0.0485 | 1.34 | 0.981 | 3.334e-16 | 1 | B1_Mode2 | 1 | 0 | 1.01 | fourier |
| re100_e1 | phase1_re_estarmap | 100 | 1 | 1 | 1 | 0 | 3 | 0.983 | 0.912 | 0.00156 | 1.34 | 0.994 | 2.904e-16 | 1 | B1_Mode2 | 1 | 0 | 0.983 | fourier |
| re100_e1 | phase1_re_estarmap | 100 | 1 | 1 | 1 | 0 | 4 | 0.981 | 0.00101 | 0.998 | -0.0445 | 0.989 | 3.261e-16 | 2 | B2_temporal | 0 | 1 | 0.981 | fourier |
| re100_e2 | phase1_re_estarmap | 100 | 2 | 1 | 1 | 0 | 0 | 1.88 | 0.228 | 0.562 | -0.927 | 0.655 | 4.715e-16 | 2 | B2_temporal | 1 | 1 | 1.88 | fourier |
| re100_e2 | phase1_re_estarmap | 100 | 2 | 1 | 1 | 0 | 1 | 1.9 | 0.571 | 0.281 | -1.47 | 0.781 | 4.547e-16 | 1 | B1_Mode2 | 1 | 0 | 1.9 | fourier |
| re100_e2 | phase1_re_estarmap | 100 | 2 | 1 | 1 | 0 | 2 | 1.99 | 0.844 | 0.0427 | 1.83 | 0.939 | 4.218e-16 | 1 | B1_Mode2 | 1 | 0 | 1.99 | fourier |
| re100_e2 | phase1_re_estarmap | 100 | 2 | 1 | 1 | 0 | 3 | 1.89 | 0.884 | 0.0111 | 1.83 | 0.982 | 4.183e-16 | 1 | B1_Mode2 | 1 | 0 | 1.89 | fourier |
| re100_e2 | phase1_re_estarmap | 100 | 2 | 1 | 1 | 0 | 4 | 1.86 | 0.135 | 0.623 | -0.711 | 0.69 | 4.478e-16 | 2 | B2_temporal | 0 | 1 | 1.86 | fourier |
| re300_e0.1 | phase1_re_estarmap | 300 | 0.1 | 1 | 1 | 0 | 0 | 0.109 | 0.0897 | 0.905 | -0.14 | 0.977 | 1.103e-16 | 2 | B2_temporal | 0 | 1 | 0.109 | fourier |
| re300_e0.1 | phase1_re_estarmap | 300 | 0.1 | 1 | 1 | 0 | 1 | 0.123 | 0.89 | 0.0117 | 0.467 | 0.984 | 1.084e-16 | 1 | B1_Mode2 | 1 | 0 | 0.123 | fourier |
| re300_e0.1 | phase1_re_estarmap | 300 | 0.1 | 1 | 1 | 0 | 2 | 0.12 | 0.878 | 0.0426 | 0.46 | 0.981 | 1.004e-16 | 1 | B1_Mode2 | 1 | 0 | 0.12 | fourier |

Table 4: Métriques par course (classification) — 185 courses, `branch_runs.csv`.

| point | étude | Re | $E^*$ | AR | $\lambda_v$ | $\lambda_b$ | s | $E_r$ | $f_2$ | $f_t$ | $A_2$ | $R_K$ | $\epsilon$ | id | branch | $i$ | $j$ | $E_t$ | base |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| re300_e0.1 | phase1_re_estarmap | 300 | 0.1 | 1 | 1 | 0 | 3 | 0.124 | 0.206 | 0.792 | 0.226 | 0.992 | 1.155e-16 | 2 | B2_temporal | 0 | 1 | 0.124 | fourier |
| re300_e0.1 | phase1_re_estarmap | 300 | 0.1 | 1 | 1 | 0 | 4 | 0.117 | 0.0913 | 0.901 | -0.146 | 0.966 | 1.101e-16 | 2 | B2_temporal | 0 | 1 | 0.117 | fourier |
| re300_e0.25 | phase1_re_estarmap | 300 | 0.25 | 1 | 1 | 0 | 0 | 0.233 | 0.0618 | 0.933 | -0.17 | 0.97 | 1.537e-16 | 2 | B2_temporal | 0 | 1 | 0.233 | fourier |
| re300_e0.25 | phase1_re_estarmap | 300 | 0.25 | 1 | 1 | 0 | 1 | 0.243 | 0.51 | 0.453 | -0.497 | 0.935 | 1.652e-16 | 3 | B3_mixed | 1 | 0 | 0.243 | fourier |
| re300_e0.25 | phase1_re_estarmap | 300 | 0.25 | 1 | 1 | 0 | 2 | 0.255 | 0.827 | 0.129 | 0.65 | 0.962 | 1.654e-16 | 1 | B1_Mode2 | 1 | 0 | 0.255 | fourier |
| re300_e0.25 | phase1_re_estarmap | 300 | 0.25 | 1 | 1 | 0 | 3 | 0.265 | 0.159 | 0.838 | 0.29 | 0.994 | 1.721e-16 | 2 | B2_temporal | 0 | 1 | 0.265 | fourier |
| re300_e0.25 | phase1_re_estarmap | 300 | 0.25 | 1 | 1 | 0 | 4 | 0.251 | 0.2 | 0.777 | -0.317 | 0.926 | 1.657e-16 | 2 | B2_temporal | 0 | 1 | 0.251 | fourier |
| re300_e0.5 | phase1_re_estarmap | 300 | 0.5 | 1 | 1 | 0 | 0 | 0.456 | 0.0526 | 0.943 | -0.219 | 0.944 | 2.283e-16 | 2 | B2_temporal | 0 | 1 | 0.456 | fourier |
| re300_e0.5 | phase1_re_estarmap | 300 | 0.5 | 1 | 1 | 0 | 1 | 0.479 | 0.738 | 0.244 | -0.841 | 0.943 | 2.285e-16 | 1 | B1_Mode2 | 1 | 0 | 0.479 | fourier |
| re300_e0.5 | phase1_re_estarmap | 300 | 0.5 | 1 | 1 | 0 | 2 | 0.499 | 0.876 | 0.0963 | 0.935 | 0.968 | 2.331e-16 | 1 | B1_Mode2 | 1 | 0 | 0.499 | fourier |
| re300_e0.5 | phase1_re_estarmap | 300 | 0.5 | 1 | 1 | 0 | 3 | 0.51 | 0.16 | 0.837 | 0.404 | 0.981 | 2.717e-16 | 2 | B2_temporal | 0 | 1 | 0.51 | fourier |
| re300_e0.5 | phase1_re_estarmap | 300 | 0.5 | 1 | 1 | 0 | 4 | 0.488 | 0.0953 | 0.893 | -0.305 | 0.922 | 2.409e-16 | 2 | B2_temporal | 0 | 1 | 0.488 | fourier |
| re300_e1 | phase1_re_estarmap | 300 | 1 | 1 | 1 | 0 | 0 | 0.903 | 0.277 | 0.709 | -0.708 | 0.903 | 3.126e-16 | 2 | B2_temporal | 0 | 1 | 0.903 | fourier |
| re300_e1 | phase1_re_estarmap | 300 | 1 | 1 | 1 | 0 | 1 | 0.93 | 0.628 | 0.36 | -1.08 | 0.926 | 2.866e-16 | 3 | B3_mixed | 1 | 0 | 0.93 | fourier |
| re300_e1 | phase1_re_estarmap | 300 | 1 | 1 | 1 | 0 | 2 | 0.993 | 0.894 | 0.066 | 1.33 | 0.961 | 2.960e-16 | 1 | B1_Mode2 | 1 | 0 | 0.993 | fourier |
| re300_e1 | phase1_re_estarmap | 300 | 1 | 1 | 1 | 0 | 3 | 0.986 | 0.87 | 0.0655 | 1.31 | 0.94 | 3.458e-16 | 1 | B1_Mode2 | 1 | 0 | 0.986 | fourier |
| re300_e1 | phase1_re_estarmap | 300 | 1 | 1 | 1 | 0 | 4 | 0.966 | 0.042 | 0.954 | -0.285 | 0.949 | 3.220e-16 | 2 | B2_temporal | 0 | 1 | 0.966 | fourier |
| re300_e2 | phase1_re_estarmap | 300 | 2 | 1 | 1 | 0 | 0 | 1.8 | 0.188 | 0.772 | -0.824 | 0.818 | 4.599e-16 | 2 | B2_temporal | 0 | 1 | 1.8 | fourier |
| re300_e2 | phase1_re_estarmap | 300 | 2 | 1 | 1 | 0 | 1 | 1.97 | 0.231 | 0.552 | -0.953 | 0.592 | 4.657e-16 | 2 | B2_temporal | 1 | 0 | 1.97 | fourier |
| re300_e2 | phase1_re_estarmap | 300 | 2 | 1 | 1 | 0 | 2 | 1.97 | 0.738 | 0.0788 | 1.71 | 0.886 | 4.161e-16 | 1 | B1_Mode2 | 1 | 0 | 1.97 | fourier |
| re300_e2 | phase1_re_estarmap | 300 | 2 | 1 | 1 | 0 | 3 | 1.91 | 0.669 | 0.222 | 1.6 | 0.827 | 4.317e-16 | 1 | B1_Mode2 | 1 | 0 | 1.91 | fourier |
| re300_e2 | phase1_re_estarmap | 300 | 2 | 1 | 1 | 0 | 4 | 1.87 | 0.202 | 0.659 | -0.869 | 0.796 | 4.499e-16 | 2 | B2_temporal | 0 | 1 | 1.87 | fourier |
| re500_e0.1 | phase1_re_estarmap | 500 | 0.1 | 1 | 1 | 0 | 0 | 0.122 | 0.845 | 0.0357 | -0.454 | 0.955 | 1.016e-16 | 1 | B1_Mode2 | 1 | 0 | 0.122 | fourier |
| re500_e0.1 | phase1_re_estarmap | 500 | 0.1 | 1 | 1 | 0 | 1 | 0.123 | 0.887 | 0.00714 | 0.467 | 0.988 | 1.016e-16 | 1 | B1_Mode2 | 1 | 0 | 0.123 | fourier |
| re500_e0.1 | phase1_re_estarmap | 500 | 0.1 | 1 | 1 | 0 | 2 | 0.121 | 0.884 | 0.018 | 0.463 | 0.978 | 1.028e-16 | 1 | B1_Mode2 | 1 | 0 | 0.121 | fourier |
| re500_e0.1 | phase1_re_estarmap | 500 | 0.1 | 1 | 1 | 0 | 3 | 0.126 | 0.0167 | 0.979 | 0.0648 | 0.987 | 1.218e-16 | 2 | B2_temporal | 0 | 1 | 0.126 | fourier |
| re500_e0.1 | phase1_re_estarmap | 500 | 0.1 | 1 | 1 | 0 | 4 | 0.116 | 0.651 | 0.312 | -0.389 | 0.964 | 1.147e-16 | 1 | B1_Mode2 | 1 | 0 | 0.116 | fourier |
| re500_e0.25 | phase1_re_estarmap | 500 | 0.25 | 1 | 1 | 0 | 0 | 0.24 | 0.52 | 0.448 | -0.499 | 0.941 | 1.596e-16 | 3 | B3_mixed | 1 | 0 | 0.24 | fourier |
| re500_e0.25 | phase1_re_estarmap | 500 | 0.25 | 1 | 1 | 0 | 1 | 0.267 | 0.911 | 0.0133 | 0.698 | 0.986 | 1.671e-16 | 1 | B1_Mode2 | 1 | 0 | 0.267 | fourier |
| re500_e0.25 | phase1_re_estarmap | 500 | 0.25 | 1 | 1 | 0 | 10 | 0.293 | 0.0126 | 0.979 | 0.086 | 0.974 | 1.902e-16 | 2 | B2_temporal | 0 | 1 | 0.293 | fourier |
| re500_e0.25 | phase1_re_estarmap | 500 | 0.25 | 1 | 1 | 0 | 11 | 0.267 | 0.871 | 0.015 | 0.682 | 0.973 | 1.486e-16 | 1 | B1_Mode2 | 1 | 0 | 0.267 | fourier |
| re500_e0.25 | phase1_re_estarmap | 500 | 0.25 | 1 | 1 | 0 | 12 | 0.238 | 0.756 | 0.229 | 0.6 | 0.978 | 1.503e-16 | 1 | B1_Mode2 | 1 | 0 | 0.238 | fourier |
| re500_e0.25 | phase1_re_estarmap | 500 | 0.25 | 1 | 1 | 0 | 13 | 0.256 | 0.886 | 0.0498 | 0.673 | 0.949 | 1.691e-16 | 1 | B1_Mode2 | 1 | 0 | 0.256 | fourier |
| re500_e0.25 | phase1_re_estarmap | 500 | 0.25 | 1 | 1 | 0 | 14 | 0.25 | 0.019 | 0.975 | 0.0975 | 0.978 | 1.715e-16 | 2 | B2_temporal | 0 | 1 | 0.25 | fourier |
| re500_e0.25 | phase1_re_estarmap | 500 | 0.25 | 1 | 1 | 0 | 2 | 0.257 | 0.89 | 0.0465 | 0.677 | 0.964 | 1.473e-16 | 1 | B1_Mode2 | 1 | 0 | 0.257 | fourier |
| re500_e0.25 | phase1_re_estarmap | 500 | 0.25 | 1 | 1 | 0 | 3 | 0.268 | 0.0162 | 0.982 | 0.0931 | 0.991 | 1.634e-16 | 2 | B2_temporal | 0 | 1 | 0.268 | fourier |
| re500_e0.25 | phase1_re_estarmap | 500 | 0.25 | 1 | 1 | 0 | 4 | 0.25 | 0.343 | 0.637 | -0.414 | 0.945 | 1.766e-16 | 3 | B3_mixed | 0 | 1 | 0.25 | fourier |
| re500_e0.25 | phase1_re_estarmap | 500 | 0.25 | 1 | 1 | 0 | 5 | 0.245 | 4.164e-06 | 0.998 | -0.00143 | 0.991 | 1.568e-16 | 2 | B2_temporal | 0 | 1 | 0.245 | fourier |
| re500_e0.25 | phase1_re_estarmap | 500 | 0.25 | 1 | 1 | 0 | 6 | 0.264 | 0.868 | 0.0095 | 0.677 | 0.961 | 1.479e-16 | 1 | B1_Mode2 | 1 | 0 | 0.264 | fourier |
| re500_e0.25 | phase1_re_estarmap | 500 | 0.25 | 1 | 1 | 0 | 7 | 0.251 | 0.93 | 0.00902 | 0.683 | 0.983 | 1.466e-16 | 1 | B1_Mode2 | 1 | 0 | 0.251 | fourier |
| re500_e0.25 | phase1_re_estarmap | 500 | 0.25 | 1 | 1 | 0 | 8 | 0.251 | 0.0341 | 0.963 | 0.131 | 0.979 | 1.818e-16 | 2 | B2_temporal | 0 | 1 | 0.251 | fourier |
| re500_e0.25 | phase1_re_estarmap | 500 | 0.25 | 1 | 1 | 0 | 9 | 0.265 | 0.000245 | 0.996 | -0.0114 | 0.989 | 1.662e-16 | 2 | B2_temporal | 0 | 1 | 0.265 | fourier |
| re500_e0.5 | phase1_re_estarmap | 500 | 0.5 | 1 | 1 | 0 | 0 | 0.453 | 0.222 | 0.765 | -0.449 | 0.935 | 2.491e-16 | 2 | B2_temporal | 0 | 1 | 0.453 | fourier |
| re500_e0.5 | phase1_re_estarmap | 500 | 0.5 | 1 | 1 | 0 | 1 | 0.516 | 0.925 | 0.0254 | 0.977 | 0.974 | 2.259e-16 | 1 | B1_Mode2 | 1 | 0 | 0.516 | fourier |
| re500_e0.5 | phase1_re_estarmap | 500 | 0.5 | 1 | 1 | 0 | 2 | 0.504 | 0.925 | 0.0429 | 0.965 | 0.972 | 2.262e-16 | 1 | B1_Mode2 | 1 | 0 | 0.504 | fourier |
| re500_e0.5 | phase1_re_estarmap | 500 | 0.5 | 1 | 1 | 0 | 3 | 0.518 | 0.0703 | 0.927 | 0.27 | 0.986 | 2.732e-16 | 2 | B2_temporal | 0 | 1 | 0.518 | fourier |
| re500_e0.5 | phase1_re_estarmap | 500 | 0.5 | 1 | 1 | 0 | 4 | 0.49 | 0.0492 | 0.94 | -0.22 | 0.935 | 2.418e-16 | 2 | B2_temporal | 0 | 1 | 0.49 | fourier |
| re500_e1 | phase1_re_estarmap | 500 | 1 | 1 | 1 | 0 | 0 | 0.919 | 0.282 | 0.707 | -0.72 | 0.87 | 3.120e-16 | 2 | B2_temporal | 0 | 1 | 0.919 | fourier |
| re500_e1 | phase1_re_estarmap | 500 | 1 | 1 | 1 | 0 | 1 | 1.01 | 0.643 | 0.189 | 1.14 | 0.845 | 2.924e-16 | 1 | B1_Mode2 | 1 | 0 | 1.01 | fourier |
| re500_e1 | phase1_re_estarmap | 500 | 1 | 1 | 1 | 0 | 2 | 1 | 0.961 | 0.0189 | 1.39 | 0.984 | 3.185e-16 | 1 | B1_Mode2 | 1 | 0 | 1 | fourier |
| re500_e1 | phase1_re_estarmap | 500 | 1 | 1 | 1 | 0 | 3 | 0.972 | 0.959 | 0.0218 | 1.37 | 0.98 | 2.952e-16 | 1 | B1_Mode2 | 1 | 0 | 0.972 | fourier |
| re500_e1 | phase1_re_estarmap | 500 | 1 | 1 | 1 | 0 | 4 | 0.963 | 0.0105 | 0.98 | -0.142 | 0.943 | 3.156e-16 | 2 | B2_temporal | 0 | 1 | 0.963 | fourier |
| re500_e2 | phase1_re_estarmap | 500 | 2 | 1 | 1 | 0 | 0 | 1.84 | 0.24 | 0.728 | -0.938 | 0.814 | 4.142e-16 | 2 | B2_temporal | 0 | 1 | 1.84 | fourier |
| re500_e2 | phase1_re_estarmap | 500 | 2 | 1 | 1 | 0 | 1 | 2.04 | 0.127 | 0.617 | 0.718 | 0.72 | 4.733e-16 | 2 | B2_temporal | 1 | 1 | 2.04 | fourier |
| re500_e2 | phase1_re_estarmap | 500 | 2 | 1 | 1 | 0 | 2 | 1.97 | 0.866 | 0.0211 | 1.85 | 0.968 | 4.224e-16 | 1 | B1_Mode2 | 1 | 0 | 1.97 | fourier |
| re500_e2 | phase1_re_estarmap | 500 | 2 | 1 | 1 | 0 | 3 | 1.91 | 0.652 | 0.267 | 1.58 | 0.842 | 4.437e-16 | 1 | B1_Mode2 | 1 | 0 | 1.91 | fourier |
| re500_e2 | phase1_re_estarmap | 500 | 2 | 1 | 1 | 0 | 4 | 1.89 | 0.263 | 0.588 | -0.996 | 0.711 | 4.564e-16 | 2 | B2_temporal | 0 | 1 | 1.89 | fourier |
| re700_e0.1 | phase1_re_estarmap | 700 | 0.1 | 1 | 1 | 0 | 0 | 0.125 | 0.92 | 0.00713 | -0.48 | 0.993 | 1.010e-16 | 1 | B1_Mode2 | 1 | 0 | 0.125 | fourier |
| re700_e0.1 | phase1_re_estarmap | 700 | 0.1 | 1 | 1 | 0 | 1 | 0.124 | 0.908 | 0.00872 | 0.475 | 0.987 | 1.001e-16 | 1 | B1_Mode2 | 1 | 0 | 0.124 | fourier |
| re700_e0.1 | phase1_re_estarmap | 700 | 0.1 | 1 | 1 | 0 | 2 | 0.124 | 0.889 | 0.0065 | 0.47 | 0.991 | 1.184e-16 | 1 | B1_Mode2 | 1 | 0 | 0.124 | fourier |
| re700_e0.1 | phase1_re_estarmap | 700 | 0.1 | 1 | 1 | 0 | 3 | 0.122 | 0.00833 | 0.989 | 0.0451 | 0.988 | 1.202e-16 | 2 | B2_temporal | 0 | 1 | 0.122 | fourier |
| re700_e0.1 | phase1_re_estarmap | 700 | 0.1 | 1 | 1 | 0 | 4 | 0.123 | 0.0146 | 0.985 | 0.0599 | 0.996 | 1.211e-16 | 2 | B2_temporal | 0 | 1 | 0.123 | fourier |
| re700_e0.25 | phase1_re_estarmap | 700 | 0.25 | 1 | 1 | 0 | 0 | 0.245 | 0.538 | 0.409 | -0.513 | 0.914 | 1.456e-16 | 3 | B3_mixed | 1 | 0 | 0.245 | fourier |
| re700_e0.25 | phase1_re_estarmap | 700 | 0.25 | 1 | 1 | 0 | 1 | 0.268 | 0.92 | 0.0141 | 0.702 | 0.984 | 1.494e-16 | 1 | B1_Mode2 | 1 | 0 | 0.268 | fourier |
| re700_e0.25 | phase1_re_estarmap | 700 | 0.25 | 1 | 1 | 0 | 2 | 0.263 | 0.897 | 0.0131 | 0.687 | 0.983 | 1.462e-16 | 1 | B1_Mode2 | 1 | 0 | 0.263 | fourier |
| re700_e0.25 | phase1_re_estarmap | 700 | 0.25 | 1 | 1 | 0 | 3 | 0.266 | 0.00699 | 0.987 | 0.0609 | 0.988 | 1.650e-16 | 2 | B2_temporal | 0 | 1 | 0.266 | fourier |
| re700_e0.25 | phase1_re_estarmap | 700 | 0.25 | 1 | 1 | 0 | 4 | 0.258 | 0.00024 | 1 | -0.0111 | 0.997 | 1.640e-16 | 2 | B2_temporal | 0 | 1 | 0.258 | fourier |
| re700_e0.5 | phase1_re_estarmap | 700 | 0.5 | 1 | 1 | 0 | 0 | 0.461 | 0.375 | 0.606 | -0.588 | 0.92 | 2.441e-16 | 3 | B3_mixed | 0 | 1 | 0.461 | fourier |

Table 4: Métriques par course (classification) — 185 courses, `branch_runs.csv`.

| point | étude | Re | $E^*$ | AR | $\lambda_v$ | $\lambda_b$ | s | $E_r$ | $f_2$ | $f_t$ | $A_2$ | $R_K$ | $\epsilon$ | id | branch | $i$ | $j$ | $E_t$ | base |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| re700_e0.5 | phase1_re_estarmap | 700 | 0.5 | 1 | 1 | 0 | 1 | 0.519 | 0.939 | 0.0271 | 0.987 | 0.978 | 2.124e-16 | 1 | B1_Mode2 | 1 | 0 | 0.519 | fourier |
| re700_e0.5 | phase1_re_estarmap | 700 | 0.5 | 1 | 1 | 0 | 2 | 0.511 | 0.926 | 0.0172 | 0.972 | 0.987 | 2.010e-16 | 1 | B1_Mode2 | 1 | 0 | 0.511 | fourier |
| re700_e0.5 | phase1_re_estarmap | 700 | 0.5 | 1 | 1 | 0 | 3 | 0.52 | 0.00272 | 0.992 | -0.0532 | 0.994 | 2.322e-16 | 2 | B2_temporal | 0 | 1 | 0.52 | fourier |
| re700_e0.5 | phase1_re_estarmap | 700 | 0.5 | 1 | 1 | 0 | 4 | 0.492 | 0.0256 | 0.973 | -0.159 | 0.961 | 2.121e-16 | 2 | B2_temporal | 0 | 1 | 0.492 | fourier |
| re700_e1 | phase1_re_estarmap | 700 | 1 | 1 | 1 | 0 | 0 | 0.972 | 0.141 | 0.505 | -0.523 | 0.777 | 3.046e-16 | 2 | B2_temporal | 1 | 1 | 0.972 | fourier |
| re700_e1 | phase1_re_estarmap | 700 | 1 | 1 | 1 | 0 | 1 | 1.01 | 0.784 | 0.182 | 1.26 | 0.912 | 2.932e-16 | 1 | B1_Mode2 | 1 | 0 | 1.01 | fourier |
| re700_e1 | phase1_re_estarmap | 700 | 1 | 1 | 1 | 0 | 2 | 1.03 | 0.947 | 0.0214 | 1.4 | 0.984 | 3.317e-16 | 1 | B1_Mode2 | 1 | 0 | 1.03 | fourier |
| re700_e1 | phase1_re_estarmap | 700 | 1 | 1 | 1 | 0 | 3 | 0.935 | 0.773 | 0.136 | 1.2 | 0.87 | 2.875e-16 | 1 | B1_Mode2 | 1 | 0 | 0.935 | fourier |
| re700_e1 | phase1_re_estarmap | 700 | 1 | 1 | 1 | 0 | 4 | 0.971 | 0.0224 | 0.974 | -0.208 | 0.944 | 3.037e-16 | 2 | B2_temporal | 0 | 1 | 0.971 | fourier |
| re700_e2 | phase1_re_estarmap | 700 | 2 | 1 | 1 | 0 | 0 | 1.93 | 0.0733 | 0.432 | -0.532 | 0.748 | 4.079e-16 | 3 | B3_mixed | 2 | 0 | 1.93 | fourier |
| re700_e2 | phase1_re_estarmap | 700 | 2 | 1 | 1 | 0 | 1 | 1.97 | 0.322 | 0.476 | 1.13 | 0.708 | 4.318e-16 | 3 | B3_mixed | 1 | 0 | 1.97 | fourier |
| re700_e2 | phase1_re_estarmap | 700 | 2 | 1 | 1 | 0 | 10 | 2.01 | 0.364 | 0.591 | -1.21 | 0.817 | 4.677e-16 | 3 | B3_mixed | 0 | 1 | 2.01 | fourier |
| re700_e2 | phase1_re_estarmap | 700 | 2 | 1 | 1 | 0 | 11 | 1.91 | 0.000133 | 0.831 | 0.0226 | 0.827 | 4.750e-16 | 2 | B2_temporal | 1 | 1 | 1.91 | fourier |
| re700_e2 | phase1_re_estarmap | 700 | 2 | 1 | 1 | 0 | 12 | 2.04 | 0.316 | 0.642 | -1.14 | 0.918 | 4.665e-16 | 2 | B2_temporal | 0 | 1 | 2.04 | fourier |
| re700_e2 | phase1_re_estarmap | 700 | 2 | 1 | 1 | 0 | 13 | 1.96 | 0.943 | 0.0419 | 1.92 | 0.982 | 4.082e-16 | 1 | B1_Mode2 | 1 | 0 | 1.96 | fourier |
| re700_e2 | phase1_re_estarmap | 700 | 2 | 1 | 1 | 0 | 14 | 1.82 | 0.392 | 0.355 | 1.19 | 0.697 | 4.444e-16 | 3 | B3_mixed | 1 | 0 | 1.82 | fourier |
| re700_e2 | phase1_re_estarmap | 700 | 2 | 1 | 1 | 0 | 2 | 1.96 | 0.679 | 0.0535 | 1.63 | 0.934 | 4.261e-16 | 1 | B1_Mode2 | 1 | 0 | 1.96 | fourier |
| re700_e2 | phase1_re_estarmap | 700 | 2 | 1 | 1 | 0 | 3 | 2.04 | 0.188 | 0.655 | 0.876 | 0.61 | 4.711e-16 | 2 | B2_temporal | 1 | 1 | 2.04 | fourier |
| re700_e2 | phase1_re_estarmap | 700 | 2 | 1 | 1 | 0 | 4 | 1.91 | 0.0195 | 0.942 | 0.273 | 0.888 | 4.482e-16 | 2 | B2_temporal | 0 | 1 | 1.91 | fourier |
| re700_e2 | phase1_re_estarmap | 700 | 2 | 1 | 1 | 0 | 5 | 1.89 | 0.32 | 0.561 | 1.1 | 0.748 | 4.850e-16 | 3 | B3_mixed | 1 | 0 | 1.89 | fourier |
| re700_e2 | phase1_re_estarmap | 700 | 2 | 1 | 1 | 0 | 6 | 2.13 | 0.519 | 0.408 | 1.49 | 0.793 | 4.427e-16 | 3 | B3_mixed | 1 | 0 | 2.13 | fourier |
| re700_e2 | phase1_re_estarmap | 700 | 2 | 1 | 1 | 0 | 7 | 1.89 | 0.747 | 0.229 | 1.68 | 0.943 | 4.213e-16 | 1 | B1_Mode2 | 1 | 0 | 1.89 | fourier |
| re700_e2 | phase1_re_estarmap | 700 | 2 | 1 | 1 | 0 | 8 | 1.87 | 0.0881 | 0.607 | 0.574 | 0.625 | 4.757e-16 | 2 | B2_temporal | 1 | 1 | 1.87 | fourier |
| re700_e2 | phase1_re_estarmap | 700 | 2 | 1 | 1 | 0 | 9 | 1.94 | 0.179 | 0.53 | 0.834 | 0.644 | 4.516e-16 | 2 | B2_temporal | 1 | 1 | 1.94 | fourier |
| re900_e0.1 | phase1_re_estarmap | 900 | 0.1 | 1 | 1 | 0 | 0 | 0.118 | 0.732 | 0.182 | -0.416 | 0.964 | 1.031e-16 | 1 | B1_Mode2 | 1 | 0 | 0.118 | fourier |
| re900_e0.1 | phase1_re_estarmap | 900 | 0.1 | 1 | 1 | 0 | 1 | 0.122 | 0.903 | 0.00547 | 0.47 | 0.989 | 1.040e-16 | 1 | B1_Mode2 | 1 | 0 | 0.122 | fourier |
| re900_e0.1 | phase1_re_estarmap | 900 | 0.1 | 1 | 1 | 0 | 2 | 0.127 | 0.895 | 0.00612 | 0.477 | 0.997 | 1.095e-16 | 1 | B1_Mode2 | 1 | 0 | 0.127 | fourier |
| re900_e0.1 | phase1_re_estarmap | 900 | 0.1 | 1 | 1 | 0 | 3 | 0.123 | 0.00194 | 0.995 | 0.0219 | 0.995 | 1.179e-16 | 2 | B2_temporal | 0 | 1 | 0.123 | fourier |
| re900_e0.1 | phase1_re_estarmap | 900 | 0.1 | 1 | 1 | 0 | 4 | 0.124 | 0.0712 | 0.929 | 0.133 | 0.996 | 1.203e-16 | 2 | B2_temporal | 0 | 1 | 0.124 | fourier |
| re900_e0.25 | phase1_re_estarmap | 900 | 0.25 | 1 | 1 | 0 | 0 | 0.24 | 0.419 | 0.537 | -0.449 | 0.922 | 1.676e-16 | 3 | B3_mixed | 0 | 1 | 0.24 | fourier |
| re900_e0.25 | phase1_re_estarmap | 900 | 0.25 | 1 | 1 | 0 | 1 | 0.266 | 0.947 | 0.022 | 0.71 | 0.985 | 1.514e-16 | 1 | B1_Mode2 | 1 | 0 | 0.266 | fourier |
| re900_e0.25 | phase1_re_estarmap | 900 | 0.25 | 1 | 1 | 0 | 2 | 0.27 | 0.915 | 0.0144 | 0.703 | 0.99 | 1.461e-16 | 1 | B1_Mode2 | 1 | 0 | 0.27 | fourier |
| re900_e0.25 | phase1_re_estarmap | 900 | 0.25 | 1 | 1 | 0 | 3 | 0.269 | 0.000264 | 0.996 | 0.0119 | 0.997 | 1.614e-16 | 2 | B2_temporal | 0 | 1 | 0.269 | fourier |
| re900_e0.25 | phase1_re_estarmap | 900 | 0.25 | 1 | 1 | 0 | 4 | 0.262 | 0.0205 | 0.979 | 0.104 | 0.993 | 1.764e-16 | 2 | B2_temporal | 0 | 1 | 0.262 | fourier |
| re900_e0.5 | phase1_re_estarmap | 900 | 0.5 | 1 | 1 | 0 | 0 | 0.463 | 0.361 | 0.612 | -0.578 | 0.915 | 2.169e-16 | 3 | B3_mixed | 0 | 1 | 0.463 | fourier |
| re900_e0.5 | phase1_re_estarmap | 900 | 0.5 | 1 | 1 | 0 | 1 | 0.517 | 0.943 | 0.0481 | 0.987 | 0.993 | 2.154e-16 | 1 | B1_Mode2 | 1 | 0 | 0.517 | fourier |
| re900_e0.5 | phase1_re_estarmap | 900 | 0.5 | 1 | 1 | 0 | 2 | 0.523 | 0.925 | 0.0375 | 0.983 | 0.975 | 2.054e-16 | 1 | B1_Mode2 | 1 | 0 | 0.523 | fourier |
| re900_e0.5 | phase1_re_estarmap | 900 | 0.5 | 1 | 1 | 0 | 3 | 0.522 | 2.505e-05 | 0.996 | -0.00512 | 0.997 | 2.447e-16 | 2 | B2_temporal | 0 | 1 | 0.522 | fourier |
| re900_e0.5 | phase1_re_estarmap | 900 | 0.5 | 1 | 1 | 0 | 4 | 0.497 | 0.000115 | 0.999 | -0.0107 | 0.992 | 2.396e-16 | 2 | B2_temporal | 0 | 1 | 0.497 | fourier |
| re900_e1 | phase1_re_estarmap | 900 | 1 | 1 | 1 | 0 | 0 | 0.961 | 0.193 | 0.376 | -0.608 | 0.805 | 3.049e-16 | 3 | B3_mixed | 2 | 0 | 0.961 | fourier |
| re900_e1 | phase1_re_estarmap | 900 | 1 | 1 | 1 | 0 | 1 | 1.03 | 0.97 | 0.0179 | 1.41 | 0.988 | 3.495e-16 | 1 | B1_Mode2 | 1 | 0 | 1.03 | fourier |
| re900_e1 | phase1_re_estarmap | 900 | 1 | 1 | 1 | 0 | 2 | 1.02 | 0.877 | 0.0805 | 1.34 | 0.94 | 2.946e-16 | 1 | B1_Mode2 | 1 | 0 | 1.02 | fourier |
| re900_e1 | phase1_re_estarmap | 900 | 1 | 1 | 1 | 0 | 3 | 0.991 | 0.114 | 0.865 | -0.475 | 0.96 | 3.016e-16 | 2 | B2_temporal | 0 | 1 | 0.991 | fourier |
| re900_e1 | phase1_re_estarmap | 900 | 1 | 1 | 1 | 0 | 4 | 0.953 | 0.0545 | 0.944 | 0.322 | 0.912 | 3.063e-16 | 2 | B2_temporal | 0 | 1 | 0.953 | fourier |
| re900_e2 | phase1_re_estarmap | 900 | 2 | 1 | 1 | 0 | 0 | 1.82 | 0.0205 | 0.151 | 0.273 | 0.88 | 4.106e-16 | 3 | B3_mixed | 2 | 0 | 1.82 | fourier |
| re900_e2 | phase1_re_estarmap | 900 | 2 | 1 | 1 | 0 | 1 | 2 | 0.739 | 0.194 | 1.72 | 0.878 | 4.390e-16 | 1 | B1_Mode2 | 1 | 0 | 2 | fourier |
| re900_e2 | phase1_re_estarmap | 900 | 2 | 1 | 1 | 0 | 2 | 2.07 | 0.717 | 0.22 | 1.72 | 0.873 | 4.237e-16 | 1 | B1_Mode2 | 1 | 0 | 2.07 | fourier |
| re900_e2 | phase1_re_estarmap | 900 | 2 | 1 | 1 | 0 | 3 | 1.96 | 0.0463 | 0.64 | 0.426 | 0.726 | 4.308e-16 | 2 | B2_temporal | 1 | 1 | 1.96 | fourier |
| re900_e2 | phase1_re_estarmap | 900 | 2 | 1 | 1 | 0 | 4 | 1.9 | 0.0857 | 0.909 | 0.571 | 0.873 | 4.583e-16 | 2 | B2_temporal | 0 | 1 | 1.9 | fourier |

### 1.4 Détail complet par course — pertes, amplitudes, modes

Table 5: Détail complet des pertes et modes — 185 courses, `phase1_all_results.csv`. $L$: perte totale ; $L_{pde/bc/d/c/\lambda/Re}$: composantes ; $E_e = |E_r - E^*|/E^*$.

| Re | $E_t$ | $E^*$ | s | point | $N_{MX}$ | $N_{MT}$ | AR | $\lambda_v$ | $\lambda_b$ | $E_r$ | $E_e$ | $f_2$ | $f_t$ | $A_1$ | $A_2$ | $A_3$ | $A_4$ | $C_3$ | epsilon | $L$ | $L_{pde}$ | $L_{bc}$ | $L_d$ | $L_c$ | $L_\lambda$ | $L_{Re}$ | branch | id | $i$ | $j$ | mode | conv. |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1e+03 | 0.1 | 0.1 | 0 | re1000_e0.1 | 6 | 5 | 1 | 1 | 0 | 0.12 | 0.88 | 0.796 | 0.0718 | -0.00235 | -0.437 | 0.138 | 0.0969 | 0.931 | 1.021e-16 | 2.21 | 0.0041 | 8.163e-05 | 0.00129 | 0.000422 | 0.121 | – | B1_Mode2 | 1 | 1 | 0 | (1,0) | 1 |
| 1e+03 | 0.1 | 0.1 | 1 | re1000_e0.1 | 6 | 5 | 1 | 1 | 0 | 0.122 | 0.878 | 0.903 | 0.00619 | -0.0124 | 0.47 | 0.101 | -0.104 | 0.989 | 9.684e-17 | 2.2 | 0.00328 | 5.565e-05 | 0.00121 | 0.000423 | 0.121 | – | B1_Mode2 | 1 | 1 | 0 | (1,0) | 1 |
| 1e+03 | 0.1 | 0.1 | 2 | re1000_e0.1 | 6 | 5 | 1 | 1 | 0 | 0.125 | 0.875 | 0.897 | 0.00436 | 0.0168 | 0.474 | -0.0328 | -0.152 | 0.994 | 1.006e-16 | 2.2 | 0.00273 | 5.219e-05 | 0.001 | 0.000424 | 0.121 | – | B1_Mode2 | 1 | 1 | 0 | (1,0) | 1 |
| 1e+03 | 0.1 | 0.1 | 3 | re1000_e0.1 | 6 | 5 | 1 | 1 | 0 | 0.122 | 0.878 | 0.0492 | 0.948 | -0.0211 | 0.109 | -0.00638 | 0.007 | 0.993 | 1.326e-16 | 2.21 | 0.00599 | 0.000142 | 0.0025 | 0.000421 | 0.121 | – | B2_temporal | 2 | 0 | 1 | (0,1) | 1 |
| 1e+03 | 0.1 | 0.1 | 4 | re1000_e0.1 | 6 | 5 | 1 | 1 | 0 | 0.125 | 0.875 | 0.0832 | 0.916 | -0.00461 | 0.144 | -0.00362 | -0.00735 | 0.993 | 1.197e-16 | 2.21 | 0.00465 | 9.543e-05 | 0.00181 | 0.000423 | 0.121 | – | B2_temporal | 2 | 0 | 1 | (0,1) | 1 |
| 1e+03 | 0.25 | 0.25 | 0 | re1000_e0.25 | 6 | 5 | 1 | 1 | 0 | 0.24 | 0.76 | 0.436 | 0.516 | 0.0488 | -0.458 | -0.00872 | 0.143 | 0.91 | 1.909e-16 | 1.38 | 0.00842 | 0.000165 | 0.00387 | 9.003e-05 | 0.259 | – | B3_mixed | 3 | 1 | 0 | (1,0) | 1 |
| 1e+03 | 0.25 | 0.25 | 1 | re1000_e0.25 | 6 | 5 | 1 | 1 | 0 | 0.266 | 0.734 | 0.936 | 0.0271 | -0.00698 | 0.706 | 0.0713 | -0.119 | 0.983 | 1.472e-16 | 1.38 | 0.00736 | 0.000181 | 0.00417 | 8.922e-05 | 0.259 | – | B1_Mode2 | 1 | 1 | 0 | (1,0) | 1 |
| 1e+03 | 0.25 | 0.25 | 2 | re1000_e0.25 | 6 | 5 | 1 | 1 | 0 | 0.264 | 0.736 | 0.922 | 0.00608 | 0.0419 | 0.698 | -0.0884 | -0.167 | 0.99 | 1.550e-16 | 1.38 | 0.00778 | 0.000168 | 0.00246 | 8.898e-05 | 0.259 | – | B1_Mode2 | 1 | 1 | 0 | (1,0) | 1 |
| 1e+03 | 0.25 | 0.25 | 3 | re1000_e0.25 | 6 | 5 | 1 | 1 | 0 | 0.264 | 0.736 | 0.0317 | 0.966 | -0.0305 | 0.129 | -0.0117 | 0.0115 | 0.994 | 1.680e-16 | 1.38 | 0.00548 | 0.000203 | 0.00658 | 9.031e-05 | 0.26 | – | B2_temporal | 2 | 0 | 1 | (0,1) | 1 |
| 1e+03 | 0.25 | 0.25 | 4 | re1000_e0.25 | 6 | 5 | 1 | 1 | 0 | 0.264 | 0.736 | 0.0386 | 0.961 | -0.00931 | 0.143 | 0.013 | 0.00128 | 0.991 | 1.722e-16 | 1.38 | 0.00803 | 0.000252 | 0.00376 | 8.999e-05 | 0.259 | – | B2_temporal | 2 | 0 | 1 | (0,1) | 1 |
| 1e+03 | 0.5 | 0.5 | 0 | re1000_e0.5 | 6 | 5 | 1 | 1 | 0 | 0.461 | 0.539 | 0.346 | 0.63 | 0.0727 | -0.565 | -0.000739 | 0.128 | 0.904 | 2.126e-16 | 0.705 | 0.0117 | 0.000316 | 0.00686 | 2.335e-05 | 0.505 | – | B3_mixed | 3 | 0 | 1 | (0,1) | 1 |
| 1e+03 | 0.5 | 0.5 | 1 | re1000_e0.5 | 6 | 5 | 1 | 1 | 0 | 0.516 | 0.484 | 0.953 | 0.0368 | 0.00546 | 0.992 | 0.0275 | -0.0986 | 0.993 | 2.351e-16 | 0.708 | 0.013 | 0.000447 | 0.011 | 2.280e-05 | 0.505 | – | B1_Mode2 | 1 | 1 | 0 | (1,0) | 1 |
| 1e+03 | 0.5 | 0.5 | 2 | re1000_e0.5 | 6 | 5 | 1 | 1 | 0 | 0.513 | 0.487 | 0.949 | 0.0127 | 0.0558 | 0.986 | -0.111 | -0.152 | 0.984 | 2.085e-16 | 0.705 | 0.0109 | 0.00039 | 0.00817 | 2.290e-05 | 0.505 | – | B1_Mode2 | 1 | 1 | 0 | (1,0) | 1 |
| 1e+03 | 0.5 | 0.5 | 3 | re1000_e0.5 | 6 | 5 | 1 | 1 | 0 | 0.516 | 0.484 | 0.016 | 0.981 | -0.0484 | 0.128 | -0.0123 | 0.0134 | 0.993 | 2.703e-16 | 0.703 | 0.00858 | 0.000383 | 0.0136 | 2.339e-05 | 0.505 | – | B2_temporal | 2 | 0 | 1 | (0,1) | 1 |
| 1e+03 | 0.5 | 0.5 | 4 | re1000_e0.5 | 6 | 5 | 1 | 1 | 0 | 0.497 | 0.503 | 0.00295 | 0.997 | 0.00893 | -0.0541 | -0.0202 | 0.00621 | 0.983 | 2.225e-16 | 0.717 | 0.0175 | 0.000701 | 0.00676 | 2.307e-05 | 0.505 | – | B2_temporal | 2 | 0 | 1 | (0,1) | 1 |
| 1e+03 | 1 | 1 | 0 | re1000_e1 | 6 | 5 | 1 | 1 | 0 | 0.887 | 0.113 | 0.00228 | 0.106 | 0.299 | 0.0637 | -1.15 | -0.218 | 0.871 | 2.850e-16 | 0.0526 | 0.0305 | 0.00145 | 0.0204 | 5.258e-06 | 1 | – | B3_mixed | 3 | 2 | 0 | (2,0) | 1 |
| 1e+03 | 1 | 1 | 1 | re1000_e1 | 6 | 5 | 1 | 1 | 0 | 1.03 | 0.0315 | 0.962 | 0.0197 | 0.039 | 1.41 | -0.0189 | -0.189 | 0.985 | 3.350e-16 | 0.044 | 0.0248 | 0.00127 | 0.0262 | 5.384e-06 | 1 | – | B1_Mode2 | 1 | 1 | 0 | (1,0) | 1 |
| 1e+03 | 1 | 1 | 10 | re1000_e1 | 6 | 5 | 1 | 1 | 0 | 1.06 | 0.0634 | 0.194 | 0.472 | 0.361 | -0.643 | -0.746 | 0.135 | 0.726 | 3.289e-16 | 0.0692 | 0.0448 | 0.00157 | 0.0141 | 4.783e-06 | 1 | – | B3_mixed | 3 | 1 | 1 | (1,1) | 1 |
| 1e+03 | 1 | 1 | 11 | re1000_e1 | 6 | 5 | 1 | 1 | 0 | 0.971 | 0.0287 | 0.215 | 0.371 | 0.344 | -0.647 | -0.77 | 0.293 | 0.709 | 2.906e-16 | 0.0599 | 0.037 | 0.00128 | 0.0151 | 5.046e-06 | 0.998 | – | B3_mixed | 3 | 2 | 0 | (2,0) | 1 |
| 1e+03 | 1 | 1 | 12 | re1000_e1 | 6 | 5 | 1 | 1 | 0 | 0.971 | 0.029 | 0.000254 | 0.989 | 0.0884 | -0.0222 | -0.112 | -0.021 | 0.98 | 3.514e-16 | 0.104 | 0.0582 | 0.00284 | 0.0213 | 4.886e-06 | 0.998 | – | B2_temporal | 2 | 0 | 1 | (0,1) | 1 |
| 1e+03 | 1 | 1 | 13 | re1000_e1 | 6 | 5 | 1 | 1 | 0 | 0.987 | 0.0132 | 0.949 | 0.0231 | 0.0424 | 1.37 | 0.0759 | -0.216 | 0.985 | 3.264e-16 | 0.0362 | 0.019 | 0.00114 | 0.026 | 5.555e-06 | 1 | – | B1_Mode2 | 1 | 1 | 0 | (1,0) | 1 |
| 1e+03 | 1 | 1 | 14 | re1000_e1 | 6 | 5 | 1 | 1 | 0 | 0.986 | 0.0139 | 0.681 | 0.154 | 0.159 | 1.16 | -0.401 | -0.363 | 0.835 | 2.912e-16 | 0.0394 | 0.0277 | 0.00077 | 0.0131 | 5.292e-06 | 1 | – | B1_Mode2 | 1 | 1 | 0 | (1,0) | 1 |
| 1e+03 | 1 | 1 | 15 | re1000_e1 | 6 | 5 | 1 | 1 | 0 | 1.04 | 0.0386 | 0.802 | 0.146 | -0.0691 | 1.29 | -0.0428 | -0.309 | 0.917 | 3.408e-16 | 0.0634 | 0.0441 | 0.00131 | 0.0159 | 4.786e-06 | 1 | – | B1_Mode2 | 1 | 1 | 0 | (1,0) | 1 |
| 1e+03 | 1 | 1 | 16 | re1000_e1 | 6 | 5 | 1 | 1 | 0 | 0.979 | 0.0207 | 0.831 | 0.104 | 0.0246 | 1.28 | 0.0957 | -0.303 | 0.913 | 2.909e-16 | 0.0622 | 0.0448 | 0.0012 | 0.0125 | 4.702e-06 | 1 | – | B1_Mode2 | 1 | 1 | 0 | (1,0) | 1 |
| 1e+03 | 1 | 1 | 17 | re1000_e1 | 6 | 5 | 1 | 1 | 0 | 0.986 | 0.0137 | 0.617 | 0.315 | 0.117 | -1.1 | -0.338 | 0.0764 | 0.854 | 3.455e-16 | 0.0527 | 0.0324 | 0.00135 | 0.016 | 5.115e-06 | 1 | – | B3_mixed | 3 | 1 | 0 | (1,0) | 1 |
| 1e+03 | 1 | 1 | 18 | re1000_e1 | 6 | 5 | 1 | 1 | 0 | 1.02 | 0.0185 | 0.974 | 0.0101 | 0.0249 | -1.41 | -0.00833 | 0.168 | 0.992 | 2.931e-16 | 0.0716 | 0.0481 | 0.00157 | 0.0166 | 4.712e-06 | 1 | – | B1_Mode2 | 1 | 1 | 0 | (1,0) | 1 |
| 1e+03 | 1 | 1 | 19 | re1000_e1 | 6 | 5 | 1 | 1 | 0 | 0.993 | 0.0073 | 0.699 | 0.265 | 0.0286 | -1.18 | -0.0798 | 0.247 | 0.894 | 3.250e-16 | 0.0587 | 0.0391 | 0.00135 | 0.0122 | 4.874e-06 | 1 | – | B1_Mode2 | 1 | 1 | 0 | (1,0) | 1 |
| 1e+03 | 1 | 1 | 2 | re1000_e1 | 6 | 5 | 1 | 1 | 0 | 1.02 | 0.0234 | 0.931 | 0.0558 | 0.052 | 1.38 | -0.0897 | -0.116 | 0.963 | 3.345e-16 | 0.0289 | 0.0178 | 0.000763 | 0.0141 | 5.476e-06 | 1 | – | B1_Mode2 | 1 | 1 | 0 | (1,0) | 1 |
| 1e+03 | 1 | 1 | 3 | re1000_e1 | 6 | 5 | 1 | 1 | 0 | 0.972 | 0.0284 | 0.00572 | 0.991 | -0.0821 | 0.105 | -0.0013 | 0.00515 | 0.984 | 3.384e-16 | 0.061 | 0.0344 | 0.00176 | 0.0283 | 5.438e-06 | 1 | – | B2_temporal | 2 | 0 | 1 | (0,1) | 1 |
| 1e+03 | 1 | 1 | 4 | re1000_e1 | 6 | 5 | 1 | 1 | 0 | 0.966 | 0.0339 | 0.0197 | 0.98 | 0.0245 | -0.195 | -0.00669 | 0.0243 | 0.959 | 3.367e-16 | 0.0575 | 0.0339 | 0.00161 | 0.0149 | 5.430e-06 | 1 | – | B2_temporal | 2 | 0 | 1 | (0,1) | 1 |
| 1e+03 | 1 | 1 | 5 | re1000_e1 | 6 | 5 | 1 | 1 | 0 | 0.989 | 0.0114 | 0.65 | 0.228 | 0.176 | 1.13 | -0.322 | -0.322 | 0.797 | 3.004e-16 | 0.0622 | 0.0423 | 0.00129 | 0.0135 | 4.832e-06 | 1 | – | B1_Mode2 | 1 | 1 | 0 | (1,0) | 1 |
| 1e+03 | 1 | 1 | 6 | re1000_e1 | 6 | 5 | 1 | 1 | 0 | 0.998 | 0.00154 | 0.918 | 0.0381 | -0.132 | 1.35 | 0.224 | -0.028 | 0.956 | 3.131e-16 | 0.0357 | 0.0233 | 0.000858 | 0.0137 | 5.462e-06 | 1 | – | B1_Mode2 | 1 | 1 | 0 | (1,0) | 1 |
| 1e+03 | 1 | 1 | 7 | re1000_e1 | 6 | 5 | 1 | 1 | 0 | 0.973 | 0.0273 | 0.943 | 0.021 | -0.108 | 1.35 | 0.107 | -0.214 | 0.976 | 2.921e-16 | 0.0385 | 0.0225 | 0.00107 | 0.022 | 5.464e-06 | 1 | – | B1_Mode2 | 1 | 1 | 0 | (1,0) | 1 |
| 1e+03 | 1 | 1 | 8 | re1000_e1 | 6 | 5 | 1 | 1 | 0 | 0.944 | 0.0558 | 0.135 | 0.539 | 0.274 | 0.505 | -0.65 | -0.321 | 0.647 | 3.023e-16 | 0.04 | 0.0238 | 0.00104 | 0.00961 | 5.587e-06 | 1 | – | B2_temporal | 2 | 0 | 1 | (0,1) | 1 |
| 1e+03 | 1 | 1 | 9 | re1000_e1 | 6 | 5 | 1 | 1 | 0 | 1.01 | 0.0105 | 0.477 | 0.348 | 0.262 | 0.982 | -0.432 | -0.312 | 0.717 | 3.112e-16 | 0.0491 | 0.0343 | 0.000956 | 0.0151 | 5.116e-06 | 1 | – | B3_mixed | 3 | 1 | 0 | (1,0) | 1 |
| 1e+03 | 2 | 2 | 0 | re1000_e2 | 6 | 5 | 1 | 1 | 0 | 1.81 | 0.809 | 0.0158 | 0.129 | 0.538 | 0.239 | -1.62 | -0.0644 | 0.887 | 4.455e-16 | -0.569 | 0.0525 | 0.00432 | 0.0535 | 9.413e-07 | 1.99 | – | B3_mixed | 3 | 2 | 0 | (2,0) | 1 |
| 1e+03 | 2 | 2 | 1 | re1000_e2 | 6 | 5 | 1 | 1 | 0 | 2.04 | 1.04 | 0.936 | 0.0427 | 0.103 | 1.96 | -0.0884 | -0.25 | 0.972 | 4.064e-16 | -0.602 | 0.0446 | 0.00295 | 0.0565 | 1.206e-06 | 2 | – | B1_Mode2 | 1 | 1 | 0 | (1,0) | 1 |
| 1e+03 | 2 | 2 | 2 | re1000_e2 | 6 | 5 | 1 | 1 | 0 | 1.93 | 0.934 | 0.554 | 0.117 | 0.437 | 1.46 | -1.04 | -0.0191 | 0.882 | 4.207e-16 | -0.612 | 0.0432 | 0.00227 | 0.0371 | 1.178e-06 | 2 | – | B1_Mode2 | 1 | 1 | 0 | (1,0) | 1 |
| 1e+03 | 2 | 2 | 3 | re1000_e2 | 6 | 5 | 1 | 1 | 0 | 1.94 | 0.941 | 0.0118 | 0.97 | 0.21 | -0.214 | -0.158 | -0.0589 | 0.968 | 4.390e-16 | -0.534 | 0.0757 | 0.00461 | 0.0556 | 1.077e-06 | 1.98 | – | B2_temporal | 2 | 0 | 1 | (0,1) | 1 |
| 1e+03 | 2 | 2 | 4 | re1000_e2 | 6 | 5 | 1 | 1 | 0 | 1.94 | 0.938 | 0.164 | 0.835 | 0.0113 | 0.797 | 0.0243 | -0.0587 | 0.942 | 4.412e-16 | -0.565 | 0.0562 | 0.00468 | 0.0344 | 1.181e-06 | 2 | – | B2_temporal | 2 | 0 | 1 | (0,1) | 1 |
| 100 | 0.1 | 0.1 | 0 | re100_e0.1 | 6 | 5 | 1 | 1 | 0 | 0.11 | 0.89 | 0.0163 | 0.979 | 0.0297 | -0.06 | -0.00461 | 0.0056 | 0.978 | 1.106e-16 | 2.2 | 0.002 | 8.338e-05 | 0.00774 | 0.000425 | 0.121 | – | B2_temporal | 2 | 0 | 1 | (0,1) | 1 |
| 100 | 0.1 | 0.1 | 1 | re100_e0.1 | 6 | 5 | 1 | 1 | 0 | 0.111 | 0.889 | 0.00474 | 0.995 | -0.0081 | -0.0325 | 0.00233 | 0.0015 | 0.989 | 1.121e-16 | 2.2 | 0.00208 | 9.109e-05 | 0.00921 | 0.000425 | 0.121 | – | B2_temporal | 2 | 0 | 1 | (0,1) | 1 |
| 100 | 0.1 | 0.1 | 2 | re100_e0.1 | 6 | 5 | 1 | 1 | 0 | 0.113 | 0.887 | 0.363 | 0.618 | 0.0285 | 0.286 | 0.00519 | -0.0584 | 0.966 | 1.129e-16 | 2.2 | 0.00117 | 5.543e-05 | 0.00943 | 0.000426 | 0.121 | – | B3_mixed | 3 | 0 | 1 | (0,1) | 1 |
| 100 | 0.1 | 0.1 | 3 | re100_e0.1 | 6 | 5 | 1 | 1 | 0 | 0.125 | 0.875 | 0.00679 | 0.991 | -0.0231 | 0.0413 | 0.00126 | -0.00404 | 0.993 | 1.133e-16 | 2.21 | 0.00381 | 0.00014 | 0.0112 | 0.000422 | 0.121 | – | B2_temporal | 2 | 0 | 1 | (0,1) | 1 |
| 100 | 0.1 | 0.1 | 4 | re100_e0.1 | 6 | 5 | 1 | 1 | 0 | 0.119 | 0.881 | 1.263e-05 | 0.999 | 0.00552 | 0.00174 | -0.0111 | 0.00464 | 0.995 | 1.129e-16 | 2.21 | 0.00356 | 0.000115 | 0.00851 | 0.000423 | 0.121 | – | B2_temporal | 2 | 0 | 1 | (0,1) | 1 |
| 100 | 0.25 | 0.25 | 0 | re100_e0.25 | 6 | 5 | 1 | 1 | 0 | 0.237 | 0.763 | 0.00677 | 0.99 | 0.0379 | -0.0567 | 0.00203 | 0.00376 | 0.98 | 1.659e-16 | 1.37 | 0.00385 | 0.00016 | 0.0169 | 9.080e-05 | 0.26 | – | B2_temporal | 2 | 0 | 1 | (0,1) | 1 |
| 100 | 0.25 | 0.25 | 1 | re100_e0.25 | 6 | 5 | 1 | 1 | 0 | 0.24 | 0.76 | 0.0111 | 0.988 | -0.0205 | -0.0729 | 0.0156 | -0.00136 | 0.986 | 1.599e-16 | 1.37 | 0.00373 | 0.000138 | 0.0197 | 9.089e-05 | 0.26 | – | B2_temporal | 2 | 0 | 1 | (0,1) | 1 |
| 100 | 0.25 | 0.25 | 2 | re100_e0.25 | 6 | 5 | 1 | 1 | 0 | 0.245 | 0.755 | 0.477 | 0.498 | 0.0407 | 0.484 | -0.00271 | -0.103 | 0.953 | 1.953e-16 | 1.37 | 0.00248 | 0.000112 | 0.021 | 9.131e-05 | 0.26 | – | B3_mixed | 3 | 1 | 0 | (1,0) | 1 |
| 100 | 0.25 | 0.25 | 3 | re100_e0.25 | 6 | 5 | 1 | 1 | 0 | 0.267 | 0.733 | 0.035 | 0.963 | -0.0325 | 0.137 | 0.00541 | -0.0136 | 0.993 | 1.639e-16 | 1.38 | 0.00457 | 0.000157 | 0.0239 | 9.051e-05 | 0.26 | – | B2_temporal | 2 | 0 | 1 | (0,1) | 1 |
| 100 | 0.25 | 0.25 | 4 | re100_e0.25 | 6 | 5 | 1 | 1 | 0 | 0.256 | 0.744 | 4.352e-06 | 1 | 0.00513 | 0.00149 | -0.00274 | 0.004 | 0.993 | 1.653e-16 | 1.38 | 0.00718 | 0.000253 | 0.0181 | 8.946e-05 | 0.259 | – | B2_temporal | 2 | 0 | 1 | (0,1) | 1 |
| 100 | 0.5 | 0.5 | 0 | re100_e0.5 | 6 | 5 | 1 | 1 | 0 | 0.463 | 0.537 | 0.000109 | 0.997 | 0.0501 | -0.01 | 0.0152 | 0.000405 | 0.981 | 2.269e-16 | 0.698 | 0.00501 | 0.000202 | 0.0339 | 2.376e-05 | 0.505 | – | B2_temporal | 2 | 0 | 1 | (0,1) | 1 |
| 100 | 0.5 | 0.5 | 1 | re100_e0.5 | 6 | 5 | 1 | 1 | 0 | 0.468 | 0.532 | 0.03 | 0.966 | -0.0433 | -0.168 | 0.0388 | 0.0103 | 0.965 | 2.642e-16 | 0.7 | 0.00591 | 0.000238 | 0.0393 | 2.362e-05 | 0.505 | – | B2_temporal | 2 | 0 | 1 | (0,1) | 1 |
| 100 | 0.5 | 0.5 | 2 | re100_e0.5 | 6 | 5 | 1 | 1 | 0 | 0.495 | 0.505 | 0.794 | 0.153 | 0.0482 | 0.887 | -0.0111 | -0.224 | 0.966 | 2.278e-16 | 0.698 | 0.00407 | 0.000212 | 0.0434 | 2.382e-05 | 0.505 | – | B1_Mode2 | 1 | 1 | 0 | (1,0) | 1 |
| 100 | 0.5 | 0.5 | 3 | re100_e0.5 | 6 | 5 | 1 | 1 | 0 | 0.491 | 0.509 | 0.918 | 0.00511 | -0.00392 | 0.949 | 0.101 | -0.25 | 0.992 | 2.249e-16 | 0.701 | 0.00681 | 0.00027 | 0.0383 | 2.342e-05 | 0.505 | – | B1_Mode2 | 1 | 1 | 0 | (1,0) | 1 |
| 100 | 0.5 | 0.5 | 4 | re100_e0.5 | 6 | 5 | 1 | 1 | 0 | 0.493 | 0.507 | 0.00557 | 0.994 | 0.00673 | -0.0741 | -0.00244 | 0.023 | 0.984 | 3.285e-16 | 0.71 | 0.0126 | 0.0005 | 0.0351 | 2.286e-05 | 0.505 | – | B2_temporal | 2 | 0 | 1 | (0,1) | 1 |
| 100 | 1 | 1 | 0 | re100_e1 | 6 | 5 | 1 | 1 | 0 | 0.913 | 0.0873 | 0.047 | 0.946 | 0.0505 | -0.293 | 0.0822 | 0.059 | 0.929 | 3.292e-16 | 0.0178 | 0.0098 | 0.000394 | 0.0664 | 5.807e-06 | 1 | – | B2_temporal | 2 | 0 | 1 | (0,1) | 1 |
| 100 | 1 | 1 | 1 | re100_e1 | 6 | 5 | 1 | 1 | 0 | 0.921 | 0.0787 | 0.149 | 0.843 | -0.0696 | -0.524 | 0.0884 | 0.0454 | 0.942 | 3.091e-16 | 0.0236 | 0.0134 | 0.000511 | 0.0774 | 5.691e-06 | 1 | – | B2_temporal | 2 | 0 | 1 | (0,1) | 1 |
| 100 | 1 | 1 | 2 | re100_e1 | 6 | 5 | 1 | 1 | 0 | 1.01 | 0.0115 | 0.887 | 0.0485 | 0.071 | 1.34 | -0.0164 | -0.351 | 0.981 | 3.334e-16 | 0.0143 | 0.00642 | 0.000304 | 0.0905 | 5.919e-06 | 1 | – | B1_Mode2 | 1 | 1 | 0 | (1,0) | 1 |
| 100 | 1 | 1 | 3 | re100_e1 | 6 | 5 | 1 | 1 | 0 | 0.983 | 0.0171 | 0.912 | 0.00156 | 0.0279 | 1.34 | 0.108 | -0.386 | 0.994 | 2.904e-16 | 0.0212 | 0.011 | 0.000477 | 0.0813 | 5.726e-06 | 1 | – | B1_Mode2 | 1 | 1 | 0 | (1,0) | 1 |
| 100 | 1 | 1 | 4 | re100_e1 | 6 | 5 | 1 | 1 | 0 | 0.981 | 0.019 | 0.00101 | 0.998 | 0.0139 | -0.0445 | -0.0175 | 0.0202 | 0.989 | 3.261e-16 | 0.0354 | 0.0203 | 0.000846 | 0.072 | 5.496e-06 | 1 | – | B2_temporal | 2 | 0 | 1 | (0,1) | 1 |
| 100 | 2 | 2 | 0 | re100_e2 | 6 | 5 | 1 | 1 | 0 | 1.88 | 0.883 | 0.228 | 0.562 | 0.521 | -0.927 | -0.654 | 0.3 | 0.655 | 4.715e-16 | -0.599 | 0.0406 | 0.00276 | 0.146 | 1.148e-06 | 1.99 | – | B2_temporal | 2 | 1 | 1 | (1,1) | 1 |
| 100 | 2 | 2 | 1 | re100_e2 | 6 | 5 | 1 | 1 | 0 | 1.9 | 0.898 | 0.571 | 0.281 | 0.218 | -1.47 | -0.446 | 0.556 | 0.781 | 4.547e-16 | -0.625 | 0.0327 | 0.00154 | 0.192 | 1.228e-06 | 1.99 | – | B1_Mode2 | 1 | 1 | 0 | (1,0) | 1 |
| 100 | 2 | 2 | 2 | re100_e2 | 6 | 5 | 1 | 1 | 0 | 1.99 | 0.987 | 0.844 | 0.0427 | 0.27 | 1.83 | -0.37 | -0.488 | 0.939 | 4.218e-16 | -0.656 | 0.0172 | 0.000626 | 0.195 | 1.363e-06 | 2 | – | B1_Mode2 | 1 | 1 | 0 | (1,0) | 1 |
| 100 | 2 | 2 | 3 | re100_e2 | 6 | 5 | 1 | 1 | 0 | 1.89 | 0.895 | 0.884 | 0.0111 | 0.183 | 1.83 | -0.154 | -0.581 | 0.982 | 4.183e-16 | -0.658 | 0.0177 | 0.000559 | 0.173 | 1.349e-06 | 2 | – | B1_Mode2 | 1 | 1 | 0 | (1,0) | 1 |
| 100 | 2 | 2 | 4 | re100_e2 | 6 | 5 | 1 | 1 | 0 | 1.86 | 0.863 | 0.135 | 0.623 | 0.474 | -0.711 | -0.74 | 0.329 | 0.69 | 4.478e-16 | -0.609 | 0.0417 | 0.00176 | 0.144 | 1.155e-06 | 1.98 | – | B2_temporal | 2 | 0 | 1 | (0,1) | 1 |
| 300 | 0.1 | 0.1 | 0 | re300_e0.1 | 6 | 5 | 1 | 1 | 0 | 0.109 | 0.891 | 0.0897 | 0.905 | 0.0275 | -0.14 | 0.0054 | 0.0169 | 0.977 | 1.103e-16 | 2.2 | 0.00189 | 6.590e-05 | 0.00369 | 0.000425 | 0.121 | – | B2_temporal | 2 | 0 | 1 | (0,1) | 1 |

Table 5: Détail complet des pertes et modes — 185 courses, `phase1_all_results.csv`. $L$: perte totale ; $L_{pde/bc/d/c/\lambda/Re}$: composantes ; $E_e = |E_r - E^*|/E^*$.

| Re | $E_t$ | $E^*$ | s | point | $N_{MX}$ | $N_{MT}$ | AR | $\lambda_v$ | $\lambda_b$ | $E_r$ | $E_e$ | $f_2$ | $f_t$ | $A_1$ | $A_2$ | $A_3$ | $A_4$ | $C_3$ | epsilon | $L$ | $L_{pde}$ | $L_{bc}$ | $L_d$ | $L_c$ | $L_\lambda$ | $L_{Re}$ | branch | id | $i$ | $j$ | mode | conv. |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 300 | 0.1 | 0.1 | 1 | re300_e0.1 | 6 | 5 | 1 | 1 | 0 | 0.123 | 0.877 | 0.89 | 0.0117 | -0.00266 | 0.467 | 0.0798 | -0.129 | 0.984 | 1.084e-16 | 2.2 | 0.00238 | 4.547e-05 | 0.00345 | 0.000424 | 0.121 | – | B1_Mode2 | 1 | 1 | 0 | (1,0) | 1 |
| 300 | 0.1 | 0.1 | 2 | re300_e0.1 | 6 | 5 | 1 | 1 | 0 | 0.12 | 0.88 | 0.878 | 0.0426 | 0.0257 | 0.46 | -0.0385 | -0.13 | 0.981 | 1.004e-16 | 2.2 | 0.00182 | 5.237e-05 | 0.0034 | 0.000425 | 0.121 | – | B1_Mode2 | 1 | 1 | 0 | (1,0) | 1 |
| 300 | 0.1 | 0.1 | 3 | re300_e0.1 | 6 | 5 | 1 | 1 | 0 | 0.124 | 0.876 | 0.206 | 0.792 | -0.0125 | 0.226 | -0.00559 | -0.0224 | 0.992 | 1.155e-16 | 2.21 | 0.00402 | 0.000103 | 0.00603 | 0.000422 | 0.121 | – | B2_temporal | 2 | 0 | 1 | (0,1) | 1 |
| 300 | 0.1 | 0.1 | 4 | re300_e0.1 | 6 | 5 | 1 | 1 | 0 | 0.117 | 0.883 | 0.0913 | 0.901 | 0.0177 | -0.146 | -0.0317 | 0.0193 | 0.966 | 1.101e-16 | 2.23 | 0.0235 | 0.000325 | 0.00385 | 0.000407 | 0.12 | – | B2_temporal | 2 | 0 | 1 | (0,1) | 1 |
| 300 | 0.25 | 0.25 | 0 | re300_e0.25 | 6 | 5 | 1 | 1 | 0 | 0.233 | 0.767 | 0.0618 | 0.933 | 0.0385 | -0.17 | 0.0204 | 0.0265 | 0.97 | 1.537e-16 | 1.37 | 0.00326 | 0.000136 | 0.0085 | 9.104e-05 | 0.26 | – | B2_temporal | 2 | 0 | 1 | (0,1) | 1 |
| 300 | 0.25 | 0.25 | 1 | re300_e0.25 | 6 | 5 | 1 | 1 | 0 | 0.243 | 0.757 | 0.51 | 0.453 | -0.0441 | -0.497 | 0.0908 | 0.0796 | 0.935 | 1.652e-16 | 1.38 | 0.00585 | 0.000206 | 0.00902 | 8.982e-05 | 0.259 | – | B3_mixed | 3 | 1 | 0 | (1,0) | 1 |
| 300 | 0.25 | 0.25 | 2 | re300_e0.25 | 6 | 5 | 1 | 1 | 0 | 0.255 | 0.745 | 0.827 | 0.129 | 0.0352 | 0.65 | -0.0117 | -0.144 | 0.962 | 1.654e-16 | 1.37 | 0.00359 | 0.000129 | 0.00883 | 9.072e-05 | 0.26 | – | B1_Mode2 | 1 | 1 | 0 | (1,0) | 1 |
| 300 | 0.25 | 0.25 | 3 | re300_e0.25 | 6 | 5 | 1 | 1 | 0 | 0.265 | 0.735 | 0.159 | 0.838 | -0.0302 | 0.29 | 0.0107 | -0.0217 | 0.994 | 1.721e-16 | 1.38 | 0.00678 | 0.000215 | 0.0127 | 8.980e-05 | 0.259 | – | B2_temporal | 2 | 0 | 1 | (0,1) | 1 |
| 300 | 0.25 | 0.25 | 4 | re300_e0.25 | 6 | 5 | 1 | 1 | 0 | 0.251 | 0.749 | 0.2 | 0.777 | 0.0378 | -0.317 | -0.0757 | 0.0641 | 0.926 | 1.657e-16 | 1.4 | 0.029 | 0.000443 | 0.00861 | 8.505e-05 | 0.259 | – | B2_temporal | 2 | 0 | 1 | (0,1) | 1 |
| 300 | 0.5 | 0.5 | 0 | re300_e0.5 | 6 | 5 | 1 | 1 | 0 | 0.456 | 0.544 | 0.0526 | 0.943 | 0.0519 | -0.219 | 0.036 | 0.0145 | 0.944 | 2.283e-16 | 0.699 | 0.00587 | 0.000253 | 0.0163 | 2.366e-05 | 0.505 | – | B2_temporal | 2 | 0 | 1 | (0,1) | 1 |
| 300 | 0.5 | 0.5 | 1 | re300_e0.5 | 6 | 5 | 1 | 1 | 0 | 0.479 | 0.521 | 0.738 | 0.244 | -0.0336 | -0.841 | 0.0374 | 0.116 | 0.943 | 2.285e-16 | 0.71 | 0.0138 | 0.000492 | 0.0168 | 2.262e-05 | 0.505 | – | B1_Mode2 | 1 | 1 | 0 | (1,0) | 1 |
| 300 | 0.5 | 0.5 | 2 | re300_e0.5 | 6 | 5 | 1 | 1 | 0 | 0.499 | 0.501 | 0.876 | 0.0963 | 0.0361 | 0.935 | 0.0224 | -0.159 | 0.968 | 2.331e-16 | 0.699 | 0.00603 | 0.000245 | 0.0178 | 2.357e-05 | 0.505 | – | B1_Mode2 | 1 | 1 | 0 | (1,0) | 1 |
| 300 | 0.5 | 0.5 | 3 | re300_e0.5 | 6 | 5 | 1 | 1 | 0 | 0.51 | 0.49 | 0.16 | 0.837 | -0.043 | 0.404 | 0.0175 | -0.0274 | 0.981 | 2.717e-16 | 0.701 | 0.00654 | 0.00032 | 0.0239 | 2.350e-05 | 0.505 | – | B2_temporal | 2 | 0 | 1 | (0,1) | 1 |
| 300 | 0.5 | 0.5 | 4 | re300_e0.5 | 6 | 5 | 1 | 1 | 0 | 0.488 | 0.512 | 0.0953 | 0.893 | 0.0517 | -0.305 | -0.0759 | 0.0454 | 0.922 | 2.409e-16 | 0.752 | 0.0449 | 0.00118 | 0.0202 | 2.133e-05 | 0.504 | – | B2_temporal | 2 | 0 | 1 | (0,1) | 1 |
| 300 | 1 | 1 | 0 | re300_e1 | 6 | 5 | 1 | 1 | 0 | 0.903 | 0.0969 | 0.277 | 0.709 | 0.12 | -0.708 | 0.0151 | 0.0715 | 0.903 | 3.126e-16 | 0.0224 | 0.0122 | 0.000661 | 0.03 | 5.720e-06 | 1 | – | B2_temporal | 2 | 0 | 1 | (0,1) | 1 |
| 300 | 1 | 1 | 1 | re300_e1 | 6 | 5 | 1 | 1 | 0 | 0.93 | 0.0699 | 0.628 | 0.36 | 0.021 | -1.08 | -0.118 | 0.0834 | 0.926 | 2.866e-16 | 0.0464 | 0.0279 | 0.00119 | 0.0343 | 5.190e-06 | 1 | – | B3_mixed | 3 | 1 | 0 | (1,0) | 1 |
| 300 | 1 | 1 | 2 | re300_e1 | 6 | 5 | 1 | 1 | 0 | 0.993 | 0.00697 | 0.894 | 0.066 | 0.0938 | 1.33 | -0.0842 | -0.245 | 0.961 | 2.960e-16 | 0.0145 | 0.00894 | 0.000327 | 0.0375 | 5.832e-06 | 1 | – | B1_Mode2 | 1 | 1 | 0 | (1,0) | 1 |
| 300 | 1 | 1 | 3 | re300_e1 | 6 | 5 | 1 | 1 | 0 | 0.986 | 0.0143 | 0.87 | 0.0655 | -0.0409 | 1.31 | 0.293 | -0.139 | 0.94 | 3.458e-16 | 0.0272 | 0.0175 | 0.000618 | 0.0308 | 5.522e-06 | 1 | – | B1_Mode2 | 1 | 1 | 0 | (1,0) | 1 |
| 300 | 1 | 1 | 4 | re300_e1 | 6 | 5 | 1 | 1 | 0 | 0.966 | 0.0339 | 0.042 | 0.954 | 0.0618 | -0.285 | -0.0441 | 0.0415 | 0.949 | 3.220e-16 | 0.077 | 0.049 | 0.0018 | 0.0368 | 5.083e-06 | 1 | – | B2_temporal | 2 | 0 | 1 | (0,1) | 1 |
| 300 | 2 | 2 | 0 | re300_e2 | 6 | 5 | 1 | 1 | 0 | 1.8 | 0.8 | 0.188 | 0.772 | 0.289 | -0.824 | -0.227 | 0.0544 | 0.818 | 4.599e-16 | -0.626 | 0.0332 | 0.00205 | 0.0614 | 1.223e-06 | 2 | – | B2_temporal | 2 | 0 | 1 | (0,1) | 1 |
| 300 | 2 | 2 | 1 | re300_e2 | 6 | 5 | 1 | 1 | 0 | 1.97 | 0.968 | 0.231 | 0.552 | 0.474 | -0.953 | -0.782 | 0.134 | 0.592 | 4.657e-16 | -0.547 | 0.0812 | 0.00334 | 0.0773 | 9.254e-07 | 1.98 | – | B2_temporal | 2 | 1 | 0 | (1,0) | 1 |
| 300 | 2 | 2 | 2 | re300_e2 | 6 | 5 | 1 | 1 | 0 | 1.97 | 0.972 | 0.738 | 0.0788 | 0.438 | 1.71 | -0.625 | -0.365 | 0.886 | 4.161e-16 | -0.651 | 0.0231 | 0.000823 | 0.0959 | 1.315e-06 | 2 | – | B1_Mode2 | 1 | 1 | 0 | (1,0) | 1 |
| 300 | 2 | 2 | 3 | re300_e2 | 6 | 5 | 1 | 1 | 0 | 1.91 | 0.91 | 0.669 | 0.222 | 0.362 | 1.6 | -0.496 | -0.195 | 0.827 | 4.317e-16 | -0.63 | 0.0341 | 0.0015 | 0.0773 | 1.196e-06 | 2 | – | B1_Mode2 | 1 | 1 | 0 | (1,0) | 1 |
| 300 | 2 | 2 | 4 | re300_e2 | 6 | 5 | 1 | 1 | 0 | 1.87 | 0.867 | 0.202 | 0.659 | 0.44 | -0.869 | -0.529 | 0.197 | 0.796 | 4.499e-16 | -0.562 | 0.0668 | 0.00333 | 0.0637 | 1.067e-06 | 1.98 | – | B2_temporal | 2 | 0 | 1 | (0,1) | 1 |
| 500 | 0.1 | 0.1 | 0 | re500_e0.1 | 6 | 5 | 1 | 1 | 0 | 0.122 | 0.878 | 0.845 | 0.0357 | -0.00283 | -0.454 | 0.131 | 0.0998 | 0.955 | 1.016e-16 | 2.21 | 0.00381 | 7.498e-05 | 0.002 | 0.000422 | 0.121 | – | B1_Mode2 | 1 | 1 | 0 | (1,0) | 1 |
| 500 | 0.1 | 0.1 | 1 | re500_e0.1 | 6 | 5 | 1 | 1 | 0 | 0.123 | 0.877 | 0.887 | 0.00714 | -0.00525 | 0.467 | 0.089 | -0.129 | 0.988 | 1.016e-16 | 2.2 | 0.00236 | 4.246e-05 | 0.0022 | 0.000424 | 0.121 | – | B1_Mode2 | 1 | 1 | 0 | (1,0) | 1 |
| 500 | 0.1 | 0.1 | 2 | re500_e0.1 | 6 | 5 | 1 | 1 | 0 | 0.121 | 0.879 | 0.884 | 0.018 | 0.0234 | 0.463 | -0.0481 | -0.143 | 0.978 | 1.028e-16 | 2.2 | 0.0022 | 4.996e-05 | 0.0021 | 0.000424 | 0.121 | – | B1_Mode2 | 1 | 1 | 0 | (1,0) | 1 |
| 500 | 0.1 | 0.1 | 3 | re500_e0.1 | 6 | 5 | 1 | 1 | 0 | 0.126 | 0.874 | 0.0167 | 0.979 | -0.0291 | 0.0648 | 0.0123 | 0.00266 | 0.987 | 1.218e-16 | 2.2 | 0.00186 | 7.189e-05 | 0.00485 | 0.000425 | 0.121 | – | B2_temporal | 2 | 0 | 1 | (0,1) | 1 |
| 500 | 0.1 | 0.1 | 4 | re500_e0.1 | 6 | 5 | 1 | 1 | 0 | 0.116 | 0.884 | 0.651 | 0.312 | 0.0291 | -0.389 | -0.0551 | 0.0666 | 0.964 | 1.147e-16 | 2.22 | 0.0116 | 0.000264 | 0.00186 | 0.000413 | 0.12 | – | B1_Mode2 | 1 | 1 | 0 | (1,0) | 1 |
| 500 | 0.25 | 0.25 | 0 | re500_e0.25 | 6 | 5 | 1 | 1 | 0 | 0.24 | 0.76 | 0.52 | 0.448 | 0.0548 | -0.499 | -0.00319 | 0.109 | 0.941 | 1.596e-16 | 1.38 | 0.00694 | 0.000172 | 0.00498 | 8.985e-05 | 0.259 | – | B3_mixed | 3 | 1 | 0 | (1,0) | 1 |
| 500 | 0.25 | 0.25 | 1 | re500_e0.25 | 6 | 5 | 1 | 1 | 0 | 0.267 | 0.733 | 0.911 | 0.0133 | 0.00292 | 0.698 | 0.0698 | -0.187 | 0.986 | 1.671e-16 | 1.38 | 0.00676 | 0.000156 | 0.00492 | 8.925e-05 | 0.259 | – | B1_Mode2 | 1 | 1 | 0 | (1,0) | 1 |
| 500 | 0.25 | 0.25 | 10 | re500_e0.25 | 6 | 5 | 1 | 1 | 0 | 0.293 | 0.707 | 0.0126 | 0.979 | -0.0582 | 0.086 | 0.0333 | -0.0129 | 0.974 | 1.902e-16 | 1.37 | 0.00414 | 0.000182 | 0.00901 | 9.061e-05 | 0.26 | – | B2_temporal | 2 | 0 | 1 | (0,1) | 1 |
| 500 | 0.25 | 0.25 | 11 | re500_e0.25 | 6 | 5 | 1 | 1 | 0 | 0.267 | 0.733 | 0.871 | 0.015 | 0.0756 | 0.682 | -0.0767 | -0.221 | 0.973 | 1.486e-16 | 1.38 | 0.00698 | 0.000146 | 0.00527 | 8.919e-05 | 0.259 | – | B1_Mode2 | 1 | 1 | 0 | (1,0) | 1 |
| 500 | 0.25 | 0.25 | 12 | re500_e0.25 | 6 | 5 | 1 | 1 | 0 | 0.238 | 0.762 | 0.756 | 0.229 | -0.0165 | 0.6 | -0.0459 | -0.0693 | 0.978 | 1.503e-16 | 1.38 | 0.00662 | 0.000148 | 0.00505 | 8.964e-05 | 0.259 | – | B1_Mode2 | 1 | 1 | 0 | (1,0) | 1 |
| 500 | 0.25 | 0.25 | 13 | re500_e0.25 | 6 | 5 | 1 | 1 | 0 | 0.256 | 0.744 | 0.886 | 0.0498 | -0.0156 | 0.673 | 0.121 | -0.131 | 0.949 | 1.691e-16 | 1.37 | 0.00347 | 0.00011 | 0.00704 | 9.082e-05 | 0.26 | – | B1_Mode2 | 1 | 1 | 0 | (1,0) | 1 |
| 500 | 0.25 | 0.25 | 14 | re500_e0.25 | 6 | 5 | 1 | 1 | 0 | 0.25 | 0.75 | 0.019 | 0.975 | 0.024 | 0.0975 | 0.0423 | -0.0207 | 0.978 | 1.715e-16 | 1.37 | 0.00374 | 0.000147 | 0.00686 | 9.090e-05 | 0.26 | – | B2_temporal | 2 | 0 | 1 | (0,1) | 1 |
| 500 | 0.25 | 0.25 | 2 | re500_e0.25 | 6 | 5 | 1 | 1 | 0 | 0.257 | 0.743 | 0.89 | 0.0465 | 0.0388 | 0.677 | -0.0567 | -0.162 | 0.964 | 1.473e-16 | 1.37 | 0.00452 | 0.000119 | 0.0054 | 9.027e-05 | 0.26 | – | B1_Mode2 | 1 | 1 | 0 | (1,0) | 1 |
| 500 | 0.25 | 0.25 | 3 | re500_e0.25 | 6 | 5 | 1 | 1 | 0 | 0.268 | 0.732 | 0.0162 | 0.982 | -0.0325 | 0.0931 | -0.000253 | 0.00778 | 0.991 | 1.634e-16 | 1.37 | 0.0029 | 0.000139 | 0.0103 | 9.119e-05 | 0.26 | – | B2_temporal | 2 | 0 | 1 | (0,1) | 1 |
| 500 | 0.25 | 0.25 | 4 | re500_e0.25 | 6 | 5 | 1 | 1 | 0 | 0.25 | 0.75 | 0.343 | 0.637 | 0.0363 | -0.414 | -0.0568 | 0.0698 | 0.945 | 1.766e-16 | 1.4 | 0.0278 | 0.00052 | 0.00618 | 8.514e-05 | 0.259 | – | B3_mixed | 3 | 0 | 1 | (0,1) | 1 |
| 500 | 0.25 | 0.25 | 5 | re500_e0.25 | 6 | 5 | 1 | 1 | 0 | 0.245 | 0.755 | 4.164e-06 | 0.998 | 0.00675 | -0.00143 | 0.0314 | 0.00799 | 0.991 | 1.568e-16 | 1.37 | 0.00316 | 0.000177 | 0.0077 | 9.090e-05 | 0.26 | – | B2_temporal | 2 | 0 | 1 | (0,1) | 1 |
| 500 | 0.25 | 0.25 | 6 | re500_e0.25 | 6 | 5 | 1 | 1 | 0 | 0.264 | 0.736 | 0.868 | 0.0095 | -0.0917 | 0.677 | 0.178 | -0.132 | 0.961 | 1.479e-16 | 1.38 | 0.00649 | 0.000125 | 0.00455 | 8.952e-05 | 0.259 | – | B1_Mode2 | 1 | 1 | 0 | (1,0) | 1 |
| 500 | 0.25 | 0.25 | 7 | re500_e0.25 | 6 | 5 | 1 | 1 | 0 | 0.251 | 0.749 | 0.93 | 0.00902 | -0.0612 | 0.683 | 0.0722 | -0.147 | 0.983 | 1.466e-16 | 1.37 | 0.00539 | 9.587e-05 | 0.00475 | 9.008e-05 | 0.259 | – | B1_Mode2 | 1 | 1 | 0 | (1,0) | 1 |
| 500 | 0.25 | 0.25 | 8 | re500_e0.25 | 6 | 5 | 1 | 1 | 0 | 0.251 | 0.749 | 0.0341 | 0.963 | -0.0102 | 0.131 | -0.0306 | -0.0256 | 0.979 | 1.818e-16 | 1.37 | 0.00292 | 0.000123 | 0.0075 | 9.130e-05 | 0.26 | – | B2_temporal | 2 | 0 | 1 | (0,1) | 1 |
| 500 | 0.25 | 0.25 | 9 | re500_e0.25 | 6 | 5 | 1 | 1 | 0 | 0.265 | 0.735 | 0.000245 | 0.996 | -0.0392 | -0.0114 | -0.00832 | 0.0151 | 0.989 | 1.662e-16 | 1.37 | 0.00267 | 0.000114 | 0.0096 | 9.133e-05 | 0.26 | – | B2_temporal | 2 | 0 | 1 | (0,1) | 1 |
| 500 | 0.5 | 0.5 | 0 | re500_e0.5 | 6 | 5 | 1 | 1 | 0 | 0.453 | 0.547 | 0.222 | 0.765 | 0.0594 | -0.449 | 0.0572 | 0.0628 | 0.935 | 2.491e-16 | 0.702 | 0.00856 | 0.000295 | 0.0107 | 2.337e-05 | 0.505 | – | B2_temporal | 2 | 0 | 1 | (0,1) | 1 |
| 500 | 0.5 | 0.5 | 1 | re500_e0.5 | 6 | 5 | 1 | 1 | 0 | 0.516 | 0.484 | 0.925 | 0.0254 | -0.000468 | 0.977 | 0.097 | -0.203 | 0.974 | 2.259e-16 | 0.709 | 0.014 | 0.0004 | 0.0101 | 2.247e-05 | 0.505 | – | B1_Mode2 | 1 | 1 | 0 | (1,0) | 1 |
| 500 | 0.5 | 0.5 | 2 | re500_e0.5 | 6 | 5 | 1 | 1 | 0 | 0.504 | 0.496 | 0.925 | 0.0429 | 0.0352 | 0.965 | -0.00779 | -0.169 | 0.972 | 2.262e-16 | 0.698 | 0.00639 | 0.000219 | 0.0108 | 2.351e-05 | 0.505 | – | B1_Mode2 | 1 | 1 | 0 | (1,0) | 1 |
| 500 | 0.5 | 0.5 | 3 | re500_e0.5 | 6 | 5 | 1 | 1 | 0 | 0.518 | 0.482 | 0.0703 | 0.927 | -0.0508 | 0.27 | 0.0221 | -0.00284 | 0.986 | 2.732e-16 | 0.697 | 0.00408 | 0.00025 | 0.0198 | 2.383e-05 | 0.505 | – | B2_temporal | 2 | 0 | 1 | (0,1) | 1 |
| 500 | 0.5 | 0.5 | 4 | re500_e0.5 | 6 | 5 | 1 | 1 | 0 | 0.49 | 0.51 | 0.0492 | 0.94 | 0.0525 | -0.22 | -0.0842 | 0.0263 | 0.935 | 2.418e-16 | 0.749 | 0.0416 | 0.0012 | 0.013 | 2.151e-05 | 0.505 | – | B2_temporal | 2 | 0 | 1 | (0,1) | 1 |
| 500 | 1 | 1 | 0 | re500_e1 | 6 | 5 | 1 | 1 | 0 | 0.919 | 0.0812 | 0.282 | 0.707 | 0.0907 | -0.72 | 0.0599 | 0.0708 | 0.87 | 3.120e-16 | 0.0299 | 0.0191 | 0.000729 | 0.0209 | 5.546e-06 | 1 | – | B2_temporal | 2 | 0 | 1 | (0,1) | 1 |
| 500 | 1 | 1 | 1 | re500_e1 | 6 | 5 | 1 | 1 | 0 | 1.01 | 0.00945 | 0.643 | 0.189 | 0.198 | 1.14 | -0.501 | -0.218 | 0.845 | 2.924e-16 | 0.0325 | 0.0223 | 0.000676 | 0.0243 | 5.389e-06 | 1 | – | B1_Mode2 | 1 | 1 | 0 | (1,0) | 1 |
| 500 | 1 | 1 | 2 | re500_e1 | 6 | 5 | 1 | 1 | 0 | 1 | 0.00062 | 0.961 | 0.0189 | 0.0486 | 1.39 | -0.0269 | -0.182 | 0.984 | 3.185e-16 | 0.0194 | 0.0134 | 0.000406 | 0.0213 | 5.713e-06 | 1 | – | B1_Mode2 | 1 | 1 | 0 | (1,0) | 1 |
| 500 | 1 | 1 | 3 | re500_e1 | 6 | 5 | 1 | 1 | 0 | 0.972 | 0.028 | 0.959 | 0.0218 | 0.0137 | 1.37 | 0.104 | -0.142 | 0.98 | 2.952e-16 | 0.0331 | 0.0218 | 0.00076 | 0.0186 | 5.408e-06 | 1 | – | B1_Mode2 | 1 | 1 | 0 | (1,0) | 1 |
| 500 | 1 | 1 | 4 | re500_e1 | 6 | 5 | 1 | 1 | 0 | 0.963 | 0.037 | 0.0105 | 0.98 | 0.079 | -0.142 | -0.109 | 0.0266 | 0.943 | 3.156e-16 | 0.0946 | 0.0544 | 0.00263 | 0.0274 | 5.005e-06 | 1 | – | B2_temporal | 2 | 0 | 1 | (0,1) | 1 |
| 500 | 2 | 2 | 0 | re500_e2 | 6 | 5 | 1 | 1 | 0 | 1.84 | 0.835 | 0.24 | 0.728 | 0.274 | -0.938 | -0.146 | 0.116 | 0.814 | 4.142e-16 | -0.604 | 0.0457 | 0.00268 | 0.0497 | 1.163e-06 | 2 | – | B2_temporal | 2 | 0 | 1 | (0,1) | 1 |
| 500 | 2 | 2 | 1 | re500_e2 | 6 | 5 | 1 | 1 | 0 | 2.04 | 1.04 | 0.127 | 0.617 | 0.549 | 0.718 | -0.86 | -0.0421 | 0.72 | 4.733e-16 | -0.593 | 0.0595 | 0.00207 | 0.0496 | 1.094e-06 | 1.99 | – | B2_temporal | 2 | 1 | 1 | (1,1) | 1 |
| 500 | 2 | 2 | 2 | re500_e2 | 6 | 5 | 1 | 1 | 0 | 1.97 | 0.971 | 0.866 | 0.0211 | 0.354 | 1.85 | -0.527 | -0.191 | 0.968 | 4.224e-16 | -0.634 | 0.0365 | 0.00119 | 0.0597 | 1.229e-06 | 2 | – | B1_Mode2 | 1 | 1 | 0 | (1,0) | 1 |
| 500 | 2 | 2 | 3 | re500_e2 | 6 | 5 | 1 | 1 | 0 | 1.91 | 0.907 | 0.652 | 0.267 | 0.328 | 1.58 | -0.439 | -0.0921 | 0.842 | 4.437e-16 | -0.584 | 0.048 | 0.00372 | 0.0755 | 1.105e-06 | 2 | – | B1_Mode2 | 1 | 1 | 0 | (1,0) | 1 |
| 500 | 2 | 2 | 4 | re500_e2 | 6 | 5 | 1 | 1 | 0 | 1.89 | 0.887 | 0.263 | 0.588 | 0.401 | -0.996 | -0.583 | 0.236 | 0.711 | 4.564e-16 | -0.565 | 0.0664 | 0.00352 | 0.0496 | 1.076e-06 | 1.99 | – | B2_temporal | 2 | 0 | 1 | (0,1) | 1 |
| 700 | 0.1 | 0.1 | 0 | re700_e0.1 | 6 | 5 | 1 | 1 | 0 | 0.125 | 0.875 | 0.92 | 0.00713 | 0.0196 | -0.48 | 0.0182 | 0.132 | 0.993 | 1.010e-16 | 2.21 | 0.00416 | 7.633e-05 | 0.00146 | 0.000422 | 0.121 | – | B1_Mode2 | 1 | 1 | 0 | (1,0) | 1 |
| 700 | 0.1 | 0.1 | 1 | re700_e0.1 | 6 | 5 | 1 | 1 | 0 | 0.124 | 0.876 | 0.908 | 0.00872 | 0.00115 | 0.475 | 0.0632 | -0.124 | 0.987 | 1.001e-16 | 2.2 | 0.00243 | 4.537e-05 | 0.00156 | 0.000424 | 0.121 | – | B1_Mode2 | 1 | 1 | 0 | (1,0) | 1 |
| 700 | 0.1 | 0.1 | 2 | re700_e0.1 | 6 | 5 | 1 | 1 | 0 | 0.124 | 0.876 | 0.889 | 0.0065 | 0.017 | 0.47 | -0.0348 | -0.156 | 0.991 | 1.184e-16 | 2.2 | 0.00248 | 5.719e-05 | 0.00147 | 0.000424 | 0.121 | – | B1_Mode2 | 1 | 1 | 0 | (1,0) | 1 |
| 700 | 0.1 | 0.1 | 3 | re700_e0.1 | 6 | 5 | 1 | 1 | 0 | 0.122 | 0.878 | 0.00833 | 0.989 | -0.0244 | 0.0451 | -0.00155 | 0.00015 | 0.988 | 1.202e-16 | 2.21 | 0.0107 | 0.00016 | 0.0025 | 0.000416 | 0.12 | – | B2_temporal | 2 | 0 | 1 | (0,1) | 1 |
| 700 | 0.1 | 0.1 | 4 | re700_e0.1 | 6 | 5 | 1 | 1 | 0 | 0.123 | 0.877 | 0.0146 | 0.985 | -0.00168 | 0.0599 | -0.00196 | 0.000856 | 0.996 | 1.211e-16 | 2.21 | 0.00844 | 0.000159 | 0.00191 | 0.000419 | 0.12 | – | B2_temporal | 2 | 0 | 1 | (0,1) | 1 |
| 700 | 0.25 | 0.25 | 0 | re700_e0.25 | 6 | 5 | 1 | 1 | 0 | 0.245 | 0.755 | 0.538 | 0.409 | 0.0453 | -0.513 | 0.0146 | 0.151 | 0.914 | 1.456e-16 | 1.38 | 0.0104 | 0.000196 | 0.00488 | 8.929e-05 | 0.259 | – | B3_mixed | 3 | 1 | 0 | (1,0) | 1 |
| 700 | 0.25 | 0.25 | 1 | re700_e0.25 | 6 | 5 | 1 | 1 | 0 | 0.268 | 0.732 | 0.92 | 0.0141 | 0.00858 | 0.702 | 0.0573 | -0.176 | 0.984 | 1.494e-16 | 1.38 | 0.00771 | 0.000148 | 0.0038 | 8.902e-05 | 0.259 | – | B1_Mode2 | 1 | 1 | 0 | (1,0) | 1 |

Table 5: Détail complet des pertes et modes — 185 courses, `phase1_all_results.csv`. $L$: perte totale ; $L_{pde/bc/d/c/\lambda/Re}$: composantes ; $E_e = |E_r - E^*|/E^*$.

| Re | $E_t$ | $E^*$ | s | point | $N_{MX}$ | $N_{MT}$ | AR | $\lambda_v$ | $\lambda_b$ | $E_r$ | $E_e$ | $f_2$ | $f_t$ | $A_1$ | $A_2$ | $A_3$ | $A_4$ | $C_3$ | epsilon | $L$ | $L_{pde}$ | $L_{bc}$ | $L_d$ | $L_c$ | $L_\lambda$ | $L_{Re}$ | branch | id | $i$ | $j$ | mode | conv. |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 700 | 0.25 | 0.25 | 2 | re700_e0.25 | 6 | 5 | 1 | 1 | 0 | 0.263 | 0.737 | 0.897 | 0.0131 | 0.0376 | 0.687 | -0.0755 | -0.199 | 0.983 | 1.462e-16 | 1.38 | 0.00659 | 0.000148 | 0.00329 | 8.943e-05 | 0.259 | – | B1_Mode2 | 1 | 1 | 0 | (1,0) | 1 |
| 700 | 0.25 | 0.25 | 3 | re700_e0.25 | 6 | 5 | 1 | 1 | 0 | 0.266 | 0.734 | 0.00699 | 0.987 | -0.0484 | 0.0609 | 0.0253 | -0.00705 | 0.988 | 1.650e-16 | 1.37 | 0.00457 | 0.000185 | 0.00852 | 9.056e-05 | 0.26 | – | B2_temporal | 2 | 0 | 1 | (0,1) | 1 |
| 700 | 0.25 | 0.25 | 4 | re700_e0.25 | 6 | 5 | 1 | 1 | 0 | 0.258 | 0.742 | 0.00024 | 1 | 0.00101 | -0.0111 | -0.00329 | 0.00611 | 0.997 | 1.640e-16 | 1.38 | 0.0097 | 0.000309 | 0.00454 | 8.947e-05 | 0.259 | – | B2_temporal | 2 | 0 | 1 | (0,1) | 1 |
| 700 | 0.5 | 0.5 | 0 | re700_e0.5 | 6 | 5 | 1 | 1 | 0 | 0.461 | 0.539 | 0.375 | 0.606 | 0.0726 | -0.588 | 0.0134 | 0.103 | 0.92 | 2.441e-16 | 0.708 | 0.0142 | 0.000313 | 0.00913 | 2.318e-05 | 0.505 | – | B3_mixed | 3 | 0 | 1 | (0,1) | 1 |
| 700 | 0.5 | 0.5 | 1 | re700_e0.5 | 6 | 5 | 1 | 1 | 0 | 0.519 | 0.481 | 0.939 | 0.0271 | 0.00772 | 0.987 | 0.0608 | -0.174 | 0.978 | 2.124e-16 | 0.71 | 0.0154 | 0.00039 | 0.00919 | 2.237e-05 | 0.505 | – | B1_Mode2 | 1 | 1 | 0 | (1,0) | 1 |
| 700 | 0.5 | 0.5 | 2 | re700_e0.5 | 6 | 5 | 1 | 1 | 0 | 0.511 | 0.489 | 0.926 | 0.0172 | 0.0383 | 0.972 | -0.0409 | -0.232 | 0.987 | 2.010e-16 | 0.708 | 0.014 | 0.000365 | 0.00735 | 2.258e-05 | 0.505 | – | B1_Mode2 | 1 | 1 | 0 | (1,0) | 1 |
| 700 | 0.5 | 0.5 | 3 | re700_e0.5 | 6 | 5 | 1 | 1 | 0 | 0.52 | 0.48 | 0.00272 | 0.992 | -0.0672 | -0.0532 | 0.0181 | -0.0127 | 0.994 | 2.322e-16 | 0.701 | 0.00652 | 0.000346 | 0.0173 | 2.356e-05 | 0.505 | – | B2_temporal | 2 | 0 | 1 | (0,1) | 1 |
| 700 | 0.5 | 0.5 | 4 | re700_e0.5 | 6 | 5 | 1 | 1 | 0 | 0.492 | 0.508 | 0.0256 | 0.973 | 0.0197 | -0.159 | -0.0251 | 0.0167 | 0.961 | 2.121e-16 | 0.725 | 0.0244 | 0.000818 | 0.00883 | 2.260e-05 | 0.505 | – | B2_temporal | 2 | 0 | 1 | (0,1) | 1 |
| 700 | 1 | 1 | 0 | re700_e1 | 6 | 5 | 1 | 1 | 0 | 0.972 | 0.0282 | 0.141 | 0.505 | 0.333 | -0.523 | -0.749 | 0.0751 | 0.777 | 3.046e-16 | 0.0523 | 0.0335 | 0.00125 | 0.0156 | 5.043e-06 | 1 | – | B2_temporal | 2 | 1 | 1 | (1,1) | 1 |
| 700 | 1 | 1 | 1 | re700_e1 | 6 | 5 | 1 | 1 | 0 | 1.01 | 0.0056 | 0.784 | 0.182 | -0.00813 | 1.26 | 0.113 | -0.212 | 0.912 | 2.932e-16 | 0.0464 | 0.0306 | 0.00107 | 0.016 | 5.070e-06 | 1 | – | B1_Mode2 | 1 | 1 | 0 | (1,0) | 1 |
| 700 | 1 | 1 | 2 | re700_e1 | 6 | 5 | 1 | 1 | 0 | 1.03 | 0.0302 | 0.947 | 0.0214 | 0.0523 | 1.4 | -0.00989 | -0.247 | 0.984 | 3.317e-16 | 0.037 | 0.024 | 0.000885 | 0.0185 | 5.307e-06 | 1 | – | B1_Mode2 | 1 | 1 | 0 | (1,0) | 1 |
| 700 | 1 | 1 | 3 | re700_e1 | 6 | 5 | 1 | 1 | 0 | 0.935 | 0.0651 | 0.773 | 0.136 | 0.167 | 1.2 | -0.261 | -0.266 | 0.87 | 2.875e-16 | 0.0415 | 0.029 | 0.000788 | 0.0177 | 5.209e-06 | 1 | – | B1_Mode2 | 1 | 1 | 0 | (1,0) | 1 |
| 700 | 1 | 1 | 4 | re700_e1 | 6 | 5 | 1 | 1 | 0 | 0.971 | 0.0292 | 0.0224 | 0.974 | 0.058 | -0.208 | -0.0612 | 0.0174 | 0.944 | 3.037e-16 | 0.0823 | 0.0463 | 0.00239 | 0.0216 | 5.143e-06 | 1 | – | B2_temporal | 2 | 0 | 1 | (0,1) | 1 |
| 700 | 2 | 2 | 0 | re700_e2 | 6 | 5 | 1 | 1 | 0 | 1.93 | 0.933 | 0.0733 | 0.432 | 0.609 | -0.532 | -1.22 | 0.0979 | 0.748 | 4.079e-16 | -0.545 | 0.0586 | 0.00535 | 0.0552 | 1.022e-06 | 1.99 | – | B3_mixed | 3 | 2 | 0 | (2,0) | 1 |
| 700 | 2 | 2 | 1 | re700_e2 | 6 | 5 | 1 | 1 | 0 | 1.97 | 0.973 | 0.322 | 0.476 | 0.503 | 1.13 | -0.699 | -0.241 | 0.708 | 4.318e-16 | -0.564 | 0.0847 | 0.00219 | 0.0329 | 9.370e-07 | 1.98 | – | B3_mixed | 3 | 1 | 0 | (1,0) | 1 |
| 700 | 2 | 2 | 10 | re700_e2 | 6 | 5 | 1 | 1 | 0 | 2.01 | 1.01 | 0.364 | 0.591 | 0.299 | -1.21 | -0.253 | 0.113 | 0.817 | 4.677e-16 | -0.568 | 0.0601 | 0.0035 | 0.0434 | 1.072e-06 | 1.98 | – | B3_mixed | 3 | 0 | 1 | (0,1) | 1 |
| 700 | 2 | 2 | 11 | re700_e2 | 6 | 5 | 1 | 1 | 0 | 1.91 | 0.907 | 0.000133 | 0.831 | 0.474 | 0.0226 | -0.643 | 0.0597 | 0.827 | 4.750e-16 | -0.592 | 0.0417 | 0.00341 | 0.0575 | 1.138e-06 | 1.99 | – | B2_temporal | 2 | 1 | 1 | (1,1) | 1 |
| 700 | 2 | 2 | 12 | re700_e2 | 6 | 5 | 1 | 1 | 0 | 2.04 | 1.04 | 0.316 | 0.642 | 0.179 | -1.14 | -0.339 | 0.153 | 0.918 | 4.665e-16 | -0.605 | 0.0427 | 0.00274 | 0.0458 | 1.224e-06 | 2 | – | B2_temporal | 2 | 0 | 1 | (0,1) | 1 |
| 700 | 2 | 2 | 13 | re700_e2 | 6 | 5 | 1 | 1 | 0 | 1.96 | 0.958 | 0.943 | 0.0419 | 0.158 | 1.92 | -0.083 | -0.162 | 0.982 | 4.082e-16 | -0.609 | 0.0441 | 0.00246 | 0.0712 | 1.233e-06 | 2 | – | B1_Mode2 | 1 | 1 | 0 | (1,0) | 1 |
| 700 | 2 | 2 | 14 | re700_e2 | 6 | 5 | 1 | 1 | 0 | 1.82 | 0.818 | 0.392 | 0.355 | 0.452 | 1.19 | -0.802 | -0.266 | 0.697 | 4.444e-16 | -0.559 | 0.0736 | 0.00293 | 0.0324 | 9.635e-07 | 1.97 | – | B3_mixed | 3 | 1 | 0 | (1,0) | 1 |
| 700 | 2 | 2 | 2 | re700_e2 | 6 | 5 | 1 | 1 | 0 | 1.96 | 0.961 | 0.679 | 0.0535 | 0.507 | 1.63 | -0.86 | -0.184 | 0.934 | 4.261e-16 | -0.581 | 0.0593 | 0.00281 | 0.0441 | 1.028e-06 | 1.98 | – | B1_Mode2 | 1 | 1 | 0 | (1,0) | 1 |
| 700 | 2 | 2 | 3 | re700_e2 | 6 | 5 | 1 | 1 | 0 | 2.04 | 1.04 | 0.188 | 0.655 | 0.573 | 0.876 | -0.557 | 0.0182 | 0.61 | 4.711e-16 | -0.529 | 0.0931 | 0.00348 | 0.0338 | 8.357e-07 | 1.97 | – | B2_temporal | 2 | 1 | 1 | (1,1) | 1 |
| 700 | 2 | 2 | 4 | re700_e2 | 6 | 5 | 1 | 1 | 0 | 1.91 | 0.91 | 0.0195 | 0.942 | 0.265 | 0.273 | -0.269 | 0.0207 | 0.888 | 4.482e-16 | -0.547 | 0.0824 | 0.00364 | 0.0381 | 1.122e-06 | 1.99 | – | B2_temporal | 2 | 0 | 1 | (0,1) | 1 |
| 700 | 2 | 2 | 5 | re700_e2 | 6 | 5 | 1 | 1 | 0 | 1.89 | 0.886 | 0.32 | 0.561 | 0.401 | 1.1 | -0.51 | -0.149 | 0.748 | 4.850e-16 | -0.558 | 0.0819 | 0.00287 | 0.0358 | 9.251e-07 | 1.98 | – | B3_mixed | 3 | 1 | 0 | (1,0) | 1 |
| 700 | 2 | 2 | 6 | re700_e2 | 6 | 5 | 1 | 1 | 0 | 2.13 | 1.13 | 0.519 | 0.408 | 0.0775 | 1.49 | -0.468 | -0.281 | 0.793 | 4.427e-16 | -0.575 | 0.0534 | 0.00396 | 0.0597 | 1.093e-06 | 2 | – | B3_mixed | 3 | 1 | 0 | (1,0) | 1 |
| 700 | 2 | 2 | 7 | re700_e2 | 6 | 5 | 1 | 1 | 0 | 1.89 | 0.888 | 0.747 | 0.229 | -0.15 | 1.68 | 0.199 | -0.166 | 0.943 | 4.213e-16 | -0.619 | 0.0366 | 0.00232 | 0.0553 | 1.240e-06 | 2 | – | B1_Mode2 | 1 | 1 | 0 | (1,0) | 1 |
| 700 | 2 | 2 | 8 | re700_e2 | 6 | 5 | 1 | 1 | 0 | 1.87 | 0.87 | 0.0881 | 0.607 | 0.543 | 0.574 | -0.854 | -0.314 | 0.625 | 4.757e-16 | -0.582 | 0.0556 | 0.00273 | 0.0269 | 1.210e-06 | 1.97 | – | B2_temporal | 2 | 1 | 1 | (1,1) | 1 |
| 700 | 2 | 2 | 9 | re700_e2 | 6 | 5 | 1 | 1 | 0 | 1.94 | 0.942 | 0.179 | 0.53 | 0.516 | 0.834 | -0.928 | 0.000862 | 0.644 | 4.516e-16 | -0.604 | 0.0514 | 0.00205 | 0.0508 | 1.135e-06 | 1.99 | – | B2_temporal | 2 | 1 | 1 | (1,1) | 1 |
| 900 | 0.1 | 0.1 | 0 | re900_e0.1 | 6 | 5 | 1 | 1 | 0 | 0.118 | 0.882 | 0.732 | 0.182 | 0.0293 | -0.416 | 0.00818 | 0.138 | 0.964 | 1.031e-16 | 2.21 | 0.0051 | 9.044e-05 | 0.00164 | 0.000422 | 0.121 | – | B1_Mode2 | 1 | 1 | 0 | (1,0) | 1 |
| 900 | 0.1 | 0.1 | 1 | re900_e0.1 | 6 | 5 | 1 | 1 | 0 | 0.122 | 0.878 | 0.903 | 0.00547 | -0.0116 | 0.47 | 0.0999 | -0.105 | 0.989 | 1.040e-16 | 2.2 | 0.00317 | 5.280e-05 | 0.00123 | 0.000423 | 0.121 | – | B1_Mode2 | 1 | 1 | 0 | (1,0) | 1 |
| 900 | 0.1 | 0.1 | 2 | re900_e0.1 | 6 | 5 | 1 | 1 | 0 | 0.127 | 0.873 | 0.895 | 0.00612 | 0.00786 | 0.477 | 0.00121 | -0.158 | 0.997 | 1.095e-16 | 2.2 | 0.00272 | 5.094e-05 | 0.00112 | 0.000424 | 0.121 | – | B1_Mode2 | 1 | 1 | 0 | (1,0) | 1 |
| 900 | 0.1 | 0.1 | 3 | re900_e0.1 | 6 | 5 | 1 | 1 | 0 | 0.123 | 0.877 | 0.00194 | 0.995 | -0.0269 | 0.0219 | -0.00106 | 0.00537 | 0.995 | 1.179e-16 | 2.21 | 0.00681 | 0.000141 | 0.00257 | 0.00042 | 0.121 | – | B2_temporal | 2 | 0 | 1 | (0,1) | 1 |
| 900 | 0.1 | 0.1 | 4 | re900_e0.1 | 6 | 5 | 1 | 1 | 0 | 0.124 | 0.876 | 0.0712 | 0.929 | -0.00479 | 0.133 | 0.00271 | -0.00104 | 0.996 | 1.203e-16 | 2.21 | 0.00566 | 0.000122 | 0.00174 | 0.000422 | 0.121 | – | B2_temporal | 2 | 0 | 1 | (0,1) | 1 |
| 900 | 0.25 | 0.25 | 0 | re900_e0.25 | 6 | 5 | 1 | 1 | 0 | 0.24 | 0.76 | 0.419 | 0.537 | 0.0455 | -0.449 | -0.0142 | 0.138 | 0.922 | 1.676e-16 | 1.38 | 0.0112 | 0.000217 | 0.00479 | 8.932e-05 | 0.259 | – | B3_mixed | 3 | 0 | 1 | (0,1) | 1 |
| 900 | 0.25 | 0.25 | 1 | re900_e0.25 | 6 | 5 | 1 | 1 | 0 | 0.266 | 0.734 | 0.947 | 0.022 | -0.0045 | 0.71 | 0.0616 | -0.112 | 0.985 | 1.514e-16 | 1.38 | 0.00678 | 0.000169 | 0.00451 | 8.939e-05 | 0.259 | – | B1_Mode2 | 1 | 1 | 0 | (1,0) | 1 |
| 900 | 0.25 | 0.25 | 2 | re900_e0.25 | 6 | 5 | 1 | 1 | 0 | 0.27 | 0.73 | 0.915 | 0.0144 | 0.0205 | 0.703 | -0.0183 | -0.193 | 0.99 | 1.461e-16 | 1.38 | 0.00773 | 0.000157 | 0.00258 | 8.891e-05 | 0.259 | – | B1_Mode2 | 1 | 1 | 0 | (1,0) | 1 |
| 900 | 0.25 | 0.25 | 3 | re900_e0.25 | 6 | 5 | 1 | 1 | 0 | 0.269 | 0.731 | 0.000264 | 0.996 | -0.0415 | 0.0119 | -0.00264 | 0.00317 | 0.997 | 1.614e-16 | 1.38 | 0.00553 | 0.000197 | 0.00709 | 9.038e-05 | 0.26 | – | B2_temporal | 2 | 0 | 1 | (0,1) | 1 |
| 900 | 0.25 | 0.25 | 4 | re900_e0.25 | 6 | 5 | 1 | 1 | 0 | 0.262 | 0.738 | 0.0205 | 0.979 | -0.00151 | 0.104 | -0.000636 | 0.00183 | 0.993 | 1.764e-16 | 1.38 | 0.00878 | 0.000257 | 0.00385 | 8.976e-05 | 0.259 | – | B2_temporal | 2 | 0 | 1 | (0,1) | 1 |
| 900 | 0.5 | 0.5 | 0 | re900_e0.5 | 6 | 5 | 1 | 1 | 0 | 0.463 | 0.537 | 0.361 | 0.612 | 0.0712 | -0.578 | -0.00787 | 0.139 | 0.915 | 2.169e-16 | 0.709 | 0.0153 | 0.00034 | 0.00832 | 2.307e-05 | 0.505 | – | B3_mixed | 3 | 0 | 1 | (0,1) | 1 |
| 900 | 0.5 | 0.5 | 1 | re900_e0.5 | 6 | 5 | 1 | 1 | 0 | 0.517 | 0.483 | 0.943 | 0.0481 | 0.0172 | 0.987 | -0.00624 | -0.0914 | 0.993 | 2.154e-16 | 0.708 | 0.0126 | 0.000457 | 0.0117 | 2.287e-05 | 0.505 | – | B1_Mode2 | 1 | 1 | 0 | (1,0) | 1 |
| 900 | 0.5 | 0.5 | 2 | re900_e0.5 | 6 | 5 | 1 | 1 | 0 | 0.523 | 0.477 | 0.925 | 0.0375 | 0.0355 | 0.983 | -0.0545 | -0.186 | 0.975 | 2.054e-16 | 0.709 | 0.0139 | 0.000426 | 0.00746 | 2.245e-05 | 0.505 | – | B1_Mode2 | 1 | 1 | 0 | (1,0) | 1 |
| 900 | 0.5 | 0.5 | 3 | re900_e0.5 | 6 | 5 | 1 | 1 | 0 | 0.522 | 0.478 | 2.505e-05 | 0.996 | -0.0604 | -0.00512 | 0.00385 | 0.00117 | 0.997 | 2.447e-16 | 0.702 | 0.00775 | 0.000373 | 0.0147 | 2.350e-05 | 0.505 | – | B2_temporal | 2 | 0 | 1 | (0,1) | 1 |
| 900 | 0.5 | 0.5 | 4 | re900_e0.5 | 6 | 5 | 1 | 1 | 0 | 0.497 | 0.503 | 0.000115 | 0.999 | 0.0102 | -0.0107 | -0.012 | 0.0119 | 0.992 | 2.396e-16 | 0.718 | 0.0192 | 0.000671 | 0.00723 | 2.292e-05 | 0.505 | – | B2_temporal | 2 | 0 | 1 | (0,1) | 1 |
| 900 | 1 | 1 | 0 | re900_e1 | 6 | 5 | 1 | 1 | 0 | 0.961 | 0.0386 | 0.193 | 0.376 | 0.34 | -0.608 | -0.831 | 0.0722 | 0.805 | 3.049e-16 | 0.0618 | 0.0299 | 0.00212 | 0.0166 | 5.092e-06 | 1 | – | B3_mixed | 3 | 2 | 0 | (2,0) | 1 |
| 900 | 1 | 1 | 1 | re900_e1 | 6 | 5 | 1 | 1 | 0 | 1.03 | 0.0307 | 0.97 | 0.0179 | 0.0458 | 1.41 | -0.043 | -0.145 | 0.988 | 3.495e-16 | 0.0417 | 0.0227 | 0.00124 | 0.028 | 5.447e-06 | 1 | – | B1_Mode2 | 1 | 1 | 0 | (1,0) | 1 |
| 900 | 1 | 1 | 2 | re900_e1 | 6 | 5 | 1 | 1 | 0 | 1.02 | 0.0237 | 0.877 | 0.0805 | 0.0654 | 1.34 | -0.204 | -0.201 | 0.94 | 2.946e-16 | 0.0362 | 0.0236 | 0.00085 | 0.0166 | 5.278e-06 | 1 | – | B1_Mode2 | 1 | 1 | 0 | (1,0) | 1 |
| 900 | 1 | 1 | 3 | re900_e1 | 6 | 5 | 1 | 1 | 0 | 0.991 | 0.00901 | 0.114 | 0.865 | -0.122 | -0.475 | 0.142 | -0.0709 | 0.96 | 3.016e-16 | 0.0513 | 0.0289 | 0.00148 | 0.0261 | 5.423e-06 | 1 | – | B2_temporal | 2 | 0 | 1 | (0,1) | 1 |
| 900 | 1 | 1 | 4 | re900_e1 | 6 | 5 | 1 | 1 | 0 | 0.953 | 0.047 | 0.0545 | 0.944 | 0.00981 | 0.322 | 0.0474 | -0.0333 | 0.912 | 3.063e-16 | 0.0802 | 0.0421 | 0.00255 | 0.0174 | 5.271e-06 | 1 | – | B2_temporal | 2 | 0 | 1 | (0,1) | 1 |
| 900 | 2 | 2 | 0 | re900_e2 | 6 | 5 | 1 | 1 | 0 | 1.82 | 0.824 | 0.0205 | 0.151 | 0.581 | 0.273 | -1.59 | -0.0837 | 0.88 | 4.106e-16 | -0.562 | 0.0549 | 0.00443 | 0.0531 | 1.065e-06 | 1.99 | – | B3_mixed | 3 | 2 | 0 | (2,0) | 1 |
| 900 | 2 | 2 | 1 | re900_e2 | 6 | 5 | 1 | 1 | 0 | 2 | 1 | 0.739 | 0.194 | 0.243 | 1.72 | -0.448 | -0.0645 | 0.878 | 4.390e-16 | -0.602 | 0.0484 | 0.00266 | 0.0477 | 1.200e-06 | 2 | – | B1_Mode2 | 1 | 1 | 0 | (1,0) | 1 |
| 900 | 2 | 2 | 2 | re900_e2 | 6 | 5 | 1 | 1 | 0 | 2.07 | 1.07 | 0.717 | 0.22 | 0.228 | 1.72 | -0.421 | -0.172 | 0.873 | 4.237e-16 | -0.573 | 0.0662 | 0.00329 | 0.0318 | 9.790e-07 | 1.99 | – | B1_Mode2 | 1 | 1 | 0 | (1,0) | 1 |
| 900 | 2 | 2 | 3 | re900_e2 | 6 | 5 | 1 | 1 | 0 | 1.96 | 0.958 | 0.0463 | 0.64 | 0.692 | 0.426 | -0.824 | 0.233 | 0.726 | 4.308e-16 | -0.533 | 0.0704 | 0.00436 | 0.0565 | 9.639e-07 | 1.96 | – | B2_temporal | 2 | 1 | 1 | (1,1) | 1 |
| 900 | 2 | 2 | 4 | re900_e2 | 6 | 5 | 1 | 1 | 0 | 1.9 | 0.904 | 0.0857 | 0.909 | 0.138 | 0.571 | 0.00176 | -0.00736 | 0.873 | 4.583e-16 | -0.542 | 0.0698 | 0.00503 | 0.0345 | 1.091e-06 | 1.99 | – | B2_temporal | 2 | 0 | 1 | (0,1) | 1 |

# 2 Campagne 2 — Sensibilité à $\lambda_v$ : 90 courses, 3 cellules $\times$ 6 valeurs

## 2.1 Probabilités de bassin agrégées par point

Table 6: Probabilités $P(B_k)$ — 18 points, `branch_points_lambda_var.csv`.

| point | Re | $E^*$ | AR | $\lambda_v$ | $\lambda_b$ | $N_s$ | $n_1$ | $n_2$ | $n_3$ | $n_{NA}$ | $P_1$ | $P_2$ | $P_3$ | label | conf | affin. | esc10 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| re500_e0.25_l0 | 500 | 0.25 | 1 | 0 | 0 | 5 | 0 | 0 | 5 | 0 | 0 | 0 | 1 | P_B3 | 1 | Non | Non |
| re500_e0.25_l0.25 | 500 | 0.25 | 1 | 0.25 | 0 | 5 | 3 | 2 | 0 | 0 | 0.6 | 0.4 | 0 | P_B1 | 0.6 | Oui | Oui |
| re500_e0.25_l0.5 | 500 | 0.25 | 1 | 0.5 | 0 | 5 | 2 | 1 | 2 | 0 | 0.4 | 0.2 | 0.4 | P_B1 | 0.4 | Oui | Oui |
| re500_e0.25_l1 | 500 | 0.25 | 1 | 1 | 0 | 5 | 2 | 1 | 2 | 0 | 0.4 | 0.2 | 0.4 | P_B1 | 0.4 | Oui | Oui |
| re500_e0.25_l2 | 500 | 0.25 | 1 | 2 | 0 | 5 | 2 | 2 | 1 | 0 | 0.4 | 0.4 | 0.2 | P_B1 | 0.4 | Oui | Oui |
| re500_e0.25_l5 | 500 | 0.25 | 1 | 5 | 0 | 5 | 2 | 3 | 0 | 0 | 0.4 | 0.6 | 0 | P_B2 | 0.6 | Oui | Oui |
| re700_e2_l0 | 700 | 2 | 1 | 0 | 0 | 5 | 0 | 0 | 5 | 0 | 0 | 0 | 1 | P_B3 | 1 | Non | Non |
| re700_e2_l0.25 | 700 | 2 | 1 | 0.25 | 0 | 5 | 0 | 0 | 5 | 0 | 0 | 0 | 1 | P_B3 | 1 | Non | Non |
| re700_e2_l0.5 | 700 | 2 | 1 | 0.5 | 0 | 5 | 0 | 3 | 2 | 0 | 0 | 0.6 | 0.4 | P_B2 | 0.6 | Oui | Oui |
| re700_e2_l1 | 700 | 2 | 1 | 1 | 0 | 5 | 1 | 2 | 2 | 0 | 0.2 | 0.4 | 0.4 | P_B2 | 0.4 | Oui | Oui |
| re700_e2_l2 | 700 | 2 | 1 | 2 | 0 | 5 | 3 | 1 | 1 | 0 | 0.6 | 0.2 | 0.2 | P_B1 | 0.6 | Oui | Oui |
| re700_e2_l5 | 700 | 2 | 1 | 5 | 0 | 5 | 3 | 1 | 1 | 0 | 0.6 | 0.2 | 0.2 | P_B1 | 0.6 | Oui | Oui |
| re1000_e1 | 1e+03 | 1 | 1 | 0 | 0 | 5 | 0 | 0 | 5 | 0 | 0 | 0 | 1 | P_B3 | 1 | Non | Non |
| re1000_e1_l0.25 | 1e+03 | 1 | 1 | 0.25 | 0 | 5 | 0 | 3 | 2 | 0 | 0 | 0.6 | 0.4 | P_B2 | 0.6 | Oui | Oui |
| re1000_e1_l0.5 | 1e+03 | 1 | 1 | 0.5 | 0 | 5 | 2 | 2 | 1 | 0 | 0.4 | 0.4 | 0.2 | P_B1 | 0.4 | Oui | Oui |
| re1000_e1_l1 | 1e+03 | 1 | 1 | 1 | 0 | 5 | 2 | 2 | 1 | 0 | 0.4 | 0.4 | 0.2 | P_B1 | 0.4 | Oui | Oui |
| re1000_e1_l2 | 1e+03 | 1 | 1 | 2 | 0 | 5 | 2 | 2 | 1 | 0 | 0.4 | 0.4 | 0.2 | P_B1 | 0.4 | Oui | Oui |
| re1000_e1_l5 | 1e+03 | 1 | 1 | 5 | 0 | 5 | 2 | 3 | 0 | 0 | 0.4 | 0.6 | 0 | P_B2 | 0.6 | Oui | Oui |

## 2.2 Détail par course

Table 7: Métriques par course ($\lambda_v$) — 90 courses, `branch_runs_lambda_var.csv`.

| point | étude | Re | $E^*$ | AR | $\lambda_v$ | $\lambda_b$ | s | $E_r$ | $f_2$ | $f_t$ | $A_2$ | $R_K$ | $\epsilon$ | id | branch |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| re1000_e1 | phase1_lambda_var | 1e+03 | 1 | 1 | 0 | 0 | 0 | 0.972 | 0.0062 | 0.022 | 0.109 | *NA* | *NA* | 3 | B3_mixed |
| re1000_e1 | phase1_lambda_var | 1e+03 | 1 | 1 | 0 | 0 | 1 | 0.988 | 0.0102 | 0.0182 | -0.142 | *NA* | *NA* | 3 | B3_mixed |
| re1000_e1 | phase1_lambda_var | 1e+03 | 1 | 1 | 0 | 0 | 2 | 0.97 | 0.0012 | 0.121 | 0.0488 | *NA* | *NA* | 3 | B3_mixed |
| re1000_e1 | phase1_lambda_var | 1e+03 | 1 | 1 | 0 | 0 | 3 | 1.02 | 0.0073 | 0.0203 | 0.122 | *NA* | *NA* | 3 | B3_mixed |
| re1000_e1 | phase1_lambda_var | 1e+03 | 1 | 1 | 0 | 0 | 4 | 1.01 | 0.0005 | 0.0322 | -0.0307 | *NA* | *NA* | 3 | B3_mixed |
| re1000_e1_l0.25 | phase1_lambda_var | 1e+03 | 1 | 1 | 0.25 | 0 | 0 | 0.978 | 0.129 | 0.281 | 0.502 | *NA* | *NA* | 3 | B3_mixed |
| re1000_e1_l0.25 | phase1_lambda_var | 1e+03 | 1 | 1 | 0.25 | 0 | 1 | 1.04 | 0.0136 | 0.575 | 0.168 | *NA* | *NA* | 2 | B2_temporal |
| re1000_e1_l0.25 | phase1_lambda_var | 1e+03 | 1 | 1 | 0.25 | 0 | 2 | 0.962 | 0.193 | 0.419 | 0.609 | *NA* | *NA* | 3 | B3_mixed |
| re1000_e1_l0.25 | phase1_lambda_var | 1e+03 | 1 | 1 | 0.25 | 0 | 3 | 1.01 | 0.0182 | 0.652 | -0.191 | *NA* | *NA* | 2 | B2_temporal |
| re1000_e1_l0.25 | phase1_lambda_var | 1e+03 | 1 | 1 | 0.25 | 0 | 4 | 1.04 | 0.0358 | 0.594 | 0.273 | *NA* | *NA* | 2 | B2_temporal |
| re1000_e1_l0.5 | phase1_lambda_var | 1e+03 | 1 | 1 | 0.5 | 0 | 0 | 0.976 | 0.243 | 0.167 | 0.689 | *NA* | *NA* | 3 | B3_mixed |
| re1000_e1_l0.5 | phase1_lambda_var | 1e+03 | 1 | 1 | 0.5 | 0 | 1 | 1 | 0.69 | 0.199 | 1.18 | *NA* | *NA* | 1 | B1_Mode2 |
| re1000_e1_l0.5 | phase1_lambda_var | 1e+03 | 1 | 1 | 0.5 | 0 | 2 | 1.02 | 0.648 | 0.218 | 1.15 | *NA* | *NA* | 1 | B1_Mode2 |
| re1000_e1_l0.5 | phase1_lambda_var | 1e+03 | 1 | 1 | 0.5 | 0 | 3 | 0.949 | 0.0199 | 0.971 | -0.195 | *NA* | *NA* | 2 | B2_temporal |
| re1000_e1_l0.5 | phase1_lambda_var | 1e+03 | 1 | 1 | 0.5 | 0 | 4 | 0.959 | 0.107 | 0.785 | -0.454 | *NA* | *NA* | 2 | B2_temporal |
| re1000_e1_l1 | phase1_lambda_var | 1e+03 | 1 | 1 | 1 | 0 | 0 | 0.901 | 0.0318 | 0.0946 | 0.239 | *NA* | *NA* | 3 | B3_mixed |
| re1000_e1_l1 | phase1_lambda_var | 1e+03 | 1 | 1 | 1 | 0 | 1 | 1.03 | 0.963 | 0.0281 | 1.41 | *NA* | *NA* | 1 | B1_Mode2 |
| re1000_e1_l1 | phase1_lambda_var | 1e+03 | 1 | 1 | 1 | 0 | 2 | 1.02 | 0.902 | 0.0683 | 1.36 | *NA* | *NA* | 1 | B1_Mode2 |
| re1000_e1_l1 | phase1_lambda_var | 1e+03 | 1 | 1 | 1 | 0 | 3 | 0.978 | 0.0038 | 0.993 | 0.0857 | *NA* | *NA* | 2 | B2_temporal |
| re1000_e1_l1 | phase1_lambda_var | 1e+03 | 1 | 1 | 1 | 0 | 4 | 0.964 | 0.0166 | 0.983 | -0.179 | *NA* | *NA* | 2 | B2_temporal |
| re1000_e1_l2 | phase1_lambda_var | 1e+03 | 1 | 1 | 2 | 0 | 0 | 1.04 | 0.468 | 0.085 | 0.989 | *NA* | *NA* | 3 | B3_mixed |
| re1000_e1_l2 | phase1_lambda_var | 1e+03 | 1 | 1 | 2 | 0 | 1 | 1.03 | 0.963 | 0.0305 | 1.41 | *NA* | *NA* | 1 | B1_Mode2 |
| re1000_e1_l2 | phase1_lambda_var | 1e+03 | 1 | 1 | 2 | 0 | 2 | 1.01 | 0.963 | 0.0157 | 1.39 | *NA* | *NA* | 1 | B1_Mode2 |
| re1000_e1_l2 | phase1_lambda_var | 1e+03 | 1 | 1 | 2 | 0 | 3 | 1.01 | 0.002 | 0.994 | 0.0628 | *NA* | *NA* | 2 | B2_temporal |
| re1000_e1_l2 | phase1_lambda_var | 1e+03 | 1 | 1 | 2 | 0 | 4 | 0.983 | 0 | 1 | 0.0033 | *NA* | *NA* | 2 | B2_temporal |
| re1000_e1_l5 | phase1_lambda_var | 1e+03 | 1 | 1 | 5 | 0 | 0 | 0.971 | 0.0652 | 0.93 | 0.356 | *NA* | *NA* | 2 | B2_temporal |
| re1000_e1_l5 | phase1_lambda_var | 1e+03 | 1 | 1 | 5 | 0 | 1 | 1.04 | 0.961 | 0.0307 | 1.41 | *NA* | *NA* | 1 | B1_Mode2 |
| re1000_e1_l5 | phase1_lambda_var | 1e+03 | 1 | 1 | 5 | 0 | 2 | 1.02 | 0.964 | 0.0087 | 1.4 | *NA* | *NA* | 1 | B1_Mode2 |
| re1000_e1_l5 | phase1_lambda_var | 1e+03 | 1 | 1 | 5 | 0 | 3 | 1.04 | 0.0039 | 0.992 | 0.0905 | *NA* | *NA* | 2 | B2_temporal |
| re1000_e1_l5 | phase1_lambda_var | 1e+03 | 1 | 1 | 5 | 0 | 4 | 1 | 0.0158 | 0.984 | 0.178 | *NA* | *NA* | 2 | B2_temporal |
| re700_e2_l0 | phase1_lambda_var | 700 | 2 | 1 | 0 | 0 | 0 | 1.95 | 0.0011 | 0.0112 | 0.0669 | *NA* | *NA* | 3 | B3_mixed |
| re700_e2_l0 | phase1_lambda_var | 700 | 2 | 1 | 0 | 0 | 1 | 1.98 | 0.0041 | 0.0135 | -0.128 | *NA* | *NA* | 3 | B3_mixed |
| re700_e2_l0 | phase1_lambda_var | 700 | 2 | 1 | 0 | 0 | 2 | 1.98 | 0.0241 | 0.0446 | 0.308 | *NA* | *NA* | 3 | B3_mixed |
| re700_e2_l0 | phase1_lambda_var | 700 | 2 | 1 | 0 | 0 | 3 | 2.03 | 0.0222 | 0.0193 | 0.3 | *NA* | *NA* | 3 | B3_mixed |
| re700_e2_l0 | phase1_lambda_var | 700 | 2 | 1 | 0 | 0 | 4 | 2 | 0.002 | 0.0191 | 0.0883 | *NA* | *NA* | 3 | B3_mixed |
| re700_e2_l0.25 | phase1_lambda_var | 700 | 2 | 1 | 0.25 | 0 | 0 | 2.02 | 0.0047 | 0.171 | 0.138 | *NA* | *NA* | 3 | B3_mixed |
| re700_e2_l0.25 | phase1_lambda_var | 700 | 2 | 1 | 0.25 | 0 | 1 | 1.97 | 0.0988 | 0.268 | -0.625 | *NA* | *NA* | 3 | B3_mixed |
| re700_e2_l0.25 | phase1_lambda_var | 700 | 2 | 1 | 0.25 | 0 | 2 | 1.9 | 0.19 | 0.177 | 0.849 | *NA* | *NA* | 3 | B3_mixed |
| re700_e2_l0.25 | phase1_lambda_var | 700 | 2 | 1 | 0.25 | 0 | 3 | 2.04 | 0.0517 | 0.321 | 0.459 | *NA* | *NA* | 3 | B3_mixed |
| re700_e2_l0.25 | phase1_lambda_var | 700 | 2 | 1 | 0.25 | 0 | 4 | 2.06 | 0.0079 | 0.312 | 0.18 | *NA* | *NA* | 3 | B3_mixed |
| re700_e2_l0.5 | phase1_lambda_var | 700 | 2 | 1 | 0.5 | 0 | 0 | 1.97 | 0.0196 | 0.274 | 0.278 | *NA* | *NA* | 3 | B3_mixed |
| re700_e2_l0.5 | phase1_lambda_var | 700 | 2 | 1 | 0.5 | 0 | 1 | 2.09 | 0.0113 | 0.635 | -0.218 | *NA* | *NA* | 2 | B2_temporal |
| re700_e2_l0.5 | phase1_lambda_var | 700 | 2 | 1 | 0.5 | 0 | 2 | 1.96 | 0.422 | 0.111 | 1.29 | *NA* | *NA* | 3 | B3_mixed |
| re700_e2_l0.5 | phase1_lambda_var | 700 | 2 | 1 | 0.5 | 0 | 3 | 2.03 | 0.024 | 0.579 | 0.312 | *NA* | *NA* | 2 | B2_temporal |
| re700_e2_l0.5 | phase1_lambda_var | 700 | 2 | 1 | 0.5 | 0 | 4 | 1.95 | 0.0049 | 0.519 | -0.138 | *NA* | *NA* | 2 | B2_temporal |
| re700_e2_l1 | phase1_lambda_var | 700 | 2 | 1 | 1 | 0 | 0 | 1.92 | 0.0911 | 0.406 | -0.592 | *NA* | *NA* | 3 | B3_mixed |
| re700_e2_l1 | phase1_lambda_var | 700 | 2 | 1 | 1 | 0 | 1 | 1.97 | 0.322 | 0.476 | 1.13 | *NA* | *NA* | 3 | B3_mixed |
| re700_e2_l1 | phase1_lambda_var | 700 | 2 | 1 | 1 | 0 | 2 | 1.96 | 0.679 | 0.0535 | 1.63 | *NA* | *NA* | 1 | B1_Mode2 |
| re700_e2_l1 | phase1_lambda_var | 700 | 2 | 1 | 1 | 0 | 3 | 2.04 | 0.188 | 0.655 | 0.876 | *NA* | *NA* | 2 | B2_temporal |
| re700_e2_l1 | phase1_lambda_var | 700 | 2 | 1 | 1 | 0 | 4 | 1.91 | 0.0195 | 0.942 | 0.273 | *NA* | *NA* | 2 | B2_temporal |
| re700_e2_l2 | phase1_lambda_var | 700 | 2 | 1 | 2 | 0 | 0 | 1.87 | 0.0936 | 0.182 | 0.592 | *NA* | *NA* | 3 | B3_mixed |
| re700_e2_l2 | phase1_lambda_var | 700 | 2 | 1 | 2 | 0 | 1 | 2.01 | 0.806 | 0.149 | 1.8 | *NA* | *NA* | 1 | B1_Mode2 |
| re700_e2_l2 | phase1_lambda_var | 700 | 2 | 1 | 2 | 0 | 2 | 1.98 | 0.945 | 0.0179 | 1.94 | *NA* | *NA* | 1 | B1_Mode2 |
| re700_e2_l2 | phase1_lambda_var | 700 | 2 | 1 | 2 | 0 | 3 | 1.92 | 0.58 | 0.287 | 1.49 | *NA* | *NA* | 1 | B1_Mode2 |
| re700_e2_l2 | phase1_lambda_var | 700 | 2 | 1 | 2 | 0 | 4 | 1.93 | 0.0257 | 0.974 | 0.315 | *NA* | *NA* | 2 | B2_temporal |
| re700_e2_l5 | phase1_lambda_var | 700 | 2 | 1 | 5 | 0 | 0 | 1.93 | 0.113 | 0.42 | 0.662 | *NA* | *NA* | 3 | B3_mixed |
| re700_e2_l5 | phase1_lambda_var | 700 | 2 | 1 | 5 | 0 | 1 | 2.04 | 0.985 | 0.0118 | 2.01 | *NA* | *NA* | 1 | B1_Mode2 |
| re700_e2_l5 | phase1_lambda_var | 700 | 2 | 1 | 5 | 0 | 2 | 1.99 | 0.962 | 0.021 | 1.96 | *NA* | *NA* | 1 | B1_Mode2 |
| re700_e2_l5 | phase1_lambda_var | 700 | 2 | 1 | 5 | 0 | 3 | 1.86 | 0.927 | 0.0278 | 1.86 | *NA* | *NA* | 1 | B1_Mode2 |
| re700_e2_l5 | phase1_lambda_var | 700 | 2 | 1 | 5 | 0 | 4 | 1.95 | 0.0079 | 0.989 | 0.176 | *NA* | *NA* | 2 | B2_temporal |
| re500_e0.25_l0 | phase1_lambda_var | 500 | 0.25 | 1 | 0 | 0 | 0 | 0.25 | 0.0031 | 0.0541 | 0.0393 | *NA* | *NA* | 3 | B3_mixed |
| re500_e0.25_l0 | phase1_lambda_var | 500 | 0.25 | 1 | 0 | 0 | 1 | 0.247 | 0.0002 | 0.0301 | -0.009 | *NA* | *NA* | 3 | B3_mixed |
| re500_e0.25_l0 | phase1_lambda_var | 500 | 0.25 | 1 | 0 | 0 | 2 | 0.246 | 0.0002 | 0.0467 | 0.0097 | *NA* | *NA* | 3 | B3_mixed |
| re500_e0.25_l0 | phase1_lambda_var | 500 | 0.25 | 1 | 0 | 0 | 3 | 0.26 | 0.0007 | 0.0343 | 0.0188 | *NA* | *NA* | 3 | B3_mixed |
| re500_e0.25_l0 | phase1_lambda_var | 500 | 0.25 | 1 | 0 | 0 | 4 | 0.252 | 0.0002 | 0.0399 | 0.009 | *NA* | *NA* | 3 | B3_mixed |
| re500_e0.25_l0.25 | phase1_lambda_var | 500 | 0.25 | 1 | 0.25 | 0 | 0 | 0.252 | 0.816 | 0.0867 | -0.641 | *NA* | *NA* | 1 | B1_Mode2 |
| re500_e0.25_l0.25 | phase1_lambda_var | 500 | 0.25 | 1 | 0.25 | 0 | 1 | 0.256 | 0.883 | 0.0203 | 0.672 | *NA* | *NA* | 1 | B1_Mode2 |
| re500_e0.25_l0.25 | phase1_lambda_var | 500 | 0.25 | 1 | 0.25 | 0 | 2 | 0.254 | 0.913 | 0.0138 | 0.681 | *NA* | *NA* | 1 | B1_Mode2 |

Table 7: Métriques par course ($\lambda_v$) — 90 courses, `branch_runs_lambda_var.csv`.

| point | étude | Re | $E^*$ | AR | $\lambda_v$ | $\lambda_b$ | s | $E_r$ | $f_2$ | $f_t$ | $A_2$ | $R_K$ | $\epsilon$ | id | branch |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| re500_e0.25_l0.25 | phase1_lambda_var | 500 | 0.25 | 1 | 0.25 | 0 | 3 | 0.263 | 0.0022 | 0.995 | 0.0337 | *NA* | *NA* | 2 | B2_temporal |
| re500_e0.25_l0.25 | phase1_lambda_var | 500 | 0.25 | 1 | 0.25 | 0 | 4 | 0.247 | 0.226 | 0.739 | -0.334 | *NA* | *NA* | 2 | B2_temporal |
| re500_e0.25_l0.5 | phase1_lambda_var | 500 | 0.25 | 1 | 0.5 | 0 | 0 | 0.241 | 0.619 | 0.346 | -0.546 | *NA* | *NA* | 3 | B3_mixed |
| re500_e0.25_l0.5 | phase1_lambda_var | 500 | 0.25 | 1 | 0.5 | 0 | 1 | 0.261 | 0.904 | 0.0135 | 0.687 | *NA* | *NA* | 1 | B1_Mode2 |
| re500_e0.25_l0.5 | phase1_lambda_var | 500 | 0.25 | 1 | 0.5 | 0 | 2 | 0.254 | 0.901 | 0.0329 | 0.676 | *NA* | *NA* | 1 | B1_Mode2 |
| re500_e0.25_l0.5 | phase1_lambda_var | 500 | 0.25 | 1 | 0.5 | 0 | 3 | 0.266 | 0.0191 | 0.979 | 0.101 | *NA* | *NA* | 2 | B2_temporal |
| re500_e0.25_l0.5 | phase1_lambda_var | 500 | 0.25 | 1 | 0.5 | 0 | 4 | 0.246 | 0.334 | 0.647 | -0.406 | *NA* | *NA* | 3 | B3_mixed |
| re500_e0.25_l1 | phase1_lambda_var | 500 | 0.25 | 1 | 1 | 0 | 0 | 0.24 | 0.52 | 0.448 | -0.499 | *NA* | *NA* | 3 | B3_mixed |
| re500_e0.25_l1 | phase1_lambda_var | 500 | 0.25 | 1 | 1 | 0 | 1 | 0.267 | 0.911 | 0.0133 | 0.698 | *NA* | *NA* | 1 | B1_Mode2 |
| re500_e0.25_l1 | phase1_lambda_var | 500 | 0.25 | 1 | 1 | 0 | 2 | 0.257 | 0.89 | 0.0465 | 0.676 | *NA* | *NA* | 1 | B1_Mode2 |
| re500_e0.25_l1 | phase1_lambda_var | 500 | 0.25 | 1 | 1 | 0 | 3 | 0.268 | 0.0162 | 0.982 | 0.0931 | *NA* | *NA* | 2 | B2_temporal |
| re500_e0.25_l1 | phase1_lambda_var | 500 | 0.25 | 1 | 1 | 0 | 4 | 0.25 | 0.343 | 0.637 | -0.414 | *NA* | *NA* | 3 | B3_mixed |
| re500_e0.25_l2 | phase1_lambda_var | 500 | 0.25 | 1 | 2 | 0 | 0 | 0.243 | 0.407 | 0.562 | -0.445 | *NA* | *NA* | 3 | B3_mixed |
| re500_e0.25_l2 | phase1_lambda_var | 500 | 0.25 | 1 | 2 | 0 | 1 | 0.277 | 0.911 | 0.0141 | 0.711 | *NA* | *NA* | 1 | B1_Mode2 |
| re500_e0.25_l2 | phase1_lambda_var | 500 | 0.25 | 1 | 2 | 0 | 2 | 0.267 | 0.895 | 0.0468 | 0.691 | *NA* | *NA* | 1 | B1_Mode2 |
| re500_e0.25_l2 | phase1_lambda_var | 500 | 0.25 | 1 | 2 | 0 | 3 | 0.278 | 0.0292 | 0.969 | 0.127 | *NA* | *NA* | 2 | B2_temporal |
| re500_e0.25_l2 | phase1_lambda_var | 500 | 0.25 | 1 | 2 | 0 | 4 | 0.264 | 0.021 | 0.978 | -0.105 | *NA* | *NA* | 2 | B2_temporal |
| re500_e0.25_l5 | phase1_lambda_var | 500 | 0.25 | 1 | 5 | 0 | 0 | 0.263 | 0.325 | 0.651 | -0.414 | *NA* | *NA* | 2 | B2_temporal |
| re500_e0.25_l5 | phase1_lambda_var | 500 | 0.25 | 1 | 5 | 0 | 1 | 0.302 | 0.915 | 0.0136 | 0.744 | *NA* | *NA* | 1 | B1_Mode2 |
| re500_e0.25_l5 | phase1_lambda_var | 500 | 0.25 | 1 | 5 | 0 | 2 | 0.29 | 0.889 | 0.0569 | 0.718 | *NA* | *NA* | 1 | B1_Mode2 |
| re500_e0.25_l5 | phase1_lambda_var | 500 | 0.25 | 1 | 5 | 0 | 3 | 0.302 | 0.0263 | 0.972 | 0.126 | *NA* | *NA* | 2 | B2_temporal |
| re500_e0.25_l5 | phase1_lambda_var | 500 | 0.25 | 1 | 5 | 0 | 4 | 0.286 | 0.0008 | 0.998 | -0.0219 | *NA* | *NA* | 2 | B2_temporal |

# 3 Campagne 3 — Classification des branches

Table 8: Résumé par classe de branche — `branch_classification_summary.csv`, 3 classes.

| branch | $n$ | $\bar{f}_2$ | med $f_2$ | $\bar{f}_t$ | med $f_t$ | $\bar{C}_3$ | $\bar{E}_r$ | $\bar{E}_r^n$ | modes | $r_c$ | CFD | repr. |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| B1_Mode2 | 78 | 0.853 | 0.889 | 0.074 | 0.038 | 0.953 | 0.794 | 0.999 | (1,0)x78 | 1 | Oui | (Re=1000, E*=1.0, seed=18) |
| B2_temporal | 81 | 0.075 | 0.035 | 0.881 | 0.954 | 0.921 | 0.776 | 0.972 | (0,1)x72,(1,1)x8,(1,0)x1 | 1 | Oui | (Re=1000, E*=1.0, seed=3) |
| B3_mixed | 26 | 0.348 | 0.363 | 0.439 | 0.451 | 0.85 | 0.977 | 0.964 | (1,0)x12,(0,1)x7,(2,0)x6 | 1 | Oui | (Re=1000, E*=1.0, seed=11) |

# 4 Campagne 4 — Validation indépendante (LBM / CFD)

Table 9: Systèmes de contrôle (phase 1d) — validation — 4 lignes, `phase1d_validation.csv`.

| contrôle | Re | $E^*$ | $\epsilon$ | $K$ | $Z$ | $K_{fl}$ | $K_f/K$ | mode | $f_{dom}$ | rms | $r_{rms}$ | $n_{it}$ | T | res. | $\bar{n}$ |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| B1_mean | 1e+03 | 1 | 0.011 | 0.0123 | 11.1 | 0 | 0 | 3 | 0.479 | 1 | 0 | 2.128e+06 | 1 | 8.648e-10 | – |
| B1 | 1e+03 | 1 | 0.011 | 0.0123 | 11.1 | 3.357e-06 | 0.000272 | 3 | 0.478 | 1 | 0.792 | 2.352e+06 | 256 | 7.138e-10 | 8.19e+03 |
| B2 | 1e+03 | 1 | 0.0595 | 0.0177 | 59.5 | 0.0236 | 1.34 | 2 | 0.864 | 1 | 0.584 | 9.760e+05 | 256 | 8.600e-10 | 8.19e+03 |
| B3 | 1e+03 | 1 | 0.00801 | 0.0103 | 8.09 | 0.00016 | 0.0155 | 2 | 0.88 | 1 | 0.724 | 8.260e+05 | 256 | 9.656e-10 | 8.19e+03 |

Table 10: Systèmes de contrôle régénérés (cycles complets) — 3 lignes, `phase1d_validation_regen.csv`.

| contrôle | Re | $E^*$ | $\epsilon$ | $K$ | $Z$ | $K_{fl}$ | $K_f/K$ | mode | $f_{dom}$ | rms | $r_{rms}$ | $n_{it}$ | T | res. | $\bar{n}$ | $n_{cy}$ |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| B1 | 1e+03 | 1 | 0.011 | 0.0123 | 11.1 | 3.357e-06 | 0.000272 | 3 | 0.478 | 1 | 0.792 | 1.64e+04 | 256 | 0.000551 | 8.19e+03 | 32 |
| B2 | 1e+03 | 1 | 0.0595 | 0.0177 | 59.5 | 0.0236 | 1.34 | 2 | 0.864 | 1 | 0.584 | 1.64e+04 | 256 | 0.000765 | 8.19e+03 | 32 |
| B3 | 1e+03 | 1 | 0.00801 | 0.0103 | 8.09 | 0.00016 | 0.0155 | 2 | 0.88 | 1 | 0.724 | 1.64e+04 | 256 | 0.00164 | 8.19e+03 | 32 |

Table 11: Équivalence de performance des stimuli — 3 lignes, `perf_equivalence.csv`.

| branch | Re | $E^*$ | $\epsilon_{dis}$ | $K_{kin}$ | $Z$ | $K_f/K$ | mode | $f_{dom}$ | $r_{rms}$ | $n_{it}$ | T | lock-in | $\Delta\epsilon$ | $A_\epsilon$ | $R_K$ |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| B1 | 1e+03 | 1 | 0.011 | 0.0123 | 11.1 | 0.000272 | (3) | 0.478 | 0.792 | 2.352e+06 | 256 | 7.138e-10 | 0.000619 | 0.00151 | 0.00036 |
| B2 | 1e+03 | 1 | 0.0595 | 0.0177 | 59.5 | 1.34 | (2) | 0.864 | 0.584 | 9.760e+05 | 256 | 8.600e-10 | 0.318 | 0.36 | 1.31 |
| B3 | 1e+03 | 1 | 0.00801 | 0.0103 | 8.09 | 0.0155 | (2) | 0.88 | 0.724 | 8.260e+05 | 256 | 9.656e-10 | 0.00368 | 0.00743 | 0.019 |

Table 12: Paires d'équivalence et GCI — 3 lignes, `perf_equivalence_pairs.csv`.

| paire | $\epsilon_A$ | $\epsilon_B$ | $\Delta\epsilon_{rel}$ | GCI256 | GCI512 | verdict |
|---|---|---|---|---|---|---|
| B1-B2 | 0.011 | 0.0595 | 4.42 | 0.142 | 0.071 | distinguishable |
| B1-B3 | 0.011 | 0.00801 | 0.37 | 0.142 | 0.071 | distinguishable |
| B2-B3 | 0.0595 | 0.00801 | 6.42 | 0.142 | 0.071 | distinguishable |

Table 13: Statistiques de fenêtre — 3 lignes, `window_statistics.csv`.

| contrôle | T | lock-in | $\bar{\epsilon}$ | Delta\{}_eps | $A_\epsilon$ | $\bar{K}$ | $\Delta K$ | K\{}_fluct\{}_W2 | $R_K$ | $\Delta R_K$ | $n_{pt}$ | $n_{pw}$ |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| B1 | 256 | 7.138e-10 | 2.743e-05 | 0.000619 | 0.00151 | 3.081e-05 | 0.000124 | 1.108e-08 | 0.00036 | 0.106 | 16 | 4 |
| B2 | 256 | 8.600e-10 | 0.000185 | 0.318 | 0.36 | 4.563e-05 | 0.0353 | 5.940e-05 | 1.31 | 0.146 | 16 | 4 |
| B3 | 256 | 9.656e-10 | 2.012e-05 | 0.00368 | 0.00743 | 2.574e-05 | 0.000451 | 4.896e-07 | 0.019 | 0.343 | 16 | 4 |

Table 14: Statistiques de fenêtre régénérées — 3 lignes, `window_statistics_regen.csv`.

| contrôle | $n_{it}$ | $\bar{n}$ | $n_{cy}$ | lock-in | $\bar{\epsilon}$ | Delta\{}_eps | $A_\epsilon$ | $\bar{K}$ | $\Delta K$ | K\{}_fluct\{}_W2 | $R_K$ | $\Delta R_K$ | $\bar{\epsilon}$ | $\bar{K}$ | $\bar{Z}$ | $K_f/K$ |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| B1 | 1.64e+04 | 8.19e+03 | 32 | 0.000551 | 0.011 | 4.218e-05 | 0.00012 | 0.0123 | 2.309e-06 | 3.286e-06 | 0.000267 | 0.0929 | 0.011 | 0.0123 | 11.1 | 0.000272 |
| B2 | 1.64e+04 | 8.19e+03 | 32 | 0.000765 | 0.0601 | 0.0196 | 0.0103 | 0.0177 | 0.00141 | 0.0233 | 1.31 | 0.0336 | 0.0595 | 0.0177 | 59.5 | 1.34 |
| B3 | 1.64e+04 | 8.19e+03 | 32 | 0.00164 | 0.00801 | 0.000774 | 0.000572 | 0.0103 | 2.517e-05 | 0.000154 | 0.015 | 0.111 | 0.00801 | 0.0103 | 8.09 | 0.0155 |

# A Annexe A — Décomposition modale : 186 enregistrements

Table 15: Décomposition modale — 185 points, 1110 lignes (6 modes $i$ par point, colonnes $E^f_{iJ}$ pour $J$=0..4), `mode_analysis.csv`.

| point | Re | $E^*$ | s | id | branch | $E_t$ | cov. | $i$ | $j$ | rmse | $A_{10}$ | $A_{20}$ | $A_{30}$ | $A_{40}$ | $A_{50}$ | $A_{60}$ | $A_{70}$ | $A_{80}$ | $i$ | $E^f_{0J}$ | $E^f_{1J}$ | $E^f_{2J}$ | $E^f_{3J}$ | $E^f_{4J}$ |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| re1000_e0.1 | 1e+03 | 0.1 | 0 | 1 | B1_Mode2 | 0.12 | 1 | 1 | 0 | 1.054e-16 | -0.00235 | -0.437 | 0.138 | 0.0969 | -0.0536 | 0.015 | 0 | 0 | 0 | 2.307e-05 | 0.0554 | 4.581e-05 | 5.579e-05 | 5.020e-06 |
| re1000_e0.1 | 1e+03 | 0.1 | 0 | 1 | B1_Mode2 | 0.12 | 1 | 1 | 0 | 1.054e-16 | -0.00235 | -0.437 | 0.138 | 0.0969 | -0.0536 | 0.015 | 0 | 0 | 1 | 0.796 | 0.000703 | 0.000402 | 0.000128 | 1.845e-05 |
| re1000_e0.1 | 1e+03 | 0.1 | 0 | 1 | B1_Mode2 | 0.12 | 1 | 1 | 0 | 1.054e-16 | -0.00235 | -0.437 | 0.138 | 0.0969 | -0.0536 | 0.015 | 0 | 0 | 2 | 0.08 | 0.00452 | 0.00165 | 7.344e-11 | 4.598e-05 |
| re1000_e0.1 | 1e+03 | 0.1 | 0 | 1 | B1_Mode2 | 0.12 | 1 | 1 | 0 | 1.054e-16 | -0.00235 | -0.437 | 0.138 | 0.0969 | -0.0536 | 0.015 | 0 | 0 | 3 | 0.0392 | 0.00734 | 3.481e-06 | 1.697e-05 | 2.739e-06 |
| re1000_e0.1 | 1e+03 | 0.1 | 0 | 1 | B1_Mode2 | 0.12 | 1 | 1 | 0 | 1.054e-16 | -0.00235 | -0.437 | 0.138 | 0.0969 | -0.0536 | 0.015 | 0 | 0 | 4 | 0.012 | 0.00117 | 0.000129 | 9.321e-07 | 6.500e-06 |
| re1000_e0.1 | 1e+03 | 0.1 | 0 | 1 | B1_Mode2 | 0.12 | 1 | 1 | 0 | 1.054e-16 | -0.00235 | -0.437 | 0.138 | 0.0969 | -0.0536 | 0.015 | 0 | 0 | 5 | 0.000932 | 4.656e-05 | 0.000187 | 4.531e-08 | 4.268e-06 |
| re1000_e0.1 | 1e+03 | 0.1 | 1 | 1 | B1_Mode2 | 0.122 | 1 | 1 | 0 | 1.101e-16 | -0.0124 | 0.47 | 0.101 | -0.104 | -0.0331 | -0.00704 | 0 | 0 | 0 | 0.000631 | 0.00252 | 5.708e-05 | 3.822e-05 | 3.223e-06 |
| re1000_e0.1 | 1e+03 | 0.1 | 1 | 1 | B1_Mode2 | 0.122 | 1 | 1 | 0 | 1.101e-16 | -0.0124 | 0.47 | 0.101 | -0.104 | -0.0331 | -0.00704 | 0 | 0 | 1 | 0.903 | 0.00249 | 0.000106 | 7.612e-06 | 6.344e-06 |
| re1000_e0.1 | 1e+03 | 0.1 | 1 | 1 | B1_Mode2 | 0.122 | 1 | 1 | 0 | 1.101e-16 | -0.0124 | 0.47 | 0.101 | -0.104 | -0.0331 | -0.00704 | 0 | 0 | 2 | 0.0414 | 3.867e-05 | 6.128e-05 | 6.793e-08 | 3.028e-06 |
| re1000_e0.1 | 1e+03 | 0.1 | 1 | 1 | B1_Mode2 | 0.122 | 1 | 1 | 0 | 1.101e-16 | -0.0124 | 0.47 | 0.101 | -0.104 | -0.0331 | -0.00704 | 0 | 0 | 3 | 0.0441 | 0.000527 | 0.000133 | 6.716e-06 | 6.622e-06 |
| re1000_e0.1 | 1e+03 | 0.1 | 1 | 1 | B1_Mode2 | 0.122 | 1 | 1 | 0 | 1.101e-16 | -0.0124 | 0.47 | 0.101 | -0.104 | -0.0331 | -0.00704 | 0 | 0 | 4 | 0.00446 | 0.000152 | 1.486e-05 | 6.607e-07 | 4.180e-07 |
| re1000_e0.1 | 1e+03 | 0.1 | 1 | 1 | B1_Mode2 | 0.122 | 1 | 1 | 0 | 1.101e-16 | -0.0124 | 0.47 | 0.101 | -0.104 | -0.0331 | -0.00704 | 0 | 0 | 5 | 0.000202 | 8.719e-06 | 1.839e-06 | 3.069e-08 | 9.060e-08 |
| re1000_e0.1 | 1e+03 | 0.1 | 2 | 1 | B1_Mode2 | 0.125 | 1 | 1 | 0 | 1.231e-16 | 0.0168 | 0.474 | -0.0328 | -0.152 | 0.00751 | 0.00706 | 0 | 0 | 0 | 0.00113 | 5.814e-06 | 5.725e-07 | 1.565e-08 | 2.268e-07 |
| re1000_e0.1 | 1e+03 | 0.1 | 2 | 1 | B1_Mode2 | 0.125 | 1 | 1 | 0 | 1.231e-16 | 0.0168 | 0.474 | -0.0328 | -0.152 | 0.00751 | 0.00706 | 0 | 0 | 1 | 0.897 | 0.00271 | 3.166e-05 | 4.379e-05 | 8.596e-07 |
| re1000_e0.1 | 1e+03 | 0.1 | 2 | 1 | B1_Mode2 | 0.125 | 1 | 1 | 0 | 1.231e-16 | 0.0168 | 0.474 | -0.0328 | -0.152 | 0.00751 | 0.00706 | 0 | 0 | 2 | 0.00428 | 0.000389 | 1.488e-05 | 4.633e-07 | 1.650e-06 |
| re1000_e0.1 | 1e+03 | 0.1 | 2 | 1 | B1_Mode2 | 0.125 | 1 | 1 | 0 | 1.231e-16 | 0.0168 | 0.474 | -0.0328 | -0.152 | 0.00751 | 0.00706 | 0 | 0 | 3 | 0.0923 | 0.000881 | 0.000103 | 2.735e-06 | 7.283e-06 |
| re1000_e0.1 | 1e+03 | 0.1 | 2 | 1 | B1_Mode2 | 0.125 | 1 | 1 | 0 | 1.231e-16 | 0.0168 | 0.474 | -0.0328 | -0.152 | 0.00751 | 0.00706 | 0 | 0 | 4 | 0.000225 | 8.418e-05 | 5.329e-08 | 7.203e-07 | 4.057e-07 |
| re1000_e0.1 | 1e+03 | 0.1 | 2 | 1 | B1_Mode2 | 0.125 | 1 | 1 | 0 | 1.231e-16 | 0.0168 | 0.474 | -0.0328 | -0.152 | 0.00751 | 0.00706 | 0 | 0 | 5 | 0.000199 | 7.676e-05 | 9.914e-07 | 6.679e-07 | 4.807e-07 |
| re1000_e0.1 | 1e+03 | 0.1 | 3 | 2 | B2_temporal | 0.122 | 1 | 0 | 1 | 1.222e-16 | -0.0211 | 0.109 | -0.00638 | 0.007 | -0.0048 | 0.00317 | 0 | 0 | 0 | 0.00183 | 0.94 | 0.00243 | 0.000963 | 1.815e-07 |
| re1000_e0.1 | 1e+03 | 0.1 | 3 | 2 | B2_temporal | 0.122 | 1 | 0 | 1 | 1.222e-16 | -0.0211 | 0.109 | -0.00638 | 0.007 | -0.0048 | 0.00317 | 0 | 0 | 1 | 0.0492 | 0.00367 | 0.00023 | 0.000442 | 7.486e-05 |
| re1000_e0.1 | 1e+03 | 0.1 | 3 | 2 | B2_temporal | 0.122 | 1 | 0 | 1 | 1.222e-16 | -0.0211 | 0.109 | -0.00638 | 0.007 | -0.0048 | 0.00317 | 0 | 0 | 2 | 0.000167 | 0.000142 | 0.000405 | 3.199e-05 | 1.926e-06 |
| re1000_e0.1 | 1e+03 | 0.1 | 3 | 2 | B2_temporal | 0.122 | 1 | 0 | 1 | 1.222e-16 | -0.0211 | 0.109 | -0.00638 | 0.007 | -0.0048 | 0.00317 | 0 | 0 | 3 | 0.000202 | 0.000243 | 2.155e-05 | 2.470e-05 | 5.589e-09 |
| re1000_e0.1 | 1e+03 | 0.1 | 3 | 2 | B2_temporal | 0.122 | 1 | 0 | 1 | 1.222e-16 | -0.0211 | 0.109 | -0.00638 | 0.007 | -0.0048 | 0.00317 | 0 | 0 | 4 | 9.463e-05 | 1.365e-05 | 1.371e-06 | 5.551e-06 | 6.304e-06 |
| re1000_e0.1 | 1e+03 | 0.1 | 3 | 2 | B2_temporal | 0.122 | 1 | 0 | 1 | 1.222e-16 | -0.0211 | 0.109 | -0.00638 | 0.007 | -0.0048 | 0.00317 | 0 | 0 | 5 | 4.127e-05 | 1.324e-05 | 5.912e-07 | 5.532e-06 | 2.647e-06 |
| re1000_e0.1 | 1e+03 | 0.1 | 4 | 2 | B2_temporal | 0.125 | 1 | 0 | 1 | 1.084e-16 | -0.00461 | 0.144 | -0.00362 | -0.00735 | -0.00337 | 0.00048 | 0 | 0 | 0 | 8.467e-05 | 0.899 | 0.00123 | 0.000239 | 7.819e-05 |
| re1000_e0.1 | 1e+03 | 0.1 | 4 | 2 | B2_temporal | 0.125 | 1 | 0 | 1 | 1.084e-16 | -0.00461 | 0.144 | -0.00362 | -0.00735 | -0.00337 | 0.00048 | 0 | 0 | 1 | 0.0832 | 0.000548 | 0.00106 | 0.0017 | 4.858e-05 |
| re1000_e0.1 | 1e+03 | 0.1 | 4 | 2 | B2_temporal | 0.125 | 1 | 0 | 1 | 1.084e-16 | -0.00461 | 0.144 | -0.00362 | -0.00735 | -0.00337 | 0.00048 | 0 | 0 | 2 | 5.223e-05 | 0.0107 | 0.000491 | 0.000393 | 3.077e-06 |
| re1000_e0.1 | 1e+03 | 0.1 | 4 | 2 | B2_temporal | 0.125 | 1 | 0 | 1 | 1.084e-16 | -0.00461 | 0.144 | -0.00362 | -0.00735 | -0.00337 | 0.00048 | 0 | 0 | 3 | 0.000216 | 5.981e-05 | 0.000283 | 9.092e-05 | 1.806e-08 |
| re1000_e0.1 | 1e+03 | 0.1 | 4 | 2 | B2_temporal | 0.125 | 1 | 0 | 1 | 1.084e-16 | -0.00461 | 0.144 | -0.00362 | -0.00735 | -0.00337 | 0.00048 | 0 | 0 | 4 | 4.528e-05 | 1.902e-07 | 1.074e-05 | 5.256e-06 | 1.925e-06 |
| re1000_e0.1 | 1e+03 | 0.1 | 4 | 2 | B2_temporal | 0.125 | 1 | 0 | 1 | 1.084e-16 | -0.00461 | 0.144 | -0.00362 | -0.00735 | -0.00337 | 0.00048 | 0 | 0 | 5 | 9.213e-07 | 3.087e-06 | 9.352e-10 | 6.719e-07 | 5.460e-07 |
| re1000_e0.25 | 1e+03 | 0.25 | 0 | 3 | B3_mixed | 0.24 | 1 | 1 | 0 | 1.574e-16 | 0.0488 | -0.458 | -0.00872 | 0.143 | -0.0167 | -0.00488 | 0 | 0 | 0 | 0.00495 | 0.412 | 0.0052 | 0.0071 | 0.000823 |
| re1000_e0.25 | 1e+03 | 0.25 | 0 | 3 | B3_mixed | 0.24 | 1 | 1 | 0 | 1.574e-16 | 0.0488 | -0.458 | -0.00872 | 0.143 | -0.0167 | -0.00488 | 0 | 0 | 1 | 0.436 | 0.00422 | 0.00326 | 0.00062 | 1.059e-05 |
| re1000_e0.25 | 1e+03 | 0.25 | 0 | 3 | B3_mixed | 0.24 | 1 | 1 | 0 | 1.574e-16 | 0.0488 | -0.458 | -0.00872 | 0.143 | -0.0167 | -0.00488 | 0 | 0 | 2 | 0.000158 | 0.0621 | 0.00998 | 0.00151 | 0.000108 |
| re1000_e0.25 | 1e+03 | 0.25 | 0 | 3 | B3_mixed | 0.24 | 1 | 1 | 0 | 1.574e-16 | 0.0488 | -0.458 | -0.00872 | 0.143 | -0.0167 | -0.00488 | 0 | 0 | 3 | 0.0426 | 0.00197 | 0.00201 | 0.000232 | 9.876e-06 |
| re1000_e0.25 | 1e+03 | 0.25 | 0 | 3 | B3_mixed | 0.24 | 1 | 1 | 0 | 1.574e-16 | 0.0488 | -0.458 | -0.00872 | 0.143 | -0.0167 | -0.00488 | 0 | 0 | 4 | 0.000584 | 0.00205 | 0.0017 | 2.472e-05 | 4.306e-05 |
| re1000_e0.25 | 1e+03 | 0.25 | 0 | 3 | B3_mixed | 0.24 | 1 | 1 | 0 | 1.574e-16 | 0.0488 | -0.458 | -0.00872 | 0.143 | -0.0167 | -0.00488 | 0 | 0 | 5 | 4.953e-05 | 0.000354 | 6.252e-06 | 9.691e-05 | 4.210e-06 |
| re1000_e0.25 | 1e+03 | 0.25 | 1 | 1 | B1_Mode2 | 0.266 | 1 | 1 | 0 | 1.391e-16 | -0.00698 | 0.706 | 0.0713 | -0.119 | -0.0128 | -0.00339 | 0 | 0 | 0 | 9.172e-05 | 0.0204 | 0.00023 | 0.000268 | 9.754e-06 |
| re1000_e0.25 | 1e+03 | 0.25 | 1 | 1 | B1_Mode2 | 0.266 | 1 | 1 | 0 | 1.391e-16 | -0.00698 | 0.706 | 0.0713 | -0.119 | -0.0128 | -0.00339 | 0 | 0 | 1 | 0.936 | 0.00327 | 0.000298 | 4.147e-06 | 1.835e-05 |
| re1000_e0.25 | 1e+03 | 0.25 | 1 | 1 | B1_Mode2 | 0.266 | 1 | 1 | 0 | 1.391e-16 | -0.00698 | 0.706 | 0.0713 | -0.119 | -0.0128 | -0.00339 | 0 | 0 | 2 | 0.00956 | 0.00112 | 2.451e-05 | 9.857e-06 | 1.815e-06 |
| re1000_e0.25 | 1e+03 | 0.25 | 1 | 1 | B1_Mode2 | 0.266 | 1 | 1 | 0 | 1.391e-16 | -0.00698 | 0.706 | 0.0713 | -0.119 | -0.0128 | -0.00339 | 0 | 0 | 3 | 0.0265 | 0.00101 | 0.000223 | 1.205e-05 | 1.096e-05 |
| re1000_e0.25 | 1e+03 | 0.25 | 1 | 1 | B1_Mode2 | 0.266 | 1 | 1 | 0 | 1.391e-16 | -0.00698 | 0.706 | 0.0713 | -0.119 | -0.0128 | -0.00339 | 0 | 0 | 4 | 0.000309 | 0.000168 | 1.502e-06 | 6.150e-07 | 1.374e-07 |
| re1000_e0.25 | 1e+03 | 0.25 | 1 | 1 | B1_Mode2 | 0.266 | 1 | 1 | 0 | 1.391e-16 | -0.00698 | 0.706 | 0.0713 | -0.119 | -0.0128 | -0.00339 | 0 | 0 | 5 | 2.162e-05 | 4.036e-07 | 7.672e-08 | 6.962e-08 | 1.129e-07 |
| re1000_e0.25 | 1e+03 | 0.25 | 2 | 1 | B1_Mode2 | 0.264 | 1 | 1 | 0 | 1.454e-16 | 0.0419 | 0.698 | -0.0884 | -0.167 | 0.0223 | 0.000786 | 0 | 0 | 0 | 0.00332 | 1.320e-05 | 7.103e-07 | 1.154e-06 | 1.699e-08 |
| re1000_e0.25 | 1e+03 | 0.25 | 2 | 1 | B1_Mode2 | 0.264 | 1 | 1 | 0 | 1.454e-16 | 0.0419 | 0.698 | -0.0884 | -0.167 | 0.0223 | 0.000786 | 0 | 0 | 1 | 0.922 | 0.00122 | 0.000157 | 2.399e-05 | 5.137e-06 |
| re1000_e0.25 | 1e+03 | 0.25 | 2 | 1 | B1_Mode2 | 0.264 | 1 | 1 | 0 | 1.454e-16 | 0.0419 | 0.698 | -0.0884 | -0.167 | 0.0223 | 0.000786 | 0 | 0 | 2 | 0.0148 | 0.000691 | 1.677e-05 | 5.983e-07 | 1.493e-06 |
| re1000_e0.25 | 1e+03 | 0.25 | 2 | 1 | B1_Mode2 | 0.264 | 1 | 1 | 0 | 1.454e-16 | 0.0419 | 0.698 | -0.0884 | -0.167 | 0.0223 | 0.000786 | 0 | 0 | 3 | 0.0529 | 0.00356 | 9.150e-05 | 1.980e-05 | 6.209e-06 |
| re1000_e0.25 | 1e+03 | 0.25 | 2 | 1 | B1_Mode2 | 0.264 | 1 | 1 | 0 | 1.454e-16 | 0.0419 | 0.698 | -0.0884 | -0.167 | 0.0223 | 0.000786 | 0 | 0 | 4 | 0.000938 | 0.000165 | 1.412e-06 | 2.535e-06 | 2.488e-07 |
| re1000_e0.25 | 1e+03 | 0.25 | 2 | 1 | B1_Mode2 | 0.264 | 1 | 1 | 0 | 1.454e-16 | 0.0419 | 0.698 | -0.0884 | -0.167 | 0.0223 | 0.000786 | 0 | 0 | 5 | 1.168e-06 | 8.720e-05 | 1.336e-05 | 2.948e-07 | 2.748e-06 |
| re1000_e0.25 | 1e+03 | 0.25 | 3 | 2 | B2_temporal | 0.264 | 1 | 0 | 1 | 1.530e-16 | -0.0305 | 0.129 | -0.0117 | 0.0115 | -0.00587 | 0.00354 | 0 | 0 | 0 | 0.00176 | 0.96 | 0.000278 | 0.000356 | 4.990e-05 |
| re1000_e0.25 | 1e+03 | 0.25 | 3 | 2 | B2_temporal | 0.264 | 1 | 0 | 1 | 1.530e-16 | -0.0305 | 0.129 | -0.0117 | 0.0115 | -0.00587 | 0.00354 | 0 | 0 | 1 | 0.0317 | 0.00183 | 0.000747 | 0.00154 | 4.079e-05 |
| re1000_e0.25 | 1e+03 | 0.25 | 3 | 2 | B2_temporal | 0.264 | 1 | 0 | 1 | 1.530e-16 | -0.0305 | 0.129 | -0.0117 | 0.0115 | -0.00587 | 0.00354 | 0 | 0 | 2 | 0.000258 | 0.000336 | 3.054e-05 | 7.753e-05 | 2.833e-05 |
| re1000_e0.25 | 1e+03 | 0.25 | 3 | 2 | B2_temporal | 0.264 | 1 | 0 | 1 | 1.530e-16 | -0.0305 | 0.129 | -0.0117 | 0.0115 | -0.00587 | 0.00354 | 0 | 0 | 3 | 0.000251 | 3.821e-05 | 4.853e-05 | 7.178e-06 | 7.886e-06 |
| re1000_e0.25 | 1e+03 | 0.25 | 3 | 2 | B2_temporal | 0.264 | 1 | 0 | 1 | 1.530e-16 | -0.0305 | 0.129 | -0.0117 | 0.0115 | -0.00587 | 0.00354 | 0 | 0 | 4 | 6.545e-05 | 3.049e-06 | 2.704e-06 | 2.878e-06 | 1.195e-06 |
| re1000_e0.25 | 1e+03 | 0.25 | 3 | 2 | B2_temporal | 0.264 | 1 | 0 | 1 | 1.530e-16 | -0.0305 | 0.129 | -0.0117 | 0.0115 | -0.00587 | 0.00354 | 0 | 0 | 5 | 2.377e-05 | 3.013e-06 | 8.681e-06 | 2.845e-06 | 2.662e-06 |
| re1000_e0.25 | 1e+03 | 0.25 | 4 | 2 | B2_temporal | 0.264 | 1 | 0 | 1 | 1.645e-16 | -0.00931 | 0.143 | 0.013 | 0.00128 | -0.00373 | 0.00161 | 0 | 0 | 0 | 0.000165 | 0.944 | 0.00219 | 0.000141 | 5.291e-05 |
| re1000_e0.25 | 1e+03 | 0.25 | 4 | 2 | B2_temporal | 0.264 | 1 | 0 | 1 | 1.645e-16 | -0.00931 | 0.143 | 0.013 | 0.00128 | -0.00373 | 0.00161 | 0 | 0 | 1 | 0.0386 | 0.00112 | 0.000548 | 0.00358 | 4.336e-05 |
| re1000_e0.25 | 1e+03 | 0.25 | 4 | 2 | B2_temporal | 0.264 | 1 | 0 | 1 | 1.645e-16 | -0.00931 | 0.143 | 0.013 | 0.00128 | -0.00373 | 0.00161 | 0 | 0 | 2 | 0.00032 | 0.00825 | 0.000257 | 0.00048 | 4.235e-05 |
| re1000_e0.25 | 1e+03 | 0.25 | 4 | 2 | B2_temporal | 0.264 | 1 | 0 | 1 | 1.645e-16 | -0.00931 | 0.143 | 0.013 | 0.00128 | -0.00373 | 0.00161 | 0 | 0 | 3 | 3.096e-06 | 0.000181 | 0.000174 | 7.287e-05 | 5.247e-06 |
| re1000_e0.25 | 1e+03 | 0.25 | 4 | 2 | B2_temporal | 0.264 | 1 | 0 | 1 | 1.645e-16 | -0.00931 | 0.143 | 0.013 | 0.00128 | -0.00373 | 0.00161 | 0 | 0 | 4 | 2.639e-05 | 1.252e-05 | 1.435e-06 | 3.701e-06 | 3.448e-06 |
| re1000_e0.25 | 1e+03 | 0.25 | 4 | 2 | B2_temporal | 0.264 | 1 | 0 | 1 | 1.645e-16 | -0.00931 | 0.143 | 0.013 | 0.00128 | -0.00373 | 0.00161 | 0 | 0 | 5 | 4.942e-06 | 6.310e-06 | 1.106e-06 | 2.133e-06 | 3.823e-07 |
| re1000_e0.5 | 1e+03 | 0.5 | 0 | 3 | B3_mixed | 0.461 | 1 | 0 | 1 | 2.216e-16 | 0.0727 | -0.565 | -0.000739 | 0.128 | -0.0223 | 0.0165 | 0 | 0 | 0 | 0.00573 | 0.493 | 0.00536 | 0.0066 | 0.0014 |
| re1000_e0.5 | 1e+03 | 0.5 | 0 | 3 | B3_mixed | 0.461 | 1 | 0 | 1 | 2.216e-16 | 0.0727 | -0.565 | -0.000739 | 0.128 | -0.0223 | 0.0165 | 0 | 0 | 1 | 0.346 | 0.017 | 0.011 | 0.00312 | 1.281e-05 |
| re1000_e0.5 | 1e+03 | 0.5 | 0 | 3 | B3_mixed | 0.461 | 1 | 0 | 1 | 2.216e-16 | 0.0727 | -0.565 | -0.000739 | 0.128 | -0.0223 | 0.0165 | 0 | 0 | 2 | 5.918e-07 | 0.0651 | 0.00906 | 0.00355 | 0.000223 |
| re1000_e0.5 | 1e+03 | 0.5 | 0 | 3 | B3_mixed | 0.461 | 1 | 0 | 1 | 2.216e-16 | 0.0727 | -0.565 | -0.000739 | 0.128 | -0.0223 | 0.0165 | 0 | 0 | 3 | 0.0179 | 0.00111 | 0.00767 | 0.00123 | 2.849e-05 |
| re1000_e0.5 | 1e+03 | 0.5 | 0 | 3 | B3_mixed | 0.461 | 1 | 0 | 1 | 2.216e-16 | 0.0727 | -0.565 | -0.000739 | 0.128 | -0.0223 | 0.0165 | 0 | 0 | 4 | 0.000539 | 0.0014 | 0.00197 | 0.000104 | 0.000137 |
| re1000_e0.5 | 1e+03 | 0.5 | 0 | 3 | B3_mixed | 0.461 | 1 | 0 | 1 | 2.216e-16 | 0.0727 | -0.565 | -0.000739 | 0.128 | -0.0223 | 0.0165 | 0 | 0 | 5 | 0.000296 | 0.000279 | 6.583e-06 | 0.000259 | 1.173e-05 |

Table 15: Décomposition modale — 185 points, 1110 lignes (6 modes $i$ par point, colonnes $E^f_{iJ}$ pour $J$=0..4), `mode_analysis.csv`.

| point | Re | $E^*$ | s | id | branch | $E_t$ | cov. | $i$ | $j$ | rmse | $A_{10}$ | $A_{20}$ | $A_{30}$ | $A_{40}$ | $A_{50}$ | $A_{60}$ | $A_{70}$ | $A_{80}$ | $i$ | $E^f_{0J}$ | $E^f_{1J}$ | $E^f_{2J}$ | $E^f_{3J}$ | $E^f_{4J}$ |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| re1000_e0.5 | 1e+03 | 0.5 | 1 | 1 | B1_Mode2 | 0.516 | 1 | 1 | 0 | 2.222e-16 | 0.00546 | 0.992 | 0.0275 | -0.0986 | -0.0055 | -0.00692 | 0 | 0 | 0 | 2.891e-05 | 0.0302 | 0.00077 | 0.000417 | 3.327e-05 |
| re1000_e0.5 | 1e+03 | 0.5 | 1 | 1 | B1_Mode2 | 0.516 | 1 | 1 | 0 | 2.222e-16 | 0.00546 | 0.992 | 0.0275 | -0.0986 | -0.0055 | -0.00692 | 0 | 0 | 1 | 0.953 | 0.0021 | 0.000204 | 2.613e-06 | 1.405e-05 |
| re1000_e0.5 | 1e+03 | 0.5 | 1 | 1 | B1_Mode2 | 0.516 | 1 | 1 | 0 | 2.222e-16 | 0.00546 | 0.992 | 0.0275 | -0.0986 | -0.0055 | -0.00692 | 0 | 0 | 2 | 0.000732 | 0.00015 | 0.000302 | 2.725e-07 | 1.692e-05 |
| re1000_e0.5 | 1e+03 | 0.5 | 1 | 1 | B1_Mode2 | 0.516 | 1 | 1 | 0 | 2.222e-16 | 0.00546 | 0.992 | 0.0275 | -0.0986 | -0.0055 | -0.00692 | 0 | 0 | 3 | 0.00941 | 0.00232 | 0.000189 | 3.044e-05 | 9.252e-06 |
| re1000_e0.5 | 1e+03 | 0.5 | 1 | 1 | B1_Mode2 | 0.516 | 1 | 1 | 0 | 2.222e-16 | 0.00546 | 0.992 | 0.0275 | -0.0986 | -0.0055 | -0.00692 | 0 | 0 | 4 | 2.927e-05 | 1.644e-05 | 4.691e-07 | 6.064e-07 | 1.045e-08 |
| re1000_e0.5 | 1e+03 | 0.5 | 1 | 1 | B1_Mode2 | 0.516 | 1 | 1 | 0 | 2.222e-16 | 0.00546 | 0.992 | 0.0275 | -0.0986 | -0.0055 | -0.00692 | 0 | 0 | 5 | 4.632e-05 | 2.860e-06 | 4.497e-06 | 2.647e-07 | 2.573e-09 |
| re1000_e0.5 | 1e+03 | 0.5 | 2 | 1 | B1_Mode2 | 0.513 | 1 | 1 | 0 | 2.192e-16 | 0.0558 | 0.986 | -0.111 | -0.152 | 0.0253 | 0.00708 | 0 | 0 | 0 | 0.00303 | 0.00187 | 6.071e-05 | 8.877e-06 | 3.077e-06 |
| re1000_e0.5 | 1e+03 | 0.5 | 2 | 1 | B1_Mode2 | 0.513 | 1 | 1 | 0 | 2.192e-16 | 0.0558 | 0.986 | -0.111 | -0.152 | 0.0253 | 0.00708 | 0 | 0 | 1 | 0.949 | 0.000669 | 0.000797 | 2.836e-07 | 3.580e-05 |
| re1000_e0.5 | 1e+03 | 0.5 | 2 | 1 | B1_Mode2 | 0.513 | 1 | 1 | 0 | 2.192e-16 | 0.0558 | 0.986 | -0.111 | -0.152 | 0.0253 | 0.00708 | 0 | 0 | 2 | 0.012 | 0.00235 | 0.000287 | 1.446e-05 | 1.656e-05 |
| re1000_e0.5 | 1e+03 | 0.5 | 2 | 1 | B1_Mode2 | 0.513 | 1 | 1 | 0 | 2.192e-16 | 0.0558 | 0.986 | -0.111 | -0.152 | 0.0253 | 0.00708 | 0 | 0 | 3 | 0.0226 | 0.00602 | 4.862e-05 | 2.236e-05 | 6.510e-06 |
| re1000_e0.5 | 1e+03 | 0.5 | 2 | 1 | B1_Mode2 | 0.513 | 1 | 1 | 0 | 2.192e-16 | 0.0558 | 0.986 | -0.111 | -0.152 | 0.0253 | 0.00708 | 0 | 0 | 4 | 0.000624 | 0.000369 | 3.443e-05 | 1.351e-05 | 8.498e-07 |
| re1000_e0.5 | 1e+03 | 0.5 | 2 | 1 | B1_Mode2 | 0.513 | 1 | 1 | 0 | 2.192e-16 | 0.0558 | 0.986 | -0.111 | -0.152 | 0.0253 | 0.00708 | 0 | 0 | 5 | 4.889e-05 | 5.830e-05 | 7.575e-06 | 1.317e-08 | 2.017e-06 |
| re1000_e0.5 | 1e+03 | 0.5 | 3 | 2 | B2_temporal | 0.516 | 1 | 0 | 1 | 2.164e-16 | -0.0484 | 0.128 | -0.0123 | 0.0134 | -0.00599 | 0.00429 | 0 | 0 | 0 | 0.00227 | 0.975 | 6.464e-06 | 0.000136 | 0.000214 |
| re1000_e0.5 | 1e+03 | 0.5 | 3 | 2 | B2_temporal | 0.516 | 1 | 0 | 1 | 2.164e-16 | -0.0484 | 0.128 | -0.0123 | 0.0134 | -0.00599 | 0.00429 | 0 | 0 | 1 | 0.016 | 0.00106 | 0.00207 | 0.00215 | 8.570e-05 |
| re1000_e0.5 | 1e+03 | 0.5 | 3 | 2 | B2_temporal | 0.516 | 1 | 0 | 1 | 2.164e-16 | -0.0484 | 0.128 | -0.0123 | 0.0134 | -0.00599 | 0.00429 | 0 | 0 | 2 | 0.000147 | 0.000286 | 0.00019 | 0.000132 | 9.631e-05 |
| re1000_e0.5 | 1e+03 | 0.5 | 3 | 2 | B2_temporal | 0.516 | 1 | 0 | 1 | 2.164e-16 | -0.0484 | 0.128 | -0.0123 | 0.0134 | -0.00599 | 0.00429 | 0 | 0 | 3 | 0.000173 | 1.901e-05 | 0.000145 | 7.172e-06 | 9.134e-06 |
| re1000_e0.5 | 1e+03 | 0.5 | 3 | 2 | B2_temporal | 0.516 | 1 | 0 | 1 | 2.164e-16 | -0.0484 | 0.128 | -0.0123 | 0.0134 | -0.00599 | 0.00429 | 0 | 0 | 4 | 3.485e-05 | 7.904e-06 | 3.081e-06 | 9.582e-06 | 6.601e-08 |
| re1000_e0.5 | 1e+03 | 0.5 | 3 | 2 | B2_temporal | 0.516 | 1 | 0 | 1 | 2.164e-16 | -0.0484 | 0.128 | -0.0123 | 0.0134 | -0.00599 | 0.00429 | 0 | 0 | 5 | 1.782e-05 | 2.157e-06 | 1.764e-05 | 7.870e-07 | 2.290e-06 |
| re1000_e0.5 | 1e+03 | 0.5 | 4 | 2 | B2_temporal | 0.497 | 1 | 0 | 1 | 2.007e-16 | 0.00893 | -0.0541 | -0.0202 | 0.00621 | 0.00122 | 0.00206 | 0 | 0 | 0 | 8.037e-05 | 0.946 | 0.0196 | 0.000514 | 0.00025 |
| re1000_e0.5 | 1e+03 | 0.5 | 4 | 2 | B2_temporal | 0.497 | 1 | 0 | 1 | 2.007e-16 | 0.00893 | -0.0541 | -0.0202 | 0.00621 | 0.00122 | 0.00206 | 0 | 0 | 1 | 0.00295 | 4.008e-06 | 2.776e-05 | 0.00501 | 0.00018 |
| re1000_e0.5 | 1e+03 | 0.5 | 4 | 2 | B2_temporal | 0.497 | 1 | 0 | 1 | 2.007e-16 | 0.00893 | -0.0541 | -0.0202 | 0.00621 | 0.00122 | 0.00206 | 0 | 0 | 2 | 0.000409 | 0.0179 | 0.00602 | 6.511e-05 | 3.544e-05 |
| re1000_e0.5 | 1e+03 | 0.5 | 4 | 2 | B2_temporal | 0.497 | 1 | 0 | 1 | 2.007e-16 | 0.00893 | -0.0541 | -0.0202 | 0.00621 | 0.00122 | 0.00206 | 0 | 0 | 3 | 3.887e-05 | 0.000287 | 1.161e-06 | 0.000449 | 0.00015 |
| re1000_e0.5 | 1e+03 | 0.5 | 4 | 2 | B2_temporal | 0.497 | 1 | 0 | 1 | 2.007e-16 | 0.00893 | -0.0541 | -0.0202 | 0.00621 | 0.00122 | 0.00206 | 0 | 0 | 4 | 1.509e-06 | 3.967e-05 | 1.279e-05 | 1.949e-07 | 3.001e-05 |
| re1000_e0.5 | 1e+03 | 0.5 | 4 | 2 | B2_temporal | 0.497 | 1 | 0 | 1 | 2.007e-16 | 0.00893 | -0.0541 | -0.0202 | 0.00621 | 0.00122 | 0.00206 | 0 | 0 | 5 | 4.277e-06 | 2.255e-07 | 2.840e-06 | 3.464e-06 | 2.933e-06 |
| re1000_e1 | 1e+03 | 1 | 0 | 3 | B3_mixed | 0.887 | 1 | 2 | 0 | 2.616e-16 | 0.299 | 0.0637 | -1.15 | -0.218 | 0.34 | 0.0978 | 0 | 0 | 0 | 0.0503 | 0.0189 | 2.030e-05 | 0.000283 | 2.525e-06 |
| re1000_e1 | 1e+03 | 1 | 0 | 3 | B3_mixed | 0.887 | 1 | 2 | 0 | 2.616e-16 | 0.299 | 0.0637 | -1.15 | -0.218 | 0.34 | 0.0978 | 0 | 0 | 1 | 0.00228 | 0.0615 | 2.783e-07 | 0.000461 | 3.851e-07 |
| re1000_e1 | 1e+03 | 1 | 0 | 3 | B3_mixed | 0.887 | 1 | 2 | 0 | 2.616e-16 | 0.299 | 0.0637 | -1.15 | -0.218 | 0.34 | 0.0978 | 0 | 0 | 2 | 0.744 | 0.0132 | 7.557e-05 | 9.272e-05 | 2.326e-06 |
| re1000_e1 | 1e+03 | 1 | 0 | 3 | B3_mixed | 0.887 | 1 | 2 | 0 | 2.616e-16 | 0.299 | 0.0637 | -1.15 | -0.218 | 0.34 | 0.0978 | 0 | 0 | 3 | 0.0268 | 0.00694 | 0.000626 | 5.255e-05 | 2.905e-05 |
| re1000_e1 | 1e+03 | 1 | 0 | 3 | B3_mixed | 0.887 | 1 | 2 | 0 | 2.616e-16 | 0.299 | 0.0637 | -1.15 | -0.218 | 0.34 | 0.0978 | 0 | 0 | 4 | 0.0652 | 0.000726 | 0.00104 | 1.828e-05 | 6.643e-05 |
| re1000_e1 | 1e+03 | 1 | 0 | 3 | B3_mixed | 0.887 | 1 | 2 | 0 | 2.616e-16 | 0.299 | 0.0637 | -1.15 | -0.218 | 0.34 | 0.0978 | 0 | 0 | 5 | 0.00539 | 0.00126 | 0.000531 | 1.411e-05 | 2.358e-05 |
| re1000_e1 | 1e+03 | 1 | 1 | 1 | B1_Mode2 | 1.03 | 1 | 1 | 0 | 2.843e-16 | 0.039 | 1.41 | -0.0189 | -0.189 | -0.0143 | -0.012 | 0 | 0 | 0 | 0.000739 | 0.00469 | 0.00118 | 0.000196 | 7.107e-05 |
| re1000_e1 | 1e+03 | 1 | 1 | 1 | B1_Mode2 | 1.03 | 1 | 1 | 0 | 2.843e-16 | 0.039 | 1.41 | -0.0189 | -0.189 | -0.0143 | -0.012 | 0 | 0 | 1 | 0.962 | 0.00626 | 4.437e-05 | 2.611e-05 | 7.874e-06 |
| re1000_e1 | 1e+03 | 1 | 1 | 1 | B1_Mode2 | 1.03 | 1 | 1 | 0 | 2.843e-16 | 0.039 | 1.41 | -0.0189 | -0.189 | -0.0143 | -0.012 | 0 | 0 | 2 | 0.000174 | 0.00476 | 0.000163 | 3.527e-05 | 3.869e-06 |
| re1000_e1 | 1e+03 | 1 | 1 | 1 | B1_Mode2 | 1.03 | 1 | 1 | 0 | 2.843e-16 | 0.039 | 1.41 | -0.0189 | -0.189 | -0.0143 | -0.012 | 0 | 0 | 3 | 0.0173 | 0.00157 | 0.000296 | 4.076e-05 | 1.305e-05 |
| re1000_e1 | 1e+03 | 1 | 1 | 1 | B1_Mode2 | 1.03 | 1 | 1 | 0 | 2.843e-16 | 0.039 | 1.41 | -0.0189 | -0.189 | -0.0143 | -0.012 | 0 | 0 | 4 | 9.959e-05 | 0.000248 | 1.593e-05 | 2.380e-07 | 1.454e-06 |
| re1000_e1 | 1e+03 | 1 | 1 | 1 | B1_Mode2 | 1.03 | 1 | 1 | 0 | 2.843e-16 | 0.039 | 1.41 | -0.0189 | -0.189 | -0.0143 | -0.012 | 0 | 0 | 5 | 7.003e-05 | 6.476e-05 | 3.256e-07 | 9.019e-08 | 7.643e-08 |
| re1000_e1 | 1e+03 | 1 | 10 | 3 | B3_mixed | 1.06 | 1 | 1 | 1 | 3.431e-16 | 0.361 | -0.643 | -0.746 | 0.135 | 0.0768 | -0.0121 | 0 | 0 | 0 | 0.0611 | 0.0592 | 0.00738 | 0.00039 | 0.000146 |
| re1000_e1 | 1e+03 | 1 | 10 | 3 | B3_mixed | 1.06 | 1 | 1 | 1 | 3.431e-16 | 0.361 | -0.643 | -0.746 | 0.135 | 0.0768 | -0.0121 | 0 | 0 | 1 | 0.194 | 0.27 | 0.011 | 0.000454 | 7.080e-05 |
| re1000_e1 | 1e+03 | 1 | 10 | 3 | B3_mixed | 1.06 | 1 | 1 | 1 | 3.431e-16 | 0.361 | -0.643 | -0.746 | 0.135 | 0.0768 | -0.0121 | 0 | 0 | 2 | 0.262 | 0.0294 | 0.031 | 0.000902 | 0.000371 |
| re1000_e1 | 1e+03 | 1 | 10 | 3 | B3_mixed | 1.06 | 1 | 1 | 1 | 3.431e-16 | 0.361 | -0.643 | -0.746 | 0.135 | 0.0768 | -0.0121 | 0 | 0 | 3 | 0.00863 | 0.0325 | 0.0124 | 0.00125 | 1.562e-05 |
| re1000_e1 | 1e+03 | 1 | 10 | 3 | B3_mixed | 1.06 | 1 | 1 | 1 | 3.431e-16 | 0.361 | -0.643 | -0.746 | 0.135 | 0.0768 | -0.0121 | 0 | 0 | 4 | 0.00277 | 0.0073 | 0.00459 | 0.00184 | 1.053e-05 |
| re1000_e1 | 1e+03 | 1 | 10 | 3 | B3_mixed | 1.06 | 1 | 1 | 1 | 3.431e-16 | 0.361 | -0.643 | -0.746 | 0.135 | 0.0768 | -0.0121 | 0 | 0 | 5 | 6.922e-05 | 0.000204 | 0.00105 | 0.000153 | 8.593e-05 |
| re1000_e1 | 1e+03 | 1 | 11 | 3 | B3_mixed | 0.971 | 1 | 2 | 0 | 2.908e-16 | 0.344 | -0.647 | -0.77 | 0.293 | 0.069 | -0.0428 | 0 | 0 | 0 | 0.0608 | 0.0142 | 0.00215 | 6.872e-05 | 8.379e-05 |
| re1000_e1 | 1e+03 | 1 | 11 | 3 | B3_mixed | 0.971 | 1 | 2 | 0 | 2.908e-16 | 0.344 | -0.647 | -0.77 | 0.293 | 0.069 | -0.0428 | 0 | 0 | 1 | 0.215 | 0.189 | 0.00307 | 0.000573 | 0.000125 |
| re1000_e1 | 1e+03 | 1 | 11 | 3 | B3_mixed | 0.971 | 1 | 2 | 0 | 2.908e-16 | 0.344 | -0.647 | -0.77 | 0.293 | 0.069 | -0.0428 | 0 | 0 | 2 | 0.305 | 0.0824 | 0.008 | 6.853e-06 | 0.000147 |
| re1000_e1 | 1e+03 | 1 | 11 | 3 | B3_mixed | 0.971 | 1 | 2 | 0 | 2.908e-16 | 0.344 | -0.647 | -0.77 | 0.293 | 0.069 | -0.0428 | 0 | 0 | 3 | 0.0441 | 0.0159 | 0.0174 | 7.288e-05 | 0.000325 |
| re1000_e1 | 1e+03 | 1 | 11 | 3 | B3_mixed | 0.971 | 1 | 2 | 0 | 2.908e-16 | 0.344 | -0.647 | -0.77 | 0.293 | 0.069 | -0.0428 | 0 | 0 | 4 | 0.00245 | 0.0334 | 0.000363 | 0.000827 | 9.892e-06 |
| re1000_e1 | 1e+03 | 1 | 11 | 3 | B3_mixed | 0.971 | 1 | 2 | 0 | 2.908e-16 | 0.344 | -0.647 | -0.77 | 0.293 | 0.069 | -0.0428 | 0 | 0 | 5 | 0.000944 | 2.044e-05 | 0.00315 | 1.945e-05 | 9.062e-08 |
| re1000_e1 | 1e+03 | 1 | 12 | 2 | B2_temporal | 0.971 | 1 | 0 | 1 | 3.230e-16 | 0.0884 | -0.0222 | -0.112 | -0.021 | 0.00291 | -0.00421 | 0 | 0 | 0 | 0.00402 | 0.926 | 0.0457 | 0.000354 | 3.686e-06 |
| re1000_e1 | 1e+03 | 1 | 12 | 2 | B2_temporal | 0.971 | 1 | 0 | 1 | 3.230e-16 | 0.0884 | -0.0222 | -0.112 | -0.021 | 0.00291 | -0.00421 | 0 | 0 | 1 | 0.000254 | 0.00176 | 0.00532 | 0.00157 | 3.780e-05 |
| re1000_e1 | 1e+03 | 1 | 12 | 2 | B2_temporal | 0.971 | 1 | 0 | 1 | 3.230e-16 | 0.0884 | -0.0222 | -0.112 | -0.021 | 0.00291 | -0.00421 | 0 | 0 | 2 | 0.00649 | 0.00791 | 4.350e-05 | 6.393e-06 | 7.149e-07 |
| re1000_e1 | 1e+03 | 1 | 12 | 2 | B2_temporal | 0.971 | 1 | 0 | 1 | 3.230e-16 | 0.0884 | -0.0222 | -0.112 | -0.021 | 0.00291 | -0.00421 | 0 | 0 | 3 | 0.000228 | 2.486e-05 | 6.652e-07 | 3.038e-05 | 1.014e-06 |
| re1000_e1 | 1e+03 | 1 | 12 | 2 | B2_temporal | 0.971 | 1 | 0 | 1 | 3.230e-16 | 0.0884 | -0.0222 | -0.112 | -0.021 | 0.00291 | -0.00421 | 0 | 0 | 4 | 4.349e-06 | 9.607e-06 | 3.903e-05 | 1.604e-06 | 3.353e-06 |
| re1000_e1 | 1e+03 | 1 | 12 | 2 | B2_temporal | 0.971 | 1 | 0 | 1 | 3.230e-16 | 0.0884 | -0.0222 | -0.112 | -0.021 | 0.00291 | -0.00421 | 0 | 0 | 5 | 9.135e-06 | 2.078e-06 | 1.438e-06 | 2.600e-06 | 1.420e-06 |
| re1000_e1 | 1e+03 | 1 | 13 | 1 | B1_Mode2 | 0.987 | 1 | 1 | 0 | 2.853e-16 | 0.0424 | 1.37 | 0.0759 | -0.216 | -0.0288 | 0.0122 | 0 | 0 | 0 | 0.000911 | 0.0119 | 0.000358 | 0.000267 | 1.403e-05 |
| re1000_e1 | 1e+03 | 1 | 13 | 1 | B1_Mode2 | 0.987 | 1 | 1 | 0 | 2.853e-16 | 0.0424 | 1.37 | 0.0759 | -0.216 | -0.0288 | 0.0122 | 0 | 0 | 1 | 0.949 | 0.00487 | 0.000239 | 9.852e-08 | 4.954e-06 |
| re1000_e1 | 1e+03 | 1 | 13 | 1 | B1_Mode2 | 0.987 | 1 | 1 | 0 | 2.853e-16 | 0.0424 | 1.37 | 0.0759 | -0.216 | -0.0288 | 0.0122 | 0 | 0 | 2 | 0.00292 | 0.00488 | 4.446e-07 | 5.430e-05 | 9.778e-08 |
| re1000_e1 | 1e+03 | 1 | 13 | 1 | B1_Mode2 | 0.987 | 1 | 1 | 0 | 2.853e-16 | 0.0424 | 1.37 | 0.0759 | -0.216 | -0.0288 | 0.0122 | 0 | 0 | 3 | 0.0236 | 6.930e-05 | 0.000195 | 1.675e-06 | 1.979e-05 |
| re1000_e1 | 1e+03 | 1 | 13 | 1 | B1_Mode2 | 0.987 | 1 | 1 | 0 | 2.853e-16 | 0.0424 | 1.37 | 0.0759 | -0.216 | -0.0288 | 0.0122 | 0 | 0 | 4 | 0.000419 | 0.000143 | 9.833e-08 | 1.482e-06 | 2.732e-07 |
| re1000_e1 | 1e+03 | 1 | 13 | 1 | B1_Mode2 | 0.987 | 1 | 1 | 0 | 2.853e-16 | 0.0424 | 1.37 | 0.0759 | -0.216 | -0.0288 | 0.0122 | 0 | 0 | 5 | 7.561e-05 | 4.225e-07 | 1.900e-07 | 1.367e-09 | 2.875e-08 |
| re1000_e1 | 1e+03 | 1 | 14 | 1 | B1_Mode2 | 0.986 | 1 | 1 | 0 | 2.870e-16 | 0.159 | 1.16 | -0.401 | -0.363 | -0.0705 | 0.0452 | 0 | 0 | 0 | 0.0128 | 0.00924 | 0.000501 | 0.000124 | 1.212e-05 |
| re1000_e1 | 1e+03 | 1 | 14 | 1 | B1_Mode2 | 0.986 | 1 | 1 | 0 | 2.870e-16 | 0.159 | 1.16 | -0.401 | -0.363 | -0.0705 | 0.0452 | 0 | 0 | 1 | 0.681 | 0.0388 | 0.000581 | 0.000173 | 3.627e-05 |
| re1000_e1 | 1e+03 | 1 | 14 | 1 | B1_Mode2 | 0.986 | 1 | 1 | 0 | 2.870e-16 | 0.159 | 1.16 | -0.401 | -0.363 | -0.0705 | 0.0452 | 0 | 0 | 2 | 0.0815 | 0.0718 | 0.000439 | 0.000373 | 1.165e-05 |
| re1000_e1 | 1e+03 | 1 | 14 | 1 | B1_Mode2 | 0.986 | 1 | 1 | 0 | 2.870e-16 | 0.159 | 1.16 | -0.401 | -0.363 | -0.0705 | 0.0452 | 0 | 0 | 3 | 0.0667 | 0.00183 | 0.00618 | 8.777e-06 | 0.000256 |
| re1000_e1 | 1e+03 | 1 | 14 | 1 | B1_Mode2 | 0.986 | 1 | 1 | 0 | 2.870e-16 | 0.159 | 1.16 | -0.401 | -0.363 | -0.0705 | 0.0452 | 0 | 0 | 4 | 0.00252 | 0.0208 | 0.000966 | 1.003e-08 | 4.392e-05 |
| re1000_e1 | 1e+03 | 1 | 14 | 1 | B1_Mode2 | 0.986 | 1 | 1 | 0 | 2.870e-16 | 0.159 | 1.16 | -0.401 | -0.363 | -0.0705 | 0.0452 | 0 | 0 | 5 | 0.00104 | 0.00113 | 0.000631 | 3.301e-05 | 4.719e-06 |

Table 15: Décomposition modale — 185 points, 1110 lignes (6 modes $i$ par point, colonnes $E^f_{iJ}$ pour $J$=0..4), `mode_analysis.csv`.

| point | Re | $E^*$ | s | id | branch | $E_t$ | cov. | $i$ | $j$ | rmse | $A_{10}$ | $A_{20}$ | $A_{30}$ | $A_{40}$ | $A_{50}$ | $A_{60}$ | $A_{70}$ | $A_{80}$ | $i$ | $E^f_{0J}$ | $E^f_{1J}$ | $E^f_{2J}$ | $E^f_{3J}$ | $E^f_{4J}$ |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| re1000_e1 | 1e+03 | 1 | 15 | 1 | B1_Mode2 | 1.04 | 1 | 1 | 0 | 2.867e-16 | -0.0691 | 1.29 | -0.0428 | -0.309 | -0.0814 | 0.00622 | 0 | 0 | 0 | 0.0023 | 0.0689 | 3.071e-05 | 0.000411 | 3.112e-07 |
| re1000_e1 | 1e+03 | 1 | 15 | 1 | B1_Mode2 | 1.04 | 1 | 1 | 0 | 2.867e-16 | -0.0691 | 1.29 | -0.0428 | -0.309 | -0.0814 | 0.00622 | 0 | 0 | 1 | 0.802 | 0.0111 | 0.00437 | 0.000131 | 0.000222 |
| re1000_e1 | 1e+03 | 1 | 15 | 1 | B1_Mode2 | 1.04 | 1 | 1 | 0 | 2.867e-16 | -0.0691 | 1.29 | -0.0428 | -0.309 | -0.0814 | 0.00622 | 0 | 0 | 2 | 0.000881 | 0.0409 | 0.000399 | 0.000123 | 2.450e-05 |
| re1000_e1 | 1e+03 | 1 | 15 | 1 | B1_Mode2 | 1.04 | 1 | 1 | 0 | 2.867e-16 | -0.0691 | 1.29 | -0.0428 | -0.309 | -0.0814 | 0.00622 | 0 | 0 | 3 | 0.0461 | 0.0113 | 0.00214 | 1.676e-07 | 0.000115 |
| re1000_e1 | 1e+03 | 1 | 15 | 1 | B1_Mode2 | 1.04 | 1 | 1 | 0 | 2.867e-16 | -0.0691 | 1.29 | -0.0428 | -0.309 | -0.0814 | 0.00622 | 0 | 0 | 4 | 0.00319 | 0.00254 | 0.00138 | 6.303e-06 | 4.988e-05 |
| re1000_e1 | 1e+03 | 1 | 15 | 1 | B1_Mode2 | 1.04 | 1 | 1 | 0 | 2.867e-16 | -0.0691 | 1.29 | -0.0428 | -0.309 | -0.0814 | 0.00622 | 0 | 0 | 5 | 1.862e-05 | 0.00175 | 7.574e-08 | 2.000e-05 | 1.683e-07 |
| re1000_e1 | 1e+03 | 1 | 16 | 1 | B1_Mode2 | 0.979 | 1 | 1 | 0 | 2.923e-16 | 0.0246 | 1.28 | 0.0957 | -0.303 | -0.164 | -0.0181 | 0 | 0 | 0 | 0.000308 | 0.0321 | 2.336e-05 | 0.000306 | 9.141e-09 |
| re1000_e1 | 1e+03 | 1 | 16 | 1 | B1_Mode2 | 0.979 | 1 | 1 | 0 | 2.923e-16 | 0.0246 | 1.28 | 0.0957 | -0.303 | -0.164 | -0.0181 | 0 | 0 | 1 | 0.831 | 0.013 | 0.000478 | 0.000221 | 2.972e-05 |
| re1000_e1 | 1e+03 | 1 | 16 | 1 | B1_Mode2 | 0.979 | 1 | 1 | 0 | 2.923e-16 | 0.0246 | 1.28 | 0.0957 | -0.303 | -0.164 | -0.0181 | 0 | 0 | 2 | 0.00468 | 0.0356 | 0.000181 | 0.000218 | 2.019e-05 |
| re1000_e1 | 1e+03 | 1 | 16 | 1 | B1_Mode2 | 0.979 | 1 | 1 | 0 | 2.923e-16 | 0.0246 | 1.28 | 0.0957 | -0.303 | -0.164 | -0.0181 | 0 | 0 | 3 | 0.0468 | 0.0154 | 0.00176 | 8.239e-05 | 7.363e-05 |
| re1000_e1 | 1e+03 | 1 | 16 | 1 | B1_Mode2 | 0.979 | 1 | 1 | 0 | 2.923e-16 | 0.0246 | 1.28 | 0.0957 | -0.303 | -0.164 | -0.0181 | 0 | 0 | 4 | 0.0137 | 0.000782 | 0.00121 | 2.025e-06 | 8.624e-05 |
| re1000_e1 | 1e+03 | 1 | 16 | 1 | B1_Mode2 | 0.979 | 1 | 1 | 0 | 2.923e-16 | 0.0246 | 1.28 | 0.0957 | -0.303 | -0.164 | -0.0181 | 0 | 0 | 5 | 0.000168 | 0.0021 | 5.368e-05 | 1.796e-05 | 1.036e-05 |
| re1000_e1 | 1e+03 | 1 | 17 | 3 | B3_mixed | 0.986 | 1 | 1 | 0 | 2.982e-16 | 0.117 | -1.1 | -0.338 | 0.0764 | -0.00832 | 0.0217 | 0 | 0 | 0 | 0.00691 | 0.171 | 0.000721 | 8.736e-07 | 1.900e-05 |
| re1000_e1 | 1e+03 | 1 | 17 | 3 | B3_mixed | 0.986 | 1 | 1 | 0 | 2.982e-16 | 0.117 | -1.1 | -0.338 | 0.0764 | -0.00832 | 0.0217 | 0 | 0 | 1 | 0.617 | 0.0657 | 0.025 | 0.00166 | 0.000263 |
| re1000_e1 | 1e+03 | 1 | 17 | 3 | B3_mixed | 0.986 | 1 | 1 | 0 | 2.982e-16 | 0.117 | -1.1 | -0.338 | 0.0764 | -0.00832 | 0.0217 | 0 | 0 | 2 | 0.058 | 0.0347 | 9.885e-05 | 0.00125 | 0.000159 |
| re1000_e1 | 1e+03 | 1 | 17 | 3 | B3_mixed | 0.986 | 1 | 1 | 0 | 2.982e-16 | 0.117 | -1.1 | -0.338 | 0.0764 | -0.00832 | 0.0217 | 0 | 0 | 3 | 0.00296 | 0.000575 | 0.00985 | 0.000394 | 1.226e-05 |
| re1000_e1 | 1e+03 | 1 | 17 | 3 | B3_mixed | 0.986 | 1 | 1 | 0 | 2.982e-16 | 0.117 | -1.1 | -0.338 | 0.0764 | -0.00832 | 0.0217 | 0 | 0 | 4 | 3.509e-05 | 0.00285 | 0.000135 | 0.00054 | 5.370e-05 |
| re1000_e1 | 1e+03 | 1 | 17 | 3 | B3_mixed | 0.986 | 1 | 1 | 0 | 2.982e-16 | 0.117 | -1.1 | -0.338 | 0.0764 | -0.00832 | 0.0217 | 0 | 0 | 5 | 0.000239 | 2.306e-06 | 2.410e-05 | 3.080e-05 | 2.330e-05 |
| re1000_e1 | 1e+03 | 1 | 18 | 1 | B1_Mode2 | 1.02 | 1 | 1 | 0 | 2.917e-16 | 0.0249 | -1.41 | -0.00833 | 0.168 | -0.0464 | 0.0398 | 0 | 0 | 0 | 0.000304 | 0.00291 | 9.940e-05 | 2.822e-05 | 7.772e-06 |
| re1000_e1 | 1e+03 | 1 | 18 | 1 | B1_Mode2 | 1.02 | 1 | 1 | 0 | 2.917e-16 | 0.0249 | -1.41 | -0.00833 | 0.168 | -0.0464 | 0.0398 | 0 | 0 | 1 | 0.974 | 0.00102 | 1.764e-05 | 1.136e-06 | 2.493e-06 |
| re1000_e1 | 1e+03 | 1 | 18 | 1 | B1_Mode2 | 1.02 | 1 | 1 | 0 | 2.917e-16 | 0.0249 | -1.41 | -0.00833 | 0.168 | -0.0464 | 0.0398 | 0 | 0 | 2 | 3.406e-05 | 0.000943 | 1.308e-05 | 1.570e-05 | 5.157e-09 |
| re1000_e1 | 1e+03 | 1 | 18 | 1 | B1_Mode2 | 1.02 | 1 | 1 | 0 | 2.917e-16 | 0.0249 | -1.41 | -0.00833 | 0.168 | -0.0464 | 0.0398 | 0 | 0 | 3 | 0.0138 | 0.00437 | 0.000194 | 6.739e-05 | 8.195e-06 |
| re1000_e1 | 1e+03 | 1 | 18 | 1 | B1_Mode2 | 1.02 | 1 | 1 | 0 | 2.917e-16 | 0.0249 | -1.41 | -0.00833 | 0.168 | -0.0464 | 0.0398 | 0 | 0 | 4 | 0.00106 | 0.000306 | 1.205e-05 | 4.889e-06 | 4.276e-07 |
| re1000_e1 | 1e+03 | 1 | 18 | 1 | B1_Mode2 | 1.02 | 1 | 1 | 0 | 2.917e-16 | 0.0249 | -1.41 | -0.00833 | 0.168 | -0.0464 | 0.0398 | 0 | 0 | 5 | 0.000777 | 4.268e-06 | 5.817e-05 | 1.691e-06 | 5.439e-06 |
| re1000_e1 | 1e+03 | 1 | 19 | 1 | B1_Mode2 | 0.993 | 1 | 1 | 0 | 2.738e-16 | 0.0286 | -1.18 | -0.0798 | 0.247 | -0.0606 | 0.0107 | 0 | 0 | 0 | 0.000413 | 0.131 | 0.000118 | 2.754e-05 | 5.157e-06 |
| re1000_e1 | 1e+03 | 1 | 19 | 1 | B1_Mode2 | 0.993 | 1 | 1 | 0 | 2.738e-16 | 0.0286 | -1.18 | -0.0798 | 0.247 | -0.0606 | 0.0107 | 0 | 0 | 1 | 0.699 | 0.0328 | 0.0132 | 0.000303 | 0.000365 |
| re1000_e1 | 1e+03 | 1 | 19 | 1 | B1_Mode2 | 0.993 | 1 | 1 | 0 | 2.738e-16 | 0.0286 | -1.18 | -0.0798 | 0.247 | -0.0606 | 0.0107 | 0 | 0 | 2 | 0.00321 | 0.0647 | 0.00283 | 1.919e-05 | 7.311e-05 |
| re1000_e1 | 1e+03 | 1 | 19 | 1 | B1_Mode2 | 0.993 | 1 | 1 | 0 | 2.738e-16 | 0.0286 | -1.18 | -0.0798 | 0.247 | -0.0606 | 0.0107 | 0 | 0 | 3 | 0.0307 | 0.00825 | 0.00483 | 0.00016 | 0.000167 |
| re1000_e1 | 1e+03 | 1 | 19 | 1 | B1_Mode2 | 0.993 | 1 | 1 | 0 | 2.738e-16 | 0.0286 | -1.18 | -0.0798 | 0.247 | -0.0606 | 0.0107 | 0 | 0 | 4 | 0.00185 | 0.00275 | 0.00258 | 1.963e-06 | 7.999e-06 |
| re1000_e1 | 1e+03 | 1 | 19 | 1 | B1_Mode2 | 0.993 | 1 | 1 | 0 | 2.738e-16 | 0.0286 | -1.18 | -0.0798 | 0.247 | -0.0606 | 0.0107 | 0 | 0 | 5 | 5.770e-05 | 0.000686 | 3.041e-06 | 0.000202 | 1.080e-07 |
| re1000_e1 | 1e+03 | 1 | 2 | 1 | B1_Mode2 | 1.02 | 1 | 1 | 0 | 2.885e-16 | 0.052 | 1.38 | -0.0897 | -0.116 | -0.0422 | 0.015 | 0 | 0 | 0 | 0.00132 | 0.0154 | 0.000166 | 4.551e-05 | 9.475e-06 |
| re1000_e1 | 1e+03 | 1 | 2 | 1 | B1_Mode2 | 1.02 | 1 | 1 | 0 | 2.885e-16 | 0.052 | 1.38 | -0.0897 | -0.116 | -0.0422 | 0.015 | 0 | 0 | 1 | 0.931 | 0.00143 | 0.00273 | 3.350e-05 | 0.000123 |
| re1000_e1 | 1e+03 | 1 | 2 | 1 | B1_Mode2 | 1.02 | 1 | 1 | 0 | 2.885e-16 | 0.052 | 1.38 | -0.0897 | -0.116 | -0.0422 | 0.015 | 0 | 0 | 2 | 0.00393 | 0.0166 | 0.00134 | 5.215e-05 | 5.908e-05 |
| re1000_e1 | 1e+03 | 1 | 2 | 1 | B1_Mode2 | 1.02 | 1 | 1 | 0 | 2.885e-16 | 0.052 | 1.38 | -0.0897 | -0.116 | -0.0422 | 0.015 | 0 | 0 | 3 | 0.00652 | 0.0132 | 0.000296 | 9.715e-06 | 2.508e-05 |
| re1000_e1 | 1e+03 | 1 | 2 | 1 | B1_Mode2 | 1.02 | 1 | 1 | 0 | 2.885e-16 | 0.052 | 1.38 | -0.0897 | -0.116 | -0.0422 | 0.015 | 0 | 0 | 4 | 0.000871 | 0.00355 | 0.000587 | 3.367e-05 | 1.824e-05 |
| re1000_e1 | 1e+03 | 1 | 2 | 1 | B1_Mode2 | 1.02 | 1 | 1 | 0 | 2.885e-16 | 0.052 | 1.38 | -0.0897 | -0.116 | -0.0422 | 0.015 | 0 | 0 | 5 | 0.00011 | 2.556e-05 | 3.074e-05 | 1.050e-05 | 5.943e-07 |
| re1000_e1 | 1e+03 | 1 | 3 | 2 | B2_temporal | 0.972 | 1 | 0 | 1 | 3.127e-16 | -0.0821 | 0.105 | -0.0013 | 0.00515 | -0.00521 | 0.00308 | 0 | 0 | 0 | 0.00347 | 0.93 | 0.0422 | 0.012 | 0.00183 |
| re1000_e1 | 1e+03 | 1 | 3 | 2 | B2_temporal | 0.972 | 1 | 0 | 1 | 3.127e-16 | -0.0821 | 0.105 | -0.0013 | 0.00515 | -0.00521 | 0.00308 | 0 | 0 | 1 | 0.00572 | 4.972e-05 | 0.00272 | 0.00103 | 1.338e-05 |
| re1000_e1 | 1e+03 | 1 | 3 | 2 | B2_temporal | 0.972 | 1 | 0 | 1 | 3.127e-16 | -0.0821 | 0.105 | -0.0013 | 0.00515 | -0.00521 | 0.00308 | 0 | 0 | 2 | 8.706e-07 | 9.964e-06 | 0.000313 | 5.924e-05 | 0.000213 |
| re1000_e1 | 1e+03 | 1 | 3 | 2 | B2_temporal | 0.972 | 1 | 0 | 1 | 3.127e-16 | -0.0821 | 0.105 | -0.0013 | 0.00515 | -0.00521 | 0.00308 | 0 | 0 | 3 | 1.366e-05 | 0.000415 | 7.579e-05 | 1.107e-05 | 8.833e-07 |
| re1000_e1 | 1e+03 | 1 | 3 | 2 | B2_temporal | 0.972 | 1 | 0 | 1 | 3.127e-16 | -0.0821 | 0.105 | -0.0013 | 0.00515 | -0.00521 | 0.00308 | 0 | 0 | 4 | 1.397e-05 | 2.848e-07 | 5.636e-09 | 4.424e-07 | 1.238e-06 |
| re1000_e1 | 1e+03 | 1 | 3 | 2 | B2_temporal | 0.972 | 1 | 0 | 1 | 3.127e-16 | -0.0821 | 0.105 | -0.0013 | 0.00515 | -0.00521 | 0.00308 | 0 | 0 | 5 | 4.883e-06 | 2.342e-05 | 5.334e-06 | 1.466e-06 | 6.143e-06 |
| re1000_e1 | 1e+03 | 1 | 4 | 2 | B2_temporal | 0.966 | 1 | 0 | 1 | 3.073e-16 | 0.0245 | -0.195 | -0.00669 | 0.0243 | -0.000117 | 0.00554 | 0 | 0 | 0 | 0.000311 | 0.867 | 0.0724 | 0.00667 | 0.000116 |
| re1000_e1 | 1e+03 | 1 | 4 | 2 | B2_temporal | 0.966 | 1 | 0 | 1 | 3.073e-16 | 0.0245 | -0.195 | -0.00669 | 0.0243 | -0.000117 | 0.00554 | 0 | 0 | 1 | 0.0197 | 0.000909 | 0.000654 | 0.00337 | 0.00106 |
| re1000_e1 | 1e+03 | 1 | 4 | 2 | B2_temporal | 0.966 | 1 | 0 | 1 | 3.073e-16 | 0.0245 | -0.195 | -0.00669 | 0.0243 | -0.000117 | 0.00554 | 0 | 0 | 2 | 2.317e-05 | 0.0126 | 0.0128 | 0.000883 | 1.650e-05 |
| re1000_e1 | 1e+03 | 1 | 4 | 2 | B2_temporal | 0.966 | 1 | 0 | 1 | 3.073e-16 | 0.0245 | -0.195 | -0.00669 | 0.0243 | -0.000117 | 0.00554 | 0 | 0 | 3 | 0.000306 | 2.499e-05 | 0.000112 | 0.000308 | 0.000423 |
| re1000_e1 | 1e+03 | 1 | 4 | 2 | B2_temporal | 0.966 | 1 | 0 | 1 | 3.073e-16 | 0.0245 | -0.195 | -0.00669 | 0.0243 | -0.000117 | 0.00554 | 0 | 0 | 4 | 7.069e-09 | 0.00037 | 8.136e-08 | 3.022e-05 | 3.986e-06 |
| re1000_e1 | 1e+03 | 1 | 4 | 2 | B2_temporal | 0.966 | 1 | 0 | 1 | 3.073e-16 | 0.0245 | -0.195 | -0.00669 | 0.0243 | -0.000117 | 0.00554 | 0 | 0 | 5 | 1.587e-05 | 5.145e-07 | 3.266e-06 | 5.635e-06 | 8.666e-06 |
| re1000_e1 | 1e+03 | 1 | 5 | 1 | B1_Mode2 | 0.989 | 1 | 1 | 0 | 2.684e-16 | 0.176 | 1.13 | -0.322 | -0.322 | -0.0527 | 0.0352 | 0 | 0 | 0 | 0.0157 | 0.0439 | 0.000894 | 0.00023 | 1.773e-05 |
| re1000_e1 | 1e+03 | 1 | 5 | 1 | B1_Mode2 | 0.989 | 1 | 1 | 0 | 2.684e-16 | 0.176 | 1.13 | -0.322 | -0.322 | -0.0527 | 0.0352 | 0 | 0 | 1 | 0.65 | 0.0848 | 0.00134 | 0.000476 | 6.012e-05 |
| re1000_e1 | 1e+03 | 1 | 5 | 1 | B1_Mode2 | 0.989 | 1 | 1 | 0 | 2.684e-16 | 0.176 | 1.13 | -0.322 | -0.322 | -0.0527 | 0.0352 | 0 | 0 | 2 | 0.0523 | 0.0625 | 4.170e-05 | 2.463e-05 | 1.025e-05 |
| re1000_e1 | 1e+03 | 1 | 5 | 1 | B1_Mode2 | 0.989 | 1 | 1 | 0 | 2.684e-16 | 0.176 | 1.13 | -0.322 | -0.322 | -0.0527 | 0.0352 | 0 | 0 | 3 | 0.0526 | 0.00278 | 0.0105 | 9.294e-06 | 0.000394 |
| re1000_e1 | 1e+03 | 1 | 5 | 1 | B1_Mode2 | 0.989 | 1 | 1 | 0 | 2.684e-16 | 0.176 | 1.13 | -0.322 | -0.322 | -0.0527 | 0.0352 | 0 | 0 | 4 | 0.0014 | 0.0174 | 0.000732 | 0.00011 | 3.229e-05 |
| re1000_e1 | 1e+03 | 1 | 5 | 1 | B1_Mode2 | 0.989 | 1 | 1 | 0 | 2.684e-16 | 0.176 | 1.13 | -0.322 | -0.322 | -0.0527 | 0.0352 | 0 | 0 | 5 | 0.000626 | 0.000953 | 0.000436 | 5.598e-05 | 1.252e-07 |
| re1000_e1 | 1e+03 | 1 | 6 | 1 | B1_Mode2 | 0.998 | 1 | 1 | 0 | 2.856e-16 | -0.132 | 1.35 | 0.224 | -0.028 | -0.135 | -0.000554 | 0 | 0 | 0 | 0.00875 | 0.00983 | 0.000158 | 3.838e-05 | 3.487e-06 |
| re1000_e1 | 1e+03 | 1 | 6 | 1 | B1_Mode2 | 0.998 | 1 | 1 | 0 | 2.856e-16 | -0.132 | 1.35 | 0.224 | -0.028 | -0.135 | -0.000554 | 0 | 0 | 1 | 0.918 | 0.0081 | 0.00107 | 6.526e-05 | 4.204e-05 |
| re1000_e1 | 1e+03 | 1 | 6 | 1 | B1_Mode2 | 0.998 | 1 | 1 | 0 | 2.856e-16 | -0.132 | 1.35 | 0.224 | -0.028 | -0.135 | -0.000554 | 0 | 0 | 2 | 0.0251 | 0.00254 | 0.00165 | 3.717e-05 | 8.502e-05 |
| re1000_e1 | 1e+03 | 1 | 6 | 1 | B1_Mode2 | 0.998 | 1 | 1 | 0 | 2.856e-16 | -0.132 | 1.35 | 0.224 | -0.028 | -0.135 | -0.000554 | 0 | 0 | 3 | 0.000393 | 0.0123 | 0.000121 | 9.684e-06 | 6.145e-06 |
| re1000_e1 | 1e+03 | 1 | 6 | 1 | B1_Mode2 | 0.998 | 1 | 1 | 0 | 2.856e-16 | -0.132 | 1.35 | 0.224 | -0.028 | -0.135 | -0.000554 | 0 | 0 | 4 | 0.00917 | 0.000887 | 0.000293 | 1.278e-05 | 9.454e-06 |
| re1000_e1 | 1e+03 | 1 | 6 | 1 | B1_Mode2 | 0.998 | 1 | 1 | 0 | 2.856e-16 | -0.132 | 1.35 | 0.224 | -0.028 | -0.135 | -0.000554 | 0 | 0 | 5 | 1.534e-07 | 0.000686 | 0.000193 | 9.965e-06 | 4.410e-06 |
| re1000_e1 | 1e+03 | 1 | 7 | 1 | B1_Mode2 | 0.973 | 1 | 1 | 0 | 2.762e-16 | -0.108 | 1.35 | 0.107 | -0.214 | -0.0141 | -0.0151 | 0 | 0 | 0 | 0.00596 | 0.00793 | 4.126e-05 | 0.000104 | 1.570e-06 |
| re1000_e1 | 1e+03 | 1 | 7 | 1 | B1_Mode2 | 0.973 | 1 | 1 | 0 | 2.762e-16 | -0.108 | 1.35 | 0.107 | -0.214 | -0.0141 | -0.0151 | 0 | 0 | 1 | 0.943 | 1.603e-06 | 0.00149 | 2.969e-05 | 7.051e-05 |
| re1000_e1 | 1e+03 | 1 | 7 | 1 | B1_Mode2 | 0.973 | 1 | 1 | 0 | 2.762e-16 | -0.108 | 1.35 | 0.107 | -0.214 | -0.0141 | -0.0151 | 0 | 0 | 2 | 0.00593 | 0.00121 | 3.932e-05 | 1.145e-05 | 1.757e-06 |
| re1000_e1 | 1e+03 | 1 | 7 | 1 | B1_Mode2 | 0.973 | 1 | 1 | 0 | 2.762e-16 | -0.108 | 1.35 | 0.107 | -0.214 | -0.0141 | -0.0151 | 0 | 0 | 3 | 0.0236 | 0.00863 | 0.000326 | 3.996e-05 | 1.526e-05 |
| re1000_e1 | 1e+03 | 1 | 7 | 1 | B1_Mode2 | 0.973 | 1 | 1 | 0 | 2.762e-16 | -0.108 | 1.35 | 0.107 | -0.214 | -0.0141 | -0.0151 | 0 | 0 | 4 | 0.000102 | 0.000182 | 3.185e-07 | 3.776e-07 | 7.196e-09 |
| re1000_e1 | 1e+03 | 1 | 7 | 1 | B1_Mode2 | 0.973 | 1 | 1 | 0 | 2.762e-16 | -0.108 | 1.35 | 0.107 | -0.214 | -0.0141 | -0.0151 | 0 | 0 | 5 | 0.000117 | 0.000838 | 1.647e-05 | 5.075e-07 | 1.915e-06 |

Table 15: Décomposition modale — 185 points, 1110 lignes (6 modes $i$ par point, colonnes $E^f_{iJ}$ pour $J$=0..4), `mode_analysis.csv`.

| point | Re | $E^*$ | s | id | branch | $E_t$ | cov. | $i$ | $j$ | rmse | $A_{10}$ | $A_{20}$ | $A_{30}$ | $A_{40}$ | $A_{50}$ | $A_{60}$ | $A_{70}$ | $A_{80}$ | $i$ | $E^f_{0J}$ | $E^f_{1J}$ | $E^f_{2J}$ | $E^f_{3J}$ | $E^f_{4J}$ |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| re1000_e1 | 1e+03 | 1 | 8 | 2 | B2_temporal | 0.944 | 1 | 0 | 1 | 3.058e-16 | 0.274 | 0.505 | -0.65 | -0.321 | 0.112 | 0.0492 | 0 | 0 | 0 | 0.0397 | 0.278 | 0.0188 | 8.073e-05 | 1.947e-05 |
| re1000_e1 | 1e+03 | 1 | 8 | 2 | B2_temporal | 0.944 | 1 | 0 | 1 | 3.058e-16 | 0.274 | 0.505 | -0.65 | -0.321 | 0.112 | 0.0492 | 0 | 0 | 1 | 0.135 | 0.145 | 0.0166 | 0.00177 | 1.525e-05 |
| re1000_e1 | 1e+03 | 1 | 8 | 2 | B2_temporal | 0.944 | 1 | 0 | 1 | 3.058e-16 | 0.274 | 0.505 | -0.65 | -0.321 | 0.112 | 0.0492 | 0 | 0 | 2 | 0.224 | 0.012 | 1.479e-05 | 0.00134 | 0.000423 |
| re1000_e1 | 1e+03 | 1 | 8 | 2 | B2_temporal | 0.944 | 1 | 0 | 1 | 3.058e-16 | 0.274 | 0.505 | -0.65 | -0.321 | 0.112 | 0.0492 | 0 | 0 | 3 | 0.0545 | 0.0454 | 0.00843 | 1.363e-06 | 9.928e-05 |
| re1000_e1 | 1e+03 | 1 | 8 | 2 | B2_temporal | 0.944 | 1 | 0 | 1 | 3.058e-16 | 0.274 | 0.505 | -0.65 | -0.321 | 0.112 | 0.0492 | 0 | 0 | 4 | 0.00659 | 0.00104 | 0.00545 | 0.000839 | 9.928e-05 |
| re1000_e1 | 1e+03 | 1 | 8 | 2 | B2_temporal | 0.944 | 1 | 0 | 1 | 3.058e-16 | 0.274 | 0.505 | -0.65 | -0.321 | 0.112 | 0.0492 | 0 | 0 | 5 | 0.00128 | 0.0026 | 4.366e-07 | 0.000461 | 0.000208 |
| re1000_e1 | 1e+03 | 1 | 9 | 3 | B3_mixed | 1.01 | 1 | 1 | 0 | 2.710e-16 | 0.262 | 0.982 | -0.432 | -0.312 | -0.0326 | 0.0269 | 0 | 0 | 0 | 0.0339 | 0.0345 | 0.00134 | 0.000199 | 5.408e-05 |
| re1000_e1 | 1e+03 | 1 | 9 | 3 | B3_mixed | 1.01 | 1 | 1 | 0 | 2.710e-16 | 0.262 | 0.982 | -0.432 | -0.312 | -0.0326 | 0.0269 | 0 | 0 | 1 | 0.477 | 0.128 | 0.00664 | 0.000315 | 0.000175 |
| re1000_e1 | 1e+03 | 1 | 9 | 3 | B3_mixed | 1.01 | 1 | 1 | 0 | 2.710e-16 | 0.262 | 0.982 | -0.432 | -0.312 | -0.0326 | 0.0269 | 0 | 0 | 2 | 0.0922 | 0.112 | 0.00224 | 9.446e-05 | 0.000102 |
| re1000_e1 | 1e+03 | 1 | 9 | 3 | B3_mixed | 1.01 | 1 | 1 | 0 | 2.710e-16 | 0.262 | 0.982 | -0.432 | -0.312 | -0.0326 | 0.0269 | 0 | 0 | 3 | 0.0481 | 2.007e-05 | 0.0251 | 2.507e-05 | 0.000261 |
| re1000_e1 | 1e+03 | 1 | 9 | 3 | B3_mixed | 1.01 | 1 | 1 | 0 | 2.710e-16 | 0.262 | 0.982 | -0.432 | -0.312 | -0.0326 | 0.0269 | 0 | 0 | 4 | 0.000525 | 0.0324 | 0.000768 | 0.001 | 1.944e-07 |
| re1000_e1 | 1e+03 | 1 | 9 | 3 | B3_mixed | 1.01 | 1 | 1 | 0 | 2.710e-16 | 0.262 | 0.982 | -0.432 | -0.312 | -0.0326 | 0.0269 | 0 | 0 | 5 | 0.000359 | 0.000429 | 0.00242 | 0.000158 | 1.015e-06 |
| re1000_e2 | 1e+03 | 2 | 0 | 3 | B3_mixed | 1.81 | 1 | 2 | 0 | 3.902e-16 | 0.538 | 0.239 | -1.62 | -0.0644 | 0.403 | 0.0531 | 0 | 0 | 0 | 0.0801 | 0.0281 | 0.00113 | 0.000398 | 3.975e-05 |
| re1000_e2 | 1e+03 | 2 | 0 | 3 | B3_mixed | 1.81 | 1 | 2 | 0 | 3.902e-16 | 0.538 | 0.239 | -1.62 | -0.0644 | 0.403 | 0.0531 | 0 | 0 | 1 | 0.0158 | 0.078 | 0.000149 | 0.000541 | 1.498e-05 |
| re1000_e2 | 1e+03 | 2 | 0 | 3 | B3_mixed | 1.81 | 1 | 2 | 0 | 3.902e-16 | 0.538 | 0.239 | -1.62 | -0.0644 | 0.403 | 0.0531 | 0 | 0 | 2 | 0.729 | 0.00331 | 0.00156 | 2.149e-05 | 6.720e-05 |
| re1000_e2 | 1e+03 | 2 | 0 | 3 | B3_mixed | 1.81 | 1 | 2 | 0 | 3.902e-16 | 0.538 | 0.239 | -1.62 | -0.0644 | 0.403 | 0.0531 | 0 | 0 | 3 | 0.00115 | 0.013 | 4.959e-06 | 3.156e-05 | 6.106e-07 |
| re1000_e2 | 1e+03 | 2 | 0 | 3 | B3_mixed | 1.81 | 1 | 2 | 0 | 3.902e-16 | 0.538 | 0.239 | -1.62 | -0.0644 | 0.403 | 0.0531 | 0 | 0 | 4 | 0.0449 | 0.000182 | 0.00143 | 1.426e-05 | 7.082e-05 |
| re1000_e2 | 1e+03 | 2 | 0 | 3 | B3_mixed | 1.81 | 1 | 2 | 0 | 3.902e-16 | 0.538 | 0.239 | -1.62 | -0.0644 | 0.403 | 0.0531 | 0 | 0 | 5 | 0.000779 | 0.000523 | 2.130e-06 | 1.676e-08 | 1.247e-06 |
| re1000_e2 | 1e+03 | 2 | 1 | 1 | B1_Mode2 | 2.04 | 1 | 1 | 0 | 4.226e-16 | 0.103 | 1.96 | -0.0884 | -0.25 | -0.0548 | -0.0297 | 0 | 0 | 0 | 0.0026 | 0.00168 | 0.00192 | 3.614e-05 | 0.000133 |
| re1000_e2 | 1e+03 | 2 | 1 | 1 | B1_Mode2 | 2.04 | 1 | 1 | 0 | 4.226e-16 | 0.103 | 1.96 | -0.0884 | -0.25 | -0.0548 | -0.0297 | 0 | 0 | 1 | 0.936 | 0.0196 | 7.206e-05 | 9.698e-05 | 1.758e-05 |
| re1000_e2 | 1e+03 | 2 | 1 | 1 | B1_Mode2 | 2.04 | 1 | 1 | 0 | 4.226e-16 | 0.103 | 1.96 | -0.0884 | -0.25 | -0.0548 | -0.0297 | 0 | 0 | 2 | 0.00191 | 0.0158 | 0.000518 | 0.000109 | 3.646e-06 |
| re1000_e2 | 1e+03 | 2 | 1 | 1 | B1_Mode2 | 2.04 | 1 | 1 | 0 | 4.226e-16 | 0.103 | 1.96 | -0.0884 | -0.25 | -0.0548 | -0.0297 | 0 | 0 | 3 | 0.0153 | 0.00035 | 0.00067 | 3.787e-05 | 2.122e-05 |
| re1000_e2 | 1e+03 | 2 | 1 | 1 | B1_Mode2 | 2.04 | 1 | 1 | 0 | 4.226e-16 | 0.103 | 1.96 | -0.0884 | -0.25 | -0.0548 | -0.0297 | 0 | 0 | 4 | 0.000736 | 0.00151 | 3.154e-05 | 8.923e-06 | 5.841e-06 |
| re1000_e2 | 1e+03 | 2 | 1 | 1 | B1_Mode2 | 2.04 | 1 | 1 | 0 | 4.226e-16 | 0.103 | 1.96 | -0.0884 | -0.25 | -0.0548 | -0.0297 | 0 | 0 | 5 | 0.000215 | 0.000116 | 9.787e-06 | 1.291e-06 | 4.248e-08 |
| re1000_e2 | 1e+03 | 2 | 2 | 1 | B1_Mode2 | 1.93 | 1 | 1 | 0 | 4.302e-16 | 0.437 | 1.46 | -1.04 | -0.0191 | 0.0617 | -0.0192 | 0 | 0 | 0 | 0.0493 | 0.0499 | 0.00228 | 7.334e-05 | 3.930e-05 |
| re1000_e2 | 1e+03 | 2 | 2 | 1 | B1_Mode2 | 1.93 | 1 | 1 | 0 | 4.302e-16 | 0.437 | 1.46 | -1.04 | -0.0191 | 0.0617 | -0.0192 | 0 | 0 | 1 | 0.554 | 0.0309 | 0.00448 | 5.481e-06 | 0.000159 |
| re1000_e2 | 1e+03 | 2 | 2 | 1 | B1_Mode2 | 1.93 | 1 | 1 | 0 | 4.302e-16 | 0.437 | 1.46 | -1.04 | -0.0191 | 0.0617 | -0.0192 | 0 | 0 | 2 | 0.278 | 0.0139 | 0.0011 | 5.661e-06 | 3.034e-05 |
| re1000_e2 | 1e+03 | 2 | 2 | 1 | B1_Mode2 | 1.93 | 1 | 1 | 0 | 4.302e-16 | 0.437 | 1.46 | -1.04 | -0.0191 | 0.0617 | -0.0192 | 0 | 0 | 3 | 9.463e-05 | 0.000483 | 0.00155 | 1.066e-07 | 4.322e-05 |
| re1000_e2 | 1e+03 | 2 | 2 | 1 | B1_Mode2 | 1.93 | 1 | 1 | 0 | 4.302e-16 | 0.437 | 1.46 | -1.04 | -0.0191 | 0.0617 | -0.0192 | 0 | 0 | 4 | 0.000983 | 0.00954 | 6.493e-05 | 7.312e-06 | 4.327e-06 |
| re1000_e2 | 1e+03 | 2 | 2 | 1 | B1_Mode2 | 1.93 | 1 | 1 | 0 | 4.302e-16 | 0.437 | 1.46 | -1.04 | -0.0191 | 0.0617 | -0.0192 | 0 | 0 | 5 | 9.532e-05 | 0.00207 | 0.000273 | 2.630e-05 | 9.759e-06 |
| re1000_e2 | 1e+03 | 2 | 3 | 2 | B2_temporal | 1.94 | 1 | 0 | 1 | 4.224e-16 | 0.21 | -0.214 | -0.158 | -0.0589 | -0.0118 | -0.00594 | 0 | 0 | 0 | 0.0113 | 0.891 | 0.0649 | 0.00122 | 0.000352 |
| re1000_e2 | 1e+03 | 2 | 3 | 2 | B2_temporal | 1.94 | 1 | 0 | 1 | 4.224e-16 | 0.21 | -0.214 | -0.158 | -0.0589 | -0.0118 | -0.00594 | 0 | 0 | 1 | 0.0118 | 0.00269 | 0.00213 | 0.00177 | 0.000822 |
| re1000_e2 | 1e+03 | 2 | 3 | 2 | B2_temporal | 1.94 | 1 | 0 | 1 | 4.224e-16 | 0.21 | -0.214 | -0.158 | -0.0589 | -0.0118 | -0.00594 | 0 | 0 | 2 | 0.00641 | 0.00127 | 1.293e-05 | 0.000469 | 0.000195 |
| re1000_e2 | 1e+03 | 2 | 3 | 2 | B2_temporal | 1.94 | 1 | 0 | 1 | 4.224e-16 | 0.21 | -0.214 | -0.158 | -0.0589 | -0.0118 | -0.00594 | 0 | 0 | 3 | 0.000892 | 0.00186 | 5.960e-05 | 3.939e-05 | 5.572e-06 |
| re1000_e2 | 1e+03 | 2 | 3 | 2 | B2_temporal | 1.94 | 1 | 0 | 1 | 4.224e-16 | 0.21 | -0.214 | -0.158 | -0.0589 | -0.0118 | -0.00594 | 0 | 0 | 4 | 3.558e-05 | 0.000188 | 0.000274 | 2.497e-07 | 8.228e-06 |
| re1000_e2 | 1e+03 | 2 | 3 | 2 | B2_temporal | 1.94 | 1 | 0 | 1 | 4.224e-16 | 0.21 | -0.214 | -0.158 | -0.0589 | -0.0118 | -0.00594 | 0 | 0 | 5 | 9.087e-06 | 0.00013 | 2.738e-05 | 2.080e-06 | 1.042e-06 |
| re1000_e2 | 1e+03 | 2 | 4 | 2 | B2_temporal | 1.94 | 1 | 0 | 1 | 4.806e-16 | 0.0113 | 0.797 | 0.0243 | -0.0587 | -0.000874 | -0.0142 | 0 | 0 | 0 | 3.275e-05 | 0.64 | 0.0393 | 0.00125 | 2.069e-05 |
| re1000_e2 | 1e+03 | 2 | 4 | 2 | B2_temporal | 1.94 | 1 | 0 | 1 | 4.806e-16 | 0.0113 | 0.797 | 0.0243 | -0.0587 | -0.000874 | -0.0142 | 0 | 0 | 1 | 0.164 | 0.138 | 0.00609 | 0.00332 | 5.841e-05 |
| re1000_e2 | 1e+03 | 2 | 4 | 2 | B2_temporal | 1.94 | 1 | 0 | 1 | 4.806e-16 | 0.0113 | 0.797 | 0.0243 | -0.0587 | -0.000874 | -0.0142 | 0 | 0 | 2 | 0.000152 | 0.00287 | 6.749e-05 | 0.000943 | 2.449e-05 |
| re1000_e2 | 1e+03 | 2 | 4 | 2 | B2_temporal | 1.94 | 1 | 0 | 1 | 4.806e-16 | 0.0113 | 0.797 | 0.0243 | -0.0587 | -0.000874 | -0.0142 | 0 | 0 | 3 | 0.00089 | 0.00181 | 0.000185 | 0.000198 | 2.672e-05 |
| re1000_e2 | 1e+03 | 2 | 4 | 2 | B2_temporal | 1.94 | 1 | 0 | 1 | 4.806e-16 | 0.0113 | 0.797 | 0.0243 | -0.0587 | -0.000874 | -0.0142 | 0 | 0 | 4 | 1.972e-07 | 9.934e-06 | 1.343e-05 | 1.360e-05 | 2.212e-05 |
| re1000_e2 | 1e+03 | 2 | 4 | 2 | B2_temporal | 1.94 | 1 | 0 | 1 | 4.806e-16 | 0.0113 | 0.797 | 0.0243 | -0.0587 | -0.000874 | -0.0142 | 0 | 0 | 5 | 5.179e-05 | 0.000148 | 4.404e-06 | 6.628e-07 | 8.926e-06 |
| re100_e0.1 | 100 | 0.1 | 0 | 2 | B2_temporal | 0.11 | 1 | 0 | 1 | 1.135e-16 | 0.0297 | -0.06 | -0.00461 | 0.0056 | 0.000238 | 0.00137 | 0 | 0 | 0 | 0.00401 | 0.944 | 0.00329 | 0.00027 | 0.00027 |
| re100_e0.1 | 100 | 0.1 | 0 | 2 | B2_temporal | 0.11 | 1 | 0 | 1 | 1.135e-16 | 0.0297 | -0.06 | -0.00461 | 0.0056 | 0.000238 | 0.00137 | 0 | 0 | 1 | 0.0163 | 0.00597 | 0.00629 | 0.000264 | 3.872e-05 |
| re100_e0.1 | 100 | 0.1 | 0 | 2 | B2_temporal | 0.11 | 1 | 0 | 1 | 1.135e-16 | 0.0297 | -0.06 | -0.00461 | 0.0056 | 0.000238 | 0.00137 | 0 | 0 | 2 | 9.673e-05 | 0.0176 | 0.000244 | 0.000567 | 1.179e-06 |
| re100_e0.1 | 100 | 0.1 | 0 | 2 | B2_temporal | 0.11 | 1 | 0 | 1 | 1.135e-16 | 0.0297 | -0.06 | -0.00461 | 0.0056 | 0.000238 | 0.00137 | 0 | 0 | 3 | 0.000143 | 3.488e-05 | 0.000294 | 2.731e-05 | 6.095e-07 |
| re100_e0.1 | 100 | 0.1 | 0 | 2 | B2_temporal | 0.11 | 1 | 0 | 1 | 1.135e-16 | 0.0297 | -0.06 | -0.00461 | 0.0056 | 0.000238 | 0.00137 | 0 | 0 | 4 | 2.569e-07 | 2.781e-05 | 2.406e-08 | 7.635e-06 | 2.280e-06 |
| re100_e0.1 | 100 | 0.1 | 0 | 2 | B2_temporal | 0.11 | 1 | 0 | 1 | 1.135e-16 | 0.0297 | -0.06 | -0.00461 | 0.0056 | 0.000238 | 0.00137 | 0 | 0 | 5 | 8.591e-06 | 9.582e-06 | 2.705e-06 | 2.350e-07 | 1.300e-07 |
| re100_e0.1 | 100 | 0.1 | 1 | 2 | B2_temporal | 0.111 | 1 | 0 | 1 | 1.037e-16 | -0.0081 | -0.0325 | 0.00233 | 0.0015 | -0.000471 | -4.066e-05 | 0 | 0 | 0 | 0.000295 | 0.97 | 0.00279 | 0.00014 | 0.000394 |
| re100_e0.1 | 100 | 0.1 | 1 | 2 | B2_temporal | 0.111 | 1 | 0 | 1 | 1.037e-16 | -0.0081 | -0.0325 | 0.00233 | 0.0015 | -0.000471 | -4.066e-05 | 0 | 0 | 1 | 0.00474 | 0.00384 | 0.00305 | 0.000288 | 6.930e-07 |
| re100_e0.1 | 100 | 0.1 | 1 | 2 | B2_temporal | 0.111 | 1 | 0 | 1 | 1.037e-16 | -0.0081 | -0.0325 | 0.00233 | 0.0015 | -0.000471 | -4.066e-05 | 0 | 0 | 2 | 2.441e-05 | 0.0135 | 1.221e-05 | 0.000505 | 1.195e-05 |
| re100_e0.1 | 100 | 0.1 | 1 | 2 | B2_temporal | 0.111 | 1 | 0 | 1 | 1.037e-16 | -0.0081 | -0.0325 | 0.00233 | 0.0015 | -0.000471 | -4.066e-05 | 0 | 0 | 3 | 1.015e-05 | 5.656e-05 | 1.725e-05 | 6.537e-06 | 2.924e-06 |
| re100_e0.1 | 100 | 0.1 | 1 | 2 | B2_temporal | 0.111 | 1 | 0 | 1 | 1.037e-16 | -0.0081 | -0.0325 | 0.00233 | 0.0015 | -0.000471 | -4.066e-05 | 0 | 0 | 4 | 9.934e-07 | 5.967e-06 | 4.382e-07 | 2.881e-06 | 5.584e-08 |
| re100_e0.1 | 100 | 0.1 | 1 | 2 | B2_temporal | 0.111 | 1 | 0 | 1 | 1.037e-16 | -0.0081 | -0.0325 | 0.00233 | 0.0015 | -0.000471 | -4.066e-05 | 0 | 0 | 5 | 7.419e-09 | 1.076e-06 | 2.586e-06 | 1.377e-06 | 4.094e-07 |
| re100_e0.1 | 100 | 0.1 | 2 | 3 | B3_mixed | 0.113 | 1 | 0 | 1 | 1.067e-16 | 0.0285 | 0.286 | 0.00519 | -0.0584 | -0.008 | -0.00286 | 0 | 0 | 0 | 0.00361 | 0.565 | 0.000112 | 0.000353 | 5.060e-09 |
| re100_e0.1 | 100 | 0.1 | 2 | 3 | B3_mixed | 0.113 | 1 | 0 | 1 | 1.067e-16 | 0.0285 | 0.286 | 0.00519 | -0.0584 | -0.008 | -0.00286 | 0 | 0 | 1 | 0.363 | 0.00291 | 0.0092 | 4.378e-05 | 0.000248 |
| re100_e0.1 | 100 | 0.1 | 2 | 3 | B3_mixed | 0.113 | 1 | 0 | 1 | 1.067e-16 | 0.0285 | 0.286 | 0.00519 | -0.0584 | -0.008 | -0.00286 | 0 | 0 | 2 | 0.000119 | 0.0382 | 8.622e-05 | 2.442e-05 | 3.176e-06 |
| re100_e0.1 | 100 | 0.1 | 2 | 3 | B3_mixed | 0.113 | 1 | 0 | 1 | 1.067e-16 | 0.0285 | 0.286 | 0.00519 | -0.0584 | -0.008 | -0.00286 | 0 | 0 | 3 | 0.0151 | 0.000414 | 0.00159 | 5.927e-08 | 1.962e-05 |
| re100_e0.1 | 100 | 0.1 | 2 | 3 | B3_mixed | 0.113 | 1 | 0 | 1 | 1.067e-16 | 0.0285 | 0.286 | 0.00519 | -0.0584 | -0.008 | -0.00286 | 0 | 0 | 4 | 0.000283 | 0.000112 | 2.964e-05 | 2.697e-05 | 1.036e-07 |
| re100_e0.1 | 100 | 0.1 | 2 | 3 | B3_mixed | 0.113 | 1 | 0 | 1 | 1.067e-16 | 0.0285 | 0.286 | 0.00519 | -0.0584 | -0.008 | -0.00286 | 0 | 0 | 5 | 3.627e-05 | 1.256e-05 | 4.763e-06 | 1.186e-07 | 1.093e-06 |
| re100_e0.1 | 100 | 0.1 | 3 | 2 | B2_temporal | 0.125 | 1 | 0 | 1 | 1.042e-16 | -0.0231 | 0.0413 | 0.00126 | -0.00404 | -0.00361 | 0.000958 | 0 | 0 | 0 | 0.00212 | 0.976 | 0.000345 | 7.593e-06 | 9.264e-05 |
| re100_e0.1 | 100 | 0.1 | 3 | 2 | B2_temporal | 0.125 | 1 | 0 | 1 | 1.042e-16 | -0.0231 | 0.0413 | 0.00126 | -0.00404 | -0.00361 | 0.000958 | 0 | 0 | 1 | 0.00679 | 0.0024 | 1.648e-05 | 0.00138 | 3.595e-07 |
| re100_e0.1 | 100 | 0.1 | 3 | 2 | B2_temporal | 0.125 | 1 | 0 | 1 | 1.042e-16 | -0.0231 | 0.0413 | 0.00126 | -0.00404 | -0.00361 | 0.000958 | 0 | 0 | 2 | 6.297e-06 | 0.0103 | 0.000426 | 0.000232 | 1.058e-06 |
| re100_e0.1 | 100 | 0.1 | 3 | 2 | B2_temporal | 0.125 | 1 | 0 | 1 | 1.042e-16 | -0.0231 | 0.0413 | 0.00126 | -0.00404 | -0.00361 | 0.000958 | 0 | 0 | 3 | 6.495e-05 | 0.000116 | 4.385e-06 | 2.167e-06 | 1.149e-06 |
| re100_e0.1 | 100 | 0.1 | 3 | 2 | B2_temporal | 0.125 | 1 | 0 | 1 | 1.042e-16 | -0.0231 | 0.0413 | 0.00126 | -0.00404 | -0.00361 | 0.000958 | 0 | 0 | 4 | 5.199e-05 | 3.403e-07 | 2.244e-05 | 2.892e-06 | 4.376e-07 |
| re100_e0.1 | 100 | 0.1 | 3 | 2 | B2_temporal | 0.125 | 1 | 0 | 1 | 1.042e-16 | -0.0231 | 0.0413 | 0.00126 | -0.00404 | -0.00361 | 0.000958 | 0 | 0 | 5 | 3.661e-06 | 4.722e-06 | 2.327e-07 | 6.195e-07 | 1.350e-06 |

Table 15: Décomposition modale — 185 points, 1110 lignes (6 modes $i$ par point, colonnes $E^f_{iJ}$ pour $J$=0..4), `mode_analysis.csv`.

| point | Re | $E^*$ | s | id | branch | $E_t$ | cov. | $i$ | $j$ | rmse | $A_{10}$ | $A_{20}$ | $A_{30}$ | $A_{40}$ | $A_{50}$ | $A_{60}$ | $A_{70}$ | $A_{80}$ | $i$ | $E^f_{0J}$ | $E^f_{1J}$ | $E^f_{2J}$ | $E^f_{3J}$ | $E^f_{4J}$ |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| re100_e0.1 | 100 | 0.1 | 4 | 2 | B2_temporal | 0.119 | 1 | 0 | 1 | 1.186e-16 | 0.00552 | 0.00174 | -0.0111 | 0.00464 | -0.00218 | 0.00129 | 0 | 0 | 0 | 0.000128 | 0.979 | 3.468e-05 | 0.000128 | 0.000145 |
| re100_e0.1 | 100 | 0.1 | 4 | 2 | B2_temporal | 0.119 | 1 | 0 | 1 | 1.186e-16 | 0.00552 | 0.00174 | -0.0111 | 0.00464 | -0.00218 | 0.00129 | 0 | 0 | 1 | 1.263e-05 | 0.00347 | 0.00214 | 0.000293 | 1.710e-06 |
| re100_e0.1 | 100 | 0.1 | 4 | 2 | B2_temporal | 0.119 | 1 | 0 | 1 | 1.186e-16 | 0.00552 | 0.00174 | -0.0111 | 0.00464 | -0.00218 | 0.00129 | 0 | 0 | 2 | 0.000514 | 0.0125 | 0.000562 | 0.000319 | 1.164e-05 |
| re100_e0.1 | 100 | 0.1 | 4 | 2 | B2_temporal | 0.119 | 1 | 0 | 1 | 1.186e-16 | 0.00552 | 0.00174 | -0.0111 | 0.00464 | -0.00218 | 0.00129 | 0 | 0 | 3 | 8.995e-05 | 0.000131 | 5.046e-05 | 9.820e-07 | 1.636e-06 |
| re100_e0.1 | 100 | 0.1 | 4 | 2 | B2_temporal | 0.119 | 1 | 0 | 1 | 1.186e-16 | 0.00552 | 0.00174 | -0.0111 | 0.00464 | -0.00218 | 0.00129 | 0 | 0 | 4 | 1.982e-05 | 3.492e-07 | 6.310e-06 | 3.762e-07 | 1.980e-06 |
| re100_e0.1 | 100 | 0.1 | 4 | 2 | B2_temporal | 0.119 | 1 | 0 | 1 | 1.186e-16 | 0.00552 | 0.00174 | -0.0111 | 0.00464 | -0.00218 | 0.00129 | 0 | 0 | 5 | 6.940e-06 | 2.662e-06 | 9.565e-08 | 1.811e-06 | 2.002e-07 |
| re100_e0.25 | 100 | 0.25 | 0 | 2 | B2_temporal | 0.237 | 1 | 0 | 1 | 1.817e-16 | 0.0379 | -0.0567 | 0.00203 | 0.00376 | 0.000663 | 0.0025 | 0 | 0 | 0 | 0.00303 | 0.955 | 0.00626 | 0.00182 | 0.000135 |
| re100_e0.25 | 100 | 0.25 | 0 | 2 | B2_temporal | 0.237 | 1 | 0 | 1 | 1.817e-16 | 0.0379 | -0.0567 | 0.00203 | 0.00376 | 0.000663 | 0.0025 | 0 | 0 | 1 | 0.00677 | 0.00355 | 0.00414 | 0.000301 | 3.532e-05 |
| re100_e0.25 | 100 | 0.25 | 0 | 2 | B2_temporal | 0.237 | 1 | 0 | 1 | 1.817e-16 | 0.0379 | -0.0567 | 0.00203 | 0.00376 | 0.000663 | 0.0025 | 0 | 0 | 2 | 8.656e-06 | 0.018 | 0.000276 | 0.000599 | 2.369e-11 |
| re100_e0.25 | 100 | 0.25 | 0 | 2 | B2_temporal | 0.237 | 1 | 0 | 1 | 1.817e-16 | 0.0379 | -0.0567 | 0.00203 | 0.00376 | 0.000663 | 0.0025 | 0 | 0 | 3 | 2.984e-05 | 1.185e-06 | 0.000133 | 5.077e-05 | 9.782e-08 |
| re100_e0.25 | 100 | 0.25 | 0 | 2 | B2_temporal | 0.237 | 1 | 0 | 1 | 1.817e-16 | 0.0379 | -0.0567 | 0.00203 | 0.00376 | 0.000663 | 0.0025 | 0 | 0 | 4 | 9.278e-07 | 3.487e-05 | 7.494e-06 | 3.475e-06 | 1.027e-06 |
| re100_e0.25 | 100 | 0.25 | 0 | 2 | B2_temporal | 0.237 | 1 | 0 | 1 | 1.817e-16 | 0.0379 | -0.0567 | 0.00203 | 0.00376 | 0.000663 | 0.0025 | 0 | 0 | 5 | 1.313e-05 | 1.321e-05 | 1.854e-06 | 3.538e-07 | 3.691e-09 |
| re100_e0.25 | 100 | 0.25 | 1 | 2 | B2_temporal | 0.24 | 1 | 0 | 1 | 1.435e-16 | -0.0205 | -0.0729 | 0.0156 | -0.00136 | -0.00131 | 0.0024 | 0 | 0 | 0 | 0.000873 | 0.955 | 0.0019 | 0.00116 | 0.000162 |
| re100_e0.25 | 100 | 0.25 | 1 | 2 | B2_temporal | 0.24 | 1 | 0 | 1 | 1.435e-16 | -0.0205 | -0.0729 | 0.0156 | -0.00136 | -0.00131 | 0.0024 | 0 | 0 | 1 | 0.0111 | 0.000923 | 0.0054 | 0.000744 | 8.349e-06 |
| re100_e0.25 | 100 | 0.25 | 1 | 2 | B2_temporal | 0.24 | 1 | 0 | 1 | 1.435e-16 | -0.0205 | -0.0729 | 0.0156 | -0.00136 | -0.00131 | 0.0024 | 0 | 0 | 2 | 0.00051 | 0.0197 | 0.00143 | 0.000399 | 1.224e-05 |
| re100_e0.25 | 100 | 0.25 | 1 | 2 | B2_temporal | 0.24 | 1 | 0 | 1 | 1.435e-16 | -0.0205 | -0.0729 | 0.0156 | -0.00136 | -0.00131 | 0.0024 | 0 | 0 | 3 | 3.843e-06 | 6.627e-05 | 5.508e-08 | 0.000114 | 5.285e-07 |
| re100_e0.25 | 100 | 0.25 | 1 | 2 | B2_temporal | 0.24 | 1 | 0 | 1 | 1.435e-16 | -0.0205 | -0.0729 | 0.0156 | -0.00136 | -0.00131 | 0.0024 | 0 | 0 | 4 | 3.576e-06 | 0.000116 | 9.817e-06 | 9.803e-07 | 5.434e-06 |
| re100_e0.25 | 100 | 0.25 | 1 | 2 | B2_temporal | 0.24 | 1 | 0 | 1 | 1.435e-16 | -0.0205 | -0.0729 | 0.0156 | -0.00136 | -0.00131 | 0.0024 | 0 | 0 | 5 | 1.204e-05 | 7.541e-07 | 1.156e-08 | 3.294e-06 | 1.571e-07 |
| re100_e0.25 | 100 | 0.25 | 2 | 3 | B3_mixed | 0.245 | 1 | 1 | 0 | 1.654e-16 | 0.0407 | 0.484 | -0.00271 | -0.103 | -0.011 | -0.00429 | 0 | 0 | 0 | 0.00338 | 0.44 | 0.00042 | 0.00028 | 4.647e-08 |
| re100_e0.25 | 100 | 0.25 | 2 | 3 | B3_mixed | 0.245 | 1 | 1 | 0 | 1.654e-16 | 0.0407 | 0.484 | -0.00271 | -0.103 | -0.011 | -0.00429 | 0 | 0 | 1 | 0.477 | 0.00645 | 0.0105 | 1.602e-05 | 0.000275 |
| re100_e0.25 | 100 | 0.25 | 2 | 3 | B3_mixed | 0.245 | 1 | 1 | 0 | 1.654e-16 | 0.0407 | 0.484 | -0.00271 | -0.103 | -0.011 | -0.00429 | 0 | 0 | 2 | 1.496e-05 | 0.0361 | 6.075e-05 | 1.790e-05 | 4.451e-07 |
| re100_e0.25 | 100 | 0.25 | 2 | 3 | B3_mixed | 0.245 | 1 | 1 | 0 | 1.654e-16 | 0.0407 | 0.484 | -0.00271 | -0.103 | -0.011 | -0.00429 | 0 | 0 | 3 | 0.0217 | 0.000649 | 0.00227 | 1.424e-06 | 3.068e-05 |
| re100_e0.25 | 100 | 0.25 | 2 | 3 | B3_mixed | 0.245 | 1 | 1 | 0 | 1.654e-16 | 0.0407 | 0.484 | -0.00271 | -0.103 | -0.011 | -0.00429 | 0 | 0 | 4 | 0.000245 | 0.000388 | 2.758e-05 | 4.468e-05 | 2.725e-07 |
| re100_e0.25 | 100 | 0.25 | 2 | 3 | B3_mixed | 0.245 | 1 | 1 | 0 | 1.654e-16 | 0.0407 | 0.484 | -0.00271 | -0.103 | -0.011 | -0.00429 | 0 | 0 | 5 | 3.745e-05 | 3.099e-06 | 2.442e-05 | 1.375e-07 | 1.219e-06 |
| re100_e0.25 | 100 | 0.25 | 3 | 2 | B2_temporal | 0.267 | 1 | 0 | 1 | 1.628e-16 | -0.0325 | 0.137 | 0.00541 | -0.0136 | -0.00134 | -0.000911 | 0 | 0 | 0 | 0.00198 | 0.95 | 0.000225 | 0.000242 | 7.323e-05 |
| re100_e0.25 | 100 | 0.25 | 3 | 2 | B2_temporal | 0.267 | 1 | 0 | 1 | 1.628e-16 | -0.0325 | 0.137 | 0.00541 | -0.0136 | -0.00134 | -0.000911 | 0 | 0 | 1 | 0.035 | 0.00239 | 4.126e-05 | 0.000958 | 1.781e-05 |
| re100_e0.25 | 100 | 0.25 | 3 | 2 | B2_temporal | 0.267 | 1 | 0 | 1 | 1.628e-16 | -0.0325 | 0.137 | 0.00541 | -0.0136 | -0.00134 | -0.000911 | 0 | 0 | 2 | 5.479e-05 | 0.00802 | 2.798e-06 | 0.000288 | 3.025e-07 |
| re100_e0.25 | 100 | 0.25 | 3 | 2 | B2_temporal | 0.267 | 1 | 0 | 1 | 1.628e-16 | -0.0325 | 0.137 | 0.00541 | -0.0136 | -0.00134 | -0.000911 | 0 | 0 | 3 | 0.000347 | 1.166e-05 | 1.427e-05 | 4.184e-06 | 4.592e-08 |
| re100_e0.25 | 100 | 0.25 | 3 | 2 | B2_temporal | 0.267 | 1 | 0 | 1 | 1.628e-16 | -0.0325 | 0.137 | 0.00541 | -0.0136 | -0.00134 | -0.000911 | 0 | 0 | 4 | 3.370e-06 | 6.197e-05 | 6.475e-06 | 1.925e-06 | 2.540e-07 |
| re100_e0.25 | 100 | 0.25 | 3 | 2 | B2_temporal | 0.267 | 1 | 0 | 1 | 1.628e-16 | -0.0325 | 0.137 | 0.00541 | -0.0136 | -0.00134 | -0.000911 | 0 | 0 | 5 | 1.556e-06 | 3.124e-06 | 1.056e-07 | 1.716e-07 | 4.555e-07 |
| re100_e0.25 | 100 | 0.25 | 4 | 2 | B2_temporal | 0.256 | 1 | 0 | 1 | 1.457e-16 | 0.00513 | 0.00149 | -0.00274 | 0.004 | -0.00277 | 0.00211 | 0 | 0 | 0 | 5.132e-05 | 0.981 | 0.00147 | 4.785e-06 | 0.000229 |
| re100_e0.25 | 100 | 0.25 | 4 | 2 | B2_temporal | 0.256 | 1 | 0 | 1 | 1.457e-16 | 0.00513 | 0.00149 | -0.00274 | 0.004 | -0.00277 | 0.00211 | 0 | 0 | 1 | 4.352e-06 | 0.00288 | 0.00258 | 0.00132 | 7.744e-06 |
| re100_e0.25 | 100 | 0.25 | 4 | 2 | B2_temporal | 0.256 | 1 | 0 | 1 | 1.457e-16 | 0.00513 | 0.00149 | -0.00274 | 0.004 | -0.00277 | 0.00211 | 0 | 0 | 2 | 1.465e-05 | 0.00884 | 0.00104 | 0.000251 | 6.179e-06 |
| re100_e0.25 | 100 | 0.25 | 4 | 2 | B2_temporal | 0.256 | 1 | 0 | 1 | 1.457e-16 | 0.00513 | 0.00149 | -0.00274 | 0.004 | -0.00277 | 0.00211 | 0 | 0 | 3 | 3.123e-05 | 0.000249 | 4.361e-06 | 2.875e-06 | 7.763e-06 |
| re100_e0.25 | 100 | 0.25 | 4 | 2 | B2_temporal | 0.256 | 1 | 0 | 1 | 1.457e-16 | 0.00513 | 0.00149 | -0.00274 | 0.004 | -0.00277 | 0.00211 | 0 | 0 | 4 | 1.496e-05 | 2.773e-05 | 1.765e-05 | 5.334e-07 | 3.723e-06 |
| re100_e0.25 | 100 | 0.25 | 4 | 2 | B2_temporal | 0.256 | 1 | 0 | 1 | 1.457e-16 | 0.00513 | 0.00149 | -0.00274 | 0.004 | -0.00277 | 0.00211 | 0 | 0 | 5 | 8.673e-06 | 6.957e-06 | 5.543e-07 | 5.822e-08 | 5.426e-07 |
| re100_e0.5 | 100 | 0.5 | 0 | 2 | B2_temporal | 0.463 | 1 | 0 | 1 | 2.097e-16 | 0.0501 | -0.01 | 0.0152 | 0.000405 | -0.00201 | 0.00392 | 0 | 0 | 0 | 0.00272 | 0.942 | 0.0192 | 0.00352 | 1.303e-07 |
| re100_e0.5 | 100 | 0.5 | 0 | 2 | B2_temporal | 0.463 | 1 | 0 | 1 | 2.097e-16 | 0.0501 | -0.01 | 0.0152 | 0.000405 | -0.00201 | 0.00392 | 0 | 0 | 1 | 0.000109 | 0.00358 | 0.00607 | 0.000989 | 8.472e-06 |
| re100_e0.5 | 100 | 0.5 | 0 | 2 | B2_temporal | 0.463 | 1 | 0 | 1 | 2.097e-16 | 0.0501 | -0.01 | 0.0152 | 0.000405 | -0.00201 | 0.00392 | 0 | 0 | 2 | 0.000251 | 0.0198 | 6.368e-05 | 0.000718 | 4.936e-06 |
| re100_e0.5 | 100 | 0.5 | 0 | 2 | B2_temporal | 0.463 | 1 | 0 | 1 | 2.097e-16 | 0.0501 | -0.01 | 0.0152 | 0.000405 | -0.00201 | 0.00392 | 0 | 0 | 3 | 1.769e-07 | 2.265e-05 | 4.578e-05 | 7.609e-05 | 2.902e-05 |
| re100_e0.5 | 100 | 0.5 | 0 | 2 | B2_temporal | 0.463 | 1 | 0 | 1 | 2.097e-16 | 0.0501 | -0.01 | 0.0152 | 0.000405 | -0.00201 | 0.00392 | 0 | 0 | 4 | 4.380e-06 | 0.000281 | 2.379e-07 | 2.035e-06 | 1.197e-07 |
| re100_e0.5 | 100 | 0.5 | 0 | 2 | B2_temporal | 0.463 | 1 | 0 | 1 | 2.097e-16 | 0.0501 | -0.01 | 0.0152 | 0.000405 | -0.00201 | 0.00392 | 0 | 0 | 5 | 1.660e-05 | 1.540e-05 | 1.678e-06 | 2.346e-06 | 1.044e-06 |
| re100_e0.5 | 100 | 0.5 | 1 | 2 | B2_temporal | 0.468 | 1 | 0 | 1 | 2.396e-16 | -0.0433 | -0.168 | 0.0388 | 0.0103 | -0.00255 | 0.00452 | 0 | 0 | 0 | 0.002 | 0.909 | 0.00174 | 0.00311 | 0.00027 |
| re100_e0.5 | 100 | 0.5 | 1 | 2 | B2_temporal | 0.468 | 1 | 0 | 1 | 2.396e-16 | -0.0433 | -0.168 | 0.0388 | 0.0103 | -0.00255 | 0.00452 | 0 | 0 | 1 | 0.03 | 0.000481 | 0.0177 | 0.00182 | 4.318e-08 |
| re100_e0.5 | 100 | 0.5 | 1 | 2 | B2_temporal | 0.468 | 1 | 0 | 1 | 2.396e-16 | -0.0433 | -0.168 | 0.0388 | 0.0103 | -0.00255 | 0.00452 | 0 | 0 | 2 | 0.00161 | 0.026 | 0.00402 | 0.0011 | 5.409e-05 |
| re100_e0.5 | 100 | 0.5 | 1 | 2 | B2_temporal | 0.468 | 1 | 0 | 1 | 2.396e-16 | -0.0433 | -0.168 | 0.0388 | 0.0103 | -0.00255 | 0.00452 | 0 | 0 | 3 | 0.000114 | 0.000415 | 0.000122 | 0.000469 | 7.750e-07 |
| re100_e0.5 | 100 | 0.5 | 1 | 2 | B2_temporal | 0.468 | 1 | 0 | 1 | 2.396e-16 | -0.0433 | -0.168 | 0.0388 | 0.0103 | -0.00255 | 0.00452 | 0 | 0 | 4 | 6.957e-06 | 0.000201 | 0.00014 | 1.200e-05 | 2.330e-05 |
| re100_e0.5 | 100 | 0.5 | 1 | 2 | B2_temporal | 0.468 | 1 | 0 | 1 | 2.396e-16 | -0.0433 | -0.168 | 0.0388 | 0.0103 | -0.00255 | 0.00452 | 0 | 0 | 5 | 2.181e-05 | 2.268e-07 | 4.899e-06 | 2.835e-08 | 1.610e-06 |
| re100_e0.5 | 100 | 0.5 | 2 | 1 | B1_Mode2 | 0.495 | 1 | 1 | 0 | 2.093e-16 | 0.0482 | 0.887 | -0.0111 | -0.224 | -0.0159 | 0.0022 | 0 | 0 | 0 | 0.00234 | 0.122 | 1.704e-06 | 0.00051 | 2.412e-07 |
| re100_e0.5 | 100 | 0.5 | 2 | 1 | B1_Mode2 | 0.495 | 1 | 1 | 0 | 2.093e-16 | 0.0482 | 0.887 | -0.0111 | -0.224 | -0.0159 | 0.0022 | 0 | 0 | 1 | 0.794 | 0.00281 | 0.00379 | 3.507e-05 | 0.00013 |
| re100_e0.5 | 100 | 0.5 | 2 | 1 | B1_Mode2 | 0.495 | 1 | 1 | 0 | 2.093e-16 | 0.0482 | 0.887 | -0.0111 | -0.224 | -0.0159 | 0.0022 | 0 | 0 | 2 | 0.000125 | 0.0201 | 9.474e-06 | 2.687e-05 | 1.752e-06 |
| re100_e0.5 | 100 | 0.5 | 2 | 1 | B1_Mode2 | 0.495 | 1 | 1 | 0 | 2.093e-16 | 0.0482 | 0.887 | -0.0111 | -0.224 | -0.0159 | 0.0022 | 0 | 0 | 3 | 0.0505 | 0.000315 | 0.00146 | 9.176e-06 | 4.140e-05 |
| re100_e0.5 | 100 | 0.5 | 2 | 1 | B1_Mode2 | 0.495 | 1 | 1 | 0 | 2.093e-16 | 0.0482 | 0.887 | -0.0111 | -0.224 | -0.0159 | 0.0022 | 0 | 0 | 4 | 0.000256 | 0.00161 | 7.891e-06 | 3.274e-06 | 5.956e-08 |
| re100_e0.5 | 100 | 0.5 | 2 | 1 | B1_Mode2 | 0.495 | 1 | 1 | 0 | 2.093e-16 | 0.0482 | 0.887 | -0.0111 | -0.224 | -0.0159 | 0.0022 | 0 | 0 | 5 | 4.894e-06 | 7.845e-07 | 5.499e-05 | 6.067e-07 | 6.950e-08 |
| re100_e0.5 | 100 | 0.5 | 3 | 1 | B1_Mode2 | 0.491 | 1 | 1 | 0 | 1.971e-16 | -0.00392 | 0.949 | 0.101 | -0.25 | -0.0481 | 0.0143 | 0 | 0 | 0 | 1.568e-05 | 0.000358 | 0.000119 | 3.209e-06 | 1.215e-05 |
| re100_e0.5 | 100 | 0.5 | 3 | 1 | B1_Mode2 | 0.491 | 1 | 1 | 0 | 1.971e-16 | -0.00392 | 0.949 | 0.101 | -0.25 | -0.0481 | 0.0143 | 0 | 0 | 1 | 0.918 | 0.00312 | 0.000119 | 2.521e-05 | 7.161e-06 |
| re100_e0.5 | 100 | 0.5 | 3 | 1 | B1_Mode2 | 0.491 | 1 | 1 | 0 | 1.971e-16 | -0.00392 | 0.949 | 0.101 | -0.25 | -0.0481 | 0.0143 | 0 | 0 | 2 | 0.0103 | 0.00102 | 2.561e-05 | 4.170e-06 | 2.350e-06 |
| re100_e0.5 | 100 | 0.5 | 3 | 1 | B1_Mode2 | 0.491 | 1 | 1 | 0 | 1.971e-16 | -0.00392 | 0.949 | 0.101 | -0.25 | -0.0481 | 0.0143 | 0 | 0 | 3 | 0.0638 | 2.687e-05 | 0.00017 | 3.058e-06 | 1.177e-05 |
| re100_e0.5 | 100 | 0.5 | 3 | 1 | B1_Mode2 | 0.491 | 1 | 1 | 0 | 1.971e-16 | -0.00392 | 0.949 | 0.101 | -0.25 | -0.0481 | 0.0143 | 0 | 0 | 4 | 0.00236 | 5.593e-05 | 2.120e-06 | 4.528e-07 | 2.870e-08 |
| re100_e0.5 | 100 | 0.5 | 3 | 1 | B1_Mode2 | 0.491 | 1 | 1 | 0 | 1.971e-16 | -0.00392 | 0.949 | 0.101 | -0.25 | -0.0481 | 0.0143 | 0 | 0 | 5 | 0.000209 | 2.413e-05 | 1.464e-07 | 1.703e-07 | 8.330e-09 |
| re100_e0.5 | 100 | 0.5 | 4 | 2 | B2_temporal | 0.493 | 1 | 0 | 1 | 2.156e-16 | 0.00673 | -0.0741 | -0.00244 | 0.023 | -0.00117 | 0.00406 | 0 | 0 | 0 | 4.592e-05 | 0.968 | 0.00554 | 9.775e-06 | 0.000351 |
| re100_e0.5 | 100 | 0.5 | 4 | 2 | B2_temporal | 0.493 | 1 | 0 | 1 | 2.156e-16 | 0.00673 | -0.0741 | -0.00244 | 0.023 | -0.00117 | 0.00406 | 0 | 0 | 1 | 0.00557 | 0.00551 | 1.195e-06 | 0.00183 | 6.100e-06 |
| re100_e0.5 | 100 | 0.5 | 4 | 2 | B2_temporal | 0.493 | 1 | 0 | 1 | 2.156e-16 | 0.00673 | -0.0741 | -0.00244 | 0.023 | -0.00117 | 0.00406 | 0 | 0 | 2 | 6.031e-06 | 0.00987 | 0.00207 | 3.873e-05 | 2.746e-06 |
| re100_e0.5 | 100 | 0.5 | 4 | 2 | B2_temporal | 0.493 | 1 | 0 | 1 | 2.156e-16 | 0.00673 | -0.0741 | -0.00244 | 0.023 | -0.00117 | 0.00406 | 0 | 0 | 3 | 0.000534 | 0.000148 | 8.935e-06 | 1.915e-05 | 7.417e-06 |
| re100_e0.5 | 100 | 0.5 | 4 | 2 | B2_temporal | 0.493 | 1 | 0 | 1 | 2.156e-16 | 0.00673 | -0.0741 | -0.00244 | 0.023 | -0.00117 | 0.00406 | 0 | 0 | 4 | 1.381e-06 | 1.613e-05 | 2.327e-06 | 2.596e-07 | 1.661e-06 |
| re100_e0.5 | 100 | 0.5 | 4 | 2 | B2_temporal | 0.493 | 1 | 0 | 1 | 2.156e-16 | 0.00673 | -0.0741 | -0.00244 | 0.023 | -0.00117 | 0.00406 | 0 | 0 | 5 | 1.671e-05 | 5.220e-06 | 2.656e-06 | 4.822e-07 | 8.930e-07 |

Table 15: Décomposition modale — 185 points, 1110 lignes (6 modes $i$ par point, colonnes $E^f_{iJ}$ pour $J$=0..4), `mode_analysis.csv`.

| point | Re | $E^*$ | s | id | branch | $E_t$ | cov. | $i$ | $j$ | rmse | $A_{10}$ | $A_{20}$ | $A_{30}$ | $A_{40}$ | $A_{50}$ | $A_{60}$ | $A_{70}$ | $A_{80}$ | $i$ | $E^f_{0J}$ | $E^f_{1J}$ | $E^f_{2J}$ | $E^f_{3J}$ | $E^f_{4J}$ |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| re100_e1 | 100 | 1 | 0 | 2 | B2_temporal | 0.913 | 1 | 0 | 1 | 2.747e-16 | 0.0505 | -0.293 | 0.0822 | 0.059 | -0.00403 | 0.01 | 0 | 0 | 0 | 0.0014 | 0.84 | 0.00332 | 0.0123 | 0.000455 |
| re100_e1 | 100 | 1 | 0 | 2 | B2_temporal | 0.913 | 1 | 0 | 1 | 2.747e-16 | 0.0505 | -0.293 | 0.0822 | 0.059 | -0.00403 | 0.01 | 0 | 0 | 1 | 0.047 | 0.00961 | 0.0284 | 0.00136 | 2.110e-05 |
| re100_e1 | 100 | 1 | 0 | 2 | B2_temporal | 0.913 | 1 | 0 | 1 | 2.747e-16 | 0.0505 | -0.293 | 0.0822 | 0.059 | -0.00403 | 0.01 | 0 | 0 | 2 | 0.00371 | 0.0419 | 0.00287 | 0.00328 | 0.00013 |
| re100_e1 | 100 | 1 | 0 | 2 | B2_temporal | 0.913 | 1 | 0 | 1 | 2.747e-16 | 0.0505 | -0.293 | 0.0822 | 0.059 | -0.00403 | 0.01 | 0 | 0 | 3 | 0.00191 | 1.162e-05 | 0.00109 | 0.000445 | 3.432e-06 |
| re100_e1 | 100 | 1 | 0 | 2 | B2_temporal | 0.913 | 1 | 0 | 1 | 2.747e-16 | 0.0505 | -0.293 | 0.0822 | 0.059 | -0.00403 | 0.01 | 0 | 0 | 4 | 8.912e-06 | 0.000364 | 0.000234 | 1.037e-05 | 9.929e-06 |
| re100_e1 | 100 | 1 | 0 | 2 | B2_temporal | 0.913 | 1 | 0 | 1 | 2.747e-16 | 0.0505 | -0.293 | 0.0822 | 0.059 | -0.00403 | 0.01 | 0 | 0 | 5 | 5.494e-05 | 2.887e-05 | 4.721e-05 | 2.954e-07 | 2.421e-06 |
| re100_e1 | 100 | 1 | 1 | 2 | B2_temporal | 0.921 | 1 | 0 | 1 | 2.961e-16 | -0.0696 | -0.524 | 0.0884 | 0.0454 | -0.0116 | 0.0232 | 0 | 0 | 0 | 0.00263 | 0.744 | 0.000563 | 0.00121 | 0.000327 |
| re100_e1 | 100 | 1 | 1 | 2 | B2_temporal | 0.921 | 1 | 0 | 1 | 2.961e-16 | -0.0696 | -0.524 | 0.0884 | 0.0454 | -0.0116 | 0.0232 | 0 | 0 | 1 | 0.149 | 0.000712 | 0.03 | 0.00214 | 4.919e-05 |
| re100_e1 | 100 | 1 | 1 | 2 | B2_temporal | 0.921 | 1 | 0 | 1 | 2.961e-16 | -0.0696 | -0.524 | 0.0884 | 0.0454 | -0.0116 | 0.0232 | 0 | 0 | 2 | 0.00424 | 0.0485 | 0.00761 | 0.00107 | 0.000124 |
| re100_e1 | 100 | 1 | 1 | 2 | B2_temporal | 0.921 | 1 | 0 | 1 | 2.961e-16 | -0.0696 | -0.524 | 0.0884 | 0.0454 | -0.0116 | 0.0232 | 0 | 0 | 3 | 0.00112 | 0.00251 | 0.00113 | 0.00138 | 3.215e-06 |
| re100_e1 | 100 | 1 | 1 | 2 | B2_temporal | 0.921 | 1 | 0 | 1 | 2.961e-16 | -0.0696 | -0.524 | 0.0884 | 0.0454 | -0.0116 | 0.0232 | 0 | 0 | 4 | 7.337e-05 | 0.000193 | 0.000632 | 3.528e-05 | 8.461e-05 |
| re100_e1 | 100 | 1 | 1 | 2 | B2_temporal | 0.921 | 1 | 0 | 1 | 2.961e-16 | -0.0696 | -0.524 | 0.0884 | 0.0454 | -0.0116 | 0.0232 | 0 | 0 | 5 | 0.000291 | 7.027e-05 | 8.447e-05 | 6.729e-06 | 1.037e-05 |
| re100_e1 | 100 | 1 | 2 | 1 | B1_Mode2 | 1.01 | 1 | 1 | 0 | 3.133e-16 | 0.071 | 1.34 | -0.0164 | -0.351 | -0.0282 | 0.00598 | 0 | 0 | 0 | 0.00249 | 0.0325 | 1.414e-05 | 0.000255 | 3.533e-09 |
| re100_e1 | 100 | 1 | 2 | 1 | B1_Mode2 | 1.01 | 1 | 1 | 0 | 3.133e-16 | 0.071 | 1.34 | -0.0164 | -0.351 | -0.0282 | 0.00598 | 0 | 0 | 1 | 0.887 | 0.000202 | 0.00103 | 6.104e-07 | 4.095e-05 |
| re100_e1 | 100 | 1 | 2 | 1 | B1_Mode2 | 1.01 | 1 | 1 | 0 | 3.133e-16 | 0.071 | 1.34 | -0.0164 | -0.351 | -0.0282 | 0.00598 | 0 | 0 | 2 | 0.000132 | 0.0118 | 4.085e-07 | 2.863e-05 | 6.094e-07 |
| re100_e1 | 100 | 1 | 2 | 1 | B1_Mode2 | 1.01 | 1 | 1 | 0 | 3.133e-16 | 0.071 | 1.34 | -0.0164 | -0.351 | -0.0282 | 0.00598 | 0 | 0 | 3 | 0.061 | 2.596e-06 | 0.000743 | 2.239e-07 | 2.477e-05 |
| re100_e1 | 100 | 1 | 2 | 1 | B1_Mode2 | 1.01 | 1 | 1 | 0 | 3.133e-16 | 0.071 | 1.34 | -0.0164 | -0.351 | -0.0282 | 0.00598 | 0 | 0 | 4 | 0.000393 | 0.00182 | 4.386e-08 | 4.912e-07 | 6.181e-07 |
| re100_e1 | 100 | 1 | 2 | 1 | B1_Mode2 | 1.01 | 1 | 1 | 0 | 3.133e-16 | 0.071 | 1.34 | -0.0164 | -0.351 | -0.0282 | 0.00598 | 0 | 0 | 5 | 1.771e-05 | 4.667e-07 | 2.829e-05 | 4.363e-08 | 4.602e-08 |
| re100_e1 | 100 | 1 | 3 | 1 | B1_Mode2 | 0.983 | 1 | 1 | 0 | 3.221e-16 | 0.0279 | 1.34 | 0.108 | -0.386 | -0.0858 | 0.0341 | 0 | 0 | 0 | 0.000396 | 0.000219 | 4.366e-05 | 9.223e-06 | 4.528e-08 |
| re100_e1 | 100 | 1 | 3 | 1 | B1_Mode2 | 0.983 | 1 | 1 | 0 | 3.221e-16 | 0.0279 | 1.34 | 0.108 | -0.386 | -0.0858 | 0.0341 | 0 | 0 | 1 | 0.912 | 0.000864 | 0.000101 | 7.070e-06 | 5.389e-06 |
| re100_e1 | 100 | 1 | 3 | 1 | B1_Mode2 | 0.983 | 1 | 1 | 0 | 3.221e-16 | 0.0279 | 1.34 | 0.108 | -0.386 | -0.0858 | 0.0341 | 0 | 0 | 2 | 0.00593 | 0.000162 | 1.140e-05 | 2.197e-06 | 5.397e-08 |
| re100_e1 | 100 | 1 | 3 | 1 | B1_Mode2 | 0.983 | 1 | 1 | 0 | 3.221e-16 | 0.0279 | 1.34 | 0.108 | -0.386 | -0.0858 | 0.0341 | 0 | 0 | 3 | 0.0756 | 8.988e-07 | 8.103e-05 | 6.558e-07 | 5.968e-06 |
| re100_e1 | 100 | 1 | 3 | 1 | B1_Mode2 | 0.983 | 1 | 1 | 0 | 3.221e-16 | 0.0279 | 1.34 | 0.108 | -0.386 | -0.0858 | 0.0341 | 0 | 0 | 4 | 0.00375 | 1.838e-05 | 1.832e-05 | 8.200e-07 | 2.837e-08 |
| re100_e1 | 100 | 1 | 3 | 1 | B1_Mode2 | 0.983 | 1 | 1 | 0 | 3.221e-16 | 0.0279 | 1.34 | 0.108 | -0.386 | -0.0858 | 0.0341 | 0 | 0 | 5 | 0.000591 | 2.261e-06 | 1.649e-06 | 3.907e-07 | 3.076e-07 |
| re100_e1 | 100 | 1 | 4 | 2 | B2_temporal | 0.981 | 1 | 0 | 1 | 3.161e-16 | 0.0139 | -0.0445 | -0.0175 | 0.0202 | 0.00095 | 0.00783 | 0 | 0 | 0 | 9.805e-05 | 0.963 | 0.0184 | 6.977e-07 | 6.091e-07 |
| re100_e1 | 100 | 1 | 4 | 2 | B2_temporal | 0.981 | 1 | 0 | 1 | 3.161e-16 | 0.0139 | -0.0445 | -0.0175 | 0.0202 | 0.00095 | 0.00783 | 0 | 0 | 1 | 0.00101 | 0.00422 | 4.108e-06 | 0.00274 | 4.054e-05 |
| re100_e1 | 100 | 1 | 4 | 2 | B2_temporal | 0.981 | 1 | 0 | 1 | 3.161e-16 | 0.0139 | -0.0445 | -0.0175 | 0.0202 | 0.00095 | 0.00783 | 0 | 0 | 2 | 0.000157 | 0.00667 | 0.00249 | 4.207e-06 | 4.662e-07 |
| re100_e1 | 100 | 1 | 4 | 2 | B2_temporal | 0.981 | 1 | 0 | 1 | 3.161e-16 | 0.0139 | -0.0445 | -0.0175 | 0.0202 | 0.00095 | 0.00783 | 0 | 0 | 3 | 0.000208 | 0.000169 | 5.465e-05 | 8.983e-05 | 5.774e-05 |
| re100_e1 | 100 | 1 | 4 | 2 | B2_temporal | 0.981 | 1 | 0 | 1 | 3.161e-16 | 0.0139 | -0.0445 | -0.0175 | 0.0202 | 0.00095 | 0.00783 | 0 | 0 | 4 | 4.598e-07 | 4.950e-05 | 1.443e-05 | 6.299e-07 | 3.951e-06 |
| re100_e1 | 100 | 1 | 4 | 2 | B2_temporal | 0.981 | 1 | 0 | 1 | 3.161e-16 | 0.0139 | -0.0445 | -0.0175 | 0.0202 | 0.00095 | 0.00783 | 0 | 0 | 5 | 3.127e-05 | 8.246e-06 | 9.518e-06 | 6.904e-07 | 1.476e-06 |
| re100_e2 | 100 | 2 | 0 | 2 | B2_temporal | 1.88 | 1 | 1 | 1 | 4.571e-16 | 0.521 | -0.927 | -0.654 | 0.3 | 0.0406 | 0.00152 | 0 | 0 | 0 | 0.0721 | 0.146 | 0.0174 | 8.181e-05 | 0.000558 |
| re100_e2 | 100 | 2 | 0 | 2 | B2_temporal | 1.88 | 1 | 1 | 1 | 4.571e-16 | 0.521 | -0.927 | -0.654 | 0.3 | 0.0406 | 0.00152 | 0 | 0 | 1 | 0.228 | 0.282 | 0.00582 | 0.000117 | 1.933e-05 |
| re100_e2 | 100 | 2 | 0 | 2 | B2_temporal | 1.88 | 1 | 1 | 1 | 4.571e-16 | 0.521 | -0.927 | -0.654 | 0.3 | 0.0406 | 0.00152 | 0 | 0 | 2 | 0.114 | 0.0527 | 0.0162 | 0.000893 | 4.631e-05 |
| re100_e2 | 100 | 2 | 0 | 2 | B2_temporal | 1.88 | 1 | 1 | 1 | 4.571e-16 | 0.521 | -0.927 | -0.654 | 0.3 | 0.0406 | 0.00152 | 0 | 0 | 3 | 0.0239 | 0.0137 | 0.015 | 0.00119 | 6.627e-05 |
| re100_e2 | 100 | 2 | 0 | 2 | B2_temporal | 1.88 | 1 | 1 | 1 | 4.571e-16 | 0.521 | -0.927 | -0.654 | 0.3 | 0.0406 | 0.00152 | 0 | 0 | 4 | 0.000437 | 0.00795 | 0.000714 | 0.000697 | 9.442e-05 |
| re100_e2 | 100 | 2 | 0 | 2 | B2_temporal | 1.88 | 1 | 1 | 1 | 4.571e-16 | 0.521 | -0.927 | -0.654 | 0.3 | 0.0406 | 0.00152 | 0 | 0 | 5 | 6.132e-07 | 0.000369 | 0.000763 | 0.000125 | 6.679e-08 |
| re100_e2 | 100 | 2 | 1 | 1 | B1_Mode2 | 1.9 | 1 | 1 | 0 | 4.285e-16 | 0.218 | -1.47 | -0.446 | 0.556 | -0.0724 | 0.0322 | 0 | 0 | 0 | 0.0125 | 0.125 | 0.000725 | 0.000294 | 9.500e-05 |
| re100_e2 | 100 | 2 | 1 | 1 | B1_Mode2 | 1.9 | 1 | 1 | 0 | 4.285e-16 | 0.218 | -1.47 | -0.446 | 0.556 | -0.0724 | 0.0322 | 0 | 0 | 1 | 0.571 | 0.0385 | 0.00535 | 0.000385 | 0.000133 |
| re100_e2 | 100 | 2 | 1 | 1 | B1_Mode2 | 1.9 | 1 | 1 | 0 | 4.285e-16 | 0.218 | -1.47 | -0.446 | 0.556 | -0.0724 | 0.0322 | 0 | 0 | 2 | 0.0524 | 0.0847 | 0.000138 | 4.939e-07 | 6.095e-05 |
| re100_e2 | 100 | 2 | 1 | 1 | B1_Mode2 | 1.9 | 1 | 1 | 0 | 4.285e-16 | 0.218 | -1.47 | -0.446 | 0.556 | -0.0724 | 0.0322 | 0 | 0 | 3 | 0.0815 | 8.654e-05 | 0.00951 | 0.000115 | 0.000206 |
| re100_e2 | 100 | 2 | 1 | 1 | B1_Mode2 | 1.9 | 1 | 1 | 0 | 4.285e-16 | 0.218 | -1.47 | -0.446 | 0.556 | -0.0724 | 0.0322 | 0 | 0 | 4 | 0.00138 | 0.014 | 0.000263 | 0.000122 | 6.296e-06 |
| re100_e2 | 100 | 2 | 1 | 1 | B1_Mode2 | 1.9 | 1 | 1 | 0 | 4.285e-16 | 0.218 | -1.47 | -0.446 | 0.556 | -0.0724 | 0.0322 | 0 | 0 | 5 | 0.000273 | 0.000919 | 0.000546 | 9.529e-06 | 9.084e-08 |
| re100_e2 | 100 | 2 | 2 | 1 | B1_Mode2 | 1.99 | 1 | 1 | 0 | 4.048e-16 | 0.27 | 1.83 | -0.37 | -0.488 | 0.0337 | 0.00923 | 0 | 0 | 0 | 0.0183 | 0.0233 | 4.468e-06 | 0.000217 | 4.139e-06 |
| re100_e2 | 100 | 2 | 2 | 1 | B1_Mode2 | 1.99 | 1 | 1 | 0 | 4.048e-16 | 0.27 | 1.83 | -0.37 | -0.488 | 0.0337 | 0.00923 | 0 | 0 | 1 | 0.844 | 0.00156 | 0.000761 | 8.690e-06 | 3.350e-05 |
| re100_e2 | 100 | 2 | 2 | 1 | B1_Mode2 | 1.99 | 1 | 1 | 0 | 4.048e-16 | 0.27 | 1.83 | -0.37 | -0.488 | 0.0337 | 0.00923 | 0 | 0 | 2 | 0.0344 | 0.0124 | 1.193e-05 | 7.318e-05 | 2.800e-06 |
| re100_e2 | 100 | 2 | 2 | 1 | B1_Mode2 | 1.99 | 1 | 1 | 0 | 4.048e-16 | 0.27 | 1.83 | -0.37 | -0.488 | 0.0337 | 0.00923 | 0 | 0 | 3 | 0.06 | 0.00138 | 0.000594 | 1.105e-05 | 1.998e-05 |
| re100_e2 | 100 | 2 | 2 | 1 | B1_Mode2 | 1.99 | 1 | 1 | 0 | 4.048e-16 | 0.27 | 1.83 | -0.37 | -0.488 | 0.0337 | 0.00923 | 0 | 0 | 4 | 0.000286 | 0.00193 | 5.523e-05 | 4.371e-06 | 5.873e-06 |
| re100_e2 | 100 | 2 | 2 | 1 | B1_Mode2 | 1.99 | 1 | 1 | 0 | 4.048e-16 | 0.27 | 1.83 | -0.37 | -0.488 | 0.0337 | 0.00923 | 0 | 0 | 5 | 2.145e-05 | 0.000213 | 1.647e-05 | 2.902e-08 | 4.220e-08 |
| re100_e2 | 100 | 2 | 3 | 1 | B1_Mode2 | 1.89 | 1 | 1 | 0 | 3.939e-16 | 0.183 | 1.83 | -0.154 | -0.581 | -0.0282 | 0.0347 | 0 | 0 | 0 | 0.00887 | 0.00385 | 8.867e-05 | 2.035e-05 | 4.121e-06 |
| re100_e2 | 100 | 2 | 3 | 1 | B1_Mode2 | 1.89 | 1 | 1 | 0 | 3.939e-16 | 0.183 | 1.83 | -0.154 | -0.581 | -0.0282 | 0.0347 | 0 | 0 | 1 | 0.884 | 0.000219 | 0.000204 | 3.730e-07 | 9.331e-06 |
| re100_e2 | 100 | 2 | 3 | 1 | B1_Mode2 | 1.89 | 1 | 1 | 0 | 3.939e-16 | 0.183 | 1.83 | -0.154 | -0.581 | -0.0282 | 0.0347 | 0 | 0 | 2 | 0.00623 | 0.00502 | 2.427e-05 | 3.766e-05 | 1.239e-06 |
| re100_e2 | 100 | 2 | 3 | 1 | B1_Mode2 | 1.89 | 1 | 1 | 0 | 3.939e-16 | 0.183 | 1.83 | -0.154 | -0.581 | -0.0282 | 0.0347 | 0 | 0 | 3 | 0.0891 | 6.296e-05 | 0.000217 | 9.749e-08 | 1.596e-05 |
| re100_e2 | 100 | 2 | 3 | 1 | B1_Mode2 | 1.89 | 1 | 1 | 0 | 3.939e-16 | 0.183 | 1.83 | -0.154 | -0.581 | -0.0282 | 0.0347 | 0 | 0 | 4 | 0.00021 | 0.00129 | 2.286e-05 | 5.326e-06 | 4.149e-07 |
| re100_e2 | 100 | 2 | 3 | 1 | B1_Mode2 | 1.89 | 1 | 1 | 0 | 3.939e-16 | 0.183 | 1.83 | -0.154 | -0.581 | -0.0282 | 0.0347 | 0 | 0 | 5 | 0.000318 | 1.991e-05 | 8.050e-07 | 3.274e-07 | 3.459e-08 |
| re100_e2 | 100 | 2 | 4 | 2 | B2_temporal | 1.86 | 1 | 0 | 1 | 4.289e-16 | 0.474 | -0.711 | -0.74 | 0.329 | 0.115 | -0.0801 | 0 | 0 | 0 | 0.0603 | 0.408 | 0.00128 | 4.332e-06 | 4.108e-05 |
| re100_e2 | 100 | 2 | 4 | 2 | B2_temporal | 1.86 | 1 | 0 | 1 | 4.289e-16 | 0.474 | -0.711 | -0.74 | 0.329 | 0.115 | -0.0801 | 0 | 0 | 1 | 0.135 | 0.107 | 0.012 | 1.599e-05 | 5.807e-05 |
| re100_e2 | 100 | 2 | 4 | 2 | B2_temporal | 1.86 | 1 | 0 | 1 | 4.289e-16 | 0.474 | -0.711 | -0.74 | 0.329 | 0.115 | -0.0801 | 0 | 0 | 2 | 0.147 | 0.0568 | 0.00421 | 0.00136 | 2.073e-05 |
| re100_e2 | 100 | 2 | 4 | 2 | B2_temporal | 1.86 | 1 | 0 | 1 | 4.289e-16 | 0.474 | -0.711 | -0.74 | 0.329 | 0.115 | -0.0801 | 0 | 0 | 3 | 0.029 | 0.0192 | 0.00672 | 3.447e-05 | 8.664e-06 |
| re100_e2 | 100 | 2 | 4 | 2 | B2_temporal | 1.86 | 1 | 0 | 1 | 4.289e-16 | 0.474 | -0.711 | -0.74 | 0.329 | 0.115 | -0.0801 | 0 | 0 | 4 | 0.00357 | 0.00495 | 0.000901 | 0.000295 | 6.101e-06 |
| re100_e2 | 100 | 2 | 4 | 2 | B2_temporal | 1.86 | 1 | 0 | 1 | 4.289e-16 | 0.474 | -0.711 | -0.74 | 0.329 | 0.115 | -0.0801 | 0 | 0 | 5 | 0.00172 | 0.000303 | 0.000301 | 1.758e-05 | 5.375e-07 |
| re300_e0.1 | 300 | 0.1 | 0 | 2 | B2_temporal | 0.109 | 1 | 0 | 1 | 1.194e-16 | 0.0275 | -0.14 | 0.0054 | 0.0169 | -0.00133 | 0.00115 | 0 | 0 | 0 | 0.00348 | 0.861 | 0.000898 | 0.000149 | 0.000153 |
| re300_e0.1 | 300 | 0.1 | 0 | 2 | B2_temporal | 0.109 | 1 | 0 | 1 | 1.194e-16 | 0.0275 | -0.14 | 0.0054 | 0.0169 | -0.00133 | 0.00115 | 0 | 0 | 1 | 0.0897 | 0.00193 | 0.0121 | 0.000506 | 8.152e-05 |
| re300_e0.1 | 300 | 0.1 | 0 | 2 | B2_temporal | 0.109 | 1 | 0 | 1 | 1.194e-16 | 0.0275 | -0.14 | 0.0054 | 0.0169 | -0.00133 | 0.00115 | 0 | 0 | 2 | 0.000134 | 0.0265 | 0.000624 | 0.000511 | 4.041e-07 |
| re300_e0.1 | 300 | 0.1 | 0 | 2 | B2_temporal | 0.109 | 1 | 0 | 1 | 1.194e-16 | 0.0275 | -0.14 | 0.0054 | 0.0169 | -0.00133 | 0.00115 | 0 | 0 | 3 | 0.00132 | 3.413e-05 | 0.00103 | 0.000101 | 8.127e-06 |
| re300_e0.1 | 300 | 0.1 | 0 | 2 | B2_temporal | 0.109 | 1 | 0 | 1 | 1.194e-16 | 0.0275 | -0.14 | 0.0054 | 0.0169 | -0.00133 | 0.00115 | 0 | 0 | 4 | 8.107e-06 | 5.148e-05 | 3.965e-05 | 1.412e-05 | 4.307e-06 |
| re300_e0.1 | 300 | 0.1 | 0 | 2 | B2_temporal | 0.109 | 1 | 0 | 1 | 1.194e-16 | 0.0275 | -0.14 | 0.0054 | 0.0169 | -0.00133 | 0.00115 | 0 | 0 | 5 | 6.045e-06 | 3.977e-06 | 9.427e-06 | 1.694e-06 | 7.510e-07 |

Table 15: Décomposition modale — 185 points, 1110 lignes (6 modes $i$ par point, colonnes $E^f_{iJ}$ pour $J$=0..4), `mode_analysis.csv`.

| point | Re | $E^*$ | s | id | branch | $E_t$ | cov. | $i$ | $j$ | rmse | $A_{10}$ | $A_{20}$ | $A_{30}$ | $A_{40}$ | $A_{50}$ | $A_{60}$ | $A_{70}$ | $A_{80}$ | $i$ | $E^f_{0J}$ | $E^f_{1J}$ | $E^f_{2J}$ | $E^f_{3J}$ | $E^f_{4J}$ |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| re300_e0.1 | 300 | 0.1 | 1 | 1 | B1_Mode2 | 0.123 | 1 | 1 | 0 | 9.839e-17 | -0.00266 | 0.467 | 0.0798 | -0.129 | -0.0316 | -0.00239 | 0 | 0 | 0 | 2.882e-05 | 0.000778 | 1.485e-05 | 1.461e-05 | 7.682e-07 |
| re300_e0.1 | 300 | 0.1 | 1 | 1 | B1_Mode2 | 0.123 | 1 | 1 | 0 | 9.839e-17 | -0.00266 | 0.467 | 0.0798 | -0.129 | -0.0316 | -0.00239 | 0 | 0 | 1 | 0.89 | 0.00621 | 5.109e-05 | 6.866e-05 | 5.901e-06 |
| re300_e0.1 | 300 | 0.1 | 1 | 1 | B1_Mode2 | 0.123 | 1 | 1 | 0 | 9.839e-17 | -0.00266 | 0.467 | 0.0798 | -0.129 | -0.0316 | -0.00239 | 0 | 0 | 2 | 0.026 | 0.00374 | 2.201e-05 | 3.398e-05 | 1.865e-06 |
| re300_e0.1 | 300 | 0.1 | 1 | 1 | B1_Mode2 | 0.123 | 1 | 1 | 0 | 9.839e-17 | -0.00266 | 0.467 | 0.0798 | -0.129 | -0.0316 | -0.00239 | 0 | 0 | 3 | 0.0681 | 0.000437 | 0.000139 | 6.058e-06 | 7.378e-06 |
| re300_e0.1 | 300 | 0.1 | 1 | 1 | B1_Mode2 | 0.123 | 1 | 1 | 0 | 9.839e-17 | -0.00266 | 0.467 | 0.0798 | -0.129 | -0.0316 | -0.00239 | 0 | 0 | 4 | 0.00406 | 8.343e-05 | 1.178e-05 | 7.174e-11 | 6.328e-07 |
| re300_e0.1 | 300 | 0.1 | 1 | 1 | B1_Mode2 | 0.123 | 1 | 1 | 0 | 9.839e-17 | -0.00266 | 0.467 | 0.0798 | -0.129 | -0.0316 | -0.00239 | 0 | 0 | 5 | 2.330e-05 | 8.772e-05 | 4.310e-10 | 3.901e-08 | 4.699e-08 |
| re300_e0.1 | 300 | 0.1 | 2 | 1 | B1_Mode2 | 0.12 | 1 | 1 | 0 | 9.627e-17 | 0.0257 | 0.46 | -0.0385 | -0.13 | 0.011 | 0.00538 | 0 | 0 | 0 | 0.00274 | 0.0331 | 1.127e-05 | 0.000259 | 4.435e-10 |
| re300_e0.1 | 300 | 0.1 | 2 | 1 | B1_Mode2 | 0.12 | 1 | 1 | 0 | 9.627e-17 | 0.0257 | 0.46 | -0.0385 | -0.13 | 0.011 | 0.00538 | 0 | 0 | 1 | 0.878 | 0.00255 | 0.000323 | 7.720e-05 | 1.174e-05 |
| re300_e0.1 | 300 | 0.1 | 2 | 1 | B1_Mode2 | 0.12 | 1 | 1 | 0 | 9.627e-17 | 0.0257 | 0.46 | -0.0385 | -0.13 | 0.011 | 0.00538 | 0 | 0 | 2 | 0.00616 | 0.0051 | 8.642e-05 | 3.797e-05 | 6.893e-06 |
| re300_e0.1 | 300 | 0.1 | 2 | 1 | B1_Mode2 | 0.12 | 1 | 1 | 0 | 9.627e-17 | 0.0257 | 0.46 | -0.0385 | -0.13 | 0.011 | 0.00538 | 0 | 0 | 3 | 0.0699 | 0.000476 | 0.000219 | 1.548e-09 | 9.379e-06 |
| re300_e0.1 | 300 | 0.1 | 2 | 1 | B1_Mode2 | 0.12 | 1 | 1 | 0 | 9.627e-17 | 0.0257 | 0.46 | -0.0385 | -0.13 | 0.011 | 0.00538 | 0 | 0 | 4 | 0.000499 | 0.000185 | 7.177e-05 | 2.484e-06 | 7.575e-06 |
| re300_e0.1 | 300 | 0.1 | 2 | 1 | B1_Mode2 | 0.12 | 1 | 1 | 0 | 9.627e-17 | 0.0257 | 0.46 | -0.0385 | -0.13 | 0.011 | 0.00538 | 0 | 0 | 5 | 0.00012 | 0.00014 | 7.963e-07 | 3.413e-07 | 8.533e-07 |
| re300_e0.1 | 300 | 0.1 | 3 | 2 | B2_temporal | 0.124 | 1 | 0 | 1 | 1.104e-16 | -0.0125 | 0.226 | -0.00559 | -0.0224 | -0.000628 | -0.00092 | 0 | 0 | 0 | 0.000628 | 0.782 | 0.00251 | 6.356e-05 | 0.000191 |
| re300_e0.1 | 300 | 0.1 | 3 | 2 | B2_temporal | 0.124 | 1 | 0 | 1 | 1.104e-16 | -0.0125 | 0.226 | -0.00559 | -0.0224 | -0.000628 | -0.00092 | 0 | 0 | 1 | 0.206 | 0.000618 | 0.000146 | 0.000517 | 6.167e-05 |
| re300_e0.1 | 300 | 0.1 | 3 | 2 | B2_temporal | 0.124 | 1 | 0 | 1 | 1.104e-16 | -0.0125 | 0.226 | -0.00559 | -0.0224 | -0.000628 | -0.00092 | 0 | 0 | 2 | 0.000126 | 0.00485 | 0.000564 | 3.821e-05 | 1.757e-05 |
| re300_e0.1 | 300 | 0.1 | 3 | 2 | B2_temporal | 0.124 | 1 | 0 | 1 | 1.104e-16 | -0.0125 | 0.226 | -0.00559 | -0.0224 | -0.000628 | -0.00092 | 0 | 0 | 3 | 0.00203 | 0.000125 | 8.127e-05 | 6.214e-07 | 8.208e-06 |
| re300_e0.1 | 300 | 0.1 | 3 | 2 | B2_temporal | 0.124 | 1 | 0 | 1 | 1.104e-16 | -0.0125 | 0.226 | -0.00559 | -0.0224 | -0.000628 | -0.00092 | 0 | 0 | 4 | 1.590e-06 | 7.387e-05 | 8.096e-06 | 3.298e-06 | 1.147e-09 |
| re300_e0.1 | 300 | 0.1 | 3 | 2 | B2_temporal | 0.124 | 1 | 0 | 1 | 1.104e-16 | -0.0125 | 0.226 | -0.00559 | -0.0224 | -0.000628 | -0.00092 | 0 | 0 | 5 | 3.413e-06 | 2.115e-06 | 1.300e-06 | 2.269e-06 | 5.740e-08 |
| re300_e0.1 | 300 | 0.1 | 4 | 2 | B2_temporal | 0.117 | 1 | 0 | 1 | 1.192e-16 | 0.0177 | -0.146 | -0.0317 | 0.0193 | 0.005 | -0.00154 | 0 | 0 | 0 | 0.00134 | 0.85 | 0.017 | 0.00108 | 8.948e-05 |
| re300_e0.1 | 300 | 0.1 | 4 | 2 | B2_temporal | 0.117 | 1 | 0 | 1 | 1.192e-16 | 0.0177 | -0.146 | -0.0317 | 0.0193 | 0.005 | -0.00154 | 0 | 0 | 1 | 0.0913 | 0.00332 | 0.00253 | 0.000261 | 7.984e-05 |
| re300_e0.1 | 300 | 0.1 | 4 | 2 | B2_temporal | 0.117 | 1 | 0 | 1 | 1.192e-16 | 0.0177 | -0.146 | -0.0317 | 0.0193 | 0.005 | -0.00154 | 0 | 0 | 2 | 0.00428 | 0.0243 | 0.00179 | 0.00064 | 1.869e-05 |
| re300_e0.1 | 300 | 0.1 | 4 | 2 | B2_temporal | 0.117 | 1 | 0 | 1 | 1.192e-16 | 0.0177 | -0.146 | -0.0317 | 0.0193 | 0.005 | -0.00154 | 0 | 0 | 3 | 0.00159 | 0.000138 | 2.729e-05 | 2.051e-05 | 2.236e-05 |
| re300_e0.1 | 300 | 0.1 | 4 | 2 | B2_temporal | 0.117 | 1 | 0 | 1 | 1.192e-16 | 0.0177 | -0.146 | -0.0317 | 0.0193 | 0.005 | -0.00154 | 0 | 0 | 4 | 0.000107 | 6.584e-05 | 7.906e-07 | 6.397e-06 | 1.563e-06 |
| re300_e0.1 | 300 | 0.1 | 4 | 2 | B2_temporal | 0.117 | 1 | 0 | 1 | 1.192e-16 | 0.0177 | -0.146 | -0.0317 | 0.0193 | 0.005 | -0.00154 | 0 | 0 | 5 | 1.014e-05 | 1.393e-06 | 1.963e-06 | 4.154e-07 | 5.314e-08 |
| re300_e0.25 | 300 | 0.25 | 0 | 2 | B2_temporal | 0.233 | 1 | 0 | 1 | 1.401e-16 | 0.0385 | -0.17 | 0.0204 | 0.0265 | -0.00526 | 0.0027 | 0 | 0 | 0 | 0.00318 | 0.881 | 0.00171 | 0.000615 | 0.000193 |
| re300_e0.25 | 300 | 0.25 | 0 | 2 | B2_temporal | 0.233 | 1 | 0 | 1 | 1.401e-16 | 0.0385 | -0.17 | 0.0204 | 0.0265 | -0.00526 | 0.0027 | 0 | 0 | 1 | 0.0618 | 0.00656 | 0.0096 | 0.00117 | 6.539e-06 |
| re300_e0.25 | 300 | 0.25 | 0 | 2 | B2_temporal | 0.233 | 1 | 0 | 1 | 1.401e-16 | 0.0385 | -0.17 | 0.0204 | 0.0265 | -0.00526 | 0.0027 | 0 | 0 | 2 | 0.000892 | 0.0272 | 0.00163 | 0.00112 | 3.684e-06 |
| re300_e0.25 | 300 | 0.25 | 0 | 2 | B2_temporal | 0.233 | 1 | 0 | 1 | 1.401e-16 | 0.0385 | -0.17 | 0.0204 | 0.0265 | -0.00526 | 0.0027 | 0 | 0 | 3 | 0.00151 | 2.511e-08 | 0.000948 | 0.000222 | 4.298e-06 |
| re300_e0.25 | 300 | 0.25 | 0 | 2 | B2_temporal | 0.233 | 1 | 0 | 1 | 1.401e-16 | 0.0385 | -0.17 | 0.0204 | 0.0265 | -0.00526 | 0.0027 | 0 | 0 | 4 | 5.938e-05 | 1.234e-06 | 7.970e-05 | 2.984e-05 | 1.607e-05 |
| re300_e0.25 | 300 | 0.25 | 0 | 2 | B2_temporal | 0.233 | 1 | 0 | 1 | 1.401e-16 | 0.0385 | -0.17 | 0.0204 | 0.0265 | -0.00526 | 0.0027 | 0 | 0 | 5 | 1.557e-05 | 6.392e-06 | 1.074e-05 | 1.306e-06 | 1.322e-06 |
| re300_e0.25 | 300 | 0.25 | 1 | 3 | B3_mixed | 0.243 | 1 | 1 | 0 | 1.412e-16 | -0.0441 | -0.497 | 0.0908 | 0.0796 | -0.0323 | 0.0118 | 0 | 0 | 0 | 0.00401 | 0.389 | 0.000126 | 0.000159 | 9.643e-06 |
| re300_e0.25 | 300 | 0.25 | 1 | 3 | B3_mixed | 0.243 | 1 | 1 | 0 | 1.412e-16 | -0.0441 | -0.497 | 0.0908 | 0.0796 | -0.0323 | 0.0118 | 0 | 0 | 1 | 0.51 | 8.875e-05 | 0.00613 | 0.000994 | 0.00022 |
| re300_e0.25 | 300 | 0.25 | 1 | 3 | B3_mixed | 0.243 | 1 | 1 | 0 | 1.412e-16 | -0.0441 | -0.497 | 0.0908 | 0.0796 | -0.0323 | 0.0118 | 0 | 0 | 2 | 0.017 | 0.0351 | 0.00991 | 7.396e-05 | 0.00016 |
| re300_e0.25 | 300 | 0.25 | 1 | 3 | B3_mixed | 0.243 | 1 | 1 | 0 | 1.412e-16 | -0.0441 | -0.497 | 0.0908 | 0.0796 | -0.0323 | 0.0118 | 0 | 0 | 3 | 0.0131 | 0.00857 | 4.005e-07 | 0.000309 | 0.000103 |
| re300_e0.25 | 300 | 0.25 | 1 | 3 | B3_mixed | 0.243 | 1 | 1 | 0 | 1.412e-16 | -0.0441 | -0.497 | 0.0908 | 0.0796 | -0.0323 | 0.0118 | 0 | 0 | 4 | 0.00215 | 0.000162 | 0.00103 | 0.000238 | 1.099e-05 |
| re300_e0.25 | 300 | 0.25 | 1 | 3 | B3_mixed | 0.243 | 1 | 1 | 0 | 1.412e-16 | -0.0441 | -0.497 | 0.0908 | 0.0796 | -0.0323 | 0.0118 | 0 | 0 | 5 | 0.000287 | 0.000188 | 0.000341 | 1.211e-07 | 5.660e-06 |
| re300_e0.25 | 300 | 0.25 | 2 | 1 | B1_Mode2 | 0.255 | 1 | 1 | 0 | 1.389e-16 | 0.0352 | 0.65 | -0.0117 | -0.144 | 0.00872 | -0.00253 | 0 | 0 | 0 | 0.00242 | 0.0943 | 7.770e-05 | 0.00031 | 1.139e-06 |
| re300_e0.25 | 300 | 0.25 | 2 | 1 | B1_Mode2 | 0.255 | 1 | 1 | 0 | 1.389e-16 | 0.0352 | 0.65 | -0.0117 | -0.144 | 0.00872 | -0.00253 | 0 | 0 | 1 | 0.827 | 0.000434 | 0.00443 | 3.224e-05 | 0.000171 |
| re300_e0.25 | 300 | 0.25 | 2 | 1 | B1_Mode2 | 0.255 | 1 | 1 | 0 | 1.389e-16 | 0.0352 | 0.65 | -0.0117 | -0.144 | 0.00872 | -0.00253 | 0 | 0 | 2 | 0.000267 | 0.0237 | 0.000138 | 2.024e-05 | 4.761e-06 |
| re300_e0.25 | 300 | 0.25 | 2 | 1 | B1_Mode2 | 0.255 | 1 | 1 | 0 | 1.389e-16 | 0.0352 | 0.65 | -0.0117 | -0.144 | 0.00872 | -0.00253 | 0 | 0 | 3 | 0.0407 | 0.00181 | 0.00159 | 4.668e-06 | 4.707e-05 |
| re300_e0.25 | 300 | 0.25 | 2 | 1 | B1_Mode2 | 0.255 | 1 | 1 | 0 | 1.389e-16 | 0.0352 | 0.65 | -0.0117 | -0.144 | 0.00872 | -0.00253 | 0 | 0 | 4 | 0.000149 | 0.00185 | 0.000238 | 1.648e-06 | 8.540e-06 |
| re300_e0.25 | 300 | 0.25 | 2 | 1 | B1_Mode2 | 0.255 | 1 | 1 | 0 | 1.389e-16 | 0.0352 | 0.65 | -0.0117 | -0.144 | 0.00872 | -0.00253 | 0 | 0 | 5 | 1.249e-05 | 0.000181 | 2.828e-05 | 1.429e-05 | 1.465e-07 |
| re300_e0.25 | 300 | 0.25 | 3 | 2 | B2_temporal | 0.265 | 1 | 0 | 1 | 1.567e-16 | -0.0302 | 0.29 | 0.0107 | -0.0217 | 0.00494 | -0.00279 | 0 | 0 | 0 | 0.00172 | 0.832 | 0.000238 | 3.817e-05 | 0.000144 |
| re300_e0.25 | 300 | 0.25 | 3 | 2 | B2_temporal | 0.265 | 1 | 0 | 1 | 1.567e-16 | -0.0302 | 0.29 | 0.0107 | -0.0217 | 0.00494 | -0.00279 | 0 | 0 | 1 | 0.159 | 7.046e-06 | 0.000921 | 0.00105 | 5.300e-05 |
| re300_e0.25 | 300 | 0.25 | 3 | 2 | B2_temporal | 0.265 | 1 | 0 | 1 | 1.567e-16 | -0.0302 | 0.29 | 0.0107 | -0.0217 | 0.00494 | -0.00279 | 0 | 0 | 2 | 0.000215 | 0.00246 | 0.000529 | 2.872e-05 | 2.447e-05 |
| re300_e0.25 | 300 | 0.25 | 3 | 2 | B2_temporal | 0.265 | 1 | 0 | 1 | 1.567e-16 | -0.0302 | 0.29 | 0.0107 | -0.0217 | 0.00494 | -0.00279 | 0 | 0 | 3 | 0.000887 | 0.000163 | 2.392e-05 | 2.756e-07 | 1.742e-06 |
| re300_e0.25 | 300 | 0.25 | 3 | 2 | B2_temporal | 0.265 | 1 | 0 | 1 | 1.567e-16 | -0.0302 | 0.29 | 0.0107 | -0.0217 | 0.00494 | -0.00279 | 0 | 0 | 4 | 4.602e-05 | 5.637e-05 | 5.198e-06 | 2.792e-06 | 1.308e-09 |
| re300_e0.25 | 300 | 0.25 | 3 | 2 | B2_temporal | 0.265 | 1 | 0 | 1 | 1.567e-16 | -0.0302 | 0.29 | 0.0107 | -0.0217 | 0.00494 | -0.00279 | 0 | 0 | 5 | 1.465e-05 | 4.951e-06 | 1.602e-06 | 1.732e-06 | 7.672e-07 |
| re300_e0.25 | 300 | 0.25 | 4 | 2 | B2_temporal | 0.251 | 1 | 0 | 1 | 1.520e-16 | 0.0378 | -0.317 | -0.0757 | 0.0641 | 0.016 | -0.000954 | 0 | 0 | 0 | 0.00284 | 0.689 | 0.0372 | 0.000772 | 0.000184 |
| re300_e0.25 | 300 | 0.25 | 4 | 2 | B2_temporal | 0.251 | 1 | 0 | 1 | 1.520e-16 | 0.0378 | -0.317 | -0.0757 | 0.0641 | 0.016 | -0.000954 | 0 | 0 | 1 | 0.2 | 0.00444 | 0.000161 | 0.00137 | 0.000248 |
| re300_e0.25 | 300 | 0.25 | 4 | 2 | B2_temporal | 0.251 | 1 | 0 | 1 | 1.520e-16 | 0.0378 | -0.317 | -0.0757 | 0.0641 | 0.016 | -0.000954 | 0 | 0 | 2 | 0.0114 | 0.0326 | 0.00915 | 0.000507 | 1.718e-05 |
| re300_e0.25 | 300 | 0.25 | 4 | 2 | B2_temporal | 0.251 | 1 | 0 | 1 | 1.520e-16 | 0.0378 | -0.317 | -0.0757 | 0.0641 | 0.016 | -0.000954 | 0 | 0 | 3 | 0.00817 | 0.00147 | 1.853e-07 | 7.084e-06 | 0.000129 |
| re300_e0.25 | 300 | 0.25 | 4 | 2 | B2_temporal | 0.251 | 1 | 0 | 1 | 1.520e-16 | 0.0378 | -0.317 | -0.0757 | 0.0641 | 0.016 | -0.000954 | 0 | 0 | 4 | 0.000511 | 0.000185 | 6.118e-05 | 2.926e-05 | 5.742e-07 |
| re300_e0.25 | 300 | 0.25 | 4 | 2 | B2_temporal | 0.251 | 1 | 0 | 1 | 1.520e-16 | 0.0378 | -0.317 | -0.0757 | 0.0641 | 0.016 | -0.000954 | 0 | 0 | 5 | 1.812e-06 | 1.073e-05 | 5.155e-07 | 3.812e-06 | 5.580e-07 |
| re300_e0.5 | 300 | 0.5 | 0 | 2 | B2_temporal | 0.456 | 1 | 0 | 1 | 2.150e-16 | 0.0519 | -0.219 | 0.036 | 0.0145 | -0.0054 | 0.00894 | 0 | 0 | 0 | 0.00295 | 0.865 | 0.0102 | 0.00787 | 0.000275 |
| re300_e0.5 | 300 | 0.5 | 0 | 2 | B2_temporal | 0.456 | 1 | 0 | 1 | 2.150e-16 | 0.0519 | -0.219 | 0.036 | 0.0145 | -0.0054 | 0.00894 | 0 | 0 | 1 | 0.0526 | 0.0103 | 0.0146 | 0.00179 | 1.012e-06 |
| re300_e0.5 | 300 | 0.5 | 0 | 2 | B2_temporal | 0.456 | 1 | 0 | 1 | 2.150e-16 | 0.0519 | -0.219 | 0.036 | 0.0145 | -0.0054 | 0.00894 | 0 | 0 | 2 | 0.00142 | 0.0259 | 0.00292 | 0.00191 | 2.833e-05 |
| re300_e0.5 | 300 | 0.5 | 0 | 2 | B2_temporal | 0.456 | 1 | 0 | 1 | 2.150e-16 | 0.0519 | -0.219 | 0.036 | 0.0145 | -0.0054 | 0.00894 | 0 | 0 | 3 | 0.00023 | 0.000461 | 0.000261 | 0.000532 | 1.015e-07 |
| re300_e0.5 | 300 | 0.5 | 0 | 2 | B2_temporal | 0.456 | 1 | 0 | 1 | 2.150e-16 | 0.0519 | -0.219 | 0.036 | 0.0145 | -0.0054 | 0.00894 | 0 | 0 | 4 | 3.195e-05 | 0.000239 | 0.000217 | 1.273e-05 | 1.914e-05 |
| re300_e0.5 | 300 | 0.5 | 0 | 2 | B2_temporal | 0.456 | 1 | 0 | 1 | 2.150e-16 | 0.0519 | -0.219 | 0.036 | 0.0145 | -0.0054 | 0.00894 | 0 | 0 | 5 | 8.763e-05 | 1.696e-05 | 3.321e-05 | 3.833e-06 | 3.744e-07 |
| re300_e0.5 | 300 | 0.5 | 1 | 1 | B1_Mode2 | 0.479 | 1 | 1 | 0 | 2.258e-16 | -0.0336 | -0.841 | 0.0374 | 0.116 | -0.0273 | 0.0208 | 0 | 0 | 0 | 0.00117 | 0.176 | 0.000187 | 0.000332 | 4.593e-06 |
| re300_e0.5 | 300 | 0.5 | 1 | 1 | B1_Mode2 | 0.479 | 1 | 1 | 0 | 2.258e-16 | -0.0336 | -0.841 | 0.0374 | 0.116 | -0.0273 | 0.0208 | 0 | 0 | 1 | 0.738 | 0.00269 | 0.00692 | 0.000579 | 0.000299 |
| re300_e0.5 | 300 | 0.5 | 1 | 1 | B1_Mode2 | 0.479 | 1 | 1 | 0 | 2.258e-16 | -0.0336 | -0.841 | 0.0374 | 0.116 | -0.0273 | 0.0208 | 0 | 0 | 2 | 0.00146 | 0.0285 | 0.00856 | 1.273e-05 | 0.00018 |
| re300_e0.5 | 300 | 0.5 | 1 | 1 | B1_Mode2 | 0.479 | 1 | 1 | 0 | 2.258e-16 | -0.0336 | -0.841 | 0.0374 | 0.116 | -0.0273 | 0.0208 | 0 | 0 | 3 | 0.014 | 0.0165 | 0.000523 | 0.000154 | 9.624e-05 |
| re300_e0.5 | 300 | 0.5 | 1 | 1 | B1_Mode2 | 0.479 | 1 | 1 | 0 | 2.258e-16 | -0.0336 | -0.841 | 0.0374 | 0.116 | -0.0273 | 0.0208 | 0 | 0 | 4 | 0.000776 | 2.043e-05 | 0.00136 | 0.000138 | 1.786e-05 |
| re300_e0.5 | 300 | 0.5 | 1 | 1 | B1_Mode2 | 0.479 | 1 | 1 | 0 | 2.258e-16 | -0.0336 | -0.841 | 0.0374 | 0.116 | -0.0273 | 0.0208 | 0 | 0 | 5 | 0.00045 | 0.000325 | 0.000309 | 4.764e-06 | 1.447e-06 |

Table 15: Décomposition modale — 185 points, 1110 lignes (6 modes $i$ par point, colonnes $E^f_{iJ}$ pour $J$=0..4), `mode_analysis.csv`.

| point | Re | $E^*$ | s | id | branch | $E_t$ | cov. | $i$ | $j$ | rmse | $A_{10}$ | $A_{20}$ | $A_{30}$ | $A_{40}$ | $A_{50}$ | $A_{60}$ | $A_{70}$ | $A_{80}$ | $i$ | $E^f_{0J}$ | $E^f_{1J}$ | $E^f_{2J}$ | $E^f_{3J}$ | $E^f_{4J}$ |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| re300_e0.5 | 300 | 0.5 | 2 | 1 | B1_Mode2 | 0.499 | 1 | 1 | 0 | 2.186e-16 | 0.0361 | 0.935 | 0.0224 | -0.159 | 0.0177 | -0.00676 | 0 | 0 | 0 | 0.0013 | 0.0667 | 1.243e-05 | 0.000343 | 7.941e-09 |
| re300_e0.5 | 300 | 0.5 | 2 | 1 | B1_Mode2 | 0.499 | 1 | 1 | 0 | 2.186e-16 | 0.0361 | 0.935 | 0.0224 | -0.159 | 0.0177 | -0.00676 | 0 | 0 | 1 | 0.876 | 1.008e-08 | 0.00353 | 7.661e-06 | 0.000126 |
| re300_e0.5 | 300 | 0.5 | 2 | 1 | B1_Mode2 | 0.499 | 1 | 1 | 0 | 2.186e-16 | 0.0361 | 0.935 | 0.0224 | -0.159 | 0.0177 | -0.00676 | 0 | 0 | 2 | 0.000503 | 0.02 | 2.394e-06 | 2.806e-05 | 1.116e-06 |
| re300_e0.5 | 300 | 0.5 | 2 | 1 | B1_Mode2 | 0.499 | 1 | 1 | 0 | 2.186e-16 | 0.0361 | 0.935 | 0.0224 | -0.159 | 0.0177 | -0.00676 | 0 | 0 | 3 | 0.0254 | 0.00209 | 0.000912 | 1.200e-06 | 2.387e-05 |
| re300_e0.5 | 300 | 0.5 | 2 | 1 | B1_Mode2 | 0.499 | 1 | 1 | 0 | 2.186e-16 | 0.0361 | 0.935 | 0.0224 | -0.159 | 0.0177 | -0.00676 | 0 | 0 | 4 | 0.000316 | 0.00226 | 0.000162 | 1.929e-09 | 9.586e-06 |
| re300_e0.5 | 300 | 0.5 | 2 | 1 | B1_Mode2 | 0.499 | 1 | 1 | 0 | 2.186e-16 | 0.0361 | 0.935 | 0.0224 | -0.159 | 0.0177 | -0.00676 | 0 | 0 | 5 | 4.570e-05 | 0.000112 | 4.089e-05 | 7.378e-06 | 3.651e-08 |
| re300_e0.5 | 300 | 0.5 | 3 | 2 | B2_temporal | 0.51 | 1 | 0 | 1 | 2.256e-16 | -0.043 | 0.404 | 0.0175 | -0.0274 | 0.0126 | -0.00543 | 0 | 0 | 0 | 0.00181 | 0.809 | 0.00174 | 9.413e-05 | 5.457e-05 |
| re300_e0.5 | 300 | 0.5 | 3 | 2 | B2_temporal | 0.51 | 1 | 0 | 1 | 2.256e-16 | -0.043 | 0.404 | 0.0175 | -0.0274 | 0.0126 | -0.00543 | 0 | 0 | 1 | 0.16 | 0.0116 | 0.00521 | 0.00157 | 4.850e-05 |
| re300_e0.5 | 300 | 0.5 | 3 | 2 | B2_temporal | 0.51 | 1 | 0 | 1 | 2.256e-16 | -0.043 | 0.404 | 0.0175 | -0.0274 | 0.0126 | -0.00543 | 0 | 0 | 2 | 0.000299 | 0.0031 | 0.00205 | 0.000807 | 1.922e-05 |
| re300_e0.5 | 300 | 0.5 | 3 | 2 | B2_temporal | 0.51 | 1 | 0 | 1 | 2.256e-16 | -0.043 | 0.404 | 0.0175 | -0.0274 | 0.0126 | -0.00543 | 0 | 0 | 3 | 0.000735 | 0.000358 | 0.000697 | 1.781e-05 | 6.714e-09 |
| re300_e0.5 | 300 | 0.5 | 3 | 2 | B2_temporal | 0.51 | 1 | 0 | 1 | 2.256e-16 | -0.043 | 0.404 | 0.0175 | -0.0274 | 0.0126 | -0.00543 | 0 | 0 | 4 | 0.000157 | 7.554e-06 | 6.947e-05 | 5.809e-06 | 1.744e-06 |
| re300_e0.5 | 300 | 0.5 | 3 | 2 | B2_temporal | 0.51 | 1 | 0 | 1 | 2.256e-16 | -0.043 | 0.404 | 0.0175 | -0.0274 | 0.0126 | -0.00543 | 0 | 0 | 5 | 2.896e-05 | 6.525e-06 | 2.702e-06 | 5.520e-08 | 8.095e-07 |
| re300_e0.5 | 300 | 0.5 | 4 | 2 | B2_temporal | 0.488 | 1 | 0 | 1 | 2.128e-16 | 0.0517 | -0.305 | -0.0759 | 0.0454 | 0.0258 | 0.000971 | 0 | 0 | 0 | 0.00274 | 0.744 | 0.0827 | 0.00249 | 4.935e-05 |
| re300_e0.5 | 300 | 0.5 | 4 | 2 | B2_temporal | 0.488 | 1 | 0 | 1 | 2.128e-16 | 0.0517 | -0.305 | -0.0759 | 0.0454 | 0.0258 | 0.000971 | 0 | 0 | 1 | 0.0953 | 0.0021 | 0.00198 | 0.001 | 0.000867 |
| re300_e0.5 | 300 | 0.5 | 4 | 2 | B2_temporal | 0.488 | 1 | 0 | 1 | 2.128e-16 | 0.0517 | -0.305 | -0.0759 | 0.0454 | 0.0258 | 0.000971 | 0 | 0 | 2 | 0.00591 | 0.0352 | 0.0199 | 0.00124 | 3.476e-05 |
| re300_e0.5 | 300 | 0.5 | 4 | 2 | B2_temporal | 0.488 | 1 | 0 | 1 | 2.128e-16 | 0.0517 | -0.305 | -0.0759 | 0.0454 | 0.0258 | 0.000971 | 0 | 0 | 3 | 0.00212 | 0.000145 | 1.528e-05 | 3.056e-05 | 0.000321 |
| re300_e0.5 | 300 | 0.5 | 4 | 2 | B2_temporal | 0.488 | 1 | 0 | 1 | 2.128e-16 | 0.0517 | -0.305 | -0.0759 | 0.0454 | 0.0258 | 0.000971 | 0 | 0 | 4 | 0.000683 | 0.000356 | 0.000308 | 9.208e-05 | 6.926e-07 |
| re300_e0.5 | 300 | 0.5 | 4 | 2 | B2_temporal | 0.488 | 1 | 0 | 1 | 2.128e-16 | 0.0517 | -0.305 | -0.0759 | 0.0454 | 0.0258 | 0.000971 | 0 | 0 | 5 | 9.664e-07 | 3.812e-06 | 3.388e-07 | 3.638e-07 | 1.425e-05 |
| re300_e1 | 300 | 1 | 0 | 2 | B2_temporal | 0.903 | 1 | 0 | 1 | 3.157e-16 | 0.12 | -0.708 | 0.0151 | 0.0715 | -0.0589 | 0.0385 | 0 | 0 | 0 | 0.00796 | 0.57 | 0.000162 | 0.000924 | 0.00027 |
| re300_e1 | 300 | 1 | 0 | 2 | B2_temporal | 0.903 | 1 | 0 | 1 | 3.157e-16 | 0.12 | -0.708 | 0.0151 | 0.0715 | -0.0589 | 0.0385 | 0 | 0 | 1 | 0.277 | 0.0549 | 0.0161 | 0.000732 | 1.347e-06 |
| re300_e1 | 300 | 1 | 0 | 2 | B2_temporal | 0.903 | 1 | 0 | 1 | 3.157e-16 | 0.12 | -0.708 | 0.0151 | 0.0715 | -0.0589 | 0.0385 | 0 | 0 | 2 | 0.000126 | 0.0499 | 0.00213 | 0.00157 | 6.543e-06 |
| re300_e1 | 300 | 1 | 0 | 2 | B2_temporal | 0.903 | 1 | 0 | 1 | 3.157e-16 | 0.12 | -0.708 | 0.0151 | 0.0715 | -0.0589 | 0.0385 | 0 | 0 | 3 | 0.00283 | 0.00339 | 0.0059 | 0.00019 | 9.048e-06 |
| re300_e1 | 300 | 1 | 0 | 2 | B2_temporal | 0.903 | 1 | 0 | 1 | 3.157e-16 | 0.12 | -0.708 | 0.0151 | 0.0715 | -0.0589 | 0.0385 | 0 | 0 | 4 | 0.00192 | 0.000505 | 0.00107 | 5.496e-05 | 1.857e-06 |
| re300_e1 | 300 | 1 | 0 | 2 | B2_temporal | 0.903 | 1 | 0 | 1 | 3.157e-16 | 0.12 | -0.708 | 0.0151 | 0.0715 | -0.0589 | 0.0385 | 0 | 0 | 5 | 0.000819 | 0.000753 | 3.788e-06 | 9.215e-05 | 5.890e-06 |
| re300_e1 | 300 | 1 | 1 | 3 | B3_mixed | 0.93 | 1 | 1 | 0 | 3.169e-16 | 0.021 | -1.08 | -0.118 | 0.0834 | -0.018 | 0.0295 | 0 | 0 | 0 | 0.000236 | 0.24 | 9.673e-05 | 0.000211 | 6.222e-06 |
| re300_e1 | 300 | 1 | 1 | 3 | B3_mixed | 0.93 | 1 | 1 | 0 | 3.169e-16 | 0.021 | -1.08 | -0.118 | 0.0834 | -0.018 | 0.0295 | 0 | 0 | 1 | 0.628 | 0.0116 | 0.0154 | 0.00176 | 0.000478 |
| re300_e1 | 300 | 1 | 1 | 3 | B3_mixed | 0.93 | 1 | 1 | 0 | 3.169e-16 | 0.021 | -1.08 | -0.118 | 0.0834 | -0.018 | 0.0295 | 0 | 0 | 2 | 0.00752 | 0.058 | 0.00983 | 6.433e-05 | 2.834e-05 |
| re300_e1 | 300 | 1 | 1 | 3 | B3_mixed | 0.93 | 1 | 1 | 0 | 3.169e-16 | 0.021 | -1.08 | -0.118 | 0.0834 | -0.018 | 0.0295 | 0 | 0 | 3 | 0.00374 | 0.0131 | 0.00384 | 0.00106 | 0.000307 |
| re300_e1 | 300 | 1 | 1 | 3 | B3_mixed | 0.93 | 1 | 1 | 0 | 3.169e-16 | 0.021 | -1.08 | -0.118 | 0.0834 | -0.018 | 0.0295 | 0 | 0 | 4 | 0.000175 | 0.00181 | 0.0024 | 6.281e-05 | 5.278e-06 |
| re300_e1 | 300 | 1 | 1 | 3 | B3_mixed | 0.93 | 1 | 1 | 0 | 3.169e-16 | 0.021 | -1.08 | -0.118 | 0.0834 | -0.018 | 0.0295 | 0 | 0 | 5 | 0.000467 | 2.177e-06 | 1.459e-05 | 0.000191 | 3.098e-05 |
| re300_e1 | 300 | 1 | 2 | 1 | B1_Mode2 | 0.993 | 1 | 1 | 0 | 2.941e-16 | 0.0938 | 1.33 | -0.0842 | -0.245 | 0.057 | -0.0145 | 0 | 0 | 0 | 0.00443 | 0.0363 | 4.797e-06 | 0.000257 | 3.568e-06 |
| re300_e1 | 300 | 1 | 2 | 1 | B1_Mode2 | 0.993 | 1 | 1 | 0 | 2.941e-16 | 0.0938 | 1.33 | -0.0842 | -0.245 | 0.057 | -0.0145 | 0 | 0 | 1 | 0.894 | 0.00123 | 0.00238 | 8.454e-08 | 0.0001 |
| re300_e1 | 300 | 1 | 2 | 1 | B1_Mode2 | 0.993 | 1 | 1 | 0 | 2.941e-16 | 0.0938 | 1.33 | -0.0842 | -0.245 | 0.057 | -0.0145 | 0 | 0 | 2 | 0.00357 | 0.0173 | 2.972e-06 | 3.980e-05 | 1.984e-06 |
| re300_e1 | 300 | 1 | 2 | 1 | B1_Mode2 | 0.993 | 1 | 1 | 0 | 2.941e-16 | 0.0938 | 1.33 | -0.0842 | -0.245 | 0.057 | -0.0145 | 0 | 0 | 3 | 0.0302 | 0.00508 | 0.00063 | 2.078e-05 | 1.806e-05 |
| re300_e1 | 300 | 1 | 2 | 1 | B1_Mode2 | 0.993 | 1 | 1 | 0 | 2.941e-16 | 0.0938 | 1.33 | -0.0842 | -0.245 | 0.057 | -0.0145 | 0 | 0 | 4 | 0.00163 | 0.00164 | 0.000311 | 2.196e-06 | 1.998e-05 |
| re300_e1 | 300 | 1 | 2 | 1 | B1_Mode2 | 0.993 | 1 | 1 | 0 | 2.941e-16 | 0.0938 | 1.33 | -0.0842 | -0.245 | 0.057 | -0.0145 | 0 | 0 | 5 | 0.000106 | 0.000665 | 1.111e-06 | 4.007e-09 | 1.570e-06 |
| re300_e1 | 300 | 1 | 3 | 1 | B1_Mode2 | 0.986 | 1 | 1 | 0 | 2.775e-16 | -0.0409 | 1.31 | 0.293 | -0.139 | -0.135 | -0.0315 | 0 | 0 | 0 | 0.000848 | 0.026 | 7.563e-06 | 0.000173 | 1.465e-06 |
| re300_e1 | 300 | 1 | 3 | 1 | B1_Mode2 | 0.986 | 1 | 1 | 0 | 2.775e-16 | -0.0409 | 1.31 | 0.293 | -0.139 | -0.135 | -0.0315 | 0 | 0 | 1 | 0.87 | 0.000486 | 0.00119 | 8.198e-06 | 6.493e-05 |
| re300_e1 | 300 | 1 | 3 | 1 | B1_Mode2 | 0.986 | 1 | 1 | 0 | 2.775e-16 | -0.0409 | 1.31 | 0.293 | -0.139 | -0.135 | -0.0315 | 0 | 0 | 2 | 0.0437 | 0.0188 | 0.000701 | 8.153e-05 | 2.481e-05 |
| re300_e1 | 300 | 1 | 3 | 1 | B1_Mode2 | 0.986 | 1 | 1 | 0 | 2.775e-16 | -0.0409 | 1.31 | 0.293 | -0.139 | -0.135 | -0.0315 | 0 | 0 | 3 | 0.00976 | 0.0151 | 0.000179 | 2.693e-05 | 1.620e-05 |
| re300_e1 | 300 | 1 | 3 | 1 | B1_Mode2 | 0.986 | 1 | 1 | 0 | 2.775e-16 | -0.0409 | 1.31 | 0.293 | -0.139 | -0.135 | -0.0315 | 0 | 0 | 4 | 0.00922 | 6.580e-06 | 0.000999 | 2.103e-06 | 3.925e-05 |
| re300_e1 | 300 | 1 | 3 | 1 | B1_Mode2 | 0.986 | 1 | 1 | 0 | 2.775e-16 | -0.0409 | 1.31 | 0.293 | -0.139 | -0.135 | -0.0315 | 0 | 0 | 5 | 0.000503 | 0.00142 | 0.000135 | 3.821e-06 | 5.535e-06 |
| re300_e1 | 300 | 1 | 4 | 2 | B2_temporal | 0.966 | 1 | 0 | 1 | 3.264e-16 | 0.0618 | -0.285 | -0.0441 | 0.0415 | 0.018 | 0.00187 | 0 | 0 | 0 | 0.00198 | 0.81 | 0.0969 | 0.00679 | 0.000179 |
| re300_e1 | 300 | 1 | 4 | 2 | B2_temporal | 0.966 | 1 | 0 | 1 | 3.264e-16 | 0.0618 | -0.285 | -0.0441 | 0.0415 | 0.018 | 0.00187 | 0 | 0 | 1 | 0.042 | 4.633e-05 | 0.00172 | 0.00155 | 0.000267 |
| re300_e1 | 300 | 1 | 4 | 2 | B2_temporal | 0.966 | 1 | 0 | 1 | 3.264e-16 | 0.0618 | -0.285 | -0.0441 | 0.0415 | 0.018 | 0.00187 | 0 | 0 | 2 | 0.00101 | 0.0174 | 0.0165 | 0.00197 | 0.000232 |
| re300_e1 | 300 | 1 | 4 | 2 | B2_temporal | 0.966 | 1 | 0 | 1 | 3.264e-16 | 0.0618 | -0.285 | -0.0441 | 0.0415 | 0.018 | 0.00187 | 0 | 0 | 3 | 0.000889 | 1.153e-07 | 7.860e-06 | 0.000101 | 0.000193 |
| re300_e1 | 300 | 1 | 4 | 2 | B2_temporal | 0.966 | 1 | 0 | 1 | 3.264e-16 | 0.0618 | -0.285 | -0.0441 | 0.0415 | 0.018 | 0.00187 | 0 | 0 | 4 | 0.000168 | 0.000133 | 2.865e-05 | 7.117e-06 | 2.471e-06 |
| re300_e1 | 300 | 1 | 4 | 2 | B2_temporal | 0.966 | 1 | 0 | 1 | 3.264e-16 | 0.0618 | -0.285 | -0.0441 | 0.0415 | 0.018 | 0.00187 | 0 | 0 | 5 | 1.815e-06 | 8.042e-07 | 1.296e-05 | 1.076e-06 | 1.557e-06 |
| re300_e2 | 300 | 2 | 0 | 2 | B2_temporal | 1.8 | 1 | 0 | 1 | 4.403e-16 | 0.289 | -0.824 | -0.227 | 0.0544 | -0.059 | 0.0225 | 0 | 0 | 0 | 0.0232 | 0.419 | 0.033 | 0.00258 | 0.000156 |
| re300_e2 | 300 | 2 | 0 | 2 | B2_temporal | 1.8 | 1 | 0 | 1 | 4.403e-16 | 0.289 | -0.824 | -0.227 | 0.0544 | -0.059 | 0.0225 | 0 | 0 | 1 | 0.188 | 0.21 | 0.0174 | 4.068e-05 | 6.314e-05 |
| re300_e2 | 300 | 2 | 0 | 2 | B2_temporal | 1.8 | 1 | 0 | 1 | 4.403e-16 | 0.289 | -0.824 | -0.227 | 0.0544 | -0.059 | 0.0225 | 0 | 0 | 2 | 0.0144 | 0.0553 | 0.00126 | 0.007 | 1.945e-05 |
| re300_e2 | 300 | 2 | 0 | 2 | B2_temporal | 1.8 | 1 | 0 | 1 | 4.403e-16 | 0.289 | -0.824 | -0.227 | 0.0544 | -0.059 | 0.0225 | 0 | 0 | 3 | 0.000821 | 0.000522 | 0.0206 | 0.000215 | 0.000135 |
| re300_e2 | 300 | 2 | 0 | 2 | B2_temporal | 1.8 | 1 | 0 | 1 | 4.403e-16 | 0.289 | -0.824 | -0.227 | 0.0544 | -0.059 | 0.0225 | 0 | 0 | 4 | 0.000965 | 0.00237 | 0.000231 | 0.00141 | 3.671e-05 |
| re300_e2 | 300 | 2 | 0 | 2 | B2_temporal | 1.8 | 1 | 0 | 1 | 4.403e-16 | 0.289 | -0.824 | -0.227 | 0.0544 | -0.059 | 0.0225 | 0 | 0 | 5 | 0.00014 | 0.000392 | 0.000118 | 3.412e-05 | 0.000101 |
| re300_e2 | 300 | 2 | 1 | 2 | B2_temporal | 1.97 | 1 | 1 | 0 | 4.308e-16 | 0.474 | -0.953 | -0.782 | 0.134 | -0.0458 | -0.002 | 0 | 0 | 0 | 0.057 | 0.125 | 0.00556 | 3.880e-05 | 0.000177 |
| re300_e2 | 300 | 2 | 1 | 2 | B2_temporal | 1.97 | 1 | 1 | 0 | 4.308e-16 | 0.474 | -0.953 | -0.782 | 0.134 | -0.0458 | -0.002 | 0 | 0 | 1 | 0.231 | 0.206 | 0.0296 | 0.000387 | 4.878e-05 |
| re300_e2 | 300 | 2 | 1 | 2 | B2_temporal | 1.97 | 1 | 1 | 0 | 4.308e-16 | 0.474 | -0.953 | -0.782 | 0.134 | -0.0458 | -0.002 | 0 | 0 | 2 | 0.155 | 0.119 | 0.00257 | 0.00562 | 0.000239 |
| re300_e2 | 300 | 2 | 1 | 2 | B2_temporal | 1.97 | 1 | 1 | 0 | 4.308e-16 | 0.474 | -0.953 | -0.782 | 0.134 | -0.0458 | -0.002 | 0 | 0 | 3 | 0.00454 | 0.000656 | 0.0368 | 0.000304 | 3.005e-06 |
| re300_e2 | 300 | 2 | 1 | 2 | B2_temporal | 1.97 | 1 | 1 | 0 | 4.308e-16 | 0.474 | -0.953 | -0.782 | 0.134 | -0.0458 | -0.002 | 0 | 0 | 4 | 0.000534 | 0.0143 | 0.00127 | 0.00269 | 0.000158 |
| re300_e2 | 300 | 2 | 1 | 2 | B2_temporal | 1.97 | 1 | 1 | 0 | 4.308e-16 | 0.474 | -0.953 | -0.782 | 0.134 | -0.0458 | -0.002 | 0 | 0 | 5 | 1.011e-06 | 0.000627 | 0.000949 | 0.000547 | 9.123e-05 |
| re300_e2 | 300 | 2 | 2 | 1 | B1_Mode2 | 1.97 | 1 | 1 | 0 | 4.994e-16 | 0.438 | 1.71 | -0.625 | -0.365 | 0.0448 | -0.0495 | 0 | 0 | 0 | 0.0487 | 0.0361 | 4.978e-05 | 0.000379 | 1.713e-05 |
| re300_e2 | 300 | 2 | 2 | 1 | B1_Mode2 | 1.97 | 1 | 1 | 0 | 4.994e-16 | 0.438 | 1.71 | -0.625 | -0.365 | 0.0448 | -0.0495 | 0 | 0 | 1 | 0.738 | 0.00661 | 0.00148 | 5.148e-05 | 6.161e-05 |
| re300_e2 | 300 | 2 | 2 | 1 | B1_Mode2 | 1.97 | 1 | 1 | 0 | 4.994e-16 | 0.438 | 1.71 | -0.625 | -0.365 | 0.0448 | -0.0495 | 0 | 0 | 2 | 0.0992 | 0.0202 | 0.000244 | 2.445e-05 | 1.943e-05 |
| re300_e2 | 300 | 2 | 2 | 1 | B1_Mode2 | 1.97 | 1 | 1 | 0 | 4.994e-16 | 0.438 | 1.71 | -0.625 | -0.365 | 0.0448 | -0.0495 | 0 | 0 | 3 | 0.0339 | 0.0106 | 0.00138 | 5.098e-05 | 4.139e-05 |
| re300_e2 | 300 | 2 | 2 | 1 | B1_Mode2 | 1.97 | 1 | 1 | 0 | 4.994e-16 | 0.438 | 1.71 | -0.625 | -0.365 | 0.0448 | -0.0495 | 0 | 0 | 4 | 0.000508 | 0.000364 | 0.000423 | 1.071e-05 | 2.206e-05 |
| re300_e2 | 300 | 2 | 2 | 1 | B1_Mode2 | 1.97 | 1 | 1 | 0 | 4.994e-16 | 0.438 | 1.71 | -0.625 | -0.365 | 0.0448 | -0.0495 | 0 | 0 | 5 | 0.000622 | 0.00065 | 1.824e-05 | 4.006e-06 | 9.301e-06 |

Table 15: Décomposition modale — 185 points, 1110 lignes (6 modes $i$ par point, colonnes $E^f_{iJ}$ pour $J$=0..4), `mode_analysis.csv`.

| point | Re | $E^*$ | s | id | branch | $E_t$ | cov. | $i$ | $j$ | rmse | $A_{10}$ | $A_{20}$ | $A_{30}$ | $A_{40}$ | $A_{50}$ | $A_{60}$ | $A_{70}$ | $A_{80}$ | $i$ | $E^f_{0J}$ | $E^f_{1J}$ | $E^f_{2J}$ | $E^f_{3J}$ | $E^f_{4J}$ |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| re300_e2 | 300 | 2 | 3 | 1 | B1_Mode2 | 1.91 | 1 | 1 | 0 | 4.161e-16 | 0.362 | 1.6 | -0.496 | -0.195 | 0.00101 | -0.0313 | 0 | 0 | 0 | 0.0344 | 0.0936 | 0.000551 | 3.543e-05 | 1.329e-06 |
| re300_e2 | 300 | 2 | 3 | 1 | B1_Mode2 | 1.91 | 1 | 1 | 0 | 4.161e-16 | 0.362 | 1.6 | -0.496 | -0.195 | 0.00101 | -0.0313 | 0 | 0 | 1 | 0.669 | 0.0541 | 0.0109 | 1.986e-05 | 0.000344 |
| re300_e2 | 300 | 2 | 3 | 1 | B1_Mode2 | 1.91 | 1 | 1 | 0 | 4.161e-16 | 0.362 | 1.6 | -0.496 | -0.195 | 0.00101 | -0.0313 | 0 | 0 | 2 | 0.0645 | 0.0442 | 0.00267 | 3.646e-05 | 5.312e-05 |
| re300_e2 | 300 | 2 | 3 | 1 | B1_Mode2 | 1.91 | 1 | 1 | 0 | 4.161e-16 | 0.362 | 1.6 | -0.496 | -0.195 | 0.00101 | -0.0313 | 0 | 0 | 3 | 0.00992 | 0.00152 | 0.00678 | 0.00015 | 0.000103 |
| re300_e2 | 300 | 2 | 3 | 1 | B1_Mode2 | 1.91 | 1 | 1 | 0 | 4.161e-16 | 0.362 | 1.6 | -0.496 | -0.195 | 0.00101 | -0.0313 | 0 | 0 | 4 | 2.646e-07 | 0.00537 | 0.000527 | 0.000256 | 2.285e-07 |
| re300_e2 | 300 | 2 | 3 | 1 | B1_Mode2 | 1.91 | 1 | 1 | 0 | 4.161e-16 | 0.362 | 1.6 | -0.496 | -0.195 | 0.00101 | -0.0313 | 0 | 0 | 5 | 0.000257 | 0.000259 | 0.000264 | 1.649e-05 | 2.389e-07 |
| re300_e2 | 300 | 2 | 4 | 2 | B2_temporal | 1.87 | 1 | 0 | 1 | 4.255e-16 | 0.44 | -0.869 | -0.529 | 0.197 | 0.0706 | -0.0185 | 0 | 0 | 0 | 0.0518 | 0.469 | 0.000908 | 0.000563 | 8.016e-05 |
| re300_e2 | 300 | 2 | 4 | 2 | B2_temporal | 1.87 | 1 | 0 | 1 | 4.255e-16 | 0.44 | -0.869 | -0.529 | 0.197 | 0.0706 | -0.0185 | 0 | 0 | 1 | 0.202 | 0.125 | 0.00774 | 1.467e-05 | 9.420e-05 |
| re300_e2 | 300 | 2 | 4 | 2 | B2_temporal | 1.87 | 1 | 0 | 1 | 4.255e-16 | 0.44 | -0.869 | -0.529 | 0.197 | 0.0706 | -0.0185 | 0 | 0 | 2 | 0.0748 | 0.0362 | 0.00174 | 0.00071 | 5.365e-05 |
| re300_e2 | 300 | 2 | 4 | 2 | B2_temporal | 1.87 | 1 | 0 | 1 | 4.255e-16 | 0.44 | -0.869 | -0.529 | 0.197 | 0.0706 | -0.0185 | 0 | 0 | 3 | 0.0104 | 0.00646 | 0.00603 | 8.686e-08 | 1.931e-05 |
| re300_e2 | 300 | 2 | 4 | 2 | B2_temporal | 1.87 | 1 | 0 | 1 | 4.255e-16 | 0.44 | -0.869 | -0.529 | 0.197 | 0.0706 | -0.0185 | 0 | 0 | 4 | 0.00133 | 0.00333 | 0.000147 | 0.00057 | 1.355e-05 |
| re300_e2 | 300 | 2 | 4 | 2 | B2_temporal | 1.87 | 1 | 0 | 1 | 4.255e-16 | 0.44 | -0.869 | -0.529 | 0.197 | 0.0706 | -0.0185 | 0 | 0 | 5 | 9.213e-05 | 9.130e-05 | 0.000455 | 2.394e-06 | 2.198e-05 |
| re500_e0.1 | 500 | 0.1 | 0 | 1 | B1_Mode2 | 0.122 | 1 | 1 | 0 | 1.038e-16 | -0.00283 | -0.454 | 0.131 | 0.0998 | -0.0446 | 0.0128 | 0 | 0 | 0 | 3.279e-05 | 0.0251 | 8.177e-05 | 2.247e-07 | 5.539e-07 |
| re500_e0.1 | 500 | 0.1 | 0 | 1 | B1_Mode2 | 0.122 | 1 | 1 | 0 | 1.038e-16 | -0.00283 | -0.454 | 0.131 | 0.0998 | -0.0446 | 0.0128 | 0 | 0 | 1 | 0.845 | 0.000298 | 0.000444 | 3.067e-05 | 2.017e-05 |
| re500_e0.1 | 500 | 0.1 | 0 | 1 | B1_Mode2 | 0.122 | 1 | 1 | 0 | 1.038e-16 | -0.00283 | -0.454 | 0.131 | 0.0998 | -0.0446 | 0.0128 | 0 | 0 | 2 | 0.0701 | 0.00176 | 0.00125 | 2.717e-06 | 6.170e-05 |
| re500_e0.1 | 500 | 0.1 | 0 | 1 | B1_Mode2 | 0.122 | 1 | 1 | 0 | 1.038e-16 | -0.00283 | -0.454 | 0.131 | 0.0998 | -0.0446 | 0.0128 | 0 | 0 | 3 | 0.0408 | 0.00538 | 7.764e-06 | 2.974e-05 | 2.184e-07 |
| re500_e0.1 | 500 | 0.1 | 0 | 1 | B1_Mode2 | 0.122 | 1 | 1 | 0 | 1.038e-16 | -0.00283 | -0.454 | 0.131 | 0.0998 | -0.0446 | 0.0128 | 0 | 0 | 4 | 0.00815 | 0.000958 | 4.945e-05 | 3.753e-06 | 4.825e-06 |
| re500_e0.1 | 500 | 0.1 | 0 | 1 | B1_Mode2 | 0.122 | 1 | 1 | 0 | 1.038e-16 | -0.00283 | -0.454 | 0.131 | 0.0998 | -0.0446 | 0.0128 | 0 | 0 | 5 | 0.00067 | 1.476e-05 | 0.000123 | 1.038e-06 | 4.058e-06 |
| re500_e0.1 | 500 | 0.1 | 1 | 1 | B1_Mode2 | 0.123 | 1 | 1 | 0 | 1.068e-16 | -0.00525 | 0.467 | 0.089 | -0.129 | -0.0352 | -0.00498 | 0 | 0 | 0 | 0.000112 | 5.449e-06 | 8.324e-06 | 8.191e-08 | 9.170e-07 |
| re500_e0.1 | 500 | 0.1 | 1 | 1 | B1_Mode2 | 0.123 | 1 | 1 | 0 | 1.068e-16 | -0.00525 | 0.467 | 0.089 | -0.129 | -0.0352 | -0.00498 | 0 | 0 | 1 | 0.887 | 0.00522 | 3.536e-05 | 5.973e-05 | 3.438e-06 |
| re500_e0.1 | 500 | 0.1 | 1 | 1 | B1_Mode2 | 0.123 | 1 | 1 | 0 | 1.068e-16 | -0.00525 | 0.467 | 0.089 | -0.129 | -0.0352 | -0.00498 | 0 | 0 | 2 | 0.0322 | 0.0011 | 2.979e-05 | 1.517e-05 | 2.540e-06 |
| re500_e0.1 | 500 | 0.1 | 1 | 1 | B1_Mode2 | 0.123 | 1 | 1 | 0 | 1.068e-16 | -0.00525 | 0.467 | 0.089 | -0.129 | -0.0352 | -0.00498 | 0 | 0 | 3 | 0.0682 | 0.000518 | 9.004e-05 | 2.844e-06 | 5.624e-06 |
| re500_e0.1 | 500 | 0.1 | 1 | 1 | B1_Mode2 | 0.123 | 1 | 1 | 0 | 1.068e-16 | -0.00525 | 0.467 | 0.089 | -0.129 | -0.0352 | -0.00498 | 0 | 0 | 4 | 0.00505 | 1.348e-05 | 7.355e-06 | 3.589e-07 | 3.799e-07 |
| re500_e0.1 | 500 | 0.1 | 1 | 1 | B1_Mode2 | 0.123 | 1 | 1 | 0 | 1.068e-16 | -0.00525 | 0.467 | 0.089 | -0.129 | -0.0352 | -0.00498 | 0 | 0 | 5 | 0.000101 | 1.956e-05 | 4.539e-07 | 3.842e-09 | 7.007e-08 |
| re500_e0.1 | 500 | 0.1 | 2 | 1 | B1_Mode2 | 0.121 | 1 | 1 | 0 | 9.567e-17 | 0.0234 | 0.463 | -0.0481 | -0.143 | 0.0219 | 0.00529 | 0 | 0 | 0 | 0.00226 | 0.0103 | 6.940e-07 | 0.000121 | 1.120e-06 |
| re500_e0.1 | 500 | 0.1 | 2 | 1 | B1_Mode2 | 0.121 | 1 | 1 | 0 | 9.567e-17 | 0.0234 | 0.463 | -0.0481 | -0.143 | 0.0219 | 0.00529 | 0 | 0 | 1 | 0.884 | 0.00332 | 6.368e-05 | 4.417e-05 | 2.339e-06 |
| re500_e0.1 | 500 | 0.1 | 2 | 1 | B1_Mode2 | 0.121 | 1 | 1 | 0 | 9.567e-17 | 0.0234 | 0.463 | -0.0481 | -0.143 | 0.0219 | 0.00529 | 0 | 0 | 2 | 0.00952 | 0.00233 | 6.820e-06 | 3.096e-05 | 1.177e-06 |
| re500_e0.1 | 500 | 0.1 | 2 | 1 | B1_Mode2 | 0.121 | 1 | 1 | 0 | 9.567e-17 | 0.0234 | 0.463 | -0.0481 | -0.143 | 0.0219 | 0.00529 | 0 | 0 | 3 | 0.0845 | 0.00125 | 0.000156 | 7.105e-06 | 7.825e-06 |
| re500_e0.1 | 500 | 0.1 | 2 | 1 | B1_Mode2 | 0.121 | 1 | 1 | 0 | 9.567e-17 | 0.0234 | 0.463 | -0.0481 | -0.143 | 0.0219 | 0.00529 | 0 | 0 | 4 | 0.00198 | 0.00011 | 2.939e-05 | 7.545e-07 | 3.579e-06 |
| re500_e0.1 | 500 | 0.1 | 2 | 1 | B1_Mode2 | 0.121 | 1 | 1 | 0 | 9.567e-17 | 0.0234 | 0.463 | -0.0481 | -0.143 | 0.0219 | 0.00529 | 0 | 0 | 5 | 0.000115 | 0.000145 | 1.165e-07 | 1.529e-12 | 5.793e-07 |
| re500_e0.1 | 500 | 0.1 | 3 | 2 | B2_temporal | 0.126 | 1 | 0 | 1 | 1.118e-16 | -0.0291 | 0.0648 | 0.0123 | 0.00266 | 0.000755 | 0.000799 | 0 | 0 | 0 | 0.00336 | 0.963 | 0.00552 | 0.00014 | 2.938e-05 |
| re500_e0.1 | 500 | 0.1 | 3 | 2 | B2_temporal | 0.126 | 1 | 0 | 1 | 1.118e-16 | -0.0291 | 0.0648 | 0.0123 | 0.00266 | 0.000755 | 0.000799 | 0 | 0 | 1 | 0.0167 | 0.00674 | 1.007e-07 | 0.00111 | 8.760e-05 |
| re500_e0.1 | 500 | 0.1 | 3 | 2 | B2_temporal | 0.126 | 1 | 0 | 1 | 1.118e-16 | -0.0291 | 0.0648 | 0.0123 | 0.00266 | 0.000755 | 0.000799 | 0 | 0 | 2 | 0.000598 | 0.00191 | 6.635e-05 | 0.000304 | 1.510e-06 |
| re500_e0.1 | 500 | 0.1 | 3 | 2 | B2_temporal | 0.126 | 1 | 0 | 1 | 1.118e-16 | -0.0291 | 0.0648 | 0.0123 | 0.00266 | 0.000755 | 0.000799 | 0 | 0 | 3 | 2.810e-05 | 2.478e-05 | 1.162e-05 | 1.430e-06 | 2.499e-06 |
| re500_e0.1 | 500 | 0.1 | 3 | 2 | B2_temporal | 0.126 | 1 | 0 | 1 | 1.118e-16 | -0.0291 | 0.0648 | 0.0123 | 0.00266 | 0.000755 | 0.000799 | 0 | 0 | 4 | 2.261e-06 | 1.477e-05 | 1.486e-06 | 1.148e-05 | 5.294e-09 |
| re500_e0.1 | 500 | 0.1 | 3 | 2 | B2_temporal | 0.126 | 1 | 0 | 1 | 1.118e-16 | -0.0291 | 0.0648 | 0.0123 | 0.00266 | 0.000755 | 0.000799 | 0 | 0 | 5 | 2.532e-06 | 1.270e-07 | 2.676e-08 | 7.881e-07 | 3.036e-06 |
| re500_e0.1 | 500 | 0.1 | 4 | 1 | B1_Mode2 | 0.116 | 1 | 1 | 0 | 1.034e-16 | 0.0291 | -0.389 | -0.0551 | 0.0666 | 0.0144 | -0.00517 | 0 | 0 | 0 | 0.00363 | 0.294 | 0.00104 | 0.000249 | 0.000224 |
| re500_e0.1 | 500 | 0.1 | 4 | 1 | B1_Mode2 | 0.116 | 1 | 1 | 0 | 1.034e-16 | 0.0291 | -0.389 | -0.0551 | 0.0666 | 0.0144 | -0.00517 | 0 | 0 | 1 | 0.651 | 0.00525 | 8.290e-05 | 0.00042 | 6.084e-05 |
| re500_e0.1 | 500 | 0.1 | 4 | 1 | B1_Mode2 | 0.116 | 1 | 1 | 0 | 1.034e-16 | 0.0291 | -0.389 | -0.0551 | 0.0666 | 0.0144 | -0.00517 | 0 | 0 | 2 | 0.013 | 0.00709 | 0.00119 | 7.908e-06 | 3.344e-05 |
| re500_e0.1 | 500 | 0.1 | 4 | 1 | B1_Mode2 | 0.116 | 1 | 1 | 0 | 1.034e-16 | 0.0291 | -0.389 | -0.0551 | 0.0666 | 0.0144 | -0.00517 | 0 | 0 | 3 | 0.019 | 0.00215 | 0.00018 | 6.343e-05 | 4.624e-05 |
| re500_e0.1 | 500 | 0.1 | 4 | 1 | B1_Mode2 | 0.116 | 1 | 1 | 0 | 1.034e-16 | 0.0291 | -0.389 | -0.0551 | 0.0666 | 0.0144 | -0.00517 | 0 | 0 | 4 | 0.000888 | 0.000133 | 5.635e-06 | 5.090e-08 | 3.499e-06 |
| re500_e0.1 | 500 | 0.1 | 4 | 1 | B1_Mode2 | 0.116 | 1 | 1 | 0 | 1.034e-16 | 0.0291 | -0.389 | -0.0551 | 0.0666 | 0.0144 | -0.00517 | 0 | 0 | 5 | 0.000115 | 4.118e-07 | 1.853e-06 | 2.039e-07 | 3.058e-08 |
| re500_e0.25 | 500 | 0.25 | 0 | 3 | B3_mixed | 0.24 | 1 | 1 | 0 | 1.382e-16 | 0.0548 | -0.499 | -0.00319 | 0.109 | -0.0199 | 0.00799 | 0 | 0 | 0 | 0.00625 | 0.39 | 0.00144 | 0.00107 | 2.623e-05 |
| re500_e0.25 | 500 | 0.25 | 0 | 3 | B3_mixed | 0.24 | 1 | 1 | 0 | 1.382e-16 | 0.0548 | -0.499 | -0.00319 | 0.109 | -0.0199 | 0.00799 | 0 | 0 | 1 | 0.52 | 0.00675 | 0.00233 | 0.000908 | 6.113e-05 |
| re500_e0.25 | 500 | 0.25 | 0 | 3 | B3_mixed | 0.24 | 1 | 1 | 0 | 1.382e-16 | 0.0548 | -0.499 | -0.00319 | 0.109 | -0.0199 | 0.00799 | 0 | 0 | 2 | 2.124e-05 | 0.0315 | 0.00788 | 0.000107 | 7.412e-05 |
| re500_e0.25 | 500 | 0.25 | 0 | 3 | B3_mixed | 0.24 | 1 | 1 | 0 | 1.382e-16 | 0.0548 | -0.499 | -0.00319 | 0.109 | -0.0199 | 0.00799 | 0 | 0 | 3 | 0.0248 | 0.0035 | 0.000278 | 0.000109 | 8.675e-05 |
| re500_e0.25 | 500 | 0.25 | 0 | 3 | B3_mixed | 0.24 | 1 | 1 | 0 | 1.382e-16 | 0.0548 | -0.499 | -0.00319 | 0.109 | -0.0199 | 0.00799 | 0 | 0 | 4 | 0.000824 | 0.000129 | 0.00093 | 9.768e-05 | 2.733e-06 |
| re500_e0.25 | 500 | 0.25 | 0 | 3 | B3_mixed | 0.24 | 1 | 1 | 0 | 1.382e-16 | 0.0548 | -0.499 | -0.00319 | 0.109 | -0.0199 | 0.00799 | 0 | 0 | 5 | 0.000133 | 0.000501 | 0.000176 | 2.816e-05 | 8.578e-08 |
| re500_e0.25 | 500 | 0.25 | 1 | 1 | B1_Mode2 | 0.267 | 1 | 1 | 0 | 1.428e-16 | 0.00292 | 0.698 | 0.0698 | -0.187 | -0.0205 | 0.00632 | 0 | 0 | 0 | 1.591e-05 | 1.287e-06 | 3.130e-05 | 1.341e-06 | 3.016e-06 |
| re500_e0.25 | 500 | 0.25 | 1 | 1 | B1_Mode2 | 0.267 | 1 | 1 | 0 | 1.428e-16 | 0.00292 | 0.698 | 0.0698 | -0.187 | -0.0205 | 0.00632 | 0 | 0 | 1 | 0.911 | 0.00835 | 5.926e-05 | 7.700e-05 | 4.890e-06 |
| re500_e0.25 | 500 | 0.25 | 1 | 1 | B1_Mode2 | 0.267 | 1 | 1 | 0 | 1.428e-16 | 0.00292 | 0.698 | 0.0698 | -0.187 | -0.0205 | 0.00632 | 0 | 0 | 2 | 0.00912 | 0.00222 | 9.600e-06 | 3.023e-05 | 1.193e-06 |
| re500_e0.25 | 500 | 0.25 | 1 | 1 | B1_Mode2 | 0.267 | 1 | 1 | 0 | 1.428e-16 | 0.00292 | 0.698 | 0.0698 | -0.187 | -0.0205 | 0.00632 | 0 | 0 | 3 | 0.0655 | 0.00209 | 0.000194 | 1.449e-05 | 9.857e-06 |
| re500_e0.25 | 500 | 0.25 | 1 | 1 | B1_Mode2 | 0.267 | 1 | 1 | 0 | 1.428e-16 | 0.00292 | 0.698 | 0.0698 | -0.187 | -0.0205 | 0.00632 | 0 | 0 | 4 | 0.000784 | 3.435e-06 | 5.956e-06 | 1.219e-07 | 4.544e-07 |
| re500_e0.25 | 500 | 0.25 | 1 | 1 | B1_Mode2 | 0.267 | 1 | 1 | 0 | 1.428e-16 | 0.00292 | 0.698 | 0.0698 | -0.187 | -0.0205 | 0.00632 | 0 | 0 | 5 | 7.477e-05 | 0.000138 | 5.135e-06 | 4.350e-07 | 4.961e-08 |
| re500_e0.25 | 500 | 0.25 | 10 | 2 | B2_temporal | 0.293 | 1 | 0 | 1 | 1.663e-16 | -0.0582 | 0.086 | 0.0333 | -0.0129 | -0.000468 | 0.000349 | 0 | 0 | 0 | 0.00578 | 0.924 | 0.0377 | 0.000804 | 5.388e-07 |
| re500_e0.25 | 500 | 0.25 | 10 | 2 | B2_temporal | 0.293 | 1 | 0 | 1 | 1.663e-16 | -0.0582 | 0.086 | 0.0333 | -0.0129 | -0.000468 | 0.000349 | 0 | 0 | 1 | 0.0126 | 0.00218 | 0.000548 | 0.00323 | 0.000119 |
| re500_e0.25 | 500 | 0.25 | 10 | 2 | B2_temporal | 0.293 | 1 | 0 | 1 | 1.663e-16 | -0.0582 | 0.086 | 0.0333 | -0.0129 | -0.000468 | 0.000349 | 0 | 0 | 2 | 0.00189 | 0.00898 | 0.00131 | 0.000448 | 9.026e-05 |
| re500_e0.25 | 500 | 0.25 | 10 | 2 | B2_temporal | 0.293 | 1 | 0 | 1 | 1.663e-16 | -0.0582 | 0.086 | 0.0333 | -0.0129 | -0.000468 | 0.000349 | 0 | 0 | 3 | 0.000285 | 0.000105 | 0.000123 | 0.000243 | 7.989e-06 |
| re500_e0.25 | 500 | 0.25 | 10 | 2 | B2_temporal | 0.293 | 1 | 0 | 1 | 1.663e-16 | -0.0582 | 0.086 | 0.0333 | -0.0129 | -0.000468 | 0.000349 | 0 | 0 | 4 | 3.738e-07 | 1.960e-05 | 1.459e-05 | 1.723e-05 | 1.749e-06 |
| re500_e0.25 | 500 | 0.25 | 10 | 2 | B2_temporal | 0.293 | 1 | 0 | 1 | 1.663e-16 | -0.0582 | 0.086 | 0.0333 | -0.0129 | -0.000468 | 0.000349 | 0 | 0 | 5 | 2.071e-07 | 1.430e-07 | 2.516e-06 | 1.540e-08 | 9.549e-08 |
| re500_e0.25 | 500 | 0.25 | 11 | 1 | B1_Mode2 | 0.267 | 1 | 1 | 0 | 1.354e-16 | 0.0756 | 0.682 | -0.0767 | -0.221 | 0.0189 | 0.0125 | 0 | 0 | 0 | 0.0107 | 0.000221 | 6.886e-05 | 1.960e-06 | 3.053e-06 |
| re500_e0.25 | 500 | 0.25 | 11 | 1 | B1_Mode2 | 0.267 | 1 | 1 | 0 | 1.354e-16 | 0.0756 | 0.682 | -0.0767 | -0.221 | 0.0189 | 0.0125 | 0 | 0 | 1 | 0.871 | 0.00716 | 0.000244 | 5.512e-05 | 1.414e-05 |
| re500_e0.25 | 500 | 0.25 | 11 | 1 | B1_Mode2 | 0.267 | 1 | 1 | 0 | 1.354e-16 | 0.0756 | 0.682 | -0.0767 | -0.221 | 0.0189 | 0.0125 | 0 | 0 | 2 | 0.011 | 0.00139 | 5.935e-05 | 1.576e-05 | 4.130e-06 |
| re500_e0.25 | 500 | 0.25 | 11 | 1 | B1_Mode2 | 0.267 | 1 | 1 | 0 | 1.354e-16 | 0.0756 | 0.682 | -0.0767 | -0.221 | 0.0189 | 0.0125 | 0 | 0 | 3 | 0.0917 | 0.00382 | 0.000429 | 2.446e-05 | 2.629e-05 |
| re500_e0.25 | 500 | 0.25 | 11 | 1 | B1_Mode2 | 0.267 | 1 | 1 | 0 | 1.354e-16 | 0.0756 | 0.682 | -0.0767 | -0.221 | 0.0189 | 0.0125 | 0 | 0 | 4 | 0.00067 | 0.000727 | 1.413e-06 | 3.677e-06 | 2.069e-07 |
| re500_e0.25 | 500 | 0.25 | 11 | 1 | B1_Mode2 | 0.267 | 1 | 1 | 0 | 1.354e-16 | 0.0756 | 0.682 | -0.0767 | -0.221 | 0.0189 | 0.0125 | 0 | 0 | 5 | 0.000294 | 0.000703 | 2.431e-05 | 1.599e-07 | 1.790e-06 |

Table 15: Décomposition modale — 185 points, 1110 lignes (6 modes $i$ par point, colonnes $E^f_{iJ}$ pour $J$=0..4), `mode_analysis.csv`.

| point | Re | $E^*$ | s | id | branch | $E_t$ | cov. | $i$ | $j$ | rmse | $A_{10}$ | $A_{20}$ | $A_{30}$ | $A_{40}$ | $A_{50}$ | $A_{60}$ | $A_{70}$ | $A_{80}$ | $i$ | $E^f_{0J}$ | $E^f_{1J}$ | $E^f_{2J}$ | $E^f_{3J}$ | $E^f_{4J}$ |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| re500_e0.25 | 500 | 0.25 | 12 | 1 | B1_Mode2 | 0.238 | 1 | 1 | 0 | 1.438e-16 | -0.0165 | 0.6 | -0.0459 | -0.0693 | -0.00059 | -0.00888 | 0 | 0 | 0 | 0.000573 | 0.212 | 0.00204 | 1.182e-05 | 1.677e-05 |
| re500_e0.25 | 500 | 0.25 | 12 | 1 | B1_Mode2 | 0.238 | 1 | 1 | 0 | 1.438e-16 | -0.0165 | 0.6 | -0.0459 | -0.0693 | -0.00059 | -0.00888 | 0 | 0 | 1 | 0.756 | 0.00497 | 0.00169 | 0.00025 | 9.474e-05 |
| re500_e0.25 | 500 | 0.25 | 12 | 1 | B1_Mode2 | 0.238 | 1 | 1 | 0 | 1.438e-16 | -0.0165 | 0.6 | -0.0459 | -0.0693 | -0.00059 | -0.00888 | 0 | 0 | 2 | 0.00442 | 6.095e-05 | 0.00449 | 8.823e-05 | 0.000123 |
| re500_e0.25 | 500 | 0.25 | 12 | 1 | B1_Mode2 | 0.238 | 1 | 1 | 0 | 1.438e-16 | -0.0165 | 0.6 | -0.0459 | -0.0693 | -0.00059 | -0.00888 | 0 | 0 | 3 | 0.0101 | 0.00265 | 4.298e-05 | 2.394e-06 | 1.827e-05 |
| re500_e0.25 | 500 | 0.25 | 12 | 1 | B1_Mode2 | 0.238 | 1 | 1 | 0 | 1.438e-16 | -0.0165 | 0.6 | -0.0459 | -0.0693 | -0.00059 | -0.00888 | 0 | 0 | 4 | 7.321e-07 | 0.000144 | 3.759e-05 | 6.114e-06 | 2.567e-07 |
| re500_e0.25 | 500 | 0.25 | 12 | 1 | B1_Mode2 | 0.238 | 1 | 1 | 0 | 1.438e-16 | -0.0165 | 0.6 | -0.0459 | -0.0693 | -0.00059 | -0.00888 | 0 | 0 | 5 | 0.000166 | 1.398e-10 | 7.724e-06 | 4.250e-06 | 9.955e-08 |
| re500_e0.25 | 500 | 0.25 | 13 | 1 | B1_Mode2 | 0.256 | 1 | 1 | 0 | 1.639e-16 | -0.0156 | 0.673 | 0.121 | -0.131 | -0.0232 | -0.00377 | 0 | 0 | 0 | 0.000473 | 0.0278 | 0.000228 | 0.000282 | 5.788e-06 |
| re500_e0.25 | 500 | 0.25 | 13 | 1 | B1_Mode2 | 0.256 | 1 | 1 | 0 | 1.639e-16 | -0.0156 | 0.673 | 0.121 | -0.131 | -0.0232 | -0.00377 | 0 | 0 | 1 | 0.886 | 0.00368 | 0.000352 | 1.432e-05 | 2.281e-05 |
| re500_e0.25 | 500 | 0.25 | 13 | 1 | B1_Mode2 | 0.256 | 1 | 1 | 0 | 1.639e-16 | -0.0156 | 0.673 | 0.121 | -0.131 | -0.0232 | -0.00377 | 0 | 0 | 2 | 0.0286 | 0.0148 | 9.150e-07 | 7.835e-05 | 1.001e-07 |
| re500_e0.25 | 500 | 0.25 | 13 | 1 | B1_Mode2 | 0.256 | 1 | 1 | 0 | 1.639e-16 | -0.0156 | 0.673 | 0.121 | -0.131 | -0.0232 | -0.00377 | 0 | 0 | 3 | 0.0337 | 4.869e-05 | 0.000506 | 2.503e-08 | 3.369e-05 |
| re500_e0.25 | 500 | 0.25 | 13 | 1 | B1_Mode2 | 0.256 | 1 | 1 | 0 | 1.639e-16 | -0.0156 | 0.673 | 0.121 | -0.131 | -0.0232 | -0.00377 | 0 | 0 | 4 | 0.00105 | 0.00181 | 2.007e-06 | 2.826e-06 | 1.258e-07 |
| re500_e0.25 | 500 | 0.25 | 13 | 1 | B1_Mode2 | 0.256 | 1 | 1 | 0 | 1.639e-16 | -0.0156 | 0.673 | 0.121 | -0.131 | -0.0232 | -0.00377 | 0 | 0 | 5 | 2.773e-05 | 6.818e-05 | 2.337e-05 | 8.272e-07 | 2.251e-06 |
| re500_e0.25 | 500 | 0.25 | 14 | 2 | B2_temporal | 0.25 | 1 | 0 | 1 | 1.971e-16 | 0.024 | 0.0975 | 0.0423 | -0.0207 | -0.00179 | -0.00243 | 0 | 0 | 0 | 0.00116 | 0.946 | 0.00228 | 0.00132 | 2.151e-05 |
| re500_e0.25 | 500 | 0.25 | 14 | 2 | B2_temporal | 0.25 | 1 | 0 | 1 | 1.971e-16 | 0.024 | 0.0975 | 0.0423 | -0.0207 | -0.00179 | -0.00243 | 0 | 0 | 1 | 0.019 | 0.0131 | 2.473e-06 | 0.00117 | 8.980e-05 |
| re500_e0.25 | 500 | 0.25 | 14 | 2 | B2_temporal | 0.25 | 1 | 0 | 1 | 1.971e-16 | 0.024 | 0.0975 | 0.0423 | -0.0207 | -0.00179 | -0.00243 | 0 | 0 | 2 | 0.00358 | 0.0112 | 2.259e-05 | 2.961e-07 | 8.793e-06 |
| re500_e0.25 | 500 | 0.25 | 14 | 2 | B2_temporal | 0.25 | 1 | 0 | 1 | 1.971e-16 | 0.024 | 0.0975 | 0.0423 | -0.0207 | -0.00179 | -0.00243 | 0 | 0 | 3 | 0.000862 | 6.466e-10 | 5.221e-06 | 6.306e-05 | 1.364e-06 |
| re500_e0.25 | 500 | 0.25 | 14 | 2 | B2_temporal | 0.25 | 1 | 0 | 1 | 1.971e-16 | 0.024 | 0.0975 | 0.0423 | -0.0207 | -0.00179 | -0.00243 | 0 | 0 | 4 | 6.439e-06 | 9.217e-05 | 1.296e-07 | 1.488e-06 | 6.049e-08 |
| re500_e0.25 | 500 | 0.25 | 14 | 2 | B2_temporal | 0.25 | 1 | 0 | 1 | 1.971e-16 | 0.024 | 0.0975 | 0.0423 | -0.0207 | -0.00179 | -0.00243 | 0 | 0 | 5 | 1.186e-05 | 7.841e-07 | 6.405e-06 | 8.947e-07 | 3.320e-07 |
| re500_e0.25 | 500 | 0.25 | 2 | 1 | B1_Mode2 | 0.257 | 1 | 1 | 0 | 1.475e-16 | 0.0388 | 0.677 | -0.0567 | -0.162 | 0.0419 | -0.005 | 0 | 0 | 0 | 0.00292 | 0.0232 | 1.390e-07 | 0.000222 | 1.623e-06 |
| re500_e0.25 | 500 | 0.25 | 2 | 1 | B1_Mode2 | 0.257 | 1 | 1 | 0 | 1.475e-16 | 0.0388 | 0.677 | -0.0567 | -0.162 | 0.0419 | -0.005 | 0 | 0 | 1 | 0.89 | 0.000784 | 0.000586 | 1.745e-05 | 2.607e-05 |
| re500_e0.25 | 500 | 0.25 | 2 | 1 | B1_Mode2 | 0.257 | 1 | 1 | 0 | 1.475e-16 | 0.0388 | 0.677 | -0.0567 | -0.162 | 0.0419 | -0.005 | 0 | 0 | 2 | 0.00625 | 0.0136 | 4.159e-05 | 9.747e-05 | 4.796e-06 |
| re500_e0.25 | 500 | 0.25 | 2 | 1 | B1_Mode2 | 0.257 | 1 | 1 | 0 | 1.475e-16 | 0.0388 | 0.677 | -0.0567 | -0.162 | 0.0419 | -0.005 | 0 | 0 | 3 | 0.0509 | 0.006 | 0.000386 | 2.644e-05 | 1.594e-05 |
| re500_e0.25 | 500 | 0.25 | 2 | 1 | B1_Mode2 | 0.257 | 1 | 1 | 0 | 1.475e-16 | 0.0388 | 0.677 | -0.0567 | -0.162 | 0.0419 | -0.005 | 0 | 0 | 4 | 0.00341 | 0.000609 | 0.000325 | 7.234e-06 | 2.050e-05 |
| re500_e0.25 | 500 | 0.25 | 2 | 1 | B1_Mode2 | 0.257 | 1 | 1 | 0 | 1.475e-16 | 0.0388 | 0.677 | -0.0567 | -0.162 | 0.0419 | -0.005 | 0 | 0 | 5 | 4.861e-05 | 0.000549 | 1.006e-05 | 8.412e-07 | 3.595e-06 |
| re500_e0.25 | 500 | 0.25 | 3 | 2 | B2_temporal | 0.268 | 1 | 0 | 1 | 1.567e-16 | -0.0325 | 0.0931 | -0.000253 | 0.00778 | -0.00149 | 0.00126 | 0 | 0 | 0 | 0.00197 | 0.972 | 0.000897 | 0.000603 | 2.831e-06 |
| re500_e0.25 | 500 | 0.25 | 3 | 2 | B2_temporal | 0.268 | 1 | 0 | 1 | 1.567e-16 | -0.0325 | 0.0931 | -0.000253 | 0.00778 | -0.00149 | 0.00126 | 0 | 0 | 1 | 0.0162 | 0.000501 | 0.00128 | 0.00259 | 3.760e-05 |
| re500_e0.25 | 500 | 0.25 | 3 | 2 | B2_temporal | 0.268 | 1 | 0 | 1 | 1.567e-16 | -0.0325 | 0.0931 | -0.000253 | 0.00778 | -0.00149 | 0.00126 | 0 | 0 | 2 | 1.191e-07 | 0.00284 | 0.000118 | 0.000354 | 3.361e-05 |
| re500_e0.25 | 500 | 0.25 | 3 | 2 | B2_temporal | 0.268 | 1 | 0 | 1 | 1.567e-16 | -0.0325 | 0.0931 | -0.000253 | 0.00778 | -0.00149 | 0.00126 | 0 | 0 | 3 | 0.000113 | 2.482e-06 | 6.120e-06 | 9.322e-06 | 1.794e-05 |
| re500_e0.25 | 500 | 0.25 | 3 | 2 | B2_temporal | 0.268 | 1 | 0 | 1 | 1.567e-16 | -0.0325 | 0.0931 | -0.000253 | 0.00778 | -0.00149 | 0.00126 | 0 | 0 | 4 | 4.124e-06 | 1.908e-05 | 5.585e-07 | 1.858e-06 | 1.880e-06 |
| re500_e0.25 | 500 | 0.25 | 3 | 2 | B2_temporal | 0.268 | 1 | 0 | 1 | 1.567e-16 | -0.0325 | 0.0931 | -0.000253 | 0.00778 | -0.00149 | 0.00126 | 0 | 0 | 5 | 2.963e-06 | 4.113e-07 | 2.948e-06 | 2.247e-07 | 3.366e-06 |
| re500_e0.25 | 500 | 0.25 | 4 | 3 | B3_mixed | 0.25 | 1 | 0 | 1 | 1.616e-16 | 0.0363 | -0.414 | -0.0568 | 0.0698 | 0.0199 | -0.000562 | 0 | 0 | 0 | 0.00263 | 0.579 | 0.0229 | 0.000737 | 0.000313 |
| re500_e0.25 | 500 | 0.25 | 4 | 3 | B3_mixed | 0.25 | 1 | 0 | 1 | 1.616e-16 | 0.0363 | -0.414 | -0.0568 | 0.0698 | 0.0199 | -0.000562 | 0 | 0 | 1 | 0.343 | 0.00569 | 0.00114 | 0.000522 | 0.00017 |
| re500_e0.25 | 500 | 0.25 | 4 | 3 | B3_mixed | 0.25 | 1 | 0 | 1 | 1.616e-16 | 0.0363 | -0.414 | -0.0568 | 0.0698 | 0.0199 | -0.000562 | 0 | 0 | 2 | 0.00646 | 0.0169 | 0.00869 | 2.298e-05 | 2.815e-06 |
| re500_e0.25 | 500 | 0.25 | 4 | 3 | B3_mixed | 0.25 | 1 | 0 | 1 | 1.616e-16 | 0.0363 | -0.414 | -0.0568 | 0.0698 | 0.0199 | -0.000562 | 0 | 0 | 3 | 0.00974 | 0.00072 | 7.054e-06 | 1.895e-05 | 4.452e-05 |
| re500_e0.25 | 500 | 0.25 | 4 | 3 | B3_mixed | 0.25 | 1 | 0 | 1 | 1.616e-16 | 0.0363 | -0.414 | -0.0568 | 0.0698 | 0.0199 | -0.000562 | 0 | 0 | 4 | 0.000793 | 9.169e-05 | 3.435e-06 | 1.839e-06 | 8.180e-07 |
| re500_e0.25 | 500 | 0.25 | 4 | 3 | B3_mixed | 0.25 | 1 | 0 | 1 | 1.616e-16 | 0.0363 | -0.414 | -0.0568 | 0.0698 | 0.0199 | -0.000562 | 0 | 0 | 5 | 6.307e-07 | 8.349e-06 | 4.542e-06 | 2.364e-06 | 3.981e-07 |
| re500_e0.25 | 500 | 0.25 | 5 | 2 | B2_temporal | 0.245 | 1 | 0 | 1 | 1.600e-16 | 0.00675 | -0.00143 | 0.0314 | 0.00799 | -0.00538 | 0.0032 | 0 | 0 | 0 | 9.284e-05 | 0.963 | 0.0157 | 0.000589 | 5.533e-09 |
| re500_e0.25 | 500 | 0.25 | 5 | 2 | B2_temporal | 0.245 | 1 | 0 | 1 | 1.600e-16 | 0.00675 | -0.00143 | 0.0314 | 0.00799 | -0.00538 | 0.0032 | 0 | 0 | 1 | 4.164e-06 | 0.00207 | 0.000729 | 0.00199 | 4.026e-05 |
| re500_e0.25 | 500 | 0.25 | 5 | 2 | B2_temporal | 0.245 | 1 | 0 | 1 | 1.600e-16 | 0.00675 | -0.00143 | 0.0314 | 0.00799 | -0.00538 | 0.0032 | 0 | 0 | 2 | 0.00201 | 0.0123 | 0.00084 | 0.000316 | 7.449e-05 |
| re500_e0.25 | 500 | 0.25 | 5 | 2 | B2_temporal | 0.245 | 1 | 0 | 1 | 1.600e-16 | 0.00675 | -0.00143 | 0.0314 | 0.00799 | -0.00538 | 0.0032 | 0 | 0 | 3 | 0.00013 | 1.215e-06 | 8.494e-06 | 4.714e-05 | 3.727e-06 |
| re500_e0.25 | 500 | 0.25 | 5 | 2 | B2_temporal | 0.245 | 1 | 0 | 1 | 1.600e-16 | 0.00675 | -0.00143 | 0.0314 | 0.00799 | -0.00538 | 0.0032 | 0 | 0 | 4 | 5.896e-05 | 3.230e-06 | 7.971e-07 | 2.895e-06 | 1.011e-06 |
| re500_e0.25 | 500 | 0.25 | 5 | 2 | B2_temporal | 0.245 | 1 | 0 | 1 | 1.600e-16 | 0.00675 | -0.00143 | 0.0314 | 0.00799 | -0.00538 | 0.0032 | 0 | 0 | 5 | 2.085e-05 | 9.467e-08 | 3.335e-07 | 9.809e-08 | 1.515e-07 |
| re500_e0.25 | 500 | 0.25 | 6 | 1 | B1_Mode2 | 0.264 | 1 | 1 | 0 | 1.433e-16 | -0.0917 | 0.677 | 0.178 | -0.132 | -0.0807 | -0.0262 | 0 | 0 | 0 | 0.0159 | 0.000247 | 0.000101 | 9.343e-08 | 3.994e-06 |
| re500_e0.25 | 500 | 0.25 | 6 | 1 | B1_Mode2 | 0.264 | 1 | 1 | 0 | 1.433e-16 | -0.0917 | 0.677 | 0.178 | -0.132 | -0.0807 | -0.0262 | 0 | 0 | 1 | 0.868 | 0.0011 | 0.000188 | 1.351e-06 | 1.041e-05 |
| re500_e0.25 | 500 | 0.25 | 6 | 1 | B1_Mode2 | 0.264 | 1 | 1 | 0 | 1.433e-16 | -0.0917 | 0.677 | 0.178 | -0.132 | -0.0807 | -0.0262 | 0 | 0 | 2 | 0.0601 | 0.00193 | 2.019e-05 | 9.895e-06 | 1.833e-06 |
| re500_e0.25 | 500 | 0.25 | 6 | 1 | B1_Mode2 | 0.264 | 1 | 1 | 0 | 1.433e-16 | -0.0917 | 0.677 | 0.178 | -0.132 | -0.0807 | -0.0262 | 0 | 0 | 3 | 0.0332 | 0.00519 | 0.000179 | 3.397e-05 | 6.457e-06 |
| re500_e0.25 | 500 | 0.25 | 6 | 1 | B1_Mode2 | 0.264 | 1 | 1 | 0 | 1.433e-16 | -0.0917 | 0.677 | 0.178 | -0.132 | -0.0807 | -0.0262 | 0 | 0 | 4 | 0.0123 | 0.000276 | 6.399e-05 | 2.549e-06 | 2.042e-06 |
| re500_e0.25 | 500 | 0.25 | 6 | 1 | B1_Mode2 | 0.264 | 1 | 1 | 0 | 1.433e-16 | -0.0917 | 0.677 | 0.178 | -0.132 | -0.0807 | -0.0262 | 0 | 0 | 5 | 0.0013 | 0.000101 | 3.206e-05 | 8.488e-08 | 1.795e-06 |
| re500_e0.25 | 500 | 0.25 | 7 | 1 | B1_Mode2 | 0.251 | 1 | 1 | 0 | 1.628e-16 | -0.0612 | 0.683 | 0.0722 | -0.147 | -0.00927 | -0.00633 | 0 | 0 | 0 | 0.00748 | 0.00278 | 0.000187 | 2.124e-05 | 1.057e-05 |
| re500_e0.25 | 500 | 0.25 | 7 | 1 | B1_Mode2 | 0.251 | 1 | 1 | 0 | 1.628e-16 | -0.0612 | 0.683 | 0.0722 | -0.147 | -0.00927 | -0.00633 | 0 | 0 | 1 | 0.93 | 0.00053 | 0.000484 | 2.986e-06 | 2.447e-05 |
| re500_e0.25 | 500 | 0.25 | 7 | 1 | B1_Mode2 | 0.251 | 1 | 1 | 0 | 1.628e-16 | -0.0612 | 0.683 | 0.0722 | -0.147 | -0.00927 | -0.00633 | 0 | 0 | 2 | 0.0104 | 0.000653 | 0.000287 | 2.793e-06 | 1.082e-05 |
| re500_e0.25 | 500 | 0.25 | 7 | 1 | B1_Mode2 | 0.251 | 1 | 1 | 0 | 1.628e-16 | -0.0612 | 0.683 | 0.0722 | -0.147 | -0.00927 | -0.00633 | 0 | 0 | 3 | 0.0432 | 0.00341 | 0.000107 | 2.306e-05 | 6.419e-06 |
| re500_e0.25 | 500 | 0.25 | 7 | 1 | B1_Mode2 | 0.251 | 1 | 1 | 0 | 1.628e-16 | -0.0612 | 0.683 | 0.0722 | -0.147 | -0.00927 | -0.00633 | 0 | 0 | 4 | 0.000171 | 0.000157 | 7.653e-06 | 1.506e-08 | 3.419e-07 |
| re500_e0.25 | 500 | 0.25 | 7 | 1 | B1_Mode2 | 0.251 | 1 | 1 | 0 | 1.628e-16 | -0.0612 | 0.683 | 0.0722 | -0.147 | -0.00927 | -0.00633 | 0 | 0 | 5 | 7.996e-05 | 0.000283 | 2.419e-05 | 2.094e-07 | 2.090e-06 |
| re500_e0.25 | 500 | 0.25 | 8 | 2 | B2_temporal | 0.251 | 1 | 0 | 1 | 1.733e-16 | -0.0102 | 0.131 | -0.0306 | -0.0256 | -0.00201 | -0.00213 | 0 | 0 | 0 | 0.000208 | 0.925 | 0.02 | 0.000153 | 2.360e-05 |
| re500_e0.25 | 500 | 0.25 | 8 | 2 | B2_temporal | 0.251 | 1 | 0 | 1 | 1.733e-16 | -0.0102 | 0.131 | -0.0306 | -0.0256 | -0.00201 | -0.00213 | 0 | 0 | 1 | 0.0341 | 0.000936 | 0.00462 | 0.00219 | 6.676e-05 |
| re500_e0.25 | 500 | 0.25 | 8 | 2 | B2_temporal | 0.251 | 1 | 0 | 1 | 1.733e-16 | -0.0102 | 0.131 | -0.0306 | -0.0256 | -0.00201 | -0.00213 | 0 | 0 | 2 | 0.00186 | 0.00836 | 0.000555 | 9.915e-05 | 7.363e-06 |
| re500_e0.25 | 500 | 0.25 | 8 | 2 | B2_temporal | 0.251 | 1 | 0 | 1 | 1.733e-16 | -0.0102 | 0.131 | -0.0306 | -0.0256 | -0.00201 | -0.00213 | 0 | 0 | 3 | 0.00131 | 6.270e-06 | 0.000159 | 0.000146 | 5.553e-06 |
| re500_e0.25 | 500 | 0.25 | 8 | 2 | B2_temporal | 0.251 | 1 | 0 | 1 | 1.733e-16 | -0.0102 | 0.131 | -0.0306 | -0.0256 | -0.00201 | -0.00213 | 0 | 0 | 4 | 8.031e-06 | 8.032e-05 | 5.717e-05 | 1.199e-05 | 3.040e-06 |
| re500_e0.25 | 500 | 0.25 | 8 | 2 | B2_temporal | 0.251 | 1 | 0 | 1 | 1.733e-16 | -0.0102 | 0.131 | -0.0306 | -0.0256 | -0.00201 | -0.00213 | 0 | 0 | 5 | 9.009e-06 | 5.478e-07 | 1.459e-06 | 5.022e-07 | 2.672e-07 |
| re500_e0.25 | 500 | 0.25 | 9 | 2 | B2_temporal | 0.265 | 1 | 0 | 1 | 1.476e-16 | -0.0392 | -0.0114 | -0.00832 | 0.0151 | -0.00227 | 0.00127 | 0 | 0 | 0 | 0.00291 | 0.953 | 0.000184 | 0.000206 | 1.801e-05 |
| re500_e0.25 | 500 | 0.25 | 9 | 2 | B2_temporal | 0.265 | 1 | 0 | 1 | 1.476e-16 | -0.0392 | -0.0114 | -0.00832 | 0.0151 | -0.00227 | 0.00127 | 0 | 0 | 1 | 0.000245 | 0.0269 | 0.00165 | 0.00401 | 5.541e-06 |
| re500_e0.25 | 500 | 0.25 | 9 | 2 | B2_temporal | 0.265 | 1 | 0 | 1 | 1.476e-16 | -0.0392 | -0.0114 | -0.00832 | 0.0151 | -0.00227 | 0.00127 | 0 | 0 | 2 | 0.000131 | 0.00947 | 0.000133 | 5.031e-05 | 4.722e-05 |
| re500_e0.25 | 500 | 0.25 | 9 | 2 | B2_temporal | 0.265 | 1 | 0 | 1 | 1.476e-16 | -0.0392 | -0.0114 | -0.00832 | 0.0151 | -0.00227 | 0.00127 | 0 | 0 | 3 | 0.000433 | 0.000214 | 9.363e-05 | 0.000182 | 1.179e-05 |
| re500_e0.25 | 500 | 0.25 | 9 | 2 | B2_temporal | 0.265 | 1 | 0 | 1 | 1.476e-16 | -0.0392 | -0.0114 | -0.00832 | 0.0151 | -0.00227 | 0.00127 | 0 | 0 | 4 | 9.732e-06 | 3.812e-05 | 2.242e-05 | 9.888e-06 | 1.269e-06 |
| re500_e0.25 | 500 | 0.25 | 9 | 2 | B2_temporal | 0.265 | 1 | 0 | 1 | 1.476e-16 | -0.0392 | -0.0114 | -0.00832 | 0.0151 | -0.00227 | 0.00127 | 0 | 0 | 5 | 3.055e-06 | 2.123e-09 | 5.047e-08 | 5.676e-07 | 1.334e-06 |

Table 15: Décomposition modale — 185 points, 1110 lignes (6 modes $i$ par point, colonnes $E^f_{iJ}$ pour $J$=0..4), `mode_analysis.csv`.

| point | Re | $E^*$ | s | id | branch | $E_t$ | cov. | $i$ | $j$ | rmse | $A_{10}$ | $A_{20}$ | $A_{30}$ | $A_{40}$ | $A_{50}$ | $A_{60}$ | $A_{70}$ | $A_{80}$ | $i$ | $E^f_{0J}$ | $E^f_{1J}$ | $E^f_{2J}$ | $E^f_{3J}$ | $E^f_{4J}$ |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| re500_e0.5 | 500 | 0.5 | 0 | 2 | B2_temporal | 0.453 | 1 | 0 | 1 | 2.134e-16 | 0.0594 | -0.449 | 0.0572 | 0.0628 | -0.0236 | 0.0167 | 0 | 0 | 0 | 0.00389 | 0.663 | 0.000337 | 0.00238 | 0.00021 |
| re500_e0.5 | 500 | 0.5 | 0 | 2 | B2_temporal | 0.453 | 1 | 0 | 1 | 2.134e-16 | 0.0594 | -0.449 | 0.0572 | 0.0628 | -0.0236 | 0.0167 | 0 | 0 | 1 | 0.222 | 0.0228 | 0.0102 | 0.00283 | 6.455e-05 |
| re500_e0.5 | 500 | 0.5 | 0 | 2 | B2_temporal | 0.453 | 1 | 0 | 1 | 2.134e-16 | 0.0594 | -0.449 | 0.0572 | 0.0628 | -0.0236 | 0.0167 | 0 | 0 | 2 | 0.00362 | 0.0492 | 0.00759 | 0.000644 | 7.014e-06 |
| re500_e0.5 | 500 | 0.5 | 0 | 2 | B2_temporal | 0.453 | 1 | 0 | 1 | 2.134e-16 | 0.0594 | -0.449 | 0.0572 | 0.0628 | -0.0236 | 0.0167 | 0 | 0 | 3 | 0.00436 | 0.00202 | 0.00122 | 0.000724 | 9.280e-05 |
| re500_e0.5 | 500 | 0.5 | 0 | 2 | B2_temporal | 0.453 | 1 | 0 | 1 | 2.134e-16 | 0.0594 | -0.449 | 0.0572 | 0.0628 | -0.0236 | 0.0167 | 0 | 0 | 4 | 0.000613 | 2.564e-05 | 0.000964 | 5.374e-05 | 1.939e-05 |
| re500_e0.5 | 500 | 0.5 | 0 | 2 | B2_temporal | 0.453 | 1 | 0 | 1 | 2.134e-16 | 0.0594 | -0.449 | 0.0572 | 0.0628 | -0.0236 | 0.0167 | 0 | 0 | 5 | 0.000307 | 0.000307 | 8.316e-05 | 8.385e-05 | 1.226e-06 |
| re500_e0.5 | 500 | 0.5 | 1 | 1 | B1_Mode2 | 0.516 | 1 | 1 | 0 | 1.987e-16 | -0.000468 | 0.977 | 0.097 | -0.203 | -0.0275 | 0.0113 | 0 | 0 | 0 | 2.119e-07 | 0.00222 | 0.000103 | 2.002e-05 | 8.212e-06 |
| re500_e0.5 | 500 | 0.5 | 1 | 1 | B1_Mode2 | 0.516 | 1 | 1 | 0 | 1.987e-16 | -0.000468 | 0.977 | 0.097 | -0.203 | -0.0275 | 0.0113 | 0 | 0 | 1 | 0.925 | 0.00742 | 0.000151 | 6.083e-05 | 8.826e-06 |
| re500_e0.5 | 500 | 0.5 | 1 | 1 | B1_Mode2 | 0.516 | 1 | 1 | 0 | 1.987e-16 | -0.000468 | 0.977 | 0.097 | -0.203 | -0.0275 | 0.0113 | 0 | 0 | 2 | 0.00912 | 0.00898 | 3.194e-05 | 8.955e-05 | 2.720e-06 |
| re500_e0.5 | 500 | 0.5 | 1 | 1 | B1_Mode2 | 0.516 | 1 | 1 | 0 | 1.987e-16 | -0.000468 | 0.977 | 0.097 | -0.203 | -0.0275 | 0.0113 | 0 | 0 | 3 | 0.0401 | 0.00511 | 0.000362 | 2.792e-05 | 1.515e-05 |
| re500_e0.5 | 500 | 0.5 | 1 | 1 | B1_Mode2 | 0.516 | 1 | 1 | 0 | 1.987e-16 | -0.000468 | 0.977 | 0.097 | -0.203 | -0.0275 | 0.0113 | 0 | 0 | 4 | 0.000731 | 0.000251 | 0.000138 | 1.274e-07 | 8.243e-06 |
| re500_e0.5 | 500 | 0.5 | 1 | 1 | B1_Mode2 | 0.516 | 1 | 1 | 0 | 1.987e-16 | -0.000468 | 0.977 | 0.097 | -0.203 | -0.0275 | 0.0113 | 0 | 0 | 5 | 0.000124 | 0.000341 | 1.295e-05 | 1.988e-07 | 3.369e-07 |
| re500_e0.5 | 500 | 0.5 | 2 | 1 | B1_Mode2 | 0.504 | 1 | 1 | 0 | 2.132e-16 | 0.0352 | 0.965 | -0.00779 | -0.169 | 0.046 | -0.00305 | 0 | 0 | 0 | 0.00123 | 0.0186 | 3.621e-06 | 0.000158 | 5.926e-07 |
| re500_e0.5 | 500 | 0.5 | 2 | 1 | B1_Mode2 | 0.504 | 1 | 1 | 0 | 2.132e-16 | 0.0352 | 0.965 | -0.00779 | -0.169 | 0.046 | -0.00305 | 0 | 0 | 1 | 0.925 | 8.453e-05 | 0.00158 | 2.612e-07 | 6.410e-05 |
| re500_e0.5 | 500 | 0.5 | 2 | 1 | B1_Mode2 | 0.504 | 1 | 1 | 0 | 2.132e-16 | 0.0352 | 0.965 | -0.00779 | -0.169 | 0.046 | -0.00305 | 0 | 0 | 2 | 6.018e-05 | 0.014 | 1.153e-05 | 5.724e-05 | 1.454e-08 |
| re500_e0.5 | 500 | 0.5 | 2 | 1 | B1_Mode2 | 0.504 | 1 | 1 | 0 | 2.132e-16 | 0.0352 | 0.965 | -0.00779 | -0.169 | 0.046 | -0.00305 | 0 | 0 | 3 | 0.0283 | 0.00598 | 0.000267 | 3.335e-05 | 8.987e-06 |
| re500_e0.5 | 500 | 0.5 | 2 | 1 | B1_Mode2 | 0.504 | 1 | 1 | 0 | 2.132e-16 | 0.0352 | 0.965 | -0.00779 | -0.169 | 0.046 | -0.00305 | 0 | 0 | 4 | 0.0021 | 0.00128 | 0.000263 | 9.576e-06 | 1.920e-05 |
| re500_e0.5 | 500 | 0.5 | 2 | 1 | B1_Mode2 | 0.504 | 1 | 1 | 0 | 2.132e-16 | 0.0352 | 0.965 | -0.00779 | -0.169 | 0.046 | -0.00305 | 0 | 0 | 5 | 9.265e-06 | 0.000457 | 4.253e-06 | 1.657e-07 | 3.353e-06 |
| re500_e0.5 | 500 | 0.5 | 3 | 2 | B2_temporal | 0.518 | 1 | 0 | 1 | 2.164e-16 | -0.0508 | 0.27 | 0.0221 | -0.00284 | 0.00259 | 0.00115 | 0 | 0 | 0 | 0.00249 | 0.911 | 0.000758 | 0.000324 | 1.935e-05 |
| re500_e0.5 | 500 | 0.5 | 3 | 2 | B2_temporal | 0.518 | 1 | 0 | 1 | 2.164e-16 | -0.0508 | 0.27 | 0.0221 | -0.00284 | 0.00259 | 0.00115 | 0 | 0 | 1 | 0.0703 | 0.00161 | 0.00462 | 0.00334 | 3.933e-05 |
| re500_e0.5 | 500 | 0.5 | 3 | 2 | B2_temporal | 0.518 | 1 | 0 | 1 | 2.164e-16 | -0.0508 | 0.27 | 0.0221 | -0.00284 | 0.00259 | 0.00115 | 0 | 0 | 2 | 0.000469 | 0.00227 | 0.000976 | 0.000684 | 4.966e-05 |
| re500_e0.5 | 500 | 0.5 | 3 | 2 | B2_temporal | 0.518 | 1 | 0 | 1 | 2.164e-16 | -0.0508 | 0.27 | 0.0221 | -0.00284 | 0.00259 | 0.00115 | 0 | 0 | 3 | 7.770e-06 | 0.00012 | 0.000407 | 1.346e-05 | 6.084e-06 |
| re500_e0.5 | 500 | 0.5 | 3 | 2 | B2_temporal | 0.518 | 1 | 0 | 1 | 2.164e-16 | -0.0508 | 0.27 | 0.0221 | -0.00284 | 0.00259 | 0.00115 | 0 | 0 | 4 | 6.452e-06 | 3.542e-05 | 1.757e-05 | 3.055e-07 | 2.258e-06 |
| re500_e0.5 | 500 | 0.5 | 3 | 2 | B2_temporal | 0.518 | 1 | 0 | 1 | 2.164e-16 | -0.0508 | 0.27 | 0.0221 | -0.00284 | 0.00259 | 0.00115 | 0 | 0 | 5 | 1.275e-06 | 8.472e-06 | 1.784e-07 | 3.227e-08 | 4.582e-09 |
| re500_e0.5 | 500 | 0.5 | 4 | 2 | B2_temporal | 0.49 | 1 | 0 | 1 | 2.221e-16 | 0.0525 | -0.22 | -0.0842 | 0.0263 | 0.0123 | 0.000963 | 0 | 0 | 0 | 0.00281 | 0.803 | 0.082 | 0.00105 | 0.000304 |
| re500_e0.5 | 500 | 0.5 | 4 | 2 | B2_temporal | 0.49 | 1 | 0 | 1 | 2.221e-16 | 0.0525 | -0.22 | -0.0842 | 0.0263 | 0.0123 | 0.000963 | 0 | 0 | 1 | 0.0492 | 0.00172 | 0.000822 | 0.00223 | 0.00104 |
| re500_e0.5 | 500 | 0.5 | 4 | 2 | B2_temporal | 0.49 | 1 | 0 | 1 | 2.221e-16 | 0.0525 | -0.22 | -0.0842 | 0.0263 | 0.0123 | 0.000963 | 0 | 0 | 2 | 0.00723 | 0.0333 | 0.0129 | 0.00028 | 4.740e-06 |
| re500_e0.5 | 500 | 0.5 | 4 | 2 | B2_temporal | 0.49 | 1 | 0 | 1 | 2.221e-16 | 0.0525 | -0.22 | -0.0842 | 0.0263 | 0.0123 | 0.000963 | 0 | 0 | 3 | 0.000706 | 4.551e-05 | 5.807e-06 | 0.000147 | 0.000125 |
| re500_e0.5 | 500 | 0.5 | 4 | 2 | B2_temporal | 0.49 | 1 | 0 | 1 | 2.221e-16 | 0.0525 | -0.22 | -0.0842 | 0.0263 | 0.0123 | 0.000963 | 0 | 0 | 4 | 0.000154 | 0.00028 | 1.709e-05 | 6.245e-07 | 2.707e-05 |
| re500_e0.5 | 500 | 0.5 | 4 | 2 | B2_temporal | 0.49 | 1 | 0 | 1 | 2.221e-16 | 0.0525 | -0.22 | -0.0842 | 0.0263 | 0.0123 | 0.000963 | 0 | 0 | 5 | 9.456e-07 | 3.780e-07 | 2.085e-06 | 4.847e-06 | 6.965e-06 |
| re500_e1 | 500 | 1 | 0 | 2 | B2_temporal | 0.919 | 1 | 0 | 1 | 2.910e-16 | 0.0907 | -0.72 | 0.0599 | 0.0708 | -0.0412 | 0.0321 | 0 | 0 | 0 | 0.00448 | 0.518 | 0.00302 | 0.00245 | 0.000411 |
| re500_e1 | 500 | 1 | 0 | 2 | B2_temporal | 0.919 | 1 | 0 | 1 | 2.910e-16 | 0.0907 | -0.72 | 0.0599 | 0.0708 | -0.0412 | 0.0321 | 0 | 0 | 1 | 0.282 | 0.0507 | 0.0296 | 0.00429 | 6.550e-06 |
| re500_e1 | 500 | 1 | 0 | 2 | B2_temporal | 0.919 | 1 | 0 | 1 | 2.910e-16 | 0.0907 | -0.72 | 0.0599 | 0.0708 | -0.0412 | 0.0321 | 0 | 0 | 2 | 0.00195 | 0.0702 | 0.00747 | 0.00249 | 0.000234 |
| re500_e1 | 500 | 1 | 0 | 2 | B2_temporal | 0.919 | 1 | 0 | 1 | 2.910e-16 | 0.0907 | -0.72 | 0.0599 | 0.0708 | -0.0412 | 0.0321 | 0 | 0 | 3 | 0.00273 | 0.00577 | 0.00842 | 0.00162 | 1.788e-05 |
| re500_e1 | 500 | 1 | 0 | 2 | B2_temporal | 0.919 | 1 | 0 | 1 | 2.910e-16 | 0.0907 | -0.72 | 0.0599 | 0.0708 | -0.0412 | 0.0321 | 0 | 0 | 4 | 0.000925 | 0.000273 | 0.00176 | 1.045e-05 | 9.902e-05 |
| re500_e1 | 500 | 1 | 0 | 2 | B2_temporal | 0.919 | 1 | 0 | 1 | 2.910e-16 | 0.0907 | -0.72 | 0.0599 | 0.0708 | -0.0412 | 0.0321 | 0 | 0 | 5 | 0.000562 | 0.000508 | 5.787e-05 | 0.000188 | 4.374e-07 |
| re500_e1 | 500 | 1 | 1 | 1 | B1_Mode2 | 1.01 | 1 | 1 | 0 | 3.218e-16 | 0.198 | 1.14 | -0.501 | -0.218 | 0.0323 | -0.0037 | 0 | 0 | 0 | 0.0194 | 0.0238 | 0.000446 | 0.000236 | 1.329e-05 |
| re500_e1 | 500 | 1 | 1 | 1 | B1_Mode2 | 1.01 | 1 | 1 | 0 | 3.218e-16 | 0.198 | 1.14 | -0.501 | -0.218 | 0.0323 | -0.0037 | 0 | 0 | 1 | 0.643 | 0.0776 | 0.00295 | 0.000152 | 0.0002 |
| re500_e1 | 500 | 1 | 1 | 1 | B1_Mode2 | 1.01 | 1 | 1 | 0 | 3.218e-16 | 0.198 | 1.14 | -0.501 | -0.218 | 0.0323 | -0.0037 | 0 | 0 | 2 | 0.124 | 0.0554 | 0.00497 | 0.000187 | 0.000121 |
| re500_e1 | 500 | 1 | 1 | 1 | B1_Mode2 | 1.01 | 1 | 1 | 0 | 3.218e-16 | 0.198 | 1.14 | -0.501 | -0.218 | 0.0323 | -0.0037 | 0 | 0 | 3 | 0.0236 | 0.003 | 0.00686 | 4.238e-05 | 0.000244 |
| re500_e1 | 500 | 1 | 1 | 1 | B1_Mode2 | 1.01 | 1 | 1 | 0 | 3.218e-16 | 0.198 | 1.14 | -0.501 | -0.218 | 0.0323 | -0.0037 | 0 | 0 | 4 | 0.000517 | 0.0119 | 5.144e-05 | 0.000132 | 3.324e-07 |
| re500_e1 | 500 | 1 | 1 | 1 | B1_Mode2 | 1.01 | 1 | 1 | 0 | 3.218e-16 | 0.198 | 1.14 | -0.501 | -0.218 | 0.0323 | -0.0037 | 0 | 0 | 5 | 6.797e-06 | 4.694e-05 | 0.000666 | 4.204e-06 | 9.514e-06 |
| re500_e1 | 500 | 1 | 2 | 1 | B1_Mode2 | 1 | 1 | 1 | 0 | 2.969e-16 | 0.0486 | 1.39 | -0.0269 | -0.182 | 0.0537 | 0.0324 | 0 | 0 | 0 | 0.00118 | 0.00537 | 3.729e-05 | 5.170e-05 | 1.492e-08 |
| re500_e1 | 500 | 1 | 2 | 1 | B1_Mode2 | 1 | 1 | 1 | 0 | 2.969e-16 | 0.0486 | 1.39 | -0.0269 | -0.182 | 0.0537 | 0.0324 | 0 | 0 | 1 | 0.961 | 0.00201 | 0.00118 | 9.040e-06 | 4.885e-05 |
| re500_e1 | 500 | 1 | 2 | 1 | B1_Mode2 | 1 | 1 | 1 | 0 | 2.969e-16 | 0.0486 | 1.39 | -0.0269 | -0.182 | 0.0537 | 0.0324 | 0 | 0 | 2 | 0.000361 | 0.00206 | 0.00056 | 2.904e-08 | 2.244e-05 |
| re500_e1 | 500 | 1 | 2 | 1 | B1_Mode2 | 1 | 1 | 1 | 0 | 2.969e-16 | 0.0486 | 1.39 | -0.0269 | -0.182 | 0.0537 | 0.0324 | 0 | 0 | 3 | 0.0165 | 0.00675 | 4.673e-05 | 5.829e-05 | 1.356e-06 |
| re500_e1 | 500 | 1 | 2 | 1 | B1_Mode2 | 1 | 1 | 1 | 0 | 2.969e-16 | 0.0486 | 1.39 | -0.0269 | -0.182 | 0.0537 | 0.0324 | 0 | 0 | 4 | 0.00144 | 0.000233 | 6.878e-05 | 1.182e-06 | 3.818e-06 |
| re500_e1 | 500 | 1 | 2 | 1 | B1_Mode2 | 1 | 1 | 1 | 0 | 2.969e-16 | 0.0486 | 1.39 | -0.0269 | -0.182 | 0.0537 | 0.0324 | 0 | 0 | 5 | 0.000526 | 0.000416 | 6.220e-06 | 1.699e-06 | 2.682e-06 |
| re500_e1 | 500 | 1 | 3 | 1 | B1_Mode2 | 0.972 | 1 | 1 | 0 | 2.950e-16 | 0.0137 | 1.37 | 0.104 | -0.142 | -0.061 | 0.0413 | 0 | 0 | 0 | 9.658e-05 | 0.00283 | 5.099e-07 | 6.333e-07 | 3.021e-06 |
| re500_e1 | 500 | 1 | 3 | 1 | B1_Mode2 | 0.972 | 1 | 1 | 0 | 2.950e-16 | 0.0137 | 1.37 | 0.104 | -0.142 | -0.061 | 0.0413 | 0 | 0 | 1 | 0.959 | 2.977e-05 | 0.00224 | 2.355e-05 | 0.000113 |
| re500_e1 | 500 | 1 | 3 | 1 | B1_Mode2 | 0.972 | 1 | 1 | 0 | 2.950e-16 | 0.0137 | 1.37 | 0.104 | -0.142 | -0.061 | 0.0413 | 0 | 0 | 2 | 0.00559 | 0.0056 | 0.000197 | 3.655e-06 | 2.910e-06 |
| re500_e1 | 500 | 1 | 3 | 1 | B1_Mode2 | 0.972 | 1 | 1 | 0 | 2.950e-16 | 0.0137 | 1.37 | 0.104 | -0.142 | -0.061 | 0.0413 | 0 | 0 | 3 | 0.0104 | 0.01 | 2.637e-06 | 1.367e-05 | 1.321e-06 |
| re500_e1 | 500 | 1 | 3 | 1 | B1_Mode2 | 0.972 | 1 | 1 | 0 | 2.950e-16 | 0.0137 | 1.37 | 0.104 | -0.142 | -0.061 | 0.0413 | 0 | 0 | 4 | 0.00191 | 2.322e-05 | 0.000328 | 2.783e-06 | 4.365e-06 |
| re500_e1 | 500 | 1 | 3 | 1 | B1_Mode2 | 0.972 | 1 | 1 | 0 | 2.950e-16 | 0.0137 | 1.37 | 0.104 | -0.142 | -0.061 | 0.0413 | 0 | 0 | 5 | 0.000878 | 0.00023 | 8.672e-05 | 5.962e-06 | 1.977e-06 |
| re500_e1 | 500 | 1 | 4 | 2 | B2_temporal | 0.963 | 1 | 0 | 1 | 2.901e-16 | 0.079 | -0.142 | -0.109 | 0.0266 | 0.0129 | 0.00511 | 0 | 0 | 0 | 0.00324 | 0.774 | 0.139 | 0.0101 | 0.000104 |
| re500_e1 | 500 | 1 | 4 | 2 | B2_temporal | 0.963 | 1 | 0 | 1 | 2.901e-16 | 0.079 | -0.142 | -0.109 | 0.0266 | 0.0129 | 0.00511 | 0 | 0 | 1 | 0.0105 | 0.00194 | 0.000377 | 0.002 | 0.00113 |
| re500_e1 | 500 | 1 | 4 | 2 | B2_temporal | 0.963 | 1 | 0 | 1 | 2.901e-16 | 0.079 | -0.142 | -0.109 | 0.0266 | 0.0129 | 0.00511 | 0 | 0 | 2 | 0.00613 | 0.0301 | 0.0188 | 0.00139 | 1.914e-05 |
| re500_e1 | 500 | 1 | 4 | 2 | B2_temporal | 0.963 | 1 | 0 | 1 | 2.901e-16 | 0.079 | -0.142 | -0.109 | 0.0266 | 0.0129 | 0.00511 | 0 | 0 | 3 | 0.000367 | 5.620e-08 | 4.797e-06 | 0.000185 | 0.000285 |
| re500_e1 | 500 | 1 | 4 | 2 | B2_temporal | 0.963 | 1 | 0 | 1 | 2.901e-16 | 0.079 | -0.142 | -0.109 | 0.0266 | 0.0129 | 0.00511 | 0 | 0 | 4 | 8.628e-05 | 9.308e-05 | 3.270e-05 | 1.701e-05 | 1.480e-05 |
| re500_e1 | 500 | 1 | 4 | 2 | B2_temporal | 0.963 | 1 | 0 | 1 | 2.901e-16 | 0.079 | -0.142 | -0.109 | 0.0266 | 0.0129 | 0.00511 | 0 | 0 | 5 | 1.356e-05 | 1.120e-07 | 1.891e-05 | 4.129e-06 | 1.288e-05 |
| re500_e2 | 500 | 2 | 0 | 2 | B2_temporal | 1.84 | 1 | 0 | 1 | 4.812e-16 | 0.274 | -0.938 | -0.146 | 0.116 | -0.0798 | 0.0561 | 0 | 0 | 0 | 0.0204 | 0.49 | 2.892e-05 | 0.00337 | 0.000899 |
| re500_e2 | 500 | 2 | 0 | 2 | B2_temporal | 1.84 | 1 | 0 | 1 | 4.812e-16 | 0.274 | -0.938 | -0.146 | 0.116 | -0.0798 | 0.0561 | 0 | 0 | 1 | 0.24 | 0.0839 | 0.0315 | 0.00233 | 0.000118 |
| re500_e2 | 500 | 2 | 0 | 2 | B2_temporal | 1.84 | 1 | 0 | 1 | 4.812e-16 | 0.274 | -0.938 | -0.146 | 0.116 | -0.0798 | 0.0561 | 0 | 0 | 2 | 0.00581 | 0.0825 | 0.00111 | 0.00544 | 0.000394 |
| re500_e2 | 500 | 2 | 0 | 2 | B2_temporal | 1.84 | 1 | 0 | 1 | 4.812e-16 | 0.274 | -0.938 | -0.146 | 0.116 | -0.0798 | 0.0561 | 0 | 0 | 3 | 0.00369 | 0.00674 | 0.0161 | 0.000385 | 6.155e-05 |
| re500_e2 | 500 | 2 | 0 | 2 | B2_temporal | 1.84 | 1 | 0 | 1 | 4.812e-16 | 0.274 | -0.938 | -0.146 | 0.116 | -0.0798 | 0.0561 | 0 | 0 | 4 | 0.00174 | 0.000342 | 0.000922 | 0.000355 | 3.906e-05 |
| re500_e2 | 500 | 2 | 0 | 2 | B2_temporal | 1.84 | 1 | 0 | 1 | 4.812e-16 | 0.274 | -0.938 | -0.146 | 0.116 | -0.0798 | 0.0561 | 0 | 0 | 5 | 0.000858 | 0.000488 | 7.110e-05 | 0.000128 | 1.788e-05 |

Table 15: Décomposition modale — 185 points, 1110 lignes (6 modes $i$ par point, colonnes $E^f_{iJ}$ pour $J$=0..4), `mode_analysis.csv`.

| point | Re | $E^*$ | s | id | branch | $E_t$ | cov. | $i$ | $j$ | rmse | $A_{10}$ | $A_{20}$ | $A_{30}$ | $A_{40}$ | $A_{50}$ | $A_{60}$ | $A_{70}$ | $A_{80}$ | $i$ | $E^f_{0J}$ | $E^f_{1J}$ | $E^f_{2J}$ | $E^f_{3J}$ | $E^f_{4J}$ |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| re500_e2 | 500 | 2 | 1 | 2 | B2_temporal | 2.04 | 1 | 1 | 1 | 4.199e-16 | 0.549 | 0.718 | -0.86 | -0.0421 | -0.0437 | 0.00232 | 0 | 0 | 0 | 0.0741 | 0.022 | 0.00977 | 2.941e-05 | 0.0003 |
| re500_e2 | 500 | 2 | 1 | 2 | B2_temporal | 2.04 | 1 | 1 | 1 | 4.199e-16 | 0.549 | 0.718 | -0.86 | -0.0421 | -0.0437 | 0.00232 | 0 | 0 | 1 | 0.127 | 0.411 | 0.00142 | 0.000325 | 1.306e-05 |
| re500_e2 | 500 | 2 | 1 | 2 | B2_temporal | 2.04 | 1 | 1 | 1 | 4.199e-16 | 0.549 | 0.718 | -0.86 | -0.0421 | -0.0437 | 0.00232 | 0 | 0 | 2 | 0.182 | 0.00805 | 0.077 | 4.019e-07 | 0.000254 |
| re500_e2 | 500 | 2 | 1 | 2 | B2_temporal | 2.04 | 1 | 1 | 1 | 4.199e-16 | 0.549 | 0.718 | -0.86 | -0.0421 | -0.0437 | 0.00232 | 0 | 0 | 3 | 0.000435 | 0.0594 | 0.000933 | 0.01 | 3.322e-05 |
| re500_e2 | 500 | 2 | 1 | 2 | B2_temporal | 2.04 | 1 | 1 | 1 | 4.199e-16 | 0.549 | 0.718 | -0.86 | -0.0421 | -0.0437 | 0.00232 | 0 | 0 | 4 | 0.000469 | 0.00071 | 0.0133 | 5.112e-05 | 0.000404 |
| re500_e2 | 500 | 2 | 1 | 2 | B2_temporal | 2.04 | 1 | 1 | 1 | 4.199e-16 | 0.549 | 0.718 | -0.86 | -0.0421 | -0.0437 | 0.00232 | 0 | 0 | 5 | 1.326e-06 | 7.345e-05 | 3.549e-06 | 0.00126 | 1.608e-09 |
| re500_e2 | 500 | 2 | 2 | 1 | B1_Mode2 | 1.97 | 1 | 1 | 0 | 4.237e-16 | 0.354 | 1.85 | -0.527 | -0.191 | 0.0797 | -0.0195 | 0 | 0 | 0 | 0.0318 | 0.00137 | 0.00011 | 1.780e-05 | 1.946e-05 |
| re500_e2 | 500 | 2 | 2 | 1 | B1_Mode2 | 1.97 | 1 | 1 | 0 | 4.237e-16 | 0.354 | 1.85 | -0.527 | -0.191 | 0.0797 | -0.0195 | 0 | 0 | 1 | 0.866 | 0.013 | 1.189e-05 | 0.000172 | 8.580e-10 |
| re500_e2 | 500 | 2 | 2 | 1 | B1_Mode2 | 1.97 | 1 | 1 | 0 | 4.237e-16 | 0.354 | 1.85 | -0.527 | -0.191 | 0.0797 | -0.0195 | 0 | 0 | 2 | 0.0704 | 0.000448 | 0.000938 | 3.125e-05 | 4.901e-05 |
| re500_e2 | 500 | 2 | 2 | 1 | B1_Mode2 | 1.97 | 1 | 1 | 0 | 4.237e-16 | 0.354 | 1.85 | -0.527 | -0.191 | 0.0797 | -0.0195 | 0 | 0 | 3 | 0.00924 | 0.00363 | 3.463e-05 | 4.095e-05 | 1.419e-06 |
| re500_e2 | 500 | 2 | 2 | 1 | B1_Mode2 | 1.97 | 1 | 1 | 0 | 4.237e-16 | 0.354 | 1.85 | -0.527 | -0.191 | 0.0797 | -0.0195 | 0 | 0 | 4 | 0.00161 | 0.00113 | 1.407e-05 | 6.521e-05 | 1.957e-06 |
| re500_e2 | 500 | 2 | 2 | 1 | B1_Mode2 | 1.97 | 1 | 1 | 0 | 4.237e-16 | 0.354 | 1.85 | -0.527 | -0.191 | 0.0797 | -0.0195 | 0 | 0 | 5 | 9.600e-05 | 2.385e-05 | 5.825e-07 | 3.942e-06 | 5.920e-07 |
| re500_e2 | 500 | 2 | 3 | 1 | B1_Mode2 | 1.91 | 1 | 1 | 0 | 4.107e-16 | 0.328 | 1.58 | -0.439 | -0.0921 | 0.00602 | -0.0195 | 0 | 0 | 0 | 0.0282 | 0.127 | 0.00211 | 0.000249 | 4.181e-05 |
| re500_e2 | 500 | 2 | 3 | 1 | B1_Mode2 | 1.91 | 1 | 1 | 0 | 4.107e-16 | 0.328 | 1.58 | -0.439 | -0.0921 | 0.00602 | -0.0195 | 0 | 0 | 1 | 0.652 | 0.0633 | 0.0191 | 0.000155 | 0.000792 |
| re500_e2 | 500 | 2 | 3 | 1 | B1_Mode2 | 1.91 | 1 | 1 | 0 | 4.107e-16 | 0.328 | 1.58 | -0.439 | -0.0921 | 0.00602 | -0.0195 | 0 | 0 | 2 | 0.0505 | 0.0408 | 0.00444 | 7.074e-05 | 4.549e-05 |
| re500_e2 | 500 | 2 | 3 | 1 | B1_Mode2 | 1.91 | 1 | 1 | 0 | 4.107e-16 | 0.328 | 1.58 | -0.439 | -0.0921 | 0.00602 | -0.0195 | 0 | 0 | 3 | 0.00223 | 0.00205 | 0.00459 | 0.000202 | 4.217e-05 |
| re500_e2 | 500 | 2 | 3 | 1 | B1_Mode2 | 1.91 | 1 | 1 | 0 | 4.107e-16 | 0.328 | 1.58 | -0.439 | -0.0921 | 0.00602 | -0.0195 | 0 | 0 | 4 | 9.496e-06 | 0.000966 | 0.000353 | 0.00024 | 3.564e-06 |
| re500_e2 | 500 | 2 | 3 | 1 | B1_Mode2 | 1.91 | 1 | 1 | 0 | 4.107e-16 | 0.328 | 1.58 | -0.439 | -0.0921 | 0.00602 | -0.0195 | 0 | 0 | 5 | 9.965e-05 | 0.000124 | 4.934e-05 | 4.491e-05 | 1.566e-06 |
| re500_e2 | 500 | 2 | 4 | 2 | B2_temporal | 1.89 | 1 | 0 | 1 | 4.386e-16 | 0.401 | -0.996 | -0.583 | 0.236 | 0.078 | -0.0228 | 0 | 0 | 0 | 0.0425 | 0.273 | 0.00316 | 0.00512 | 0.000609 |
| re500_e2 | 500 | 2 | 4 | 2 | B2_temporal | 1.89 | 1 | 0 | 1 | 4.386e-16 | 0.401 | -0.996 | -0.583 | 0.236 | 0.078 | -0.0228 | 0 | 0 | 1 | 0.263 | 0.175 | 0.0171 | 0.000494 | 5.302e-05 |
| re500_e2 | 500 | 2 | 4 | 2 | B2_temporal | 1.89 | 1 | 0 | 1 | 4.386e-16 | 0.401 | -0.996 | -0.583 | 0.236 | 0.078 | -0.0228 | 0 | 0 | 2 | 0.09 | 0.0723 | 0.0048 | 0.00314 | 0.000209 |
| re500_e2 | 500 | 2 | 4 | 2 | B2_temporal | 1.89 | 1 | 0 | 1 | 4.386e-16 | 0.401 | -0.996 | -0.583 | 0.236 | 0.078 | -0.0228 | 0 | 0 | 3 | 0.0147 | 0.00856 | 0.0154 | 4.889e-09 | 9.719e-07 |
| re500_e2 | 500 | 2 | 4 | 2 | B2_temporal | 1.89 | 1 | 0 | 1 | 4.386e-16 | 0.401 | -0.996 | -0.583 | 0.236 | 0.078 | -0.0228 | 0 | 0 | 4 | 0.00161 | 0.00754 | 0.000112 | 0.00102 | 1.715e-07 |
| re500_e2 | 500 | 2 | 4 | 2 | B2_temporal | 1.89 | 1 | 0 | 1 | 4.386e-16 | 0.401 | -0.996 | -0.583 | 0.236 | 0.078 | -0.0228 | 0 | 0 | 5 | 0.000138 | 5.518e-05 | 0.000446 | 1.512e-07 | 3.331e-05 |
| re700_e0.1 | 700 | 0.1 | 0 | 1 | B1_Mode2 | 0.125 | 1 | 1 | 0 | 1.014e-16 | 0.0196 | -0.48 | 0.0182 | 0.132 | -0.00326 | 0.00475 | 0 | 0 | 0 | 0.00153 | 0.0034 | 2.343e-05 | 5.623e-05 | 4.193e-07 |
| re700_e0.1 | 700 | 0.1 | 0 | 1 | B1_Mode2 | 0.125 | 1 | 1 | 0 | 1.014e-16 | 0.0196 | -0.48 | 0.0182 | 0.132 | -0.00326 | 0.00475 | 0 | 0 | 1 | 0.92 | 0.000184 | 8.933e-05 | 3.304e-05 | 3.891e-06 |
| re700_e0.1 | 700 | 0.1 | 0 | 1 | B1_Mode2 | 0.125 | 1 | 1 | 0 | 1.014e-16 | 0.0196 | -0.48 | 0.0182 | 0.132 | -0.00326 | 0.00475 | 0 | 0 | 2 | 0.00132 | 5.431e-05 | 7.223e-05 | 2.336e-05 | 1.281e-06 |
| re700_e0.1 | 700 | 0.1 | 0 | 1 | B1_Mode2 | 0.125 | 1 | 1 | 0 | 1.014e-16 | 0.0196 | -0.48 | 0.0182 | 0.132 | -0.00326 | 0.00475 | 0 | 0 | 3 | 0.07 | 0.00277 | 0.00011 | 2.780e-05 | 7.957e-06 |
| re700_e0.1 | 700 | 0.1 | 0 | 1 | B1_Mode2 | 0.125 | 1 | 1 | 0 | 1.014e-16 | 0.0196 | -0.48 | 0.0182 | 0.132 | -0.00326 | 0.00475 | 0 | 0 | 4 | 4.254e-05 | 0.000127 | 1.305e-06 | 6.259e-07 | 2.620e-07 |
| re700_e0.1 | 700 | 0.1 | 0 | 1 | B1_Mode2 | 0.125 | 1 | 1 | 0 | 1.014e-16 | 0.0196 | -0.48 | 0.0182 | 0.132 | -0.00326 | 0.00475 | 0 | 0 | 5 | 9.017e-05 | 0.000129 | 1.610e-05 | 9.230e-09 | 7.644e-07 |
| re700_e0.1 | 700 | 0.1 | 1 | 1 | B1_Mode2 | 0.124 | 1 | 1 | 0 | 1.050e-16 | 0.00115 | 0.475 | 0.0632 | -0.124 | -0.0317 | -0.0116 | 0 | 0 | 0 | 5.366e-06 | 2.360e-05 | 4.212e-05 | 2.554e-09 | 3.267e-06 |
| re700_e0.1 | 700 | 0.1 | 1 | 1 | B1_Mode2 | 0.124 | 1 | 1 | 0 | 1.050e-16 | 0.00115 | 0.475 | 0.0632 | -0.124 | -0.0317 | -0.0116 | 0 | 0 | 1 | 0.908 | 0.00525 | 0.000111 | 5.421e-05 | 8.891e-06 |
| re700_e0.1 | 700 | 0.1 | 1 | 1 | B1_Mode2 | 0.124 | 1 | 1 | 0 | 1.050e-16 | 0.00115 | 0.475 | 0.0632 | -0.124 | -0.0317 | -0.0116 | 0 | 0 | 2 | 0.0161 | 0.00162 | 3.116e-05 | 2.021e-05 | 2.758e-06 |
| re700_e0.1 | 700 | 0.1 | 1 | 1 | B1_Mode2 | 0.124 | 1 | 1 | 0 | 1.050e-16 | 0.00115 | 0.475 | 0.0632 | -0.124 | -0.0317 | -0.0116 | 0 | 0 | 3 | 0.0625 | 0.00134 | 0.00014 | 8.273e-06 | 8.599e-06 |
| re700_e0.1 | 700 | 0.1 | 1 | 1 | B1_Mode2 | 0.124 | 1 | 1 | 0 | 1.050e-16 | 0.00115 | 0.475 | 0.0632 | -0.124 | -0.0317 | -0.0116 | 0 | 0 | 4 | 0.00405 | 1.683e-06 | 2.007e-05 | 2.119e-07 | 9.253e-07 |
| re700_e0.1 | 700 | 0.1 | 1 | 1 | B1_Mode2 | 0.124 | 1 | 1 | 0 | 1.050e-16 | 0.00115 | 0.475 | 0.0632 | -0.124 | -0.0317 | -0.0116 | 0 | 0 | 5 | 0.000543 | 2.851e-05 | 8.207e-06 | 1.158e-08 | 2.185e-09 |
| re700_e0.1 | 700 | 0.1 | 2 | 1 | B1_Mode2 | 0.124 | 1 | 1 | 0 | 9.726e-17 | 0.017 | 0.47 | -0.0348 | -0.156 | 0.0122 | 0.00772 | 0 | 0 | 0 | 0.00116 | 0.000953 | 7.609e-06 | 9.761e-06 | 4.275e-07 |
| re700_e0.1 | 700 | 0.1 | 2 | 1 | B1_Mode2 | 0.124 | 1 | 1 | 0 | 9.726e-17 | 0.017 | 0.47 | -0.0348 | -0.156 | 0.0122 | 0.00772 | 0 | 0 | 1 | 0.889 | 0.00453 | 1.772e-05 | 6.528e-05 | 4.007e-07 |
| re700_e0.1 | 700 | 0.1 | 2 | 1 | B1_Mode2 | 0.124 | 1 | 1 | 0 | 9.726e-17 | 0.017 | 0.47 | -0.0348 | -0.156 | 0.0122 | 0.00772 | 0 | 0 | 2 | 0.00487 | 2.805e-05 | 2.892e-06 | 1.217e-06 | 3.586e-07 |
| re700_e0.1 | 700 | 0.1 | 2 | 1 | B1_Mode2 | 0.124 | 1 | 1 | 0 | 9.726e-17 | 0.017 | 0.47 | -0.0348 | -0.156 | 0.0122 | 0.00772 | 0 | 0 | 3 | 0.098 | 0.000682 | 7.986e-05 | 3.533e-06 | 5.039e-06 |
| re700_e0.1 | 700 | 0.1 | 2 | 1 | B1_Mode2 | 0.124 | 1 | 1 | 0 | 9.726e-17 | 0.017 | 0.47 | -0.0348 | -0.156 | 0.0122 | 0.00772 | 0 | 0 | 4 | 0.000602 | 7.265e-06 | 1.677e-06 | 7.358e-08 | 2.367e-07 |
| re700_e0.1 | 700 | 0.1 | 2 | 1 | B1_Mode2 | 0.124 | 1 | 1 | 0 | 9.726e-17 | 0.017 | 0.47 | -0.0348 | -0.156 | 0.0122 | 0.00772 | 0 | 0 | 5 | 0.00024 | 0.000104 | 1.955e-06 | 4.428e-07 | 4.843e-07 |
| re700_e0.1 | 700 | 0.1 | 3 | 2 | B2_temporal | 0.122 | 1 | 0 | 1 | 1.220e-16 | -0.0244 | 0.0451 | -0.00155 | 0.00015 | -0.00126 | 0.00121 | 0 | 0 | 0 | 0.00243 | 0.972 | 0.00659 | 0.000121 | 1.076e-05 |
| re700_e0.1 | 700 | 0.1 | 3 | 2 | B2_temporal | 0.122 | 1 | 0 | 1 | 1.220e-16 | -0.0244 | 0.0451 | -0.00155 | 0.00015 | -0.00126 | 0.00121 | 0 | 0 | 1 | 0.00833 | 0.00732 | 0.000314 | 0.00041 | 6.220e-05 |
| re700_e0.1 | 700 | 0.1 | 3 | 2 | B2_temporal | 0.122 | 1 | 0 | 1 | 1.220e-16 | -0.0244 | 0.0451 | -0.00155 | 0.00015 | -0.00126 | 0.00121 | 0 | 0 | 2 | 9.862e-06 | 0.00185 | 0.000496 | 6.588e-05 | 4.725e-07 |
| re700_e0.1 | 700 | 0.1 | 3 | 2 | B2_temporal | 0.122 | 1 | 0 | 1 | 1.220e-16 | -0.0244 | 0.0451 | -0.00155 | 0.00015 | -0.00126 | 0.00121 | 0 | 0 | 3 | 9.234e-08 | 4.676e-05 | 1.601e-05 | 1.439e-05 | 3.945e-07 |
| re700_e0.1 | 700 | 0.1 | 3 | 2 | B2_temporal | 0.122 | 1 | 0 | 1 | 1.220e-16 | -0.0244 | 0.0451 | -0.00155 | 0.00015 | -0.00126 | 0.00121 | 0 | 0 | 4 | 6.516e-06 | 1.027e-06 | 1.320e-07 | 1.682e-05 | 6.512e-09 |
| re700_e0.1 | 700 | 0.1 | 3 | 2 | B2_temporal | 0.122 | 1 | 0 | 1 | 1.220e-16 | -0.0244 | 0.0451 | -0.00155 | 0.00015 | -0.00126 | 0.00121 | 0 | 0 | 5 | 5.963e-06 | 8.832e-06 | 1.326e-08 | 1.498e-06 | 1.650e-06 |
| re700_e0.1 | 700 | 0.1 | 4 | 2 | B2_temporal | 0.123 | 1 | 0 | 1 | 1.065e-16 | -0.00168 | 0.0599 | -0.00196 | 0.000856 | -0.00242 | 0.00165 | 0 | 0 | 0 | 1.153e-05 | 0.975 | 0.00125 | 0.000142 | 0.000181 |
| re700_e0.1 | 700 | 0.1 | 4 | 2 | B2_temporal | 0.123 | 1 | 0 | 1 | 1.065e-16 | -0.00168 | 0.0599 | -0.00196 | 0.000856 | -0.00242 | 0.00165 | 0 | 0 | 1 | 0.0146 | 8.798e-06 | 0.000301 | 0.00116 | 2.608e-05 |
| re700_e0.1 | 700 | 0.1 | 4 | 2 | B2_temporal | 0.123 | 1 | 0 | 1 | 1.065e-16 | -0.00168 | 0.0599 | -0.00196 | 0.000856 | -0.00242 | 0.00165 | 0 | 0 | 2 | 1.569e-05 | 0.00687 | 3.280e-05 | 0.000369 | 2.117e-05 |
| re700_e0.1 | 700 | 0.1 | 4 | 2 | B2_temporal | 0.123 | 1 | 0 | 1 | 1.065e-16 | -0.00168 | 0.0599 | -0.00196 | 0.000856 | -0.00242 | 0.00165 | 0 | 0 | 3 | 2.989e-06 | 0.000124 | 6.990e-05 | 1.920e-05 | 2.240e-08 |
| re700_e0.1 | 700 | 0.1 | 4 | 2 | B2_temporal | 0.123 | 1 | 0 | 1 | 1.065e-16 | -0.00168 | 0.0599 | -0.00196 | 0.000856 | -0.00242 | 0.00165 | 0 | 0 | 4 | 2.393e-05 | 2.781e-07 | 1.721e-05 | 9.363e-09 | 1.572e-06 |
| re700_e0.1 | 700 | 0.1 | 4 | 2 | B2_temporal | 0.123 | 1 | 0 | 1 | 1.065e-16 | -0.00168 | 0.0599 | -0.00196 | 0.000856 | -0.00242 | 0.00165 | 0 | 0 | 5 | 1.111e-05 | 8.544e-06 | 5.824e-07 | 5.561e-07 | 6.747e-08 |
| re700_e0.25 | 700 | 0.25 | 0 | 3 | B3_mixed | 0.245 | 1 | 1 | 0 | 1.371e-16 | 0.0453 | -0.513 | 0.0146 | 0.151 | -0.0225 | -0.00487 | 0 | 0 | 0 | 0.0042 | 0.325 | 0.00416 | 0.00308 | 0.000509 |
| re700_e0.25 | 700 | 0.25 | 0 | 3 | B3_mixed | 0.245 | 1 | 1 | 0 | 1.371e-16 | 0.0453 | -0.513 | 0.0146 | 0.151 | -0.0225 | -0.00487 | 0 | 0 | 1 | 0.538 | 0.00146 | 0.00211 | 0.000537 | 5.590e-06 |
| re700_e0.25 | 700 | 0.25 | 0 | 3 | B3_mixed | 0.245 | 1 | 1 | 0 | 1.371e-16 | 0.0453 | -0.513 | 0.0146 | 0.151 | -0.0225 | -0.00487 | 0 | 0 | 2 | 0.000439 | 0.0514 | 0.0102 | 0.000574 | 1.006e-05 |
| re700_e0.25 | 700 | 0.25 | 0 | 3 | B3_mixed | 0.245 | 1 | 1 | 0 | 1.371e-16 | 0.0453 | -0.513 | 0.0146 | 0.151 | -0.0225 | -0.00487 | 0 | 0 | 3 | 0.0469 | 0.00643 | 0.000587 | 0.00013 | 1.728e-05 |
| re700_e0.25 | 700 | 0.25 | 0 | 3 | B3_mixed | 0.245 | 1 | 1 | 0 | 1.371e-16 | 0.0453 | -0.513 | 0.0146 | 0.151 | -0.0225 | -0.00487 | 0 | 0 | 4 | 0.00104 | 0.000791 | 0.00162 | 2.478e-05 | 8.890e-07 |
| re700_e0.25 | 700 | 0.25 | 0 | 3 | B3_mixed | 0.245 | 1 | 1 | 0 | 1.371e-16 | 0.0453 | -0.513 | 0.0146 | 0.151 | -0.0225 | -0.00487 | 0 | 0 | 5 | 4.857e-05 | 0.000861 | 0.000111 | 3.416e-05 | 1.336e-09 |
| re700_e0.25 | 700 | 0.25 | 1 | 1 | B1_Mode2 | 0.268 | 1 | 1 | 0 | 1.610e-16 | 0.00858 | 0.702 | 0.0573 | -0.176 | -0.0329 | -0.00597 | 0 | 0 | 0 | 0.000137 | 8.732e-06 | 1.834e-05 | 3.135e-06 | 1.891e-06 |
| re700_e0.25 | 700 | 0.25 | 1 | 1 | B1_Mode2 | 0.268 | 1 | 1 | 0 | 1.610e-16 | 0.00858 | 0.702 | 0.0573 | -0.176 | -0.0329 | -0.00597 | 0 | 0 | 1 | 0.92 | 0.00641 | 0.000208 | 6.531e-05 | 1.531e-05 |
| re700_e0.25 | 700 | 0.25 | 1 | 1 | B1_Mode2 | 0.268 | 1 | 1 | 0 | 1.610e-16 | 0.00858 | 0.702 | 0.0573 | -0.176 | -0.0329 | -0.00597 | 0 | 0 | 2 | 0.00613 | 0.0034 | 1.461e-07 | 4.034e-05 | 2.365e-07 |
| re700_e0.25 | 700 | 0.25 | 1 | 1 | B1_Mode2 | 0.268 | 1 | 1 | 0 | 1.610e-16 | 0.00858 | 0.702 | 0.0573 | -0.176 | -0.0329 | -0.00597 | 0 | 0 | 3 | 0.0579 | 0.00353 | 0.000229 | 2.382e-05 | 1.206e-05 |
| re700_e0.25 | 700 | 0.25 | 1 | 1 | B1_Mode2 | 0.268 | 1 | 1 | 0 | 1.610e-16 | 0.00858 | 0.702 | 0.0573 | -0.176 | -0.0329 | -0.00597 | 0 | 0 | 4 | 0.00202 | 3.288e-07 | 3.393e-05 | 3.066e-07 | 1.826e-06 |
| re700_e0.25 | 700 | 0.25 | 1 | 1 | B1_Mode2 | 0.268 | 1 | 1 | 0 | 1.610e-16 | 0.00858 | 0.702 | 0.0573 | -0.176 | -0.0329 | -0.00597 | 0 | 0 | 5 | 6.648e-05 | 0.000113 | 2.004e-05 | 1.149e-07 | 1.596e-07 |

Table 15: Décomposition modale — 185 points, 1110 lignes (6 modes $i$ par point, colonnes $E^f_{iJ}$ pour $J$=0..4), `mode_analysis.csv`.

| point | Re | $E^*$ | s | id | branch | $E_t$ | cov. | $i$ | $j$ | rmse | $A_{10}$ | $A_{20}$ | $A_{30}$ | $A_{40}$ | $A_{50}$ | $A_{60}$ | $A_{70}$ | $A_{80}$ | $i$ | $E^f_{0J}$ | $E^f_{1J}$ | $E^f_{2J}$ | $E^f_{3J}$ | $E^f_{4J}$ |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| re700_e0.25 | 700 | 0.25 | 2 | 1 | B1_Mode2 | 0.263 | 1 | 1 | 0 | 1.878e-16 | 0.0376 | 0.687 | -0.0755 | -0.199 | 0.028 | 0.00347 | 0 | 0 | 0 | 0.00269 | 0.00283 | 1.228e-06 | 2.529e-05 | 3.092e-07 |
| re700_e0.25 | 700 | 0.25 | 2 | 1 | B1_Mode2 | 0.263 | 1 | 1 | 0 | 1.878e-16 | 0.0376 | 0.687 | -0.0755 | -0.199 | 0.028 | 0.00347 | 0 | 0 | 1 | 0.897 | 0.0041 | 0.000267 | 6.010e-05 | 1.224e-05 |
| re700_e0.25 | 700 | 0.25 | 2 | 1 | B1_Mode2 | 0.263 | 1 | 1 | 0 | 1.878e-16 | 0.0376 | 0.687 | -0.0755 | -0.199 | 0.028 | 0.00347 | 0 | 0 | 2 | 0.0109 | 0.000431 | 4.954e-06 | 9.481e-06 | 2.142e-07 |
| re700_e0.25 | 700 | 0.25 | 2 | 1 | B1_Mode2 | 0.263 | 1 | 1 | 0 | 1.878e-16 | 0.0376 | 0.687 | -0.0755 | -0.199 | 0.028 | 0.00347 | 0 | 0 | 3 | 0.0752 | 0.00445 | 0.000195 | 3.142e-05 | 1.053e-05 |
| re700_e0.25 | 700 | 0.25 | 2 | 1 | B1_Mode2 | 0.263 | 1 | 1 | 0 | 1.878e-16 | 0.0376 | 0.687 | -0.0755 | -0.199 | 0.028 | 0.00347 | 0 | 0 | 4 | 0.00149 | 9.619e-06 | 1.257e-06 | 1.354e-08 | 1.316e-06 |
| re700_e0.25 | 700 | 0.25 | 2 | 1 | B1_Mode2 | 0.263 | 1 | 1 | 0 | 1.878e-16 | 0.0376 | 0.687 | -0.0755 | -0.199 | 0.028 | 0.00347 | 0 | 0 | 5 | 2.285e-05 | 0.000603 | 3.122e-05 | 1.431e-06 | 4.037e-06 |
| re700_e0.25 | 700 | 0.25 | 3 | 2 | B2_temporal | 0.266 | 1 | 0 | 1 | 1.518e-16 | -0.0484 | 0.0609 | 0.0253 | -0.00705 | 0.00141 | 0.000101 | 0 | 0 | 0 | 0.00441 | 0.976 | 0.00178 | 0.000403 | 9.548e-06 |
| re700_e0.25 | 700 | 0.25 | 3 | 2 | B2_temporal | 0.266 | 1 | 0 | 1 | 1.518e-16 | -0.0484 | 0.0609 | 0.0253 | -0.00705 | 0.00141 | 0.000101 | 0 | 0 | 1 | 0.00699 | 0.00329 | 0.000177 | 0.00169 | 9.545e-06 |
| re700_e0.25 | 700 | 0.25 | 3 | 2 | B2_temporal | 0.266 | 1 | 0 | 1 | 1.518e-16 | -0.0484 | 0.0609 | 0.0253 | -0.00705 | 0.00141 | 0.000101 | 0 | 0 | 2 | 0.00121 | 0.00319 | 0.000259 | 0.000131 | 5.266e-05 |
| re700_e0.25 | 700 | 0.25 | 3 | 2 | B2_temporal | 0.266 | 1 | 0 | 1 | 1.518e-16 | -0.0484 | 0.0609 | 0.0253 | -0.00705 | 0.00141 | 0.000101 | 0 | 0 | 3 | 9.352e-05 | 2.468e-06 | 3.732e-07 | 1.167e-06 | 8.446e-06 |
| re700_e0.25 | 700 | 0.25 | 3 | 2 | B2_temporal | 0.266 | 1 | 0 | 1 | 1.518e-16 | -0.0484 | 0.0609 | 0.0253 | -0.00705 | 0.00141 | 0.000101 | 0 | 0 | 4 | 3.737e-06 | 4.093e-05 | 9.919e-09 | 1.544e-05 | 8.958e-07 |
| re700_e0.25 | 700 | 0.25 | 3 | 2 | B2_temporal | 0.266 | 1 | 0 | 1 | 1.518e-16 | -0.0484 | 0.0609 | 0.0253 | -0.00705 | 0.00141 | 0.000101 | 0 | 0 | 5 | 1.904e-08 | 1.744e-06 | 3.721e-06 | 4.642e-08 | 2.765e-06 |
| re700_e0.25 | 700 | 0.25 | 4 | 2 | B2_temporal | 0.258 | 1 | 0 | 1 | 1.403e-16 | 0.00101 | -0.0111 | -0.00329 | 0.00611 | -0.000475 | 0.00211 | 0 | 0 | 0 | 1.987e-06 | 0.982 | 0.000169 | 0.000121 | 0.000228 |
| re700_e0.25 | 700 | 0.25 | 4 | 2 | B2_temporal | 0.258 | 1 | 0 | 1 | 1.403e-16 | 0.00101 | -0.0111 | -0.00329 | 0.00611 | -0.000475 | 0.00211 | 0 | 0 | 1 | 0.00024 | 0.000912 | 0.000277 | 0.00336 | 1.253e-06 |
| re700_e0.25 | 700 | 0.25 | 4 | 2 | B2_temporal | 0.258 | 1 | 0 | 1 | 1.403e-16 | 0.00101 | -0.0111 | -0.00329 | 0.00611 | -0.000475 | 0.00211 | 0 | 0 | 2 | 2.094e-05 | 0.0118 | 0.000419 | 0.000297 | 9.406e-07 |
| re700_e0.25 | 700 | 0.25 | 4 | 2 | B2_temporal | 0.258 | 1 | 0 | 1 | 1.403e-16 | 0.00101 | -0.0111 | -0.00329 | 0.00611 | -0.000475 | 0.00211 | 0 | 0 | 3 | 7.232e-05 | 9.456e-05 | 2.986e-06 | 4.640e-05 | 1.452e-05 |
| re700_e0.25 | 700 | 0.25 | 4 | 2 | B2_temporal | 0.258 | 1 | 0 | 1 | 1.403e-16 | 0.00101 | -0.0111 | -0.00329 | 0.00611 | -0.000475 | 0.00211 | 0 | 0 | 4 | 4.371e-07 | 2.436e-05 | 1.256e-05 | 1.702e-06 | 6.089e-06 |
| re700_e0.25 | 700 | 0.25 | 4 | 2 | B2_temporal | 0.258 | 1 | 0 | 1 | 1.403e-16 | 0.00101 | -0.0111 | -0.00329 | 0.00611 | -0.000475 | 0.00211 | 0 | 0 | 5 | 8.636e-06 | 6.852e-06 | 9.836e-07 | 3.545e-06 | 5.985e-08 |
| re700_e0.5 | 700 | 0.5 | 0 | 3 | B3_mixed | 0.461 | 1 | 0 | 1 | 2.141e-16 | 0.0726 | -0.588 | 0.0134 | 0.103 | -0.0326 | 0.0265 | 0 | 0 | 0 | 0.00571 | 0.479 | 0.0022 | 0.00528 | 0.000634 |
| re700_e0.5 | 700 | 0.5 | 0 | 3 | B3_mixed | 0.461 | 1 | 0 | 1 | 2.141e-16 | 0.0726 | -0.588 | 0.0134 | 0.103 | -0.0326 | 0.0265 | 0 | 0 | 1 | 0.375 | 0.0128 | 0.00866 | 0.00257 | 6.256e-06 |
| re700_e0.5 | 700 | 0.5 | 0 | 3 | B3_mixed | 0.461 | 1 | 0 | 1 | 2.141e-16 | 0.0726 | -0.588 | 0.0134 | 0.103 | -0.0326 | 0.0265 | 0 | 0 | 2 | 0.000195 | 0.0664 | 0.0105 | 0.0027 | 0.000171 |
| re700_e0.5 | 700 | 0.5 | 0 | 3 | B3_mixed | 0.461 | 1 | 0 | 1 | 2.141e-16 | 0.0726 | -0.588 | 0.0134 | 0.103 | -0.0326 | 0.0265 | 0 | 0 | 3 | 0.0115 | 0.00424 | 0.00461 | 0.00147 | 5.212e-06 |
| re700_e0.5 | 700 | 0.5 | 0 | 3 | B3_mixed | 0.461 | 1 | 0 | 1 | 2.141e-16 | 0.0726 | -0.588 | 0.0134 | 0.103 | -0.0326 | 0.0265 | 0 | 0 | 4 | 0.00115 | 0.00122 | 0.00223 | 2.302e-05 | 0.000151 |
| re700_e0.5 | 700 | 0.5 | 0 | 3 | B3_mixed | 0.461 | 1 | 0 | 1 | 2.141e-16 | 0.0726 | -0.588 | 0.0134 | 0.103 | -0.0326 | 0.0265 | 0 | 0 | 5 | 0.000761 | 0.000511 | 1.366e-05 | 0.00027 | 5.048e-07 |
| re700_e0.5 | 700 | 0.5 | 1 | 1 | B1_Mode2 | 0.519 | 1 | 1 | 0 | 1.942e-16 | 0.00772 | 0.987 | 0.0608 | -0.174 | -0.0362 | 9.708e-06 | 0 | 0 | 0 | 5.747e-05 | 0.00205 | 1.645e-05 | 2.039e-05 | 1.776e-06 |
| re700_e0.5 | 700 | 0.5 | 1 | 1 | B1_Mode2 | 0.519 | 1 | 1 | 0 | 1.942e-16 | 0.00772 | 0.987 | 0.0608 | -0.174 | -0.0362 | 9.708e-06 | 0 | 0 | 1 | 0.939 | 0.00471 | 0.000621 | 3.721e-05 | 3.522e-05 |
| re700_e0.5 | 700 | 0.5 | 1 | 1 | B1_Mode2 | 0.519 | 1 | 1 | 0 | 1.942e-16 | 0.00772 | 0.987 | 0.0608 | -0.174 | -0.0362 | 9.708e-06 | 0 | 0 | 2 | 0.00356 | 0.0102 | 3.875e-06 | 7.536e-05 | 4.649e-07 |
| re700_e0.5 | 700 | 0.5 | 1 | 1 | B1_Mode2 | 0.519 | 1 | 1 | 0 | 1.942e-16 | 0.00772 | 0.987 | 0.0608 | -0.174 | -0.0362 | 9.708e-06 | 0 | 0 | 3 | 0.0292 | 0.00786 | 0.000475 | 3.510e-05 | 1.877e-05 |
| re700_e0.5 | 700 | 0.5 | 1 | 1 | B1_Mode2 | 0.519 | 1 | 1 | 0 | 1.942e-16 | 0.00772 | 0.987 | 0.0608 | -0.174 | -0.0362 | 9.708e-06 | 0 | 0 | 4 | 0.00126 | 0.000245 | 0.000245 | 8.182e-07 | 1.167e-05 |
| re700_e0.5 | 700 | 0.5 | 1 | 1 | B1_Mode2 | 0.519 | 1 | 1 | 0 | 1.942e-16 | 0.00772 | 0.987 | 0.0608 | -0.174 | -0.0362 | 9.708e-06 | 0 | 0 | 5 | 9.083e-11 | 0.000416 | 1.219e-05 | 6.712e-07 | 1.741e-07 |
| re700_e0.5 | 700 | 0.5 | 2 | 1 | B1_Mode2 | 0.511 | 1 | 1 | 0 | 2.139e-16 | 0.0383 | 0.972 | -0.0409 | -0.232 | 0.0403 | 0.00426 | 0 | 0 | 0 | 0.00144 | 0.00151 | 7.537e-06 | 3.808e-05 | 5.221e-07 |
| re700_e0.5 | 700 | 0.5 | 2 | 1 | B1_Mode2 | 0.511 | 1 | 1 | 0 | 2.139e-16 | 0.0383 | 0.972 | -0.0409 | -0.232 | 0.0403 | 0.00426 | 0 | 0 | 1 | 0.926 | 0.00137 | 0.000803 | 1.625e-05 | 3.392e-05 |
| re700_e0.5 | 700 | 0.5 | 2 | 1 | B1_Mode2 | 0.511 | 1 | 1 | 0 | 2.139e-16 | 0.0383 | 0.972 | -0.0409 | -0.232 | 0.0403 | 0.00426 | 0 | 0 | 2 | 0.00164 | 0.00282 | 0.000109 | 3.793e-05 | 4.257e-06 |
| re700_e0.5 | 700 | 0.5 | 2 | 1 | B1_Mode2 | 0.511 | 1 | 1 | 0 | 2.139e-16 | 0.0383 | 0.972 | -0.0409 | -0.232 | 0.0403 | 0.00426 | 0 | 0 | 3 | 0.0525 | 0.0093 | 0.000149 | 5.538e-05 | 5.861e-06 |
| re700_e0.5 | 700 | 0.5 | 2 | 1 | B1_Mode2 | 0.511 | 1 | 1 | 0 | 2.139e-16 | 0.0383 | 0.972 | -0.0409 | -0.232 | 0.0403 | 0.00426 | 0 | 0 | 4 | 0.00159 | 8.613e-05 | 7.464e-05 | 1.533e-06 | 1.014e-05 |
| re700_e0.5 | 700 | 0.5 | 2 | 1 | B1_Mode2 | 0.511 | 1 | 1 | 0 | 2.139e-16 | 0.0383 | 0.972 | -0.0409 | -0.232 | 0.0403 | 0.00426 | 0 | 0 | 5 | 1.777e-05 | 0.000703 | 4.125e-05 | 1.067e-06 | 7.125e-06 |
| re700_e0.5 | 700 | 0.5 | 3 | 2 | B2_temporal | 0.52 | 1 | 0 | 1 | 2.051e-16 | -0.0672 | -0.0532 | 0.0181 | -0.0127 | -0.000292 | 0.000682 | 0 | 0 | 0 | 0.00435 | 0.987 | 0.000124 | 0.000116 | 0.000166 |
| re700_e0.5 | 700 | 0.5 | 3 | 2 | B2_temporal | 0.52 | 1 | 0 | 1 | 2.051e-16 | -0.0672 | -0.0532 | 0.0181 | -0.0127 | -0.000292 | 0.000682 | 0 | 0 | 1 | 0.00272 | 0.00203 | 4.156e-05 | 0.00216 | 2.268e-05 |
| re700_e0.5 | 700 | 0.5 | 3 | 2 | B2_temporal | 0.52 | 1 | 0 | 1 | 2.051e-16 | -0.0672 | -0.0532 | 0.0181 | -0.0127 | -0.000292 | 0.000682 | 0 | 0 | 2 | 0.000315 | 0.000411 | 0.000155 | 6.996e-07 | 3.740e-05 |
| re700_e0.5 | 700 | 0.5 | 3 | 2 | B2_temporal | 0.52 | 1 | 0 | 1 | 2.051e-16 | -0.0672 | -0.0532 | 0.0181 | -0.0127 | -0.000292 | 0.000682 | 0 | 0 | 3 | 0.000156 | 2.340e-06 | 3.532e-06 | 4.780e-07 | 1.837e-06 |
| re700_e0.5 | 700 | 0.5 | 3 | 2 | B2_temporal | 0.52 | 1 | 0 | 1 | 2.051e-16 | -0.0672 | -0.0532 | 0.0181 | -0.0127 | -0.000292 | 0.000682 | 0 | 0 | 4 | 8.221e-08 | 4.176e-05 | 2.789e-06 | 6.866e-06 | 2.866e-08 |
| re700_e0.5 | 700 | 0.5 | 3 | 2 | B2_temporal | 0.52 | 1 | 0 | 1 | 2.051e-16 | -0.0672 | -0.0532 | 0.0181 | -0.0127 | -0.000292 | 0.000682 | 0 | 0 | 5 | 4.479e-07 | 2.909e-07 | 1.693e-06 | 2.217e-07 | 2.829e-06 |
| re700_e0.5 | 700 | 0.5 | 4 | 2 | B2_temporal | 0.492 | 1 | 0 | 1 | 2.012e-16 | 0.0197 | -0.159 | -0.0251 | 0.0167 | 0.00321 | 0.00487 | 0 | 0 | 0 | 0.000395 | 0.899 | 0.0369 | 0.000616 | 0.000189 |
| re700_e0.5 | 700 | 0.5 | 4 | 2 | B2_temporal | 0.492 | 1 | 0 | 1 | 2.012e-16 | 0.0197 | -0.159 | -0.0251 | 0.0167 | 0.00321 | 0.00487 | 0 | 0 | 1 | 0.0256 | 1.073e-05 | 0.000214 | 0.00333 | 0.000647 |
| re700_e0.5 | 700 | 0.5 | 4 | 2 | B2_temporal | 0.492 | 1 | 0 | 1 | 2.012e-16 | 0.0197 | -0.159 | -0.0251 | 0.0167 | 0.00321 | 0.00487 | 0 | 0 | 2 | 0.000643 | 0.0219 | 0.00946 | 0.000127 | 1.218e-06 |
| re700_e0.5 | 700 | 0.5 | 4 | 2 | B2_temporal | 0.492 | 1 | 0 | 1 | 2.012e-16 | 0.0197 | -0.159 | -0.0251 | 0.0167 | 0.00321 | 0.00487 | 0 | 0 | 3 | 0.000282 | 3.004e-05 | 2.706e-05 | 0.000419 | 0.000274 |
| re700_e0.5 | 700 | 0.5 | 4 | 2 | B2_temporal | 0.492 | 1 | 0 | 1 | 2.012e-16 | 0.0197 | -0.159 | -0.0251 | 0.0167 | 0.00321 | 0.00487 | 0 | 0 | 4 | 1.049e-05 | 2.849e-06 | 1.239e-08 | 4.431e-06 | 3.006e-05 |
| re700_e0.5 | 700 | 0.5 | 4 | 2 | B2_temporal | 0.492 | 1 | 0 | 1 | 2.012e-16 | 0.0197 | -0.159 | -0.0251 | 0.0167 | 0.00321 | 0.00487 | 0 | 0 | 5 | 2.414e-05 | 4.039e-07 | 7.634e-06 | 9.435e-08 | 4.784e-06 |
| re700_e1 | 700 | 1 | 0 | 2 | B2_temporal | 0.972 | 1 | 1 | 1 | 2.935e-16 | 0.333 | -0.523 | -0.749 | 0.0751 | 0.107 | 0.00488 | 0 | 0 | 0 | 0.0571 | 0.00648 | 0.0046 | 7.842e-05 | 0.000141 |
| re700_e1 | 700 | 1 | 0 | 2 | B2_temporal | 0.972 | 1 | 1 | 1 | 2.935e-16 | 0.333 | -0.523 | -0.749 | 0.0751 | 0.107 | 0.00488 | 0 | 0 | 1 | 0.141 | 0.348 | 0.00822 | 8.815e-06 | 6.044e-05 |
| re700_e1 | 700 | 1 | 0 | 2 | B2_temporal | 0.972 | 1 | 1 | 1 | 2.935e-16 | 0.333 | -0.523 | -0.749 | 0.0751 | 0.107 | 0.00488 | 0 | 0 | 2 | 0.288 | 0.0475 | 0.0217 | 0.000459 | 0.000113 |
| re700_e1 | 700 | 1 | 0 | 2 | B2_temporal | 0.972 | 1 | 1 | 1 | 2.935e-16 | 0.333 | -0.523 | -0.749 | 0.0751 | 0.107 | 0.00488 | 0 | 0 | 3 | 0.0029 | 0.0272 | 0.0187 | 0.00175 | 4.275e-05 |
| re700_e1 | 700 | 1 | 0 | 2 | B2_temporal | 0.972 | 1 | 1 | 1 | 2.935e-16 | 0.333 | -0.523 | -0.749 | 0.0751 | 0.107 | 0.00488 | 0 | 0 | 4 | 0.0059 | 0.0137 | 0.00257 | 0.00198 | 4.921e-05 |
| re700_e1 | 700 | 1 | 0 | 2 | B2_temporal | 0.972 | 1 | 1 | 1 | 2.935e-16 | 0.333 | -0.523 | -0.749 | 0.0751 | 0.107 | 0.00488 | 0 | 0 | 5 | 1.226e-05 | 0.000582 | 0.00148 | 1.512e-05 | 5.381e-05 |
| re700_e1 | 700 | 1 | 1 | 1 | B1_Mode2 | 1.01 | 1 | 1 | 0 | 3.204e-16 | -0.00813 | 1.26 | 0.113 | -0.212 | -0.1 | -0.026 | 0 | 0 | 0 | 3.284e-05 | 0.0608 | 1.437e-06 | 0.000159 | 7.760e-07 |
| re700_e1 | 700 | 1 | 1 | 1 | B1_Mode2 | 1.01 | 1 | 1 | 0 | 3.204e-16 | -0.00813 | 1.26 | 0.113 | -0.212 | -0.1 | -0.026 | 0 | 0 | 1 | 0.784 | 0.0118 | 0.00445 | 5.437e-05 | 0.000245 |
| re700_e1 | 700 | 1 | 1 | 1 | B1_Mode2 | 1.01 | 1 | 1 | 0 | 3.204e-16 | -0.00813 | 1.26 | 0.113 | -0.212 | -0.1 | -0.026 | 0 | 0 | 2 | 0.00631 | 0.0671 | 0.000714 | 0.000232 | 2.348e-05 |
| re700_e1 | 700 | 1 | 1 | 1 | B1_Mode2 | 1.01 | 1 | 1 | 0 | 3.204e-16 | -0.00813 | 1.26 | 0.113 | -0.212 | -0.1 | -0.026 | 0 | 0 | 3 | 0.0223 | 0.0233 | 0.00406 | 2.541e-05 | 0.000151 |
| re700_e1 | 700 | 1 | 1 | 1 | B1_Mode2 | 1.01 | 1 | 1 | 0 | 3.204e-16 | -0.00813 | 1.26 | 0.113 | -0.212 | -0.1 | -0.026 | 0 | 0 | 4 | 0.00502 | 0.00152 | 0.00437 | 4.421e-05 | 0.000103 |
| re700_e1 | 700 | 1 | 1 | 1 | B1_Mode2 | 1.01 | 1 | 1 | 0 | 3.204e-16 | -0.00813 | 1.26 | 0.113 | -0.212 | -0.1 | -0.026 | 0 | 0 | 5 | 0.000335 | 0.00239 | 5.268e-06 | 0.000177 | 1.290e-07 |
| re700_e1 | 700 | 1 | 2 | 1 | B1_Mode2 | 1.03 | 1 | 1 | 0 | 2.908e-16 | 0.0523 | 1.4 | -0.00989 | -0.247 | -0.0178 | 0.0268 | 0 | 0 | 0 | 0.00133 | 0.00267 | 2.730e-06 | 2.436e-05 | 8.611e-07 |
| re700_e1 | 700 | 1 | 2 | 1 | B1_Mode2 | 1.03 | 1 | 1 | 0 | 2.908e-16 | 0.0523 | 1.4 | -0.00989 | -0.247 | -0.0178 | 0.0268 | 0 | 0 | 1 | 0.947 | 0.00425 | 0.00125 | 4.638e-07 | 6.355e-05 |
| re700_e1 | 700 | 1 | 2 | 1 | B1_Mode2 | 1.03 | 1 | 1 | 0 | 2.908e-16 | 0.0523 | 1.4 | -0.00989 | -0.247 | -0.0178 | 0.0268 | 0 | 0 | 2 | 4.748e-05 | 0.00338 | 0.000911 | 3.690e-05 | 4.948e-05 |
| re700_e1 | 700 | 1 | 2 | 1 | B1_Mode2 | 1.03 | 1 | 1 | 0 | 2.908e-16 | 0.0523 | 1.4 | -0.00989 | -0.247 | -0.0178 | 0.0268 | 0 | 0 | 3 | 0.0296 | 0.00778 | 2.524e-05 | 2.529e-05 | 6.225e-06 |
| re700_e1 | 700 | 1 | 2 | 1 | B1_Mode2 | 1.03 | 1 | 1 | 0 | 2.908e-16 | 0.0523 | 1.4 | -0.00989 | -0.247 | -0.0178 | 0.0268 | 0 | 0 | 4 | 0.000154 | 0.000418 | 0.000148 | 2.049e-05 | 6.966e-06 |
| re700_e1 | 700 | 1 | 2 | 1 | B1_Mode2 | 1.03 | 1 | 1 | 0 | 2.908e-16 | 0.0523 | 1.4 | -0.00989 | -0.247 | -0.0178 | 0.0268 | 0 | 0 | 5 | 0.000349 | 0.000363 | 1.565e-05 | 3.636e-07 | 1.072e-06 |

Table 15: Décomposition modale — 185 points, 1110 lignes (6 modes $i$ par point, colonnes $E^f_{iJ}$ pour $J$=0..4), `mode_analysis.csv`.

| point | Re | $E^*$ | s | id | branch | $E_t$ | cov. | $i$ | $j$ | rmse | $A_{10}$ | $A_{20}$ | $A_{30}$ | $A_{40}$ | $A_{50}$ | $A_{60}$ | $A_{70}$ | $A_{80}$ | $i$ | $E^f_{0J}$ | $E^f_{1J}$ | $E^f_{2J}$ | $E^f_{3J}$ | $E^f_{4J}$ |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| re700_e1 | 700 | 1 | 3 | 1 | B1_Mode2 | 0.935 | 1 | 1 | 0 | 2.813e-16 | 0.167 | 1.2 | -0.261 | -0.266 | -0.0433 | -0.00643 | 0 | 0 | 0 | 0.0149 | 0.0187 | 0.000531 | 5.734e-05 | 2.420e-05 |
| re700_e1 | 700 | 1 | 3 | 1 | B1_Mode2 | 0.935 | 1 | 1 | 0 | 2.813e-16 | 0.167 | 1.2 | -0.261 | -0.266 | -0.0433 | -0.00643 | 0 | 0 | 1 | 0.773 | 0.0321 | 0.00255 | 5.439e-05 | 9.737e-05 |
| re700_e1 | 700 | 1 | 3 | 1 | B1_Mode2 | 0.935 | 1 | 1 | 0 | 2.813e-16 | 0.167 | 1.2 | -0.261 | -0.266 | -0.0433 | -0.00643 | 0 | 0 | 2 | 0.0363 | 0.0588 | 0.0005 | 9.832e-05 | 1.485e-05 |
| re700_e1 | 700 | 1 | 3 | 1 | B1_Mode2 | 0.935 | 1 | 1 | 0 | 2.813e-16 | 0.167 | 1.2 | -0.261 | -0.266 | -0.0433 | -0.00643 | 0 | 0 | 3 | 0.038 | 0.00588 | 0.00528 | 4.691e-05 | 0.000203 |
| re700_e1 | 700 | 1 | 3 | 1 | B1_Mode2 | 0.935 | 1 | 1 | 0 | 2.813e-16 | 0.167 | 1.2 | -0.261 | -0.266 | -0.0433 | -0.00643 | 0 | 0 | 4 | 0.001 | 0.00939 | 0.00116 | 1.788e-05 | 3.789e-05 |
| re700_e1 | 700 | 1 | 3 | 1 | B1_Mode2 | 0.935 | 1 | 1 | 0 | 2.813e-16 | 0.167 | 1.2 | -0.261 | -0.266 | -0.0433 | -0.00643 | 0 | 0 | 5 | 2.213e-05 | 0.000651 | 0.000137 | 6.351e-05 | 1.797e-10 |
| re700_e1 | 700 | 1 | 4 | 2 | B2_temporal | 0.971 | 1 | 0 | 1 | 2.962e-16 | 0.058 | -0.208 | -0.0612 | 0.0174 | 0.0145 | 0.003 | 0 | 0 | 0 | 0.00173 | 0.791 | 0.131 | 0.00757 | 0.000143 |
| re700_e1 | 700 | 1 | 4 | 2 | B2_temporal | 0.971 | 1 | 0 | 1 | 2.962e-16 | 0.058 | -0.208 | -0.0612 | 0.0174 | 0.0145 | 0.003 | 0 | 0 | 1 | 0.0224 | 0.000759 | 0.00232 | 0.00109 | 0.00075 |
| re700_e1 | 700 | 1 | 4 | 2 | B2_temporal | 0.971 | 1 | 0 | 1 | 2.962e-16 | 0.058 | -0.208 | -0.0612 | 0.0174 | 0.0145 | 0.003 | 0 | 0 | 2 | 0.00193 | 0.0206 | 0.017 | 0.00111 | 8.324e-06 |
| re700_e1 | 700 | 1 | 4 | 2 | B2_temporal | 0.971 | 1 | 0 | 1 | 2.962e-16 | 0.058 | -0.208 | -0.0612 | 0.0174 | 0.0145 | 0.003 | 0 | 0 | 3 | 0.000155 | 2.025e-06 | 4.248e-06 | 9.683e-05 | 0.000324 |
| re700_e1 | 700 | 1 | 4 | 2 | B2_temporal | 0.971 | 1 | 0 | 1 | 2.962e-16 | 0.058 | -0.208 | -0.0612 | 0.0174 | 0.0145 | 0.003 | 0 | 0 | 4 | 0.000108 | 1.412e-05 | 4.655e-07 | 3.080e-06 | 1.716e-05 |
| re700_e1 | 700 | 1 | 4 | 2 | B2_temporal | 0.971 | 1 | 0 | 1 | 2.962e-16 | 0.058 | -0.208 | -0.0612 | 0.0174 | 0.0145 | 0.003 | 0 | 0 | 5 | 4.630e-06 | 1.947e-07 | 4.040e-06 | 1.166e-07 | 1.411e-06 |
| re700_e2 | 700 | 2 | 0 | 3 | B3_mixed | 1.93 | 1 | 2 | 0 | 4.091e-16 | 0.609 | -0.532 | -1.22 | 0.0979 | 0.206 | 0.019 | 0 | 0 | 0 | 0.0961 | 0.000166 | 0.00637 | 0.000231 | 0.000149 |
| re700_e2 | 700 | 2 | 0 | 3 | B3_mixed | 1.93 | 1 | 2 | 0 | 4.091e-16 | 0.609 | -0.532 | -1.22 | 0.0979 | 0.206 | 0.019 | 0 | 0 | 1 | 0.0733 | 0.268 | 0.00787 | 5.445e-05 | 4.764e-05 |
| re700_e2 | 700 | 2 | 0 | 3 | B3_mixed | 1.93 | 1 | 2 | 0 | 4.091e-16 | 0.609 | -0.532 | -1.22 | 0.0979 | 0.206 | 0.019 | 0 | 0 | 2 | 0.385 | 0.0617 | 0.0278 | 0.000459 | 0.000284 |
| re700_e2 | 700 | 2 | 0 | 3 | B3_mixed | 1.93 | 1 | 2 | 0 | 4.091e-16 | 0.609 | -0.532 | -1.22 | 0.0979 | 0.206 | 0.019 | 0 | 0 | 3 | 0.00248 | 0.0236 | 0.0141 | 0.00146 | 4.185e-05 |
| re700_e2 | 700 | 2 | 0 | 3 | B3_mixed | 1.93 | 1 | 2 | 0 | 4.091e-16 | 0.609 | -0.532 | -1.22 | 0.0979 | 0.206 | 0.019 | 0 | 0 | 4 | 0.0109 | 0.0157 | 0.00214 | 0.00102 | 2.730e-05 |
| re700_e2 | 700 | 2 | 0 | 3 | B3_mixed | 1.93 | 1 | 2 | 0 | 4.091e-16 | 0.609 | -0.532 | -1.22 | 0.0979 | 0.206 | 0.019 | 0 | 0 | 5 | 9.347e-05 | 0.000497 | 0.000997 | 4.703e-05 | 8.800e-06 |
| re700_e2 | 700 | 2 | 1 | 3 | B3_mixed | 1.97 | 1 | 1 | 0 | 4.740e-16 | 0.503 | 1.13 | -0.699 | -0.241 | -0.0225 | 0.0022 | 0 | 0 | 0 | 0.0641 | 0.0513 | 0.000341 | 4.387e-05 | 6.228e-05 |
| re700_e2 | 700 | 2 | 1 | 3 | B3_mixed | 1.97 | 1 | 1 | 0 | 4.740e-16 | 0.503 | 1.13 | -0.699 | -0.241 | -0.0225 | 0.0022 | 0 | 0 | 1 | 0.322 | 0.262 | 0.0123 | 6.296e-05 | 0.000458 |
| re700_e2 | 700 | 2 | 1 | 3 | B3_mixed | 1.97 | 1 | 1 | 0 | 4.740e-16 | 0.503 | 1.13 | -0.699 | -0.241 | -0.0225 | 0.0022 | 0 | 0 | 2 | 0.124 | 0.0771 | 0.025 | 0.000133 | 0.000446 |
| re700_e2 | 700 | 2 | 1 | 3 | B3_mixed | 1.97 | 1 | 1 | 0 | 4.740e-16 | 0.503 | 1.13 | -0.699 | -0.241 | -0.0225 | 0.0022 | 0 | 0 | 3 | 0.0147 | 0.0145 | 0.0181 | 0.00111 | 0.000103 |
| re700_e2 | 700 | 2 | 1 | 3 | B3_mixed | 1.97 | 1 | 1 | 0 | 4.740e-16 | 0.503 | 1.13 | -0.699 | -0.241 | -0.0225 | 0.0022 | 0 | 0 | 4 | 0.000129 | 0.00805 | 0.00125 | 0.00219 | 3.421e-05 |
| re700_e2 | 700 | 2 | 1 | 3 | B3_mixed | 1.97 | 1 | 1 | 0 | 4.740e-16 | 0.503 | 1.13 | -0.699 | -0.241 | -0.0225 | 0.0022 | 0 | 0 | 5 | 1.232e-06 | 2.151e-05 | 0.000789 | 9.302e-05 | 0.000123 |
| re700_e2 | 700 | 2 | 10 | 3 | B3_mixed | 2.01 | 1 | 0 | 1 | 4.888e-16 | 0.299 | -1.21 | -0.253 | 0.113 | -0.108 | 0.0606 | 0 | 0 | 0 | 0.0223 | 0.39 | 0.00934 | 0.0112 | 5.485e-05 |
| re700_e2 | 700 | 2 | 10 | 3 | B3_mixed | 2.01 | 1 | 0 | 1 | 4.888e-16 | 0.299 | -1.21 | -0.253 | 0.113 | -0.108 | 0.0606 | 0 | 0 | 1 | 0.364 | 0.0266 | 0.0608 | 0.00326 | 4.577e-05 |
| re700_e2 | 700 | 2 | 10 | 3 | B3_mixed | 2.01 | 1 | 0 | 1 | 4.888e-16 | 0.299 | -1.21 | -0.253 | 0.113 | -0.108 | 0.0606 | 0 | 0 | 2 | 0.016 | 0.0634 | 0.00139 | 0.00668 | 0.000902 |
| re700_e2 | 700 | 2 | 10 | 3 | B3_mixed | 2.01 | 1 | 0 | 1 | 4.888e-16 | 0.299 | -1.21 | -0.253 | 0.113 | -0.108 | 0.0606 | 0 | 0 | 3 | 0.0032 | 0.000824 | 0.0113 | 0.00309 | 6.213e-06 |
| re700_e2 | 700 | 2 | 10 | 3 | B3_mixed | 2.01 | 1 | 0 | 1 | 4.888e-16 | 0.299 | -1.21 | -0.253 | 0.113 | -0.108 | 0.0606 | 0 | 0 | 4 | 0.0029 | 0.000189 | 0.00132 | 0.000126 | 0.000349 |
| re700_e2 | 700 | 2 | 10 | 3 | B3_mixed | 2.01 | 1 | 0 | 1 | 4.888e-16 | 0.299 | -1.21 | -0.253 | 0.113 | -0.108 | 0.0606 | 0 | 0 | 5 | 0.000915 | 0.00011 | 0.000308 | 6.926e-05 | 8.117e-06 |
| re700_e2 | 700 | 2 | 11 | 2 | B2_temporal | 1.91 | 1 | 1 | 1 | 4.472e-16 | 0.474 | 0.0226 | -0.643 | 0.0597 | -0.00442 | 0.0219 | 0 | 0 | 0 | 0.059 | 0.223 | 0.00536 | 0.00155 | 0.000458 |
| re700_e2 | 700 | 2 | 11 | 2 | B2_temporal | 1.91 | 1 | 1 | 1 | 4.472e-16 | 0.474 | 0.0226 | -0.643 | 0.0597 | -0.00442 | 0.0219 | 0 | 0 | 1 | 0.000133 | 0.496 | 0.036 | 5.558e-07 | 8.245e-05 |
| re700_e2 | 700 | 2 | 11 | 2 | B2_temporal | 1.91 | 1 | 1 | 1 | 4.472e-16 | 0.474 | 0.0226 | -0.643 | 0.0597 | -0.00442 | 0.0219 | 0 | 0 | 2 | 0.108 | 0.00156 | 0.0362 | 0.0038 | 5.214e-05 |
| re700_e2 | 700 | 2 | 11 | 2 | B2_temporal | 1.91 | 1 | 1 | 1 | 4.472e-16 | 0.474 | 0.0226 | -0.643 | 0.0597 | -0.00442 | 0.0219 | 0 | 0 | 3 | 0.000934 | 0.0175 | 0.000725 | 0.00502 | 7.850e-05 |
| re700_e2 | 700 | 2 | 11 | 2 | B2_temporal | 1.91 | 1 | 1 | 1 | 4.472e-16 | 0.474 | 0.0226 | -0.643 | 0.0597 | -0.00442 | 0.0219 | 0 | 0 | 4 | 5.122e-06 | 0.000393 | 0.00286 | 2.057e-06 | 0.000373 |
| re700_e2 | 700 | 2 | 11 | 2 | B2_temporal | 1.91 | 1 | 1 | 1 | 4.472e-16 | 0.474 | 0.0226 | -0.643 | 0.0597 | -0.00442 | 0.0219 | 0 | 0 | 5 | 0.000126 | 1.645e-06 | 1.792e-05 | 0.00026 | 4.467e-05 |
| re700_e2 | 700 | 2 | 12 | 2 | B2_temporal | 2.04 | 1 | 0 | 1 | 4.705e-16 | 0.179 | -1.14 | -0.339 | 0.153 | -0.00432 | -0.00386 | 0 | 0 | 0 | 0.00787 | 0.539 | 0.000241 | 0.000658 | 0.000275 |
| re700_e2 | 700 | 2 | 12 | 2 | B2_temporal | 2.04 | 1 | 0 | 1 | 4.705e-16 | 0.179 | -1.14 | -0.339 | 0.153 | -0.00432 | -0.00386 | 0 | 0 | 1 | 0.316 | 0.0627 | 0.0101 | 6.871e-07 | 7.290e-05 |
| re700_e2 | 700 | 2 | 12 | 2 | B2_temporal | 2.04 | 1 | 0 | 1 | 4.705e-16 | 0.179 | -1.14 | -0.339 | 0.153 | -0.00432 | -0.00386 | 0 | 0 | 2 | 0.0282 | 0.0252 | 0.000272 | 0.000139 | 3.082e-05 |
| re700_e2 | 700 | 2 | 12 | 2 | B2_temporal | 2.04 | 1 | 0 | 1 | 4.705e-16 | 0.179 | -1.14 | -0.339 | 0.153 | -0.00432 | -0.00386 | 0 | 0 | 3 | 0.00574 | 6.136e-05 | 0.00198 | 9.493e-05 | 0.000121 |
| re700_e2 | 700 | 2 | 12 | 2 | B2_temporal | 2.04 | 1 | 0 | 1 | 4.705e-16 | 0.179 | -1.14 | -0.339 | 0.153 | -0.00432 | -0.00386 | 0 | 0 | 4 | 4.566e-06 | 0.000873 | 0.000133 | 4.201e-06 | 8.448e-06 |
| re700_e2 | 700 | 2 | 12 | 2 | B2_temporal | 2.04 | 1 | 0 | 1 | 4.705e-16 | 0.179 | -1.14 | -0.339 | 0.153 | -0.00432 | -0.00386 | 0 | 0 | 5 | 3.652e-06 | 9.730e-06 | 2.680e-05 | 4.796e-05 | 2.438e-06 |
| re700_e2 | 700 | 2 | 13 | 1 | B1_Mode2 | 1.96 | 1 | 1 | 0 | 4.186e-16 | 0.158 | 1.92 | -0.083 | -0.162 | -0.0011 | 0.0388 | 0 | 0 | 0 | 0.00637 | 0.0328 | 0.000218 | 0.000377 | 1.333e-05 |
| re700_e2 | 700 | 2 | 13 | 1 | B1_Mode2 | 1.96 | 1 | 1 | 0 | 4.186e-16 | 0.158 | 1.92 | -0.083 | -0.162 | -0.0011 | 0.0388 | 0 | 0 | 1 | 0.943 | 0.00471 | 9.146e-06 | 2.942e-06 | 7.680e-06 |
| re700_e2 | 700 | 2 | 13 | 1 | B1_Mode2 | 1.96 | 1 | 1 | 0 | 4.186e-16 | 0.158 | 1.92 | -0.083 | -0.162 | -0.0011 | 0.0388 | 0 | 0 | 2 | 0.00176 | 0.00288 | 0.000281 | 2.724e-05 | 1.409e-05 |
| re700_e2 | 700 | 2 | 13 | 1 | B1_Mode2 | 1.96 | 1 | 1 | 0 | 4.186e-16 | 0.158 | 1.92 | -0.083 | -0.162 | -0.0011 | 0.0388 | 0 | 0 | 3 | 0.00673 | 6.420e-06 | 8.174e-05 | 4.413e-06 | 6.666e-06 |
| re700_e2 | 700 | 2 | 13 | 1 | B1_Mode2 | 1.96 | 1 | 1 | 0 | 4.186e-16 | 0.158 | 1.92 | -0.083 | -0.162 | -0.0011 | 0.0388 | 0 | 0 | 4 | 3.113e-07 | 0.000322 | 1.041e-05 | 1.980e-06 | 7.257e-07 |
| re700_e2 | 700 | 2 | 13 | 1 | B1_Mode2 | 1.96 | 1 | 1 | 0 | 4.186e-16 | 0.158 | 1.92 | -0.083 | -0.162 | -0.0011 | 0.0388 | 0 | 0 | 5 | 0.000384 | 0.000115 | 6.889e-08 | 2.204e-07 | 1.311e-07 |
| re700_e2 | 700 | 2 | 14 | 3 | B3_mixed | 1.82 | 1 | 1 | 0 | 4.019e-16 | 0.452 | 1.19 | -0.802 | -0.266 | -0.00305 | 0.0119 | 0 | 0 | 0 | 0.0562 | 0.0542 | 4.669e-05 | 0.00046 | 1.219e-08 |
| re700_e2 | 700 | 2 | 14 | 3 | B3_mixed | 1.82 | 1 | 1 | 0 | 4.019e-16 | 0.452 | 1.19 | -0.802 | -0.266 | -0.00305 | 0.0119 | 0 | 0 | 1 | 0.392 | 0.128 | 0.00653 | 0.000949 | 0.000335 |
| re700_e2 | 700 | 2 | 14 | 3 | B3_mixed | 1.82 | 1 | 1 | 0 | 4.019e-16 | 0.452 | 1.19 | -0.802 | -0.266 | -0.00305 | 0.0119 | 0 | 0 | 2 | 0.177 | 0.0994 | 0.00574 | 6.584e-07 | 0.000233 |
| re700_e2 | 700 | 2 | 14 | 3 | B3_mixed | 1.82 | 1 | 1 | 0 | 4.019e-16 | 0.452 | 1.19 | -0.802 | -0.266 | -0.00305 | 0.0119 | 0 | 0 | 3 | 0.0195 | 0.0152 | 0.0195 | 3.073e-07 | 0.000412 |
| re700_e2 | 700 | 2 | 14 | 3 | B3_mixed | 1.82 | 1 | 1 | 0 | 4.019e-16 | 0.452 | 1.19 | -0.802 | -0.266 | -0.00305 | 0.0119 | 0 | 0 | 4 | 2.559e-06 | 0.0207 | 0.000445 | 0.00101 | 2.778e-05 |
| re700_e2 | 700 | 2 | 14 | 3 | B3_mixed | 1.82 | 1 | 1 | 0 | 4.019e-16 | 0.452 | 1.19 | -0.802 | -0.266 | -0.00305 | 0.0119 | 0 | 0 | 5 | 3.926e-05 | 6.394e-06 | 0.00228 | 5.898e-07 | 3.151e-06 |
| re700_e2 | 700 | 2 | 2 | 1 | B1_Mode2 | 1.96 | 1 | 1 | 0 | 3.995e-16 | 0.507 | 1.63 | -0.86 | -0.184 | 0.115 | -0.0557 | 0 | 0 | 0 | 0.0656 | 0.0348 | 1.264e-07 | 0.000143 | 5.925e-07 |
| re700_e2 | 700 | 2 | 2 | 1 | B1_Mode2 | 1.96 | 1 | 1 | 0 | 3.995e-16 | 0.507 | 1.63 | -0.86 | -0.184 | 0.115 | -0.0557 | 0 | 0 | 1 | 0.679 | 0.00242 | 0.00051 | 1.059e-05 | 2.242e-05 |
| re700_e2 | 700 | 2 | 2 | 1 | B1_Mode2 | 1.96 | 1 | 1 | 0 | 3.995e-16 | 0.507 | 1.63 | -0.86 | -0.184 | 0.115 | -0.0557 | 0 | 0 | 2 | 0.189 | 0.00771 | 0.000647 | 4.779e-05 | 3.241e-05 |
| re700_e2 | 700 | 2 | 2 | 1 | B1_Mode2 | 1.96 | 1 | 1 | 0 | 3.995e-16 | 0.507 | 1.63 | -0.86 | -0.184 | 0.115 | -0.0557 | 0 | 0 | 3 | 0.00863 | 0.000363 | 0.000702 | 6.380e-07 | 4.250e-05 |
| re700_e2 | 700 | 2 | 2 | 1 | B1_Mode2 | 1.96 | 1 | 1 | 0 | 3.995e-16 | 0.507 | 1.63 | -0.86 | -0.184 | 0.115 | -0.0557 | 0 | 0 | 4 | 0.00335 | 0.00445 | 5.138e-06 | 3.625e-05 | 1.034e-09 |
| re700_e2 | 700 | 2 | 2 | 1 | B1_Mode2 | 1.96 | 1 | 1 | 0 | 3.995e-16 | 0.507 | 1.63 | -0.86 | -0.184 | 0.115 | -0.0557 | 0 | 0 | 5 | 0.000791 | 0.00138 | 0.000183 | 4.143e-05 | 3.555e-06 |
| re700_e2 | 700 | 2 | 3 | 2 | B2_temporal | 2.04 | 1 | 1 | 1 | 4.787e-16 | 0.573 | 0.876 | -0.557 | 0.0182 | -0.0635 | -0.0293 | 0 | 0 | 0 | 0.0804 | 0.0662 | 0.0184 | 0.000158 | 0.000207 |
| re700_e2 | 700 | 2 | 3 | 2 | B2_temporal | 2.04 | 1 | 1 | 1 | 4.787e-16 | 0.573 | 0.876 | -0.557 | 0.0182 | -0.0635 | -0.0293 | 0 | 0 | 1 | 0.188 | 0.342 | 0.0207 | 0.00176 | 5.524e-05 |
| re700_e2 | 700 | 2 | 3 | 2 | B2_temporal | 2.04 | 1 | 1 | 1 | 4.787e-16 | 0.573 | 0.876 | -0.557 | 0.0182 | -0.0635 | -0.0293 | 0 | 0 | 2 | 0.0758 | 0.0635 | 0.0588 | 0.00539 | 1.056e-05 |
| re700_e2 | 700 | 2 | 3 | 2 | B2_temporal | 2.04 | 1 | 1 | 1 | 4.787e-16 | 0.573 | 0.876 | -0.557 | 0.0182 | -0.0635 | -0.0293 | 0 | 0 | 3 | 8.100e-05 | 0.0271 | 0.0279 | 0.00828 | 0.000367 |
| re700_e2 | 700 | 2 | 3 | 2 | B2_temporal | 2.04 | 1 | 1 | 1 | 4.787e-16 | 0.573 | 0.876 | -0.557 | 0.0182 | -0.0635 | -0.0293 | 0 | 0 | 4 | 0.000987 | 0.0011 | 0.00505 | 0.00581 | 0.000502 |
| re700_e2 | 700 | 2 | 3 | 2 | B2_temporal | 2.04 | 1 | 1 | 1 | 4.787e-16 | 0.573 | 0.876 | -0.557 | 0.0182 | -0.0635 | -0.0293 | 0 | 0 | 5 | 0.00021 | 7.108e-05 | 0.000236 | 0.000402 | 0.000688 |

Table 15: Décomposition modale — 185 points, 1110 lignes (6 modes $i$ par point, colonnes $E^f_{iJ}$ pour $J$=0..4), `mode_analysis.csv`.

| point | Re | $E^*$ | s | id | branch | $E_t$ | cov. | $i$ | $j$ | rmse | $A_{10}$ | $A_{20}$ | $A_{30}$ | $A_{40}$ | $A_{50}$ | $A_{60}$ | $A_{70}$ | $A_{80}$ | $i$ | $E^f_{0J}$ | $E^f_{1J}$ | $E^f_{2J}$ | $E^f_{3J}$ | $E^f_{4J}$ |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| re700_e2 | 700 | 2 | 4 | 2 | B2_temporal | 1.91 | 1 | 0 | 1 | 4.241e-16 | 0.265 | 0.273 | -0.269 | 0.0207 | 0.0333 | -0.0422 | 0 | 0 | 0 | 0.0184 | 0.763 | 0.0432 | 2.965e-10 | 0.000182 |
| re700_e2 | 700 | 2 | 4 | 2 | B2_temporal | 1.91 | 1 | 0 | 1 | 4.241e-16 | 0.265 | 0.273 | -0.269 | 0.0207 | 0.0333 | -0.0422 | 0 | 0 | 1 | 0.0195 | 0.0827 | 0.013 | 0.0131 | 0.00165 |
| re700_e2 | 700 | 2 | 4 | 2 | B2_temporal | 1.91 | 1 | 0 | 1 | 4.241e-16 | 0.265 | 0.273 | -0.269 | 0.0207 | 0.0333 | -0.0422 | 0 | 0 | 2 | 0.0189 | 0.0179 | 0.000289 | 0.00414 | 0.000631 |
| re700_e2 | 700 | 2 | 4 | 2 | B2_temporal | 1.91 | 1 | 0 | 1 | 4.241e-16 | 0.265 | 0.273 | -0.269 | 0.0207 | 0.0333 | -0.0422 | 0 | 0 | 3 | 0.000112 | 8.074e-05 | 0.000264 | 0.000793 | 4.590e-05 |
| re700_e2 | 700 | 2 | 4 | 2 | B2_temporal | 1.91 | 1 | 0 | 1 | 4.241e-16 | 0.265 | 0.273 | -0.269 | 0.0207 | 0.0333 | -0.0422 | 0 | 0 | 4 | 0.000291 | 6.643e-05 | 0.000836 | 0.000147 | 8.882e-07 |
| re700_e2 | 700 | 2 | 4 | 2 | B2_temporal | 1.91 | 1 | 0 | 1 | 4.241e-16 | 0.265 | 0.273 | -0.269 | 0.0207 | 0.0333 | -0.0422 | 0 | 0 | 5 | 0.000467 | 0.000671 | 2.622e-10 | 9.648e-06 | 3.884e-09 |
| re700_e2 | 700 | 2 | 5 | 3 | B3_mixed | 1.89 | 1 | 1 | 0 | 4.200e-16 | 0.401 | 1.1 | -0.51 | -0.149 | -0.049 | 0.0137 | 0 | 0 | 0 | 0.0425 | 0.127 | 0.0118 | 0.000155 | 0.000232 |
| re700_e2 | 700 | 2 | 5 | 3 | B3_mixed | 1.89 | 1 | 1 | 0 | 4.200e-16 | 0.401 | 1.1 | -0.51 | -0.149 | -0.049 | 0.0137 | 0 | 0 | 1 | 0.32 | 0.301 | 0.00437 | 0.000639 | 5.293e-05 |
| re700_e2 | 700 | 2 | 5 | 3 | B3_mixed | 1.89 | 1 | 1 | 0 | 4.200e-16 | 0.401 | 1.1 | -0.51 | -0.149 | -0.049 | 0.0137 | 0 | 0 | 2 | 0.0689 | 0.052 | 0.0116 | 0.00234 | 0.000354 |
| re700_e2 | 700 | 2 | 5 | 3 | B3_mixed | 1.89 | 1 | 1 | 0 | 4.200e-16 | 0.401 | 1.1 | -0.51 | -0.149 | -0.049 | 0.0137 | 0 | 0 | 3 | 0.00592 | 0.00773 | 0.0259 | 0.00123 | 9.973e-06 |
| re700_e2 | 700 | 2 | 5 | 3 | B3_mixed | 1.89 | 1 | 1 | 0 | 4.200e-16 | 0.401 | 1.1 | -0.51 | -0.149 | -0.049 | 0.0137 | 0 | 0 | 4 | 0.000636 | 0.00969 | 0.00185 | 0.00296 | 8.636e-05 |
| re700_e2 | 700 | 2 | 5 | 3 | B3_mixed | 1.89 | 1 | 1 | 0 | 4.200e-16 | 0.401 | 1.1 | -0.51 | -0.149 | -0.049 | 0.0137 | 0 | 0 | 5 | 5.011e-05 | 2.839e-09 | 0.000733 | 0.000172 | 0.000257 |
| re700_e2 | 700 | 2 | 6 | 3 | B3_mixed | 2.13 | 1 | 1 | 0 | 4.698e-16 | 0.0775 | 1.49 | -0.468 | -0.281 | -0.0708 | 0.0247 | 0 | 0 | 0 | 0.00141 | 0.179 | 0.00169 | 7.925e-05 | 0.000114 |
| re700_e2 | 700 | 2 | 6 | 3 | B3_mixed | 2.13 | 1 | 1 | 0 | 4.698e-16 | 0.0775 | 1.49 | -0.468 | -0.281 | -0.0708 | 0.0247 | 0 | 0 | 1 | 0.519 | 0.0682 | 0.0173 | 0.000412 | 0.000245 |
| re700_e2 | 700 | 2 | 6 | 3 | B3_mixed | 2.13 | 1 | 1 | 0 | 4.698e-16 | 0.0775 | 1.49 | -0.468 | -0.281 | -0.0708 | 0.0247 | 0 | 0 | 2 | 0.0514 | 0.0944 | 0.00209 | 0.00138 | 0.000152 |
| re700_e2 | 700 | 2 | 6 | 3 | B3_mixed | 2.13 | 1 | 1 | 0 | 4.698e-16 | 0.0775 | 1.49 | -0.468 | -0.281 | -0.0708 | 0.0247 | 0 | 0 | 3 | 0.0185 | 0.000724 | 0.0232 | 2.374e-05 | 3.624e-05 |
| re700_e2 | 700 | 2 | 6 | 3 | B3_mixed | 2.13 | 1 | 1 | 0 | 4.698e-16 | 0.0775 | 1.49 | -0.468 | -0.281 | -0.0708 | 0.0247 | 0 | 0 | 4 | 0.00117 | 0.0148 | 0.000115 | 0.00187 | 2.428e-05 |
| re700_e2 | 700 | 2 | 6 | 3 | B3_mixed | 2.13 | 1 | 1 | 0 | 4.698e-16 | 0.0775 | 1.49 | -0.468 | -0.281 | -0.0708 | 0.0247 | 0 | 0 | 5 | 0.000143 | 0.000179 | 0.0018 | 7.444e-05 | 4.158e-05 |
| re700_e2 | 700 | 2 | 7 | 1 | B1_Mode2 | 1.89 | 1 | 1 | 0 | 4.291e-16 | -0.15 | 1.68 | 0.199 | -0.166 | -0.0325 | -0.0222 | 0 | 0 | 0 | 0.00596 | 0.179 | 0.00152 | 4.977e-05 | 2.114e-05 |
| re700_e2 | 700 | 2 | 7 | 1 | B1_Mode2 | 1.89 | 1 | 1 | 0 | 4.291e-16 | -0.15 | 1.68 | 0.199 | -0.166 | -0.0325 | -0.0222 | 0 | 0 | 1 | 0.747 | 0.0173 | 0.0124 | 1.272e-05 | 0.000293 |
| re700_e2 | 700 | 2 | 7 | 1 | B1_Mode2 | 1.89 | 1 | 1 | 0 | 4.291e-16 | -0.15 | 1.68 | 0.199 | -0.166 | -0.0325 | -0.0222 | 0 | 0 | 2 | 0.0105 | 0.00474 | 0.00428 | 6.853e-06 | 0.000106 |
| re700_e2 | 700 | 2 | 7 | 1 | B1_Mode2 | 1.89 | 1 | 1 | 0 | 4.291e-16 | -0.15 | 1.68 | 0.199 | -0.166 | -0.0325 | -0.0222 | 0 | 0 | 3 | 0.00729 | 0.00812 | 8.439e-05 | 1.036e-05 | 9.754e-06 |
| re700_e2 | 700 | 2 | 7 | 1 | B1_Mode2 | 1.89 | 1 | 1 | 0 | 4.291e-16 | -0.15 | 1.68 | 0.199 | -0.166 | -0.0325 | -0.0222 | 0 | 0 | 4 | 0.00028 | 7.613e-05 | 0.000192 | 1.852e-06 | 7.225e-07 |
| re700_e2 | 700 | 2 | 7 | 1 | B1_Mode2 | 1.89 | 1 | 1 | 0 | 4.291e-16 | -0.15 | 1.68 | 0.199 | -0.166 | -0.0325 | -0.0222 | 0 | 0 | 5 | 0.00013 | 0.000488 | 4.702e-05 | 1.976e-05 | 3.899e-06 |
| re700_e2 | 700 | 2 | 8 | 2 | B2_temporal | 1.87 | 1 | 1 | 1 | 4.189e-16 | 0.543 | 0.574 | -0.854 | -0.314 | 0.122 | 0.0468 | 0 | 0 | 0 | 0.079 | 0.184 | 0.0186 | 6.988e-05 | 0.000159 |
| re700_e2 | 700 | 2 | 8 | 2 | B2_temporal | 1.87 | 1 | 1 | 1 | 4.189e-16 | 0.543 | 0.574 | -0.854 | -0.314 | 0.122 | 0.0468 | 0 | 0 | 1 | 0.0881 | 0.247 | 0.0284 | 0.00289 | 0.000151 |
| re700_e2 | 700 | 2 | 8 | 2 | B2_temporal | 1.87 | 1 | 1 | 1 | 4.189e-16 | 0.543 | 0.574 | -0.854 | -0.314 | 0.122 | 0.0468 | 0 | 0 | 2 | 0.195 | 0.0314 | 0.000258 | 0.0017 | 0.000652 |
| re700_e2 | 700 | 2 | 8 | 2 | B2_temporal | 1.87 | 1 | 1 | 1 | 4.189e-16 | 0.543 | 0.574 | -0.854 | -0.314 | 0.122 | 0.0468 | 0 | 0 | 3 | 0.0263 | 0.0677 | 0.013 | 9.959e-06 | 0.000208 |
| re700_e2 | 700 | 2 | 8 | 2 | B2_temporal | 1.87 | 1 | 1 | 1 | 4.189e-16 | 0.543 | 0.574 | -0.854 | -0.314 | 0.122 | 0.0468 | 0 | 0 | 4 | 0.00397 | 0.000419 | 0.0058 | 0.00227 | 0.00013 |
| re700_e2 | 700 | 2 | 8 | 2 | B2_temporal | 1.87 | 1 | 1 | 1 | 4.189e-16 | 0.543 | 0.574 | -0.854 | -0.314 | 0.122 | 0.0468 | 0 | 0 | 5 | 0.000585 | 0.0018 | 0.000231 | 0.000575 | 0.00037 |
| re700_e2 | 700 | 2 | 9 | 2 | B2_temporal | 1.94 | 1 | 1 | 1 | 4.396e-16 | 0.516 | 0.834 | -0.928 | 0.000862 | -0.0374 | 0.027 | 0 | 0 | 0 | 0.0687 | 0.114 | 0.00259 | 1.844e-05 | 6.483e-07 |
| re700_e2 | 700 | 2 | 9 | 2 | B2_temporal | 1.94 | 1 | 1 | 1 | 4.396e-16 | 0.516 | 0.834 | -0.928 | 0.000862 | -0.0374 | 0.027 | 0 | 0 | 1 | 0.179 | 0.243 | 0.0156 | 2.867e-05 | 0.000259 |
| re700_e2 | 700 | 2 | 9 | 2 | B2_temporal | 1.94 | 1 | 1 | 1 | 4.396e-16 | 0.516 | 0.834 | -0.928 | 0.000862 | -0.0374 | 0.027 | 0 | 0 | 2 | 0.222 | 0.0536 | 0.0184 | 0.00167 | 0.000211 |
| re700_e2 | 700 | 2 | 9 | 2 | B2_temporal | 1.94 | 1 | 1 | 1 | 4.396e-16 | 0.516 | 0.834 | -0.928 | 0.000862 | -0.0374 | 0.027 | 0 | 0 | 3 | 1.914e-07 | 0.0402 | 0.0167 | 0.00127 | 1.274e-06 |
| re700_e2 | 700 | 2 | 9 | 2 | B2_temporal | 1.94 | 1 | 1 | 1 | 4.396e-16 | 0.516 | 0.834 | -0.928 | 0.000862 | -0.0374 | 0.027 | 0 | 0 | 4 | 0.000359 | 0.0092 | 0.00833 | 0.00224 | 4.533e-06 |
| re700_e2 | 700 | 2 | 9 | 2 | B2_temporal | 1.94 | 1 | 1 | 1 | 4.396e-16 | 0.516 | 0.834 | -0.928 | 0.000862 | -0.0374 | 0.027 | 0 | 0 | 5 | 0.000188 | 0.000982 | 0.000606 | 0.000737 | 0.000108 |
| re900_e0.1 | 900 | 0.1 | 0 | 1 | B1_Mode2 | 0.118 | 1 | 1 | 0 | 1.071e-16 | 0.0293 | -0.416 | 0.00818 | 0.138 | -0.0176 | -0.00802 | 0 | 0 | 0 | 0.00362 | 0.151 | 0.00172 | 0.000482 | 3.786e-05 |
| re900_e0.1 | 900 | 0.1 | 0 | 1 | B1_Mode2 | 0.118 | 1 | 1 | 0 | 1.071e-16 | 0.0293 | -0.416 | 0.00818 | 0.138 | -0.0176 | -0.00802 | 0 | 0 | 1 | 0.732 | 2.084e-06 | 0.00037 | 2.437e-05 | 1.149e-05 |
| re900_e0.1 | 900 | 0.1 | 0 | 1 | B1_Mode2 | 0.118 | 1 | 1 | 0 | 1.071e-16 | 0.0293 | -0.416 | 0.00818 | 0.138 | -0.0176 | -0.00802 | 0 | 0 | 2 | 0.000283 | 0.0188 | 0.00225 | 4.450e-05 | 3.170e-06 |
| re900_e0.1 | 900 | 0.1 | 0 | 1 | B1_Mode2 | 0.118 | 1 | 1 | 0 | 1.071e-16 | 0.0293 | -0.416 | 0.00818 | 0.138 | -0.0176 | -0.00802 | 0 | 0 | 3 | 0.0807 | 0.00572 | 0.000267 | 1.264e-05 | 1.924e-05 |
| re900_e0.1 | 900 | 0.1 | 0 | 1 | B1_Mode2 | 0.118 | 1 | 1 | 0 | 1.071e-16 | 0.0293 | -0.416 | 0.00818 | 0.138 | -0.0176 | -0.00802 | 0 | 0 | 4 | 0.00131 | 2.997e-05 | 0.000302 | 4.792e-06 | 2.494e-06 |
| re900_e0.1 | 900 | 0.1 | 0 | 1 | B1_Mode2 | 0.118 | 1 | 1 | 0 | 1.071e-16 | 0.0293 | -0.416 | 0.00818 | 0.138 | -0.0176 | -0.00802 | 0 | 0 | 5 | 0.000272 | 0.000403 | 2.451e-05 | 1.418e-06 | 2.006e-07 |
| re900_e0.1 | 900 | 0.1 | 1 | 1 | B1_Mode2 | 0.122 | 1 | 1 | 0 | 1.033e-16 | -0.0116 | 0.47 | 0.0999 | -0.105 | -0.0339 | -0.0101 | 0 | 0 | 0 | 0.000549 | 0.00108 | 4.818e-05 | 2.029e-05 | 3.050e-06 |
| re900_e0.1 | 900 | 0.1 | 1 | 1 | B1_Mode2 | 0.122 | 1 | 1 | 0 | 1.033e-16 | -0.0116 | 0.47 | 0.0999 | -0.105 | -0.0339 | -0.0101 | 0 | 0 | 1 | 0.903 | 0.0025 | 0.000128 | 7.989e-06 | 8.395e-06 |
| re900_e0.1 | 900 | 0.1 | 1 | 1 | B1_Mode2 | 0.122 | 1 | 1 | 0 | 1.033e-16 | -0.0116 | 0.47 | 0.0999 | -0.105 | -0.0339 | -0.0101 | 0 | 0 | 2 | 0.0408 | 0.000502 | 4.952e-05 | 3.147e-06 | 2.692e-06 |
| re900_e0.1 | 900 | 0.1 | 1 | 1 | B1_Mode2 | 0.122 | 1 | 1 | 0 | 1.033e-16 | -0.0116 | 0.47 | 0.0999 | -0.105 | -0.0339 | -0.0101 | 0 | 0 | 3 | 0.045 | 0.000878 | 0.000145 | 9.626e-06 | 7.000e-06 |
| re900_e0.1 | 900 | 0.1 | 1 | 1 | B1_Mode2 | 0.122 | 1 | 1 | 0 | 1.033e-16 | -0.0116 | 0.47 | 0.0999 | -0.105 | -0.0339 | -0.0101 | 0 | 0 | 4 | 0.00469 | 5.242e-05 | 2.658e-05 | 3.786e-07 | 9.864e-07 |
| re900_e0.1 | 900 | 0.1 | 1 | 1 | B1_Mode2 | 0.122 | 1 | 1 | 0 | 1.033e-16 | -0.0116 | 0.47 | 0.0999 | -0.105 | -0.0339 | -0.0101 | 0 | 0 | 5 | 0.00042 | 1.519e-07 | 4.848e-06 | 3.323e-08 | 1.975e-07 |
| re900_e0.1 | 900 | 0.1 | 2 | 1 | B1_Mode2 | 0.127 | 1 | 1 | 0 | 1.076e-16 | 0.00786 | 0.477 | 0.00121 | -0.158 | -0.00555 | 0.00735 | 0 | 0 | 0 | 0.000243 | 5.534e-05 | 8.164e-06 | 2.073e-07 | 2.206e-07 |
| re900_e0.1 | 900 | 0.1 | 2 | 1 | B1_Mode2 | 0.127 | 1 | 1 | 0 | 1.076e-16 | 0.00786 | 0.477 | 0.00121 | -0.158 | -0.00555 | 0.00735 | 0 | 0 | 1 | 0.895 | 0.00343 | 5.233e-05 | 4.368e-05 | 2.073e-06 |
| re900_e0.1 | 900 | 0.1 | 2 | 1 | B1_Mode2 | 0.127 | 1 | 1 | 0 | 1.076e-16 | 0.00786 | 0.477 | 0.00121 | -0.158 | -0.00555 | 0.00735 | 0 | 0 | 2 | 5.750e-06 | 0.000941 | 3.254e-06 | 3.382e-06 | 9.050e-07 |
| re900_e0.1 | 900 | 0.1 | 2 | 1 | B1_Mode2 | 0.127 | 1 | 1 | 0 | 1.076e-16 | 0.00786 | 0.477 | 0.00121 | -0.158 | -0.00555 | 0.00735 | 0 | 0 | 3 | 0.0986 | 0.0012 | 0.000155 | 7.243e-06 | 1.083e-05 |
| re900_e0.1 | 900 | 0.1 | 2 | 1 | B1_Mode2 | 0.127 | 1 | 1 | 0 | 1.076e-16 | 0.00786 | 0.477 | 0.00121 | -0.158 | -0.00555 | 0.00735 | 0 | 0 | 4 | 0.000121 | 8.656e-05 | 1.655e-06 | 3.382e-07 | 1.115e-07 |
| re900_e0.1 | 900 | 0.1 | 2 | 1 | B1_Mode2 | 0.127 | 1 | 1 | 0 | 1.076e-16 | 0.00786 | 0.477 | 0.00121 | -0.158 | -0.00555 | 0.00735 | 0 | 0 | 5 | 0.000213 | 0.000122 | 9.161e-07 | 5.088e-07 | 1.779e-07 |
| re900_e0.1 | 900 | 0.1 | 3 | 2 | B2_temporal | 0.123 | 1 | 0 | 1 | 1.115e-16 | -0.0269 | 0.0219 | -0.00106 | 0.00537 | -0.00123 | 0.00187 | 0 | 0 | 0 | 0.00295 | 0.988 | 0.00152 | 5.310e-05 | 1.147e-05 |
| re900_e0.1 | 900 | 0.1 | 3 | 2 | B2_temporal | 0.123 | 1 | 0 | 1 | 1.115e-16 | -0.0269 | 0.0219 | -0.00106 | 0.00537 | -0.00123 | 0.00187 | 0 | 0 | 1 | 0.00194 | 0.00467 | 0.000128 | 0.000758 | 1.427e-05 |
| re900_e0.1 | 900 | 0.1 | 3 | 2 | B2_temporal | 0.123 | 1 | 0 | 1 | 1.115e-16 | -0.0269 | 0.0219 | -0.00106 | 0.00537 | -0.00123 | 0.00187 | 0 | 0 | 2 | 4.575e-06 | 4.477e-06 | 0.000131 | 4.500e-05 | 1.203e-05 |
| re900_e0.1 | 900 | 0.1 | 3 | 2 | B2_temporal | 0.123 | 1 | 0 | 1 | 1.115e-16 | -0.0269 | 0.0219 | -0.00106 | 0.00537 | -0.00123 | 0.00187 | 0 | 0 | 3 | 0.000117 | 4.128e-07 | 4.303e-05 | 5.893e-07 | 6.091e-06 |
| re900_e0.1 | 900 | 0.1 | 3 | 2 | B2_temporal | 0.123 | 1 | 0 | 1 | 1.115e-16 | -0.0269 | 0.0219 | -0.00106 | 0.00537 | -0.00123 | 0.00187 | 0 | 0 | 4 | 6.145e-06 | 4.275e-07 | 7.915e-06 | 7.550e-06 | 2.976e-09 |
| re900_e0.1 | 900 | 0.1 | 3 | 2 | B2_temporal | 0.123 | 1 | 0 | 1 | 1.115e-16 | -0.0269 | 0.0219 | -0.00106 | 0.00537 | -0.00123 | 0.00187 | 0 | 0 | 5 | 1.429e-05 | 3.110e-06 | 1.801e-06 | 1.150e-07 | 2.197e-06 |
| re900_e0.1 | 900 | 0.1 | 4 | 2 | B2_temporal | 0.124 | 1 | 0 | 1 | 1.078e-16 | -0.00479 | 0.133 | 0.00271 | -0.00104 | -0.00199 | 0.00107 | 0 | 0 | 0 | 9.250e-05 | 0.92 | 0.000305 | 0.00018 | 5.930e-05 |
| re900_e0.1 | 900 | 0.1 | 4 | 2 | B2_temporal | 0.124 | 1 | 0 | 1 | 1.078e-16 | -0.00479 | 0.133 | 0.00271 | -0.00104 | -0.00199 | 0.00107 | 0 | 0 | 1 | 0.0712 | 0.000147 | 0.00158 | 0.00109 | 1.486e-05 |
| re900_e0.1 | 900 | 0.1 | 4 | 2 | B2_temporal | 0.124 | 1 | 0 | 1 | 1.078e-16 | -0.00479 | 0.133 | 0.00271 | -0.00104 | -0.00199 | 0.00107 | 0 | 0 | 2 | 2.970e-05 | 0.00515 | 1.161e-05 | 0.000249 | 1.100e-05 |
| re900_e0.1 | 900 | 0.1 | 4 | 2 | B2_temporal | 0.124 | 1 | 0 | 1 | 1.078e-16 | -0.00479 | 0.133 | 0.00271 | -0.00104 | -0.00199 | 0.00107 | 0 | 0 | 3 | 4.353e-06 | 4.659e-05 | 0.000188 | 1.029e-06 | 1.884e-06 |
| re900_e0.1 | 900 | 0.1 | 4 | 2 | B2_temporal | 0.124 | 1 | 0 | 1 | 1.078e-16 | -0.00479 | 0.133 | 0.00271 | -0.00104 | -0.00199 | 0.00107 | 0 | 0 | 4 | 1.605e-05 | 1.925e-05 | 1.947e-06 | 5.866e-07 | 1.198e-07 |
| re900_e0.1 | 900 | 0.1 | 4 | 2 | B2_temporal | 0.124 | 1 | 0 | 1 | 1.078e-16 | -0.00479 | 0.133 | 0.00271 | -0.00104 | -0.00199 | 0.00107 | 0 | 0 | 5 | 4.635e-06 | 1.970e-06 | 5.377e-06 | 1.659e-06 | 4.254e-07 |

Table 15: Décomposition modale — 185 points, 1110 lignes (6 modes $i$ par point, colonnes $E^f_{iJ}$ pour $J$=0..4), `mode_analysis.csv`.

| point | Re | $E^*$ | s | id | branch | $E_t$ | cov. | $i$ | $j$ | rmse | $A_{10}$ | $A_{20}$ | $A_{30}$ | $A_{40}$ | $A_{50}$ | $A_{60}$ | $A_{70}$ | $A_{80}$ | $i$ | $E^f_{0J}$ | $E^f_{1J}$ | $E^f_{2J}$ | $E^f_{3J}$ | $E^f_{4J}$ |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| re900_e0.25 | 900 | 0.25 | 0 | 3 | B3_mixed | 0.24 | 1 | 0 | 1 | 1.458e-16 | 0.0455 | -0.449 | -0.0142 | 0.138 | -0.0065 | -0.00777 | 0 | 0 | 0 | 0.0043 | 0.445 | 0.00685 | 0.00437 | 0.000792 |
| re900_e0.25 | 900 | 0.25 | 0 | 3 | B3_mixed | 0.24 | 1 | 0 | 1 | 1.458e-16 | 0.0455 | -0.449 | -0.0142 | 0.138 | -0.0065 | -0.00777 | 0 | 0 | 1 | 0.419 | 7.080e-05 | 0.00216 | 0.000943 | 1.671e-06 |
| re900_e0.25 | 900 | 0.25 | 0 | 3 | B3_mixed | 0.24 | 1 | 0 | 1 | 1.458e-16 | 0.0455 | -0.449 | -0.0142 | 0.138 | -0.0065 | -0.00777 | 0 | 0 | 2 | 0.000419 | 0.0575 | 0.00847 | 0.000902 | 8.070e-05 |
| re900_e0.25 | 900 | 0.25 | 0 | 3 | B3_mixed | 0.24 | 1 | 0 | 1 | 1.458e-16 | 0.0455 | -0.449 | -0.0142 | 0.138 | -0.0065 | -0.00777 | 0 | 0 | 3 | 0.0395 | 0.00392 | 0.00172 | 0.000217 | 3.805e-06 |
| re900_e0.25 | 900 | 0.25 | 0 | 3 | B3_mixed | 0.24 | 1 | 0 | 1 | 1.458e-16 | 0.0455 | -0.449 | -0.0142 | 0.138 | -0.0065 | -0.00777 | 0 | 0 | 4 | 8.778e-05 | 0.00146 | 0.00128 | 5.104e-05 | 5.630e-05 |
| re900_e0.25 | 900 | 0.25 | 0 | 3 | B3_mixed | 0.24 | 1 | 0 | 1 | 1.458e-16 | 0.0455 | -0.449 | -0.0142 | 0.138 | -0.0065 | -0.00777 | 0 | 0 | 5 | 0.000126 | 0.000521 | 2.489e-05 | 9.636e-05 | 5.029e-06 |
| re900_e0.25 | 900 | 0.25 | 1 | 1 | B1_Mode2 | 0.266 | 1 | 1 | 0 | 1.504e-16 | -0.0045 | 0.71 | 0.0616 | -0.112 | -0.011 | -0.00979 | 0 | 0 | 0 | 3.807e-05 | 0.0148 | 0.000213 | 0.000195 | 9.973e-06 |
| re900_e0.25 | 900 | 0.25 | 1 | 1 | B1_Mode2 | 0.266 | 1 | 1 | 0 | 1.504e-16 | -0.0045 | 0.71 | 0.0616 | -0.112 | -0.011 | -0.00979 | 0 | 0 | 1 | 0.947 | 0.00321 | 0.000363 | 6.447e-07 | 2.344e-05 |
| re900_e0.25 | 900 | 0.25 | 1 | 1 | B1_Mode2 | 0.266 | 1 | 1 | 0 | 1.504e-16 | -0.0045 | 0.71 | 0.0616 | -0.112 | -0.011 | -0.00979 | 0 | 0 | 2 | 0.00713 | 2.728e-05 | 4.123e-05 | 6.135e-07 | 2.164e-06 |
| re900_e0.25 | 900 | 0.25 | 1 | 1 | B1_Mode2 | 0.266 | 1 | 1 | 0 | 1.504e-16 | -0.0045 | 0.71 | 0.0616 | -0.112 | -0.011 | -0.00979 | 0 | 0 | 3 | 0.0236 | 0.0027 | 0.00022 | 2.794e-05 | 1.001e-05 |
| re900_e0.25 | 900 | 0.25 | 1 | 1 | B1_Mode2 | 0.266 | 1 | 1 | 0 | 1.504e-16 | -0.0045 | 0.71 | 0.0616 | -0.112 | -0.011 | -0.00979 | 0 | 0 | 4 | 0.000228 | 0.000136 | 8.122e-07 | 1.071e-06 | 1.693e-08 |
| re900_e0.25 | 900 | 0.25 | 1 | 1 | B1_Mode2 | 0.266 | 1 | 1 | 0 | 1.504e-16 | -0.0045 | 0.71 | 0.0616 | -0.112 | -0.011 | -0.00979 | 0 | 0 | 5 | 0.00018 | 9.093e-06 | 6.567e-06 | 1.346e-08 | 1.130e-07 |
| re900_e0.25 | 900 | 0.25 | 2 | 1 | B1_Mode2 | 0.27 | 1 | 1 | 0 | 1.476e-16 | 0.0205 | 0.703 | -0.0183 | -0.193 | 0.00288 | -0.000953 | 0 | 0 | 0 | 0.000778 | 0.000347 | 3.260e-05 | 7.854e-08 | 7.225e-07 |
| re900_e0.25 | 900 | 0.25 | 2 | 1 | B1_Mode2 | 0.27 | 1 | 1 | 0 | 1.476e-16 | 0.0205 | 0.703 | -0.0183 | -0.193 | 0.00288 | -0.000953 | 0 | 0 | 1 | 0.915 | 0.00364 | 0.000457 | 5.564e-05 | 2.005e-05 |
| re900_e0.25 | 900 | 0.25 | 2 | 1 | B1_Mode2 | 0.27 | 1 | 1 | 0 | 1.476e-16 | 0.0205 | 0.703 | -0.0183 | -0.193 | 0.00288 | -0.000953 | 0 | 0 | 2 | 0.000621 | 0.0024 | 1.456e-05 | 9.952e-06 | 1.507e-06 |
| re900_e0.25 | 900 | 0.25 | 2 | 1 | B1_Mode2 | 0.27 | 1 | 1 | 0 | 1.476e-16 | 0.0205 | 0.703 | -0.0183 | -0.193 | 0.00288 | -0.000953 | 0 | 0 | 3 | 0.0687 | 0.006 | 0.000239 | 2.470e-05 | 1.771e-05 |
| re900_e0.25 | 900 | 0.25 | 2 | 1 | B1_Mode2 | 0.27 | 1 | 1 | 0 | 1.476e-16 | 0.0205 | 0.703 | -0.0183 | -0.193 | 0.00288 | -0.000953 | 0 | 0 | 4 | 1.534e-05 | 0.000506 | 2.948e-06 | 1.977e-06 | 1.503e-07 |
| re900_e0.25 | 900 | 0.25 | 2 | 1 | B1_Mode2 | 0.27 | 1 | 1 | 0 | 1.476e-16 | 0.0205 | 0.703 | -0.0183 | -0.193 | 0.00288 | -0.000953 | 0 | 0 | 5 | 1.683e-06 | 0.00054 | 4.256e-05 | 1.088e-06 | 3.347e-06 |
| re900_e0.25 | 900 | 0.25 | 3 | 2 | B2_temporal | 0.269 | 1 | 0 | 1 | 1.468e-16 | -0.0415 | 0.0119 | -0.00264 | 0.00317 | -0.000662 | 0.00225 | 0 | 0 | 0 | 0.00321 | 0.985 | 4.600e-05 | 0.000118 | 3.164e-06 |
| re900_e0.25 | 900 | 0.25 | 3 | 2 | B2_temporal | 0.269 | 1 | 0 | 1 | 1.468e-16 | -0.0415 | 0.0119 | -0.00264 | 0.00317 | -0.000662 | 0.00225 | 0 | 0 | 1 | 0.000264 | 0.0088 | 0.000668 | 0.0013 | 5.934e-06 |
| re900_e0.25 | 900 | 0.25 | 3 | 2 | B2_temporal | 0.269 | 1 | 0 | 1 | 1.468e-16 | -0.0415 | 0.0119 | -0.00264 | 0.00317 | -0.000662 | 0.00225 | 0 | 0 | 2 | 1.293e-05 | 1.806e-05 | 2.245e-05 | 0.00011 | 1.433e-05 |
| re900_e0.25 | 900 | 0.25 | 3 | 2 | B2_temporal | 0.269 | 1 | 0 | 1 | 1.468e-16 | -0.0415 | 0.0119 | -0.00264 | 0.00317 | -0.000662 | 0.00225 | 0 | 0 | 3 | 1.870e-05 | 5.669e-05 | 9.349e-06 | 1.229e-05 | 2.118e-06 |
| re900_e0.25 | 900 | 0.25 | 3 | 2 | B2_temporal | 0.269 | 1 | 0 | 1 | 1.468e-16 | -0.0415 | 0.0119 | -0.00264 | 0.00317 | -0.000662 | 0.00225 | 0 | 0 | 4 | 8.154e-07 | 3.740e-09 | 3.964e-06 | 2.015e-06 | 9.844e-10 |
| re900_e0.25 | 900 | 0.25 | 3 | 2 | B2_temporal | 0.269 | 1 | 0 | 1 | 1.468e-16 | -0.0415 | 0.0119 | -0.00264 | 0.00317 | -0.000662 | 0.00225 | 0 | 0 | 5 | 9.439e-06 | 3.033e-06 | 2.501e-06 | 6.477e-07 | 2.424e-06 |
| re900_e0.25 | 900 | 0.25 | 4 | 2 | B2_temporal | 0.262 | 1 | 0 | 1 | 1.461e-16 | -0.00151 | 0.104 | -0.000636 | 0.00183 | -0.00336 | 0.000987 | 0 | 0 | 0 | 4.332e-06 | 0.965 | 5.664e-05 | 0.000206 | 4.927e-05 |
| re900_e0.25 | 900 | 0.25 | 4 | 2 | B2_temporal | 0.262 | 1 | 0 | 1 | 1.461e-16 | -0.00151 | 0.104 | -0.000636 | 0.00183 | -0.00336 | 0.000987 | 0 | 0 | 1 | 0.0205 | 2.901e-05 | 0.00232 | 0.00279 | 9.652e-06 |
| re900_e0.25 | 900 | 0.25 | 4 | 2 | B2_temporal | 0.262 | 1 | 0 | 1 | 1.461e-16 | -0.00151 | 0.104 | -0.000636 | 0.00183 | -0.00336 | 0.000987 | 0 | 0 | 2 | 7.702e-07 | 0.00804 | 6.057e-06 | 0.000643 | 4.692e-05 |
| re900_e0.25 | 900 | 0.25 | 4 | 2 | B2_temporal | 0.262 | 1 | 0 | 1 | 1.461e-16 | -0.00151 | 0.104 | -0.000636 | 0.00183 | -0.00336 | 0.000987 | 0 | 0 | 3 | 6.347e-06 | 0.00013 | 0.000164 | 1.311e-05 | 2.650e-05 |
| re900_e0.25 | 900 | 0.25 | 4 | 2 | B2_temporal | 0.262 | 1 | 0 | 1 | 1.461e-16 | -0.00151 | 0.104 | -0.000636 | 0.00183 | -0.00336 | 0.000987 | 0 | 0 | 4 | 2.153e-05 | 5.402e-05 | 1.568e-06 | 6.164e-06 | 8.491e-07 |
| re900_e0.25 | 900 | 0.25 | 4 | 2 | B2_temporal | 0.262 | 1 | 0 | 1 | 1.461e-16 | -0.00151 | 0.104 | -0.000636 | 0.00183 | -0.00336 | 0.000987 | 0 | 0 | 5 | 1.857e-06 | 1.067e-05 | 1.683e-06 | 3.330e-06 | 4.944e-07 |
| re900_e0.5 | 900 | 0.5 | 0 | 3 | B3_mixed | 0.463 | 1 | 0 | 1 | 2.309e-16 | 0.0712 | -0.578 | -0.00787 | 0.139 | -0.0227 | 0.00401 | 0 | 0 | 0 | 0.00547 | 0.494 | 0.00566 | 0.00747 | 0.00149 |
| re900_e0.5 | 900 | 0.5 | 0 | 3 | B3_mixed | 0.463 | 1 | 0 | 1 | 2.309e-16 | 0.0712 | -0.578 | -0.00787 | 0.139 | -0.0227 | 0.00401 | 0 | 0 | 1 | 0.361 | 0.00617 | 0.00933 | 0.00259 | 4.089e-08 |
| re900_e0.5 | 900 | 0.5 | 0 | 3 | B3_mixed | 0.463 | 1 | 0 | 1 | 2.309e-16 | 0.0712 | -0.578 | -0.00787 | 0.139 | -0.0227 | 0.00401 | 0 | 0 | 2 | 6.675e-05 | 0.0603 | 0.00995 | 0.00239 | 0.000297 |
| re900_e0.5 | 900 | 0.5 | 0 | 3 | B3_mixed | 0.463 | 1 | 0 | 1 | 2.309e-16 | 0.0712 | -0.578 | -0.00787 | 0.139 | -0.0227 | 0.00401 | 0 | 0 | 3 | 0.0209 | 0.00294 | 0.00433 | 0.00124 | 1.419e-06 |
| re900_e0.5 | 900 | 0.5 | 0 | 3 | B3_mixed | 0.463 | 1 | 0 | 1 | 2.309e-16 | 0.0712 | -0.578 | -0.00787 | 0.139 | -0.0227 | 0.00401 | 0 | 0 | 4 | 0.000558 | 0.00133 | 0.00196 | 9.102e-05 | 0.000101 |
| re900_e0.5 | 900 | 0.5 | 0 | 3 | B3_mixed | 0.463 | 1 | 0 | 1 | 2.309e-16 | 0.0712 | -0.578 | -0.00787 | 0.139 | -0.0227 | 0.00401 | 0 | 0 | 5 | 1.732e-05 | 0.000338 | 9.881e-07 | 0.000234 | 4.535e-06 |
| re900_e0.5 | 900 | 0.5 | 1 | 1 | B1_Mode2 | 0.517 | 1 | 1 | 0 | 2.187e-16 | 0.0172 | 0.987 | -0.00624 | -0.0914 | -0.00458 | -0.0145 | 0 | 0 | 0 | 0.000288 | 0.0411 | 0.000613 | 0.00063 | 2.274e-05 |
| re900_e0.5 | 900 | 0.5 | 1 | 1 | B1_Mode2 | 0.517 | 1 | 1 | 0 | 2.187e-16 | 0.0172 | 0.987 | -0.00624 | -0.0914 | -0.00458 | -0.0145 | 0 | 0 | 1 | 0.943 | 0.00165 | 0.00044 | 1.637e-05 | 3.121e-05 |
| re900_e0.5 | 900 | 0.5 | 1 | 1 | B1_Mode2 | 0.517 | 1 | 1 | 0 | 2.187e-16 | 0.0172 | 0.987 | -0.00624 | -0.0914 | -0.00458 | -0.0145 | 0 | 0 | 2 | 3.769e-05 | 1.581e-05 | 0.000454 | 2.784e-07 | 2.522e-05 |
| re900_e0.5 | 900 | 0.5 | 1 | 1 | B1_Mode2 | 0.517 | 1 | 1 | 0 | 2.187e-16 | 0.0172 | 0.987 | -0.00624 | -0.0914 | -0.00458 | -0.0145 | 0 | 0 | 3 | 0.00808 | 0.00276 | 0.000222 | 3.110e-05 | 9.647e-06 |
| re900_e0.5 | 900 | 0.5 | 1 | 1 | B1_Mode2 | 0.517 | 1 | 1 | 0 | 2.187e-16 | 0.0172 | 0.987 | -0.00624 | -0.0914 | -0.00458 | -0.0145 | 0 | 0 | 4 | 2.032e-05 | 8.496e-06 | 1.234e-06 | 1.152e-06 | 8.384e-09 |
| re900_e0.5 | 900 | 0.5 | 1 | 1 | B1_Mode2 | 0.517 | 1 | 1 | 0 | 2.187e-16 | 0.0172 | 0.987 | -0.00624 | -0.0914 | -0.00458 | -0.0145 | 0 | 0 | 5 | 0.000202 | 8.078e-06 | 7.699e-06 | 9.824e-08 | 6.222e-09 |
| re900_e0.5 | 900 | 0.5 | 2 | 1 | B1_Mode2 | 0.523 | 1 | 1 | 0 | 2.094e-16 | 0.0355 | 0.983 | -0.0545 | -0.186 | -0.0116 | -0.0032 | 0 | 0 | 0 | 0.00121 | 0.00671 | 9.542e-06 | 2.311e-05 | 5.070e-11 |
| re900_e0.5 | 900 | 0.5 | 2 | 1 | B1_Mode2 | 0.523 | 1 | 1 | 0 | 2.094e-16 | 0.0355 | 0.983 | -0.0545 | -0.186 | -0.0116 | -0.0032 | 0 | 0 | 1 | 0.925 | 0.000493 | 0.00139 | 3.680e-05 | 6.081e-05 |
| re900_e0.5 | 900 | 0.5 | 2 | 1 | B1_Mode2 | 0.523 | 1 | 1 | 0 | 2.094e-16 | 0.0355 | 0.983 | -0.0545 | -0.186 | -0.0116 | -0.0032 | 0 | 0 | 2 | 0.00284 | 0.0164 | 0.000629 | 3.812e-05 | 3.274e-05 |
| re900_e0.5 | 900 | 0.5 | 2 | 1 | B1_Mode2 | 0.523 | 1 | 1 | 0 | 2.094e-16 | 0.0355 | 0.983 | -0.0545 | -0.186 | -0.0116 | -0.0032 | 0 | 0 | 3 | 0.0331 | 0.00894 | 0.000515 | 6.233e-06 | 3.502e-05 |
| re900_e0.5 | 900 | 0.5 | 2 | 1 | B1_Mode2 | 0.523 | 1 | 1 | 0 | 2.094e-16 | 0.0355 | 0.983 | -0.0545 | -0.186 | -0.0116 | -0.0032 | 0 | 0 | 4 | 0.000128 | 0.00114 | 0.000423 | 1.403e-05 | 9.384e-06 |
| re900_e0.5 | 900 | 0.5 | 2 | 1 | B1_Mode2 | 0.523 | 1 | 1 | 0 | 2.094e-16 | 0.0355 | 0.983 | -0.0545 | -0.186 | -0.0116 | -0.0032 | 0 | 0 | 5 | 9.817e-06 | 0.000548 | 3.682e-05 | 1.220e-06 | 1.858e-06 |
| re900_e0.5 | 900 | 0.5 | 3 | 2 | B2_temporal | 0.522 | 1 | 0 | 1 | 2.491e-16 | -0.0604 | -0.00512 | 0.00385 | 0.00117 | -0.00199 | 0.00135 | 0 | 0 | 0 | 0.00349 | 0.99 | 6.258e-05 | 4.692e-05 | 6.763e-05 |
| re900_e0.5 | 900 | 0.5 | 3 | 2 | B2_temporal | 0.522 | 1 | 0 | 1 | 2.491e-16 | -0.0604 | -0.00512 | 0.00385 | 0.00117 | -0.00199 | 0.00135 | 0 | 0 | 1 | 2.505e-05 | 0.00396 | 7.928e-07 | 0.0025 | 6.636e-08 |
| re900_e0.5 | 900 | 0.5 | 3 | 2 | B2_temporal | 0.522 | 1 | 0 | 1 | 2.491e-16 | -0.0604 | -0.00512 | 0.00385 | 0.00117 | -0.00199 | 0.00135 | 0 | 0 | 2 | 1.416e-05 | 1.373e-05 | 3.744e-05 | 2.629e-05 | 8.293e-05 |
| re900_e0.5 | 900 | 0.5 | 3 | 2 | B2_temporal | 0.522 | 1 | 0 | 1 | 2.491e-16 | -0.0604 | -0.00512 | 0.00385 | 0.00117 | -0.00199 | 0.00135 | 0 | 0 | 3 | 1.307e-06 | 1.387e-05 | 4.228e-05 | 1.416e-05 | 1.005e-05 |
| re900_e0.5 | 900 | 0.5 | 3 | 2 | B2_temporal | 0.522 | 1 | 0 | 1 | 2.491e-16 | -0.0604 | -0.00512 | 0.00385 | 0.00117 | -0.00199 | 0.00135 | 0 | 0 | 4 | 3.772e-06 | 1.933e-07 | 2.683e-08 | 4.689e-06 | 2.152e-08 |
| re900_e0.5 | 900 | 0.5 | 3 | 2 | B2_temporal | 0.522 | 1 | 0 | 1 | 2.491e-16 | -0.0604 | -0.00512 | 0.00385 | 0.00117 | -0.00199 | 0.00135 | 0 | 0 | 5 | 1.748e-06 | 9.559e-07 | 5.215e-06 | 2.357e-07 | 3.010e-06 |
| re900_e0.5 | 900 | 0.5 | 4 | 2 | B2_temporal | 0.497 | 1 | 0 | 1 | 2.016e-16 | 0.0102 | -0.0107 | -0.012 | 0.0119 | -0.00485 | 0.00142 | 0 | 0 | 0 | 0.000105 | 0.959 | 0.0221 | 0.000547 | 1.794e-05 |
| re900_e0.5 | 900 | 0.5 | 4 | 2 | B2_temporal | 0.497 | 1 | 0 | 1 | 2.016e-16 | 0.0102 | -0.0107 | -0.012 | 0.0119 | -0.00485 | 0.00142 | 0 | 0 | 1 | 0.000115 | 0.000319 | 1.361e-05 | 0.00329 | 0.00015 |
| re900_e0.5 | 900 | 0.5 | 4 | 2 | B2_temporal | 0.497 | 1 | 0 | 1 | 2.016e-16 | 0.0102 | -0.0107 | -0.012 | 0.0119 | -0.00485 | 0.00142 | 0 | 0 | 2 | 0.000145 | 0.0104 | 0.00277 | 7.814e-05 | 1.085e-07 |
| re900_e0.5 | 900 | 0.5 | 4 | 2 | B2_temporal | 0.497 | 1 | 0 | 1 | 2.016e-16 | 0.0102 | -0.0107 | -0.012 | 0.0119 | -0.00485 | 0.00142 | 0 | 0 | 3 | 0.000144 | 0.000475 | 5.234e-07 | 6.466e-05 | 5.377e-05 |
| re900_e0.5 | 900 | 0.5 | 4 | 2 | B2_temporal | 0.497 | 1 | 0 | 1 | 2.016e-16 | 0.0102 | -0.0107 | -0.012 | 0.0119 | -0.00485 | 0.00142 | 0 | 0 | 4 | 2.369e-05 | 9.294e-05 | 5.014e-05 | 2.702e-06 | 1.070e-05 |
| re900_e0.5 | 900 | 0.5 | 4 | 2 | B2_temporal | 0.497 | 1 | 0 | 1 | 2.016e-16 | 0.0102 | -0.0107 | -0.012 | 0.0119 | -0.00485 | 0.00142 | 0 | 0 | 5 | 2.035e-06 | 1.615e-05 | 3.799e-06 | 8.130e-06 | 3.624e-07 |
| re900_e1 | 900 | 1 | 0 | 3 | B3_mixed | 0.961 | 1 | 2 | 0 | 3.094e-16 | 0.34 | -0.608 | -0.831 | 0.0722 | 0.132 | 0.0198 | 0 | 0 | 0 | 0.0601 | 0.00312 | 0.00306 | 0.000138 | 0.000141 |
| re900_e1 | 900 | 1 | 0 | 3 | B3_mixed | 0.961 | 1 | 2 | 0 | 3.094e-16 | 0.34 | -0.608 | -0.831 | 0.0722 | 0.132 | 0.0198 | 0 | 0 | 1 | 0.193 | 0.253 | 0.00717 | 5.959e-05 | 6.889e-05 |
| re900_e1 | 900 | 1 | 0 | 3 | B3_mixed | 0.961 | 1 | 2 | 0 | 3.094e-16 | 0.34 | -0.608 | -0.831 | 0.0722 | 0.132 | 0.0198 | 0 | 0 | 2 | 0.36 | 0.0335 | 0.0221 | 0.00044 | 0.000403 |
| re900_e1 | 900 | 1 | 0 | 3 | B3_mixed | 0.961 | 1 | 2 | 0 | 3.094e-16 | 0.34 | -0.608 | -0.831 | 0.0722 | 0.132 | 0.0198 | 0 | 0 | 3 | 0.00271 | 0.0232 | 0.0117 | 0.000913 | 7.271e-05 |
| re900_e1 | 900 | 1 | 0 | 3 | B3_mixed | 0.961 | 1 | 2 | 0 | 3.094e-16 | 0.34 | -0.608 | -0.831 | 0.0722 | 0.132 | 0.0198 | 0 | 0 | 4 | 0.00902 | 0.0128 | 0.00227 | 0.000584 | 7.287e-07 |
| re900_e1 | 900 | 1 | 0 | 3 | B3_mixed | 0.961 | 1 | 2 | 0 | 3.094e-16 | 0.34 | -0.608 | -0.831 | 0.0722 | 0.132 | 0.0198 | 0 | 0 | 5 | 0.000205 | 0.000542 | 0.000772 | 2.323e-05 | 3.198e-07 |

Table 15: Décomposition modale — 185 points, 1110 lignes (6 modes $i$ par point, colonnes $E^f_{iJ}$ pour $J$=0..4), `mode_analysis.csv`.

| point | Re | $E^*$ | s | id | branch | $E_t$ | cov. | $i$ | $j$ | rmse | $A_{10}$ | $A_{20}$ | $A_{30}$ | $A_{40}$ | $A_{50}$ | $A_{60}$ | $A_{70}$ | $A_{80}$ | $i$ | $E^f_{0J}$ | $E^f_{1J}$ | $E^f_{2J}$ | $E^f_{3J}$ | $E^f_{4J}$ |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| re900_e1 | 900 | 1 | 1 | 1 | B1_Mode2 | 1.03 | 1 | 1 | 0 | 2.759e-16 | 0.0458 | 1.41 | -0.043 | -0.145 | 0.00793 | -0.00939 | 0 | 0 | 0 | 0.00102 | 0.00746 | 0.000988 | 0.000185 | 6.473e-05 |
| re900_e1 | 900 | 1 | 1 | 1 | B1_Mode2 | 1.03 | 1 | 1 | 0 | 2.759e-16 | 0.0458 | 1.41 | -0.043 | -0.145 | 0.00793 | -0.00939 | 0 | 0 | 1 | 0.97 | 0.00319 | 0.000153 | 4.633e-05 | 1.521e-05 |
| re900_e1 | 900 | 1 | 1 | 1 | B1_Mode2 | 1.03 | 1 | 1 | 0 | 2.759e-16 | 0.0458 | 1.41 | -0.043 | -0.145 | 0.00793 | -0.00939 | 0 | 0 | 2 | 0.000899 | 0.0038 | 8.356e-05 | 2.196e-05 | 1.496e-06 |
| re900_e1 | 900 | 1 | 1 | 1 | B1_Mode2 | 1.03 | 1 | 1 | 0 | 2.759e-16 | 0.0458 | 1.41 | -0.043 | -0.145 | 0.00793 | -0.00939 | 0 | 0 | 3 | 0.0102 | 0.0014 | 0.00018 | 2.319e-05 | 6.446e-06 |
| re900_e1 | 900 | 1 | 1 | 1 | B1_Mode2 | 1.03 | 1 | 1 | 0 | 2.759e-16 | 0.0458 | 1.41 | -0.043 | -0.145 | 0.00793 | -0.00939 | 0 | 0 | 4 | 3.052e-05 | 0.000189 | 2.015e-05 | 4.874e-09 | 1.584e-06 |
| re900_e1 | 900 | 1 | 1 | 1 | B1_Mode2 | 1.03 | 1 | 1 | 0 | 2.759e-16 | 0.0458 | 1.41 | -0.043 | -0.145 | 0.00793 | -0.00939 | 0 | 0 | 5 | 4.274e-05 | 8.007e-05 | 8.396e-06 | 9.526e-08 | 8.144e-08 |
| re900_e1 | 900 | 1 | 2 | 1 | B1_Mode2 | 1.02 | 1 | 1 | 0 | 2.836e-16 | 0.0654 | 1.34 | -0.204 | -0.201 | -0.03 | -0.00829 | 0 | 0 | 0 | 0.00209 | 0.0123 | 1.756e-06 | 3.418e-05 | 1.220e-07 |
| re900_e1 | 900 | 1 | 2 | 1 | B1_Mode2 | 1.02 | 1 | 1 | 0 | 2.836e-16 | 0.0654 | 1.34 | -0.204 | -0.201 | -0.03 | -0.00829 | 0 | 0 | 1 | 0.877 | 0.00215 | 0.00288 | 5.631e-05 | 0.000122 |
| re900_e1 | 900 | 1 | 2 | 1 | B1_Mode2 | 1.02 | 1 | 1 | 0 | 2.836e-16 | 0.0654 | 1.34 | -0.204 | -0.201 | -0.03 | -0.00829 | 0 | 0 | 2 | 0.0204 | 0.0426 | 0.000616 | 4.619e-05 | 3.329e-05 |
| re900_e1 | 900 | 1 | 2 | 1 | B1_Mode2 | 1.02 | 1 | 1 | 0 | 2.836e-16 | 0.0654 | 1.34 | -0.204 | -0.201 | -0.03 | -0.00829 | 0 | 0 | 3 | 0.0198 | 0.0085 | 0.00278 | 2.905e-06 | 0.000125 |
| re900_e1 | 900 | 1 | 2 | 1 | B1_Mode2 | 1.02 | 1 | 1 | 0 | 2.836e-16 | 0.0654 | 1.34 | -0.204 | -0.201 | -0.03 | -0.00829 | 0 | 0 | 4 | 0.000441 | 0.00657 | 0.00125 | 2.399e-08 | 2.255e-05 |
| re900_e1 | 900 | 1 | 2 | 1 | B1_Mode2 | 1.02 | 1 | 1 | 0 | 2.836e-16 | 0.0654 | 1.34 | -0.204 | -0.201 | -0.03 | -0.00829 | 0 | 0 | 5 | 3.359e-05 | 0.000244 | 0.00016 | 4.650e-05 | 1.166e-06 |
| re900_e1 | 900 | 1 | 3 | 2 | B2_temporal | 0.991 | 1 | 0 | 1 | 3.002e-16 | -0.122 | -0.475 | 0.142 | -0.0709 | 0.0362 | -0.00326 | 0 | 0 | 0 | 0.00749 | 0.83 | 0.00724 | 0.000781 | 9.861e-05 |
| re900_e1 | 900 | 1 | 3 | 2 | B2_temporal | 0.991 | 1 | 0 | 1 | 3.002e-16 | -0.122 | -0.475 | 0.142 | -0.0709 | 0.0362 | -0.00326 | 0 | 0 | 1 | 0.114 | 0.017 | 0.000542 | 0.00125 | 1.677e-05 |
| re900_e1 | 900 | 1 | 3 | 2 | B2_temporal | 0.991 | 1 | 0 | 1 | 3.002e-16 | -0.122 | -0.475 | 0.142 | -0.0709 | 0.0362 | -0.00326 | 0 | 0 | 2 | 0.0102 | 0.000413 | 0.00564 | 0.000619 | 0.000253 |
| re900_e1 | 900 | 1 | 3 | 2 | B2_temporal | 0.991 | 1 | 0 | 1 | 3.002e-16 | -0.122 | -0.475 | 0.142 | -0.0709 | 0.0362 | -0.00326 | 0 | 0 | 3 | 0.00254 | 0.00011 | 0.000686 | 0.00015 | 5.646e-05 |
| re900_e1 | 900 | 1 | 3 | 2 | B2_temporal | 0.991 | 1 | 0 | 1 | 3.002e-16 | -0.122 | -0.475 | 0.142 | -0.0709 | 0.0362 | -0.00326 | 0 | 0 | 4 | 0.00066 | 0.000474 | 0.000189 | 0.000181 | 2.803e-05 |
| re900_e1 | 900 | 1 | 3 | 2 | B2_temporal | 0.991 | 1 | 0 | 1 | 3.002e-16 | -0.122 | -0.475 | 0.142 | -0.0709 | 0.0362 | -0.00326 | 0 | 0 | 5 | 5.360e-06 | 6.127e-05 | 7.231e-06 | 9.838e-06 | 4.641e-08 |
| re900_e1 | 900 | 1 | 4 | 2 | B2_temporal | 0.953 | 1 | 0 | 1 | 3.045e-16 | 0.00981 | 0.322 | 0.0474 | -0.0333 | -0.00292 | -0.005 | 0 | 0 | 0 | 5.052e-05 | 0.69 | 0.128 | 0.0118 | 0.000558 |
| re900_e1 | 900 | 1 | 4 | 2 | B2_temporal | 0.953 | 1 | 0 | 1 | 3.045e-16 | 0.00981 | 0.322 | 0.0474 | -0.0333 | -0.00292 | -0.005 | 0 | 0 | 1 | 0.0545 | 0.093 | 0.00973 | 0.00477 | 0.000523 |
| re900_e1 | 900 | 1 | 4 | 2 | B2_temporal | 0.953 | 1 | 0 | 1 | 3.045e-16 | 0.00981 | 0.322 | 0.0474 | -0.0333 | -0.00292 | -0.005 | 0 | 0 | 2 | 0.00118 | 0.00133 | 0.00141 | 0.000197 | 0.000101 |
| re900_e1 | 900 | 1 | 4 | 2 | B2_temporal | 0.953 | 1 | 0 | 1 | 3.045e-16 | 0.00981 | 0.322 | 0.0474 | -0.0333 | -0.00292 | -0.005 | 0 | 0 | 3 | 0.000583 | 0.00117 | 8.818e-10 | 1.726e-05 | 1.911e-06 |
| re900_e1 | 900 | 1 | 4 | 2 | B2_temporal | 0.953 | 1 | 0 | 1 | 3.045e-16 | 0.00981 | 0.322 | 0.0474 | -0.0333 | -0.00292 | -0.005 | 0 | 0 | 4 | 4.469e-06 | 8.063e-05 | 5.249e-05 | 2.764e-06 | 6.185e-06 |
| re900_e1 | 900 | 1 | 4 | 2 | B2_temporal | 0.953 | 1 | 0 | 1 | 3.045e-16 | 0.00981 | 0.322 | 0.0474 | -0.0333 | -0.00292 | -0.005 | 0 | 0 | 5 | 1.310e-05 | 7.973e-05 | 5.147e-06 | 2.744e-08 | 3.138e-06 |
| re900_e2 | 900 | 2 | 0 | 3 | B3_mixed | 1.82 | 1 | 2 | 0 | 3.760e-16 | 0.581 | 0.273 | -1.59 | -0.0837 | 0.395 | 0.0555 | 0 | 0 | 0 | 0.0924 | 0.0296 | 0.0036 | 0.000421 | 0.00011 |
| re900_e2 | 900 | 2 | 0 | 3 | B3_mixed | 1.82 | 1 | 2 | 0 | 3.760e-16 | 0.581 | 0.273 | -1.59 | -0.0837 | 0.395 | 0.0555 | 0 | 0 | 1 | 0.0205 | 0.0963 | 1.226e-06 | 0.000368 | 2.642e-06 |
| re900_e2 | 900 | 2 | 0 | 3 | B3_mixed | 1.82 | 1 | 2 | 0 | 3.760e-16 | 0.581 | 0.273 | -1.59 | -0.0837 | 0.395 | 0.0555 | 0 | 0 | 2 | 0.691 | 0.00073 | 0.00228 | 2.243e-05 | 0.000115 |
| re900_e2 | 900 | 2 | 0 | 3 | B3_mixed | 1.82 | 1 | 2 | 0 | 3.760e-16 | 0.581 | 0.273 | -1.59 | -0.0837 | 0.395 | 0.0555 | 0 | 0 | 3 | 0.00192 | 0.014 | 0.000284 | 4.800e-05 | 7.480e-06 |
| re900_e2 | 900 | 2 | 0 | 3 | B3_mixed | 1.82 | 1 | 2 | 0 | 3.760e-16 | 0.581 | 0.273 | -1.59 | -0.0837 | 0.395 | 0.0555 | 0 | 0 | 4 | 0.0428 | 0.000431 | 0.00122 | 6.966e-05 | 5.338e-05 |
| re900_e2 | 900 | 2 | 0 | 3 | B3_mixed | 1.82 | 1 | 2 | 0 | 3.760e-16 | 0.581 | 0.273 | -1.59 | -0.0837 | 0.395 | 0.0555 | 0 | 0 | 5 | 0.000846 | 0.000833 | 9.146e-05 | 7.460e-06 | 1.620e-09 |
| re900_e2 | 900 | 2 | 1 | 1 | B1_Mode2 | 2 | 1 | 1 | 0 | 4.560e-16 | 0.243 | 1.72 | -0.448 | -0.0645 | -0.0532 | -0.0376 | 0 | 0 | 0 | 0.0148 | 0.0629 | 0.000535 | 6.183e-05 | 3.014e-05 |
| re900_e2 | 900 | 2 | 1 | 1 | B1_Mode2 | 2 | 1 | 1 | 0 | 4.560e-16 | 0.243 | 1.72 | -0.448 | -0.0645 | -0.0532 | -0.0376 | 0 | 0 | 1 | 0.739 | 0.0762 | 0.00667 | 4.499e-05 | 0.0004 |
| re900_e2 | 900 | 2 | 1 | 1 | B1_Mode2 | 2 | 1 | 1 | 0 | 4.560e-16 | 0.243 | 1.72 | -0.448 | -0.0645 | -0.0532 | -0.0376 | 0 | 0 | 2 | 0.05 | 0.0353 | 0.00434 | 0.000206 | 6.379e-05 |
| re900_e2 | 900 | 2 | 1 | 1 | B1_Mode2 | 2 | 1 | 1 | 0 | 4.560e-16 | 0.243 | 1.72 | -0.448 | -0.0645 | -0.0532 | -0.0376 | 0 | 0 | 3 | 0.00104 | 0.000124 | 0.00412 | 4.066e-05 | 0.00015 |
| re900_e2 | 900 | 2 | 1 | 1 | B1_Mode2 | 2 | 1 | 1 | 0 | 4.560e-16 | 0.243 | 1.72 | -0.448 | -0.0645 | -0.0532 | -0.0376 | 0 | 0 | 4 | 0.000707 | 0.00237 | 1.311e-07 | 0.000259 | 4.011e-06 |
| re900_e2 | 900 | 2 | 1 | 1 | B1_Mode2 | 2 | 1 | 1 | 0 | 4.560e-16 | 0.243 | 1.72 | -0.448 | -0.0645 | -0.0532 | -0.0376 | 0 | 0 | 5 | 0.000353 | 1.016e-05 | 0.00012 | 4.297e-06 | 8.525e-07 |
| re900_e2 | 900 | 2 | 2 | 1 | B1_Mode2 | 2.07 | 1 | 1 | 0 | 4.671e-16 | 0.228 | 1.72 | -0.421 | -0.172 | -0.0346 | -0.0172 | 0 | 0 | 0 | 0.0126 | 0.105 | 1.697e-07 | 1.652e-06 | 2.033e-06 |
| re900_e2 | 900 | 2 | 2 | 1 | B1_Mode2 | 2.07 | 1 | 1 | 0 | 4.671e-16 | 0.228 | 1.72 | -0.421 | -0.172 | -0.0346 | -0.0172 | 0 | 0 | 1 | 0.717 | 0.0276 | 0.0179 | 0.000252 | 0.000458 |
| re900_e2 | 900 | 2 | 2 | 1 | B1_Mode2 | 2.07 | 1 | 1 | 0 | 4.671e-16 | 0.228 | 1.72 | -0.421 | -0.172 | -0.0346 | -0.0172 | 0 | 0 | 2 | 0.0429 | 0.0508 | 7.534e-07 | 0.000126 | 5.239e-06 |
| re900_e2 | 900 | 2 | 2 | 1 | B1_Mode2 | 2.07 | 1 | 1 | 0 | 4.671e-16 | 0.228 | 1.72 | -0.421 | -0.172 | -0.0346 | -0.0172 | 0 | 0 | 3 | 0.00716 | 0.00149 | 0.0058 | 0.000101 | 0.000138 |
| re900_e2 | 900 | 2 | 2 | 1 | B1_Mode2 | 2.07 | 1 | 1 | 0 | 4.671e-16 | 0.228 | 1.72 | -0.421 | -0.172 | -0.0346 | -0.0172 | 0 | 0 | 4 | 0.00029 | 0.00842 | 0.000309 | 0.00017 | 4.799e-06 |
| re900_e2 | 900 | 2 | 2 | 1 | B1_Mode2 | 2.07 | 1 | 1 | 0 | 4.671e-16 | 0.228 | 1.72 | -0.421 | -0.172 | -0.0346 | -0.0172 | 0 | 0 | 5 | 7.187e-05 | 3.740e-05 | 0.000776 | 3.977e-05 | 1.031e-07 |
| re900_e2 | 900 | 2 | 3 | 2 | B2_temporal | 1.96 | 1 | 1 | 1 | 3.912e-16 | 0.692 | 0.426 | -0.824 | 0.233 | -0.124 | 0.00793 | 0 | 0 | 0 | 0.122 | 0.0408 | 0.00305 | 0.00217 | 0.000299 |
| re900_e2 | 900 | 2 | 3 | 2 | B2_temporal | 1.96 | 1 | 1 | 1 | 3.912e-16 | 0.692 | 0.426 | -0.824 | 0.233 | -0.124 | 0.00793 | 0 | 0 | 1 | 0.0463 | 0.43 | 0.0209 | 0.000778 | 0.000472 |
| re900_e2 | 900 | 2 | 3 | 2 | B2_temporal | 1.96 | 1 | 1 | 1 | 3.912e-16 | 0.692 | 0.426 | -0.824 | 0.233 | -0.124 | 0.00793 | 0 | 0 | 2 | 0.174 | 0.00707 | 0.0561 | 0.00762 | 0.0001 |
| re900_e2 | 900 | 2 | 3 | 2 | B2_temporal | 1.96 | 1 | 1 | 1 | 3.912e-16 | 0.692 | 0.426 | -0.824 | 0.233 | -0.124 | 0.00793 | 0 | 0 | 3 | 0.0139 | 0.0468 | 0.0057 | 0.00535 | 0.000574 |
| re900_e2 | 900 | 2 | 3 | 2 | B2_temporal | 1.96 | 1 | 1 | 1 | 3.912e-16 | 0.692 | 0.426 | -0.824 | 0.233 | -0.124 | 0.00793 | 0 | 0 | 4 | 0.00395 | 0.00695 | 0.00301 | 0.000476 | 0.000409 |
| re900_e2 | 900 | 2 | 3 | 2 | B2_temporal | 1.96 | 1 | 1 | 1 | 3.912e-16 | 0.692 | 0.426 | -0.824 | 0.233 | -0.124 | 0.00793 | 0 | 0 | 5 | 1.608e-05 | 0.000713 | 0.000897 | 0.00011 | 1.628e-05 |
| re900_e2 | 900 | 2 | 4 | 2 | B2_temporal | 1.9 | 1 | 0 | 1 | 4.737e-16 | 0.138 | 0.571 | 0.00176 | -0.00736 | -0.00715 | -0.0257 | 0 | 0 | 0 | 0.005 | 0.622 | 0.124 | 0.0111 | 6.408e-05 |
| re900_e2 | 900 | 2 | 4 | 2 | B2_temporal | 1.9 | 1 | 0 | 1 | 4.737e-16 | 0.138 | 0.571 | 0.00176 | -0.00736 | -0.00715 | -0.0257 | 0 | 0 | 1 | 0.0857 | 0.127 | 0.0152 | 0.00353 | 0.000316 |
| re900_e2 | 900 | 2 | 4 | 2 | B2_temporal | 1.9 | 1 | 0 | 1 | 4.737e-16 | 0.138 | 0.571 | 0.00176 | -0.00736 | -0.00715 | -0.0257 | 0 | 0 | 2 | 8.094e-07 | 0.00221 | 0.00052 | 0.000872 | 0.000297 |
| re900_e2 | 900 | 2 | 4 | 2 | B2_temporal | 1.9 | 1 | 0 | 1 | 4.737e-16 | 0.138 | 0.571 | 0.00176 | -0.00736 | -0.00715 | -0.0257 | 0 | 0 | 3 | 1.424e-05 | 8.598e-07 | 0.000433 | 0.000298 | 3.630e-05 |
| re900_e2 | 900 | 2 | 4 | 2 | B2_temporal | 1.9 | 1 | 0 | 1 | 4.737e-16 | 0.138 | 0.571 | 0.00176 | -0.00736 | -0.00715 | -0.0257 | 0 | 0 | 4 | 1.343e-05 | 5.649e-05 | 7.456e-05 | 1.115e-06 | 5.337e-06 |
| re900_e2 | 900 | 2 | 4 | 2 | B2_temporal | 1.9 | 1 | 0 | 1 | 4.737e-16 | 0.138 | 0.571 | 0.00176 | -0.00736 | -0.00715 | -0.0257 | 0 | 0 | 5 | 0.000174 | 0.000454 | 4.771e-05 | 1.496e-05 | 3.637e-05 |

# B Annexe B — Séparabilité des points de repère : 10 points

Table 16: Points de repère — 10 points, `landmark_blabels.csv` (colonnes brutes $c_{ij}$, $c^*_{ij}$ omises).

| Re | $E_t$ | $\sum E_m$ | cov. | A | $A_{10}$ | $A_{20}$ | $A_{30}$ | $A_{40}$ | $A_{50}$ | $A_{60}$ | $A_{70}$ | $A_{80}$ | $m_2$ | $m_2^F$ | $m_t$ | $i$ | $j$ | rmse | $E_f$ | $E_r$ | $f_2$ | $f_t$ | $A_2$ | $\epsilon$ | $R_K$ | branch | $c_{HC}$ | $c_{DB}$ | $c_{KM}$ | PC1 | PC2 | id | base | s |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 500 | 0.24 | 0.24 | 1 | 1 | 0.0547 | -0.505 | -0.00222 | 0.104 | -0.0217 | 0.012 | 0 | 0 | 0.532 | 0.532 | 0.438 | 1 | 0 | 1.333e-16 | 0.24 | 0.24 | 0.532 | 0.438 | -0.505 | 1.333e-16 | 0.945 | B3_mixed | C3 | *NA* | C1 | -1.27 | -3.19 | 3 | fourier | 0 |
| 500 | 0.267 | 0.267 | 1 | 1 | 0.00249 | 0.698 | 0.0714 | -0.187 | -0.0213 | 0.00636 | 0 | 0 | 0.911 | 0.911 | 0.0131 | 1 | 0 | 1.548e-16 | 0.267 | 0.267 | 0.911 | 0.0131 | 0.698 | 1.548e-16 | 0.986 | B1_Mode2 | C1 | *NA* | C3 | -0.979 | 2.15 | 1 | fourier | 1 |
| 500 | 0.257 | 0.257 | 1 | 1 | 0.0408 | 0.677 | -0.0537 | -0.155 | 0.0351 | -0.0031 | 0 | 0 | 0.892 | 0.892 | 0.0504 | 1 | 0 | 1.673e-16 | 0.257 | 0.257 | 0.892 | 0.0504 | 0.677 | 1.673e-16 | 0.967 | B1_Mode2 | C1 | *NA* | C3 | -1.2 | 1.4 | 1 | fourier | 2 |
| 500 | 0.268 | 0.268 | 1 | 1 | -0.0291 | 0.105 | -0.00819 | 0.00772 | -0.00207 | 0.00115 | 0 | 0 | 0.0206 | 0.0206 | 0.978 | 0 | 1 | 1.549e-16 | 0.268 | 0.268 | 0.0206 | 0.978 | 0.105 | 1.549e-16 | 0.991 | B2_temporal | C2 | *NA* | C2 | 2.09 | 0.856 | 2 | fourier | 3 |
| 500 | 0.249 | 0.249 | 1 | 1 | 0.0364 | -0.51 | -0.058 | 0.096 | 0.0152 | -0.00371 | 0 | 0 | 0.522 | 0.522 | 0.45 | 1 | 0 | 1.451e-16 | 0.249 | 0.249 | 0.522 | 0.45 | -0.51 | 1.451e-16 | 0.965 | B3_mixed | C3 | *NA* | C1 | -0.227 | -1.67 | 3 | fourier | 4 |
| 500 | 0.245 | 0.245 | 1 | 1 | 0.00934 | 0.0419 | 0.0243 | 0.00379 | -0.00601 | 0.00265 | 0 | 0 | 0.00358 | 0.00358 | 0.995 | 0 | 1 | 1.517e-16 | 0.245 | 0.245 | 0.00358 | 0.995 | 0.0419 | 1.517e-16 | 0.987 | B2_temporal | C2 | *NA* | C2 | 1.75 | -0.779 | 2 | fourier | 5 |
| 500 | 0.265 | 0.265 | 1 | 1 | -0.0939 | 0.676 | 0.182 | -0.135 | -0.0799 | -0.0257 | 0 | 0 | 0.863 | 0.863 | 0.0102 | 1 | 0 | 1.383e-16 | 0.265 | 0.265 | 0.863 | 0.0102 | 0.676 | 1.383e-16 | 0.96 | B1_Mode2 | C1 | *NA* | C3 | -2.18 | 0.616 | 1 | fourier | 6 |
| 500 | 0.25 | 0.25 | 1 | 1 | -0.0681 | 0.675 | 0.108 | -0.133 | -0.0335 | -0.0153 | 0 | 0 | 0.913 | 0.913 | 0.0162 | 1 | 0 | 1.413e-16 | 0.25 | 0.25 | 0.913 | 0.0162 | 0.675 | 1.413e-16 | 0.972 | B1_Mode2 | C1 | *NA* | C3 | -1.92 | 0.255 | 1 | fourier | 7 |
| 500 | 0.25 | 0.25 | 1 | 1 | -0.0158 | 0.0817 | -0.0251 | -0.0196 | -0.00111 | -0.00152 | 0 | 0 | 0.0133 | 0.0133 | 0.984 | 0 | 1 | 1.666e-16 | 0.25 | 0.25 | 0.0133 | 0.984 | 0.0817 | 1.666e-16 | 0.978 | B2_temporal | C2 | *NA* | C2 | 1.9 | -0.157 | 2 | fourier | 8 |
| 500 | 0.267 | 0.267 | 1 | 1 | -0.0349 | 0.0245 | -0.0161 | 0.00997 | -0.00238 | 0.00109 | 0 | 0 | 0.00113 | 0.00113 | 0.996 | 0 | 1 | 1.529e-16 | 0.267 | 0.267 | 0.00113 | 0.996 | 0.0245 | 1.529e-16 | 0.989 | B2_temporal | C2 | *NA* | C2 | 2.04 | 0.52 | 2 | fourier | 9 |